Supplementary material: Complex Hamiltonian learning and KAM-preserving magnetic control of a catheter model

The International Journal of Robotics Research
XX(X):1–109
©The Author(s) 0000
Reprints and permission:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/ToBeAssigned
www.sagepub.com/
SAGE

Anonymous authors

Abstract

This supplement gives the distributed force accounting, complete learning-error estimates, nonlinear normal forms, invariant-torus proofs and extended numerical experiments supporting the main article. It also supplies the full magnetic-feedback lift, energy-based clearance proofs and explicit local KAM measure calculation. Numerical observations and infinite-time conclusions retain their respective domains and assumptions. It further gives the pole-free analytic domain, power identity, exact weak-loss error separation and real-to-analytic KAM residual bridge for the end-to-end deep extension. Its learning guarantees require the specified sampling, regularity and physical-error hypotheses; the correlated trajectory experiment does not establish an iid generalization bound. External theorems are attributed at their point of application; model-specific derivations and proofs are included here.

S1 Distributed mechanics and the unified magnetic model

S1.1 Rod mechanics and a variationally consistent reduction

Following the spatial rod formulation (Simo 1985; Antman 2005), let $s \in [0, L]$ denote reference arclength, $r(s,t) \in \mathbb{R}^3$ the centreline and $D(s,t) \in SO(3)$ its director frame. Define

$$\Gamma = D^{\mathsf{T}} r_s, \quad D_s = D\widehat{\kappa}, \quad D_t = D\widehat{\omega}_b. \tag{S1}$$

The hat map is defined by $\widehat{a}b = a \times b$. The extensible, unshearable Kirchhoff specialization has $\Gamma = (0, 0, 1+e)^{\mathsf{T}}$; inextensibility additionally sets $e = 0$. Transverse internal forces are then constraint reactions. They must not be set to zero by substituting zero shear into a shearable constitutive law. In spatial components the balances are

$$\rho A r_{tt} = n_s + f_{\mathrm{mag}} + f_{\mathrm{h}} + f_{\mathrm{c}} + \rho A g, \tag{S2}$$

$$\partial_t (D\mathcal{J}_b \omega_b) = m_s + r_s \times n + \tau_{\mathrm{ext}}. \tag{S3}$$

For constant body inertia, the left side of the second equation is $D(\mathcal{J}_b \dot{\omega}_b + \omega_b \times \mathcal{J}_b \omega_b)$. With stored energy density $W(\Gamma, \kappa)$, unconstrained components of $D^{\mathsf{T}} n$ and $D^{\mathsf{T}} m$ are its strain derivatives. A fixed base specifies position and orientation; a free tip specifies the appropriate force and moment resultants, including any concentrated magnetic load.

For a body-fixed magnetic moment density $\mu_b(s)$ and a static applied field $B_0(r)$, neglecting magnetic interactions between distinct material elements gives the applied-field potential

$$V_{\mathrm{mag}} = -\int_0^L (D\mu_b) \cdot B_0(r)\, ds. \tag{S4}$$

Its translational and rotational variations give $f_{\mathrm{mag}} = \nabla_r (\mu \cdot B_0)$ at fixed orientation and $\tau_{\mathrm{mag}} = \mu \times B_0$, where $\mu = D\mu_b$. Thus static magnetic loading belongs to the conservative energy. Field modulation supplies explicit time-dependent work or an input port. The permanent-dipole load is the leading term of the Maxwell-stress resultant derived below; it is counted once in the force balance.

For a complete localized magnetic element, the magnetostatic force can be evaluated on a surrounding vacuum control surface Σ:

$$F_{\mathrm{mag}} = \int_\Sigma T_M n_\Sigma \, \mathrm{d}A, \qquad T_M = \mu_0^{-1}(BB^{\mathsf{T}} - \tfrac{1}{2}|B|^2 I). \tag{S5}$$

Here B is the total field, including the element's self-field. The associated moment integrates the moment arm crossed with $T_M n_\Sigma$. To recover the permanent-dipole approximation, represent the element by its complete

localized steady current $j(\xi)$, including bound currents, with $\xi = x - r$. The prescribed magnetization is unaffected by the applied field B_{ext}, whose sources lie outside Σ. Current closure gives $\nabla \cdot j = 0$ and no flux through the boundary. The identity $\nabla \cdot T_M = j \times B$ and symmetry of T_M convert the stress resultants into Lorentz force and moment. The self-field has zero net force and moment, as seen by taking its stress surface to infinity; the applied-field-only stress also has zero resultant in this source-free volume. The remaining interaction is

$$\begin{aligned} m_{\text{d}} &= \tfrac{1}{2}\int \xi \times j(\xi)\ \mathrm{d}V, \\ F_{\text{mag}} &= \int j(\xi) \times B_{\text{ext}}(r+\xi)\ \mathrm{d}V \\ &= \nabla_r(m_{\text{d}} \cdot B_{\text{ext}}(r)) + R_F, \\ \tau_{\text{mag}} &= \int \xi \times [j(\xi) \times B_{\text{ext}}(r+\xi)]\ \mathrm{d}V \\ &= m_{\text{d}} \times B_{\text{ext}}(r) + R_\tau. \end{aligned} \tag{S6}$$

The gradient holds the moment orientation fixed. Integration by parts gives $\int j\ \mathrm{d}V = 0$ and $\int \xi_i j_j\ \mathrm{d}V = \epsilon_{ijk}(m_{\text{d}})_k$. Taylor expansion of B_{ext} and $\nabla \cdot B_{\text{ext}} = 0$ then give (S6). For $J_2 = \int |\xi|^2 |j(\xi)|\ \mathrm{d}V$, the remainders obey $|R_F| \le \frac{1}{2} J_2 \sup \|D^2 B_{\text{ext}}\|$ and $|R_\tau| \le J_2 \sup \|D B_{\text{ext}}\|$, with operator norms taken over the element and its Taylor segments. Surface-current contributions are included in these integrals when present. Neglecting the higher multipoles recovers the permanent-dipole force and torque used in magnetic robotics (Abbott et al. 2020, Section 4.1, Eqs. (12) and (16)).

A modified Morison-type local closure, stated per unit reference length, is

$$f_{\text{h}} = m_f a_f + m_a(a_f - r_{tt}) - c_1 v_r - c_2 |v_r| v_r, \tag{S7}$$

$$v_r = r_t - v_f, \qquad c_1, c_2 \ge 0. \tag{S8}$$

Here $m_f = \rho_f A_f$ and m_a are displaced-fluid and added mass per unit length; a_f is the prescribed fluid acceleration. This is a reduced closure requiring calibration, not a solved Navier–Stokes interaction. In still fluid its drag power is $-c_1|r_t|^2 - c_2|r_t|^3$. In moving fluid it is dissipative relative to the fluid velocity and can do positive work on the rod. A constant added mass may be moved to the inertial side exactly once.

For a stationary wall, define penetration $\delta(r)$ positive inside the wall and $n_c = \nabla\delta$. One nonadhesive penalty model is

$$V_c = \frac{k_c}{m+1}[\delta]_+^{m+1}, \tag{S9}$$

$$f_c = -k_c[\delta]_+^m n_c - c_c \mathbf{1}_{\{\delta>0\}}[\dot\delta]_+ n_c. \tag{S10}$$

The elastic term is included through V_c and the second term dissipates power during compression. Friction can be added as a dissipative port. This switching law is used for force accounting outside the contact-free KAM domain, not as an analytic perturbation through an impact.

Choose a configuration map $(r(s,q), D(s,q))$ that satisfies the essential boundary conditions and the rod constraints. A linear centreline ansatz alone is insufficient to specify rotational inertia. Define

$$r_i = \partial_{q_i} r, \qquad a_i = \operatorname{vee}(D^{\mathsf{T}} \partial_{q_i} D). \tag{S11}$$

Then $\omega_b = \sum_i a_i \dot q_i$ and the reduced kinetic energy is

$$T_N = \tfrac{1}{2}\dot q^{\mathsf{T}} M(q) \dot q, \tag{S12}$$

$$M_{ij}(q) = \int_0^L \left(\rho A r_i \cdot r_j + a_i^{\mathsf{T}} \mathcal{J}_b a_j\right) ds. \tag{S13}$$

An added-fluid-mass contribution may be included when its consistent kinetic metric and associated inertial terms are specified. Otherwise hydrodynamic inertia remains in the generalized force balance and is not also included in M. Galerkin projection of the virtual-work identity gives

$$M(q)\ddot q + C(q,\dot q)\dot q + \nabla V_N(q) = Q_{\text{nc}}, \tag{S14}$$

$$Q_i = \int_0^L (f \cdot r_i + \tau \cdot D a_i) ds + Q_i^{\partial}. \tag{S15}$$

Here $C\dot q$ denotes the Christoffel contribution of M, and V_N contains elastic, static magnetic and retained gravitational potentials. Microgravity reduces the last contribution; it does not eliminate fluid drag or establish a particular blood-flow profile. Relative-flow drag, moving-base work and contact losses are retained in Q_{nc}. A stationary-fluid quadratic drag, for example, has nonpositive power; a moving fluid may inject energy. Contact switching is excluded from the analytic neighbourhood used for KAM analysis.

For $M(q) \succ 0$, the Legendre transform yields

$$p = M(q)\dot q, \qquad H_N(q,p) = \tfrac{1}{2} p^{\mathsf{T}} M(q)^{-1} p + V_N(q). \tag{S16}$$

In particular, $\dot p_i = -\frac{1}{2} p^{\mathsf{T}} \partial_{q_i}(M^{-1}) p - \partial_{q_i} V_N + (Q_{\text{nc}})_i$. The configuration dependence of inertia is retained. Potential forces appear either in V_N or in Q_{nc}, never both.

S1.2 Complex energy–dissipation representation with an explicit decoder

Energy and Rayleigh dissipation have different physical units. Let E_0 and t_0 be reference energy and time, and choose diagonal position and momentum scales Q_0, P_0 with $Q_0^{\mathsf{T}} P_0 = E_0 t_0 I$. Set $q = Q_0 x$, $p = P_0 y$, $\tau = t/t_0$ and

$$h(x, y) = H_N(Q_0 x, P_0 y)/E_0, \tag{S17}$$

$$d(x, v) = \frac{t_0}{E_0} \mathcal{D}_N(Q_0 x, Q_0 v/t_0). \tag{S18}$$

Here $v = dx/d\tau$ and $\mathcal{D}_N$ is a Rayleigh dissipation potential. The two entries of $h + id$ are now dimensionless. A complex network provides coupled features and two physically decoded real readouts,

$$\mathcal{H}_\theta(x, y, v) = h_\theta(x, y) + i d_\theta(x, v), \tag{S19}$$

$$h_\theta = h_{\text{prior}} + h_{\text{res},\theta}. \tag{S20}$$

One admissible dissipative head is

$$d_\theta(x, v) = \tfrac{1}{2} v^{\mathsf{T}} D_\theta(x) v, \qquad D_\theta(x) = L_\theta(x) L_\theta(x)^{\mathsf{T}} \succeq 0. \tag{S21}$$

More general heads may be used if $d_\theta(x, \cdot)$ is differentiable, convex and minimized at zero. The decoder is

$$v = \partial_y h_\theta, \qquad x' = v, \tag{S22}$$

$$y' = -\partial_x h_\theta - \partial_v d_\theta(x, v) + B(x) u. \tag{S23}$$

The derivative of d_θ is taken with respect to its independent velocity argument before substituting $v = \partial_y h_\theta$. It is not a total derivative through this substitution.

Proposition S1. Power-consistent complex representation. *For a differentiable h_θ, the decoded system satisfies*

$$\frac{dh_\theta}{d\tau} = -v^{\mathsf{T}} \partial_v d_\theta(x, v) + u^{\mathsf{T}} B(x)^{\mathsf{T}} v + \partial_\tau h_\theta. \tag{S24}$$

For the quadratic head, the first term equals $-v^{\mathsf{T}} D_\theta(x) v \le 0$. For the convex head it is also nonpositive.

Proof. Apply the chain rule to h_θ and substitute (S23). The two canonical terms cancel. Convexity and the minimum at zero imply $v^{\mathsf{T}} \partial_v d_\theta(x, v) \ge d_\theta(x, v) - d_\theta(x, 0) \ge 0$.

With $z = (x + iy)/\sqrt{2}$, the canonical convention is $\{z_j, \bar{z}_k\} = -i\delta_{jk}$ and

$$z' = -i\partial_{\bar{z}} h_\theta - \frac{i}{\sqrt{2}} \partial_v d_\theta + \frac{i}{\sqrt{2}} B(x) u. \tag{S25}$$

This is a unified complex representation of conservative and nonconservative dynamics with a specified physical decoder.

Table S1. Declared parameters of the unified magnetic cantilever. These are illustrative mechanical parameters, not measurements from a clinical or spaceflight device.

Quantity	Symbol	Value
Length	L	$0.08\,\mathrm{m}$
Circular radius	r_c	$0.0006\,\mathrm{m}$
Young's modulus	E	$5.0 \times 10^6\,\mathrm{Pa}$
Density	ρ	$1100\,\mathrm{kg\,m^{-3}}$
Tip dipole	μ	$2.0 \times 10^{-5}\,\mathrm{A\,m^2}$
Axial bias field	B_0	$0.03\,\mathrm{T}$
Mass damping rate	α	$0.55558524\,\mathrm{s^{-1}}$
Transverse field amplitude	B_a	$0.006\,\mathrm{T}$
Drive frequency	Ω_d	$9.4449490\,\mathrm{rad\,s^{-1}}$
Ramp time	t_r	$0.25\,\mathrm{s}$
Uniform Hermite elements	N_e	32

The rule $z' = -i\partial_{\bar{z}}(h_\theta + i d_\theta)$ would generally produce a different vector field and has no automatic dissipativity guarantee. Real analyticity in (x, y), and a controlled analytic extension to a complex neighbourhood, are the relevant requirements for analytic KAM. Holomorphicity in z alone is not required of a nonconstant real energy $h_\theta(z, \bar{z})$. In particular, ReLU or modReLU features require a separate smooth approximation or a theorem of appropriate regularity; their ordinary piecewise-smooth implementation is not analytic.

The dimensional specialization below uses explicit SI units. The learning and KAM statements use dimensionless canonical variables, with their energy and time scales declared before fitting.

S1.3 One clamped–free magnetic beam throughout the computational study

We instantiate the rod reduction as a straight, planar, small-slope Euler–Bernoulli cantilever with a concentrated magnetic dipole at its free tip. The base position and slope are fixed. The elastic part is linear in curvature; the full trigonometric magnetic potential is retained. These choices define a reproducible mechanical specialization of the spatial rod, rather than a catheter-specific material identification. Table S1 states all dimensional parameters. Gravity, imposed fluid flow and field gradient are zero in this specialization. Still-fluid/structural losses are represented by a mass-proportional Rayleigh term. The more general force accounting above remains applicable when those terms are present.

Write $w(s, t)$ for transverse displacement and $\vartheta(t) = w_s(L, t)$ for the small-slope tip rotation. A uniform field $(B_0, B_y(t))$ has zero translational dipole force and the tip

moment

$$M_{\mathrm{tip}} = \mu\{B_y(t)\cos\vartheta - B_0\sin\vartheta\}. \tag{S26}$$

This is the natural moment boundary load. It does not impose a second kinematic constraint at the free tip. The magnetic potential is $-\mu(B_0\cos\vartheta + B_y\sin\vartheta)$, so that (S26) follows by variation. Keeping this potential exact does not remove the small-slope approximation in the beam kinematics.

Let $q = (w_1, \vartheta_1, \ldots, w_{N_e}, \vartheta_{N_e})^{\mathsf{T}}$ contain the unconstrained cubic-Hermite nodal coordinates, e_ϑ select its final rotation, and M, K be the consistent mass and bending stiffness matrices. The physical equations and static-bias energy are

$$M\ddot{q} + \alpha M\dot{q} + Kq = e_\vartheta M_{\mathrm{tip}}, \tag{S27}$$

$$H_0(q,p) = \tfrac{1}{2}p^{\mathsf{T}}M^{-1}p + \tfrac{1}{2}q^{\mathsf{T}}Kq + E_0\{1-\cos(e_\vartheta^{\mathsf{T}}q)\}, \tag{S28}$$

$$E_0 = \mu B_0 = 6\times10^{-7}\,\mathrm{J}, \qquad p = M\dot{q}. \tag{S29}$$

There is no axial end restraint and no artificial stretching quartic. The nonlinearity used in learning and in the KAM calculation is the magnetic nonlinearity of this same cantilever.

The static-field linearization is $K_b = K + E_0 e_\vartheta e_\vartheta^{\mathsf{T}}$. Choose $\Phi^{\mathsf{T}}M\Phi = I$ and $\Phi^{\mathsf{T}}K_b\Phi = \mathrm{diag}(\omega_j^2)$, with positive tip-slope entries fixing modal signs. For n retained modes, $q = \Phi_n a$, and the modal momentum is $\pi = \dot{a}$. Set $\omega_0 = \omega_1$, $\tau = \omega_0 t$, $\nu_j = \omega_j/\omega_0$, and

$$a_j = \sqrt{\frac{E_0}{\omega_0\omega_j}}X_j, \qquad \pi_j = \sqrt{\frac{E_0\omega_j}{\omega_0}}Y_j. \tag{S30}$$

The product of each position–momentum scale is E_0/ω_0; division of energy by E_0 and rescaling time therefore give the standard dimensionless canonical equations. Define

$$S = \Phi_n\,\mathrm{diag}\sqrt{\frac{E_0}{\omega_0\omega_j}}, \quad d = S^{\mathsf{T}}e_\vartheta, \quad \ell(s)^{\mathsf{T}} = N(s)S, \tag{S31}$$

where $N(s)$ is the Hermite displacement row. Thus the full centreline and tip slope are recovered in SI units as $w(s) = \ell(s)^{\mathsf{T}}X$ and $\vartheta = d^{\mathsf{T}}X$. The single dimensionless Hamiltonian used below is

$$h_*(X,Y) = h_2(X,Y) + R_{\mathrm{mag}}(X), \tag{S32}$$

$$h_2 = \tfrac{1}{2}\sum_{j=1}^{n}\nu_j(X_j^2+Y_j^2), \tag{S33}$$

$$R_{\mathrm{mag}} = 1 - \cos(d^{\mathsf{T}}X) - \tfrac{1}{2}(d^{\mathsf{T}}X)^2. \tag{S34}$$

The subtracted quadratic avoids counting the static magnetic stiffness twice. For the two-mode model, $(\omega_1,\omega_2) = (11.41728728, 70.24049294)\,\mathrm{rad\,s^{-1}}$. All learned models use this fixed chart; states are not independently renormalized after fitting.

With $u = B_y/B_0$, $\gamma = \alpha/\omega_0$ and $\mathsf{N} = \mathrm{diag}(\nu_j)$, the physical port decoder becomes

$$X' = \partial_Y h, \tag{S35}$$

$$Y' = -\partial_X h - \gamma\mathsf{N}^{-1}\partial_Y h + u\,d\cos(d^{\mathsf{T}}X). \tag{S36}$$

For $h = h_*$, the damping term is exactly $-\gamma Y$. The Rayleigh head is $\mathcal{D}(v) = \frac{1}{2}\gamma v^{\mathsf{T}}\mathsf{N}^{-1}v$. For a learned energy, retaining the velocity-based form in (S36) preserves nonnegative dissipation. Using $-\gamma Y$ after learning a momentum-dependent residual would, in general, lose that guarantee. The corresponding power identity is

$$h' = -\gamma(\partial_Y h)^{\mathsf{T}}\mathsf{N}^{-1}\partial_Y h + u\cos(d^{\mathsf{T}}X)d^{\mathsf{T}}\partial_Y h. \tag{S37}$$

Alternatively, including $-u\sin(d^{\mathsf{T}}X)$ in the time-dependent energy moves the input power to $-u'\sin(d^{\mathsf{T}}X)$. These are equivalent bookkeeping conventions.

S1.4 Independent COMSOL realization of the same discretization

The model was also constructed and solved in COMSOL Multiphysics 6.3.0.290 through LiveLink for MATLAB. Its two-dimensional Beam interface uses the Euler–Bernoulli formulation, 32 uniform elements, a consistent mass matrix, a fixed base, and the nonlinear tip moment (S26). Geometric nonlinearity is disabled to match the declared small-slope mechanical model. The zero axial displacement subspace decouples, leaving 64 transverse displacement/rotation degrees of freedom.

After selecting transverse degrees of freedom by their field names and coordinates, the independently assembled COMSOL and Python mass and elastic stiffness matrices have relative Frobenius differences 7.76×10^{-15} and 3.05×10^{-14}, respectively. The first six modal frequencies agree to a maximum relative difference of 1.88×10^{-9}. These checks establish agreement of the stated discretizations; they do not

establish the material validity of the chosen parameters. The native model, assembly exports, parameter values and time-domain verification records accompany the manuscript.

S1.5 A global result for the unforced physical model

Proposition S2. Unforced global stability of the full FE target. *For the full clamped–free Hermite model (S28), set $B_y = 0$. With the declared parameters, the potential is globally strongly convex and the origin is its unique equilibrium. If $\alpha > 0$, the origin is globally asymptotically stable. If $\alpha = 0$, every finite-energy trajectory remains in a bounded energy sublevel and the origin is Lyapunov stable. These are global statements for the declared small-slope mathematical model; they do not establish its physical validity at large deflections or a global property of arbitrary learned quartic residuals.*

Proof. Write $b = e_\vartheta$ and $U(q) = q^\mathsf{T} K q/2 + E_0\{1 - \cos(b^\mathsf{T} q)\}$. The clamped conditions give $|w_s(L)|^2 \le L \int_0^L |w_{ss}|^2\, ds$, including for every Hermite finite-element function. Equivalently, $bb^\mathsf{T} \preceq [L/(EI)]K$; equality of the corresponding tip compliance follows from the static quadratic bending shape. Thus, for every q,

$$D^2U(q) = K + E_0 \cos(b^\mathsf{T} q) bb^\mathsf{T} \succeq (1 - \kappa)K \succeq 0.905K, \tag{S38}$$

$$\kappa = \frac{E_0 L}{EI} = \frac{8}{27\pi} < 0.095. \tag{S39}$$

In particular, $\nabla U = 0$ only at $q = 0$. Both M and K are positive definite, and

$$H_0 \ge \tfrac{1}{2} p^\mathsf{T} M^{-1} p + \tfrac{1}{2} q^\mathsf{T} K q.$$

Its sublevels are compact. Along the unforced dynamics,

$$\dot H_0 = -\alpha p^\mathsf{T} M^{-1} p \le 0.$$

This proves boundedness, global continuation and Lyapunov stability. For $\alpha > 0$, the largest invariant set with $\dot H_0 = 0$ has $p = 0$ and $\nabla U(q) = 0$, hence consists only of the origin. LaSalle's invariance principle on each compact sublevel gives global asymptotic stability.

S2 Complex Hamiltonian learning

S2.1 A fixed canonical chart

For a positive definite constant mass M and reference stiffness K_{ref}, select U such that $U^\mathsf{T} M U = I$ and $U^\mathsf{T} K_{\mathrm{ref}} U = \Omega^2$, where $\Omega = \mathrm{diag}(\omega_1, \dots, \omega_n)$. In the magnetic specialization, $K_{\mathrm{ref}} = K_b$ includes the static-field stiffness. The transformation

$$q = U\Omega^{-1/2} Q, \qquad p = MU\Omega^{1/2} P \tag{S40}$$

is symplectic: its configuration and momentum blocks satisfy $S_q^\mathsf{T} S_p = I$. Consequently $H_2 = \frac{1}{2}\sum_j \omega_j (Q_j^2 + P_j^2)$. The computational variables in (S30) obey $(Q, P) = \sqrt{E_0/\omega_0}\,(X, Y)$, together with the stated energy and time normalization. That fixed dimensionless chart is used for training, derivative errors and normal forms. Canonical normalization requires the position and momentum scalings to preserve the symplectic form. Optional training weights change only the loss metric.

Write $x = (Q, P)$ and use one complex normalization throughout:

$$z_j = \frac{Q_j + iP_j}{\sqrt{2}}, \quad I_j = z_j \bar z_j, \qquad \{z_j, \bar z_k\} = -i\delta_{jk}. \tag{S41}$$

Wirtinger calculus gives $\partial_{\bar z_j} = (\partial_{Q_j} + i\partial_{P_j})/\sqrt{2}$ and therefore $\dot z = -i\partial_{\bar z} H$ for real H. This is a change of representation of a real symplectic system. Holomorphicity in z alone would exclude most real energy functions, including $H_2 = \sum_j \omega_j z_j \bar z_j$. The analyticity required by analytic KAM instead means a holomorphic extension of the real variables (Q, P) to a complex neighbourhood, in which z and $\bar z$ are independent linear combinations.

S2.2 Architecture and objective

A shared complex feature map $F_\theta(z, \bar z, b)$, with magnetic configuration b, drives real readouts for H_θ and a dissipation factor L_θ. The latter defines $\mathcal{D}_\theta(q, v) = \frac{1}{2}\left\|L_\theta(q)^\mathsf{T} v\right\|^2$. The physical decoder in Section S1 determines the vector field. An affine complex layer has the real block realization

$$(A + iB)(a + ib) = (Aa - Bb) + i(Ba + Ab). \tag{S42}$$

Thus complex weights impose a specific coupling, rather than furnishing an extra physical state. As a separate smooth approximation construction, split activations $\sigma(a + ib) = \tanh(a) + i\tanh(b)$ are real analytic on bounded real domains and extend to a sufficiently small pole-free complex neighbourhood when weights are bounded. Their use does not justify assuming an arbitrarily wide analytic strip.

For observed dimensionless states and velocities, a derivative-labelled objective is

$$\widehat{\mathcal{L}} = \frac{1}{N}\sum_{i=1}^{N} \left\| \widehat{\dot{x}}_i - f_\theta(x_i, u_i) \right\|^p + \lambda_E \widehat{\mathcal{L}}_E + \lambda_R \mathcal{R}(\theta). \tag{S43}$$

Energy observations, if available, fix a gauge through $\widehat{\mathcal{L}}_E$; vector-field training alone identifies H only up to a constant. To infer conservative gradient error from the full decoder loss, errors in dissipation and inputs must first be bounded or measured and included in the label-error budget. They cannot be absorbed into a Hamiltonian residual merely because their magnitudes are small. The native COMSOL joint fit instead uses the five-node weak loss in Section S3.4; it is not an empirical instance of the pointwise loss in (S43).

The finite-feature experiments specialize this architecture to a finite analytic complex feature network. The end-to-end complex $\tanh$ extension and its separate, jointly trained damping head are specified in Section S3.6. The exact magnetic feature R_{mag} in (S34) is represented by complex exponentials and a quadratic correction, with a learned real readout. Quartic features complement its degree-four Taylor coefficient; this keeps the full cosine target in the hypothesis class while avoiding a duplicate quartic direction. Products of complex affine forms and their conjugates generate all real quadratic and quartic monomials. They can be implemented by a finite network with polynomial activation, using $uv = ((u+v)^2 - (u-v)^2)/4$. A real readout combines the full magnetic analytic feature with the polynomial residual; conjugate features enforce the reality of the energy. The prescribed features exclude a spurious cubic term and, for the gray-box model, preserve the known static-field quadratic part exactly. Fitting the output coefficients from energy derivatives is linear least squares. The experiment therefore tests an explicitly specified finite-feature CHNN. The magnetic coefficient and polynomial residual readouts are estimated by linear least squares within this finite-feature architecture.

S2.3 What complex approximation theory establishes

Zhang et al. (2022, Theorem 1) establish universal approximation with the specified complex-reaction activation. Their Theorem 2 gives a separate L^2 comparison with real networks under a particular distribution and resource budget. These results have distinct hypotheses and approximation criteria. Such a theorem supplies neither higher derivative accuracy nor a universal advantage for magnetic catheter energies. In particular, piecewise reaction activations cannot be inserted unchanged into a smooth KAM proof.

There is nevertheless a direct smooth approximation route. For input $(Q+iP)/\sqrt{2}$, first-layer weights $\sqrt{2}(A_Q - iA_P)$ and real bias b give preactivation $(A_QQ + A_PP + b) + i(A_QP - A_PQ)$. Split activations and subsequent real weights keep the real channel independent of the imaginary channel, so a real-part output realizes standard real networks. For bounded, nonconstant C^m activations such as $\tanh$, Hornik (1991, Theorem 3) gives uniform approximation through derivative order m of C^m targets on compact sets, with network width allowed to grow. Alternatively, increasing the polynomial degree and network size realizes successive truncations of a convergent analytic power series, with uniform derivative control on compact polydiscs inside its convergence domain. The fixed quartic feature class used in the finite-feature experiments instead requires the quantitative coefficient certificate developed below.

S3 From nonzero training loss to KAM persistence

S3.1 A statistical bridge in the norm required by KAM theory

All Hamiltonians below are real-valued scalar functions, including when their parameters and hidden representations are complex. The realification of a complex parameter vector has dimension P. Let $\mathcal{D} = \mathbb{T}^n \times B$, where B is a bounded ball in action space, and write $d = 2n$. Lebesgue measure on $\mathcal{D}$ is denoted by dx. The following result completes the derivative-norm interface needed to combine the learning argument of Chen et al. (2022) with a finite-differentiability KAM theorem. It does not infer derivative smallness from function values alone.

Theorem S3. Uniform learning-to-C^ℓ bound. *Fix $d < p < \infty$, an integer $\ell \geq 1$, and an integer $s > \ell + d/p$. Let $X_1, \dots, X_N$ be independent samples with density $f_\mu \geq m_0 > 0$ on $\mathcal{D}$. Let $H_\star$ be the target Hamiltonian and $\{H_\vartheta : \vartheta \in \Theta\}$ the hypothesis class, with $\Theta \subset [-B_\Theta, B_\Theta]^P$. Assume that for all $x \in \mathcal{D}$ and $\vartheta, \eta \in \Theta$,*

$$\|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p \leq Q, \tag{S44}$$

$$\|\nabla H_\vartheta(x) - \nabla H_\eta(x)\|_p \leq L_g \|\vartheta - \eta\|_\infty. \tag{S45}$$

After fixing the additive gauge by

$$e_\vartheta = H_\vartheta - H_\star - \frac{1}{|\mathcal{D}|}\int_{\mathcal{D}} (H_\vartheta - H_\star)\,dx,$$
$$\sup_{\vartheta\in\Theta} \|e_\vartheta\|_{W^{s,p}(\mathcal{D})} \le M_s, \tag{S46}$$

suppose that M_s is a valid uniform bound. Define

$$\widehat{R}_N(\vartheta) = \frac{1}{N}\sum_{i=1}^{N} \|\nabla H_\vartheta(X_i) - \nabla H_\star(X_i)\|_p^p, \tag{S47}$$
$$L_{\mathrm{par}} = pQ^{p-1}L_g, \qquad L_{\max} = Q^p. \tag{S48}$$

For any fixed $0 < a \le 2B_\Theta$ and $0 < \delta < 1$, put

$$\Gamma_N(a,\delta) = 2L_{\mathrm{par}}a + \frac{L_{\max}}{\sqrt{2N}}\sqrt{P\log(1+2B_\Theta/a) + \log(1/\delta)}, \tag{S49}$$
$$S_N(\vartheta) = \widehat{R}_N(\vartheta) + \Gamma_N(a,\delta), \tag{S50}$$
$$G_N(\vartheta) = C_{\mathrm{P}}\, m_0^{-1/p} S_N(\vartheta)^{1/p}, \tag{S51}$$
$$\alpha = \frac{\ell - 1 + d/p}{s-1} \in (0,1). \tag{S52}$$

Here C_{P} is a Poincare constant chosen so that a mean-zero function satisfies $\|e\|_{W^{1,p}} \le C_{\mathrm{P}}\|\nabla e\|_{L^p}$. With probability at least $1-\delta$, simultaneously for every $\vartheta \in \Theta$,

$$\|e_\vartheta\|_{C^\ell(\overline{\mathcal{D}})} \le E_N(\vartheta) := C_{\mathrm{GN}}\left[G_N(\vartheta) + M_s^\alpha G_N(\vartheta)^{1-\alpha}\right]. \tag{S53}$$

C_{GN} depends only on the domain and (d,p,s,ℓ). We use $\|e\|_{C^\ell} = \max_{|a|\le\ell}\sup|D^a e|$; equivalent conventions only change the displayed constants. The event is uniform over the hypothesis class and therefore includes parameters selected using these same training data.

Proof. The loss $r_\vartheta(x) = \|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p^p$ lies in $[0, L_{\max}]$ and is L_{par}-Lipschitz in the ℓ^∞ parameter metric. Partitioning the parameter cube into cells of side length a and choosing a representative from each nonempty cell gives an a-net with at most $(1+2B_\Theta/a)^P$ points. The one-sided Hoeffding inequality and a union bound control $R(\eta) - \widehat{R}_N(\eta)$ at every net point by the square-root term in Γ_N, where $R(\eta) = \int r_\eta\,d\mu$. The population and empirical losses each change by at most $L_{\mathrm{par}}a$ between a parameter and its net representative. Consequently $R(\vartheta) \le S_N(\vartheta)$ uniformly.

The density lower bound gives $\|\nabla e_\vartheta\|_{L^p(dx)} \le m_0^{-1/p} S_N(\vartheta)^{1/p}$. The gauge in (S46) and Poincare's inequality then yield $\|e_\vartheta\|_{W^{1,p}} \le G_N(\vartheta)$. Apply the Gagliardo–Nirenberg inequality to $g = \nabla e_\vartheta$, with derivative orders $j = \ell - 1$, $m = s - 1$, and both endpoint Lebesgue exponents equal to p. Its interpolation identity is

$$0 = \frac{j}{d} + \alpha\left(\frac{1}{p} - \frac{m}{d}\right) + \frac{1-\alpha}{p},$$

which gives the stated α. The strict inequality $s > \ell + d/p$ ensures $\alpha < 1$ and avoids the critical Sobolev endpoint. Extension from the smooth bounded action domain, together with periodic charts in the angular variables, gives

$$\|e\|_{C^\ell} \le C_{\mathrm{GN}}\left(\|e\|_{W^{1,p}} + \|e\|_{W^{1,p}}^{1-\alpha}\|e\|_{W^{s,p}}^{\alpha}\right).$$

Lower-order derivatives are included by the inhomogeneous term and the $W^{1,p} \hookrightarrow C^0$ embedding, which uses $p > d$. Substitution proves (S53).

For instance, one may select an integer $\ell > 2(\tau+1)$, $p > 2n$, and $s = \ell + 1$. This makes every exponent explicit, although the resulting interpolation bound can be conservative. Bounds on M_s, Q, L_g follow from bounded parameters and derivatives of smooth activations on a compact domain; the corresponding bounds for $H_\star$ must also be justified. Merely declaring the functions smooth does not provide these quantitative bounds. Alternatively, a loss containing all spatial derivatives of the vector field through order ℓ controls $W^{\ell+1,p}$ directly, and Sobolev embedding then gives a C^ℓ error with exponent $1/p$, without interpolation.

Corollary S4. KAM transfer for the target and the trained Hamiltonian. *Let $n \ge 2$, $\tau > n-1$ and an integer $\ell > 2(\tau+1)$ be fixed. Suppose that, in one specified canonical action–angle chart,*

$$H_\star = h_0(I) + P_\star(I,\varphi), \qquad H_\vartheta = h_0(I) + P_\vartheta(I,\varphi).$$

Assume $h_0 \in C^{\ell+2}(\overline{B})$ with finite norm, and $\nabla h_0 : B \to \Omega$ is a diffeomorphism with bounded inverse in C^ℓ. Fix $\gamma > 0$ and set

$$\Omega_{\gamma,\tau} = \left\{\omega \in \Omega : \begin{array}{l} |k\cdot\omega| \ge \gamma|k|_1^{-\tau}\ (0 \ne k \in \mathbb{Z}^n), \\ \operatorname{dist}(\omega, \partial\Omega) \ge \gamma \end{array}\right\}. \tag{S54}$$

Let $c_{\mathrm{K}} > 0$ be the constant in Theorem A of Bounemoura (2020) for this domain and integrable Hamiltonian. If a certified bound $\beta_\vartheta \ge \|P_\vartheta - \overline{P}_\vartheta\|_{C^\ell}$ satisfies

$$\beta_\vartheta + E_N(\vartheta) < c_{\mathrm{K}}\gamma^2, \tag{S55}$$

then, on the event in Theorem S3, both $H_\star$ and H_ϑ possess invariant Lagrangian tori for every frequency in $\Omega_{\gamma,\tau}$. The flow on each torus is C^1-conjugate to the linear flow with that frequency. For a piecewise smooth frequency-domain

boundary, the complement of each family has measure at most $C\gamma$. In particular, each family has positive measure when γ is small enough.

Proof. Write $\overline{P}$ for the spatial mean over $\mathcal{D}$. The gauge-fixed difference satisfies

$$P_\star - \overline{P}_\star = P_\vartheta - \overline{P}_\vartheta - e_\vartheta.$$

Thus the two perturbation norms are at most β_ϑ and $\beta_\vartheta + E_N(\vartheta)$, respectively. Constants added to a Hamiltonian do not change its vector field. Apply the cited finite-smoothness KAM theorem separately to the two Hamiltonians with the same h_0, domain, and Diophantine set. The theorem supplies tori of class $C^{\tau+1}$ with Lipschitz dependence as invariant graphs on frequency. The two systems need not have identical invariant embeddings.

The certified budget includes the physical perturbation from h_0, any normal-form remainder, and the learning error. A small learning error alone does not establish that the physical system is close to an integrable one. Theorem A in the published 2020 version of Bounemoura (2020) gives the measure bound $C\gamma$; the $C\gamma^2$ measure bound printed in the 2018 preprint should not be used.

S3.2 A coefficient certificate for a finite analytic CHNN

The finite analytic readout used in the experiments yields a stronger certificate that avoids a dimension-dependent interpolation loss. Let $\psi_1, \dots, \psi_m$ be real-valued basis functions on the real phase domain. Complex features may be converted into this basis by taking their real and imaginary parts and collecting equal monomials. The certificate applies to the effective coefficients of the actual trained Hamiltonian, irrespective of how they are parameterized inside the network.

Proposition S5. Deterministic global certificate from sampled gradients. *Suppose*

$$H_\vartheta = H_{\mathrm{prior}} + \sum_{j=1}^{m} c_{\vartheta,j}\psi_j,$$

$$H_\star = H_{\mathrm{prior}} + \sum_{j=1}^{m} c_{\star,j}\psi_j + R.$$

The basis contains no constant gauge direction. Let

$$G_N = \frac{1}{N}\sum_{i=1}^{N} D\psi(X_i)^\top D\psi(X_i),$$

$$\lambda_N = \lambda_{\min}(G_N) > 0,$$

where the jth column of $D\psi$ is $\nabla\psi_j$. For gradient observations $Y_i = \nabla H_\star(X_i) + \xi_i$, define

$$r_N = \left(\frac{1}{N}\sum_i \|\nabla H_\vartheta(X_i) - Y_i\|_2^2\right)^{1/2},$$

$$t_N = \left(\frac{1}{N}\sum_i \|\nabla R(X_i) + \xi_i\|_2^2\right)^{1/2}.$$

Then

$$\|c_\vartheta - c_\star\|_2 \le \frac{r_N + t_N}{\sqrt{\lambda_N}}. \tag{S56}$$

For any function norm $\|\cdot\|_{\mathcal{X}}$ satisfying the triangle inequality, put $K_{\mathcal{X}} = (\sum_j \|\psi_j\|_{\mathcal{X}}^2)^{1/2}$. If $\|R\|_{\mathcal{X}} \le \varepsilon_R$, then

$$\|H_\vartheta - H_\star\|_{\mathcal{X}} \le K_{\mathcal{X}}\frac{r_N + t_N}{\sqrt{\lambda_N}} + \varepsilon_R. \tag{S57}$$

The norm may be C^ℓ on a real compact domain or the supremum norm on a prescribed complex neighborhood where every basis function and R is holomorphic. No independent sampling assumption is needed.

Proof. Let $\Delta c = c_\vartheta - c_\star$. The empirical norm of $D\psi\Delta c$ is $(\Delta c^\top G_N \Delta c)^{1/2}$. Since $D\psi\Delta c = (\nabla H_\vartheta - Y) + (\nabla R + \xi)$ at every sample, the triangle inequality gives $\sqrt{\lambda_N}\|\Delta c\|_2 \le r_N + t_N$. Finally, $\|\sum_j \Delta c_j\psi_j - R\|_{\mathcal{X}} \le K_{\mathcal{X}}\|\Delta c\|_2 + \varepsilon_R$ by Cauchy–Schwarz.

For raw monomials $\psi_\nu(x) = x^\nu$, an explicit C^ℓ bound on $[-r, r]^d$ is obtained from

$$\|\psi_\nu\|_{C^\ell} \le \max_{|a|\le\ell,\, a\le\nu} \frac{\nu!}{(\nu-a)!}\, r^{|\nu|-|a|}.$$

The analytic sup norm on $\{|x_j| \le r\} \subset \mathbb{C}^d$ is $r^{|\nu|}$. For all 35 degree-four monomials in four variables, $K_{\mathrm{an}} \le \sqrt{35}\, r^4$. If a canonical chart $x = \Phi(I, \varphi)$ maps an action–angle complex domain into this polydisc, the same analytic bound holds after composition. For a C^ℓ certificate after composition, the derivative bounds of Φ must instead be included by the chain rule. Unmodeled physical terms require the tail bounds t_N and ε_R. Their approximation-tail contribution is zero only when the target lies exactly in the declared class. The full-cosine feature class below has this property for its specified reduced target; it does not remove truncation error relative to the full beam.

The statistical result requires independent state samples with the stated coverage density. Consecutive states on a trajectory do not satisfy this assumption automatically. Accordingly, the finite-feature trajectory experiments below use the deterministic Gram certificate, which requires no sampling independence. This coefficient-linear argument does not apply unchanged to the end-to-end joint weak loss; its separate error decomposition is given in Section S3.4. The finite-feature implementation divides both squared residuals and the Gram matrix by $4N$ rather than N; the coefficient bound is unchanged provided the noise norm uses that same scalar-observation normalization.

S3.3 Native finite-element projection and its certified defect

The trajectory data used by the joint deep fit come from a geometrically linear, fixed-mass beam with a nonlinear magnetic tip moment. The transverse subsystem contains 64 displacement degrees of freedom. The saved native mass and stiffness matrices are exactly symmetric as binary64 arrays; their coupling blocks with the free axial coordinates vanish entrywise. This subsection distinguishes the ideal modal identities from the continuous ODE defined by the saved coefficients. Neither interpretation treats a numerical time history as an exact solution of that ODE.

Let $U \in \mathbb{R}^{64}$, let e select the tip rotation, and put $\theta_f = e^{\mathsf{T}}U$, $E_0 = \mu B_0 > 0$ and $u = B_y/B_0$. For fixed positive-definite M and elastic stiffness K, define $K_b = K + E_0 e e^{\mathsf{T}}$ and

$$\begin{aligned} M\ddot{U} + \alpha M\dot{U} + K_b U &= E_0 e\, g(\theta_f, u), \\ g(\theta, u) &= \theta - \sin\theta + u\cos\theta. \end{aligned} \tag{S58}$$

Dots in this subsection denote physical time; primes denote $\tau = \omega_0 t$. The proportional damping in (S58) is the mass Rayleigh term used in the data generator; the stiffness-proportional term is zero.

Proposition S6. Exact projection with an observed full-beam port. *Suppose $K_b\Phi = M\Phi\Omega^2$ and $\Phi^{\mathsf{T}}M\Phi = I$. Retain two columns Φ_L, write $\Omega_L = \operatorname{diag}(\omega_1, \omega_2)$, and set*

$$\begin{aligned} S_j^2 &= E_0/(\omega_0\omega_j), \qquad N = \Omega_L/\omega_0, \\ q &= S^{-1}\Phi_L^{\mathsf{T}}MU, \\ p &= (S\Omega_L)^{-1}\Phi_L^{\mathsf{T}}M\dot{U}, \qquad d = S\Phi_L^{\mathsf{T}}e. \end{aligned} \tag{S59}$$

Here $S = \operatorname{diag}(S_1, S_2)$. With $\theta_\ell = d^{\mathsf{T}}q$, define

$$\begin{aligned} H_* &= H_2 + 1 - \cos\theta_\ell - \tfrac{1}{2}\theta_\ell^2, \\ D_* &= \operatorname{diag}\big(\alpha/(\omega_0\nu_j)\big), \\ C_* &= d[\theta_f - \sin\theta_f - \theta_\ell + \sin\theta_\ell + u\cos\theta_f]. \end{aligned} \tag{S60}$$

Then every solution of (S58) *satisfies*

$$q' = H_{*,p}, \qquad p' = -H_{*,q} - D_* H_{*,p} + C_*. \tag{S61}$$

No zero-higher-mode assumption is needed.

Proof. Multiplication by Φ_L^{T} and the scalings in (S59) give $q' = Np$ and $p' = -Nq - (\alpha/\omega_0)p + d\,g(\theta_f, u)$. The identity $E_0/(S_j\omega_j\omega_0) = S_j$ gives the magnetic coefficient. Since $H_{*,q} = Nq + d(\sin\theta_\ell - \theta_\ell)$ and $H_{*,p} = Np$, substitution of (S60) cancels the two low-angle nonlinear terms and yields precisely the projected equation.

The full-angle correction retains the influence of discarded modes through θ_f. Consequently it is not an unknown nonlinear truncation error for this observed-port problem. It also does not make H_* the energy of the entire beam or close the dynamics as an autonomous four-state forced system. Replacing C_* by $ud\cos\theta_\ell$ changes the problem.

S3.3.1 Literal saved coefficients For the numerical certificate let M_c, K_c denote the saved native mass and elastic matrices. Every stored binary64 entry is interpreted as an exact rational number, E_0 is the exact product of the recorded μ and B_0, and $K_{b,c} = K_c + E_0 e e^{\mathsf{T}}$. The independently saved biased matrix differs from this definition by at most $1.455729\,10^{-20}$ per entry. It is not silently substituted for $K_{b,c}$.

Let $M_n, \Phi_L, S, \Omega_L, N, d$ be the saved nominal projection coefficients and define

$$P = \Phi_L^{\mathsf{T}}M_n, \quad A = \operatorname{diag}\big((S_j\omega_j\omega_0)^{-1}\big), \quad \bar{N} = \Omega_L/\omega_0. \tag{S62}$$

Use $q = S^{-1}PU$ and $p = (S\Omega_L)^{-1}P\dot{U}$. These coefficients need not satisfy an exact eigenproblem. Direct substitution into the native ODE gives the exact identities

$$\begin{aligned} q' &= Np + r_q, \qquad r_q = (\bar{N} - N)p, \\ p' &= -H_{*,q} - D_* H_{*,p} + C_* + r_{\mathrm{mat}}, \\ r_{\mathrm{mat}} &= B_Q U + b_g g(\theta_f, u), \\ B_Q &= NS^{-1}P - APM_c^{-1}K_{b,c}, \\ b_g &= E_0 APM_c^{-1}e - d. \end{aligned} \tag{S63}$$

Here H_*, D_*, C_* use the saved N, d. The damping term remains exactly $-(\alpha/\omega_0)p$ because $M_c^{-1}(\alpha M_c) = \alpha I$, independently of P. In particular, no separate damping-projection defect is required. The second diagonal coefficient of $\bar{N} - N$ is approximately $-1.990145\,10^{-16}$ and is retained in the certificate.

S3.3.2 Continuous-time bounds from input work Define the nonnegative mechanical energy

$$\mathcal{H}_0 = \tfrac{1}{2}\dot{U}^{\mathsf{T}} M_c \dot{U} + \tfrac{1}{2} U^{\mathsf{T}} K_c U + E_0(1 - \cos\theta_f). \quad \text{(S64)}$$

Along (S58),

$$\frac{d}{dt}(\mathcal{H}_0 - E_0 u \sin\theta_f) = -\alpha \dot{U}^{\mathsf{T}} M_c \dot{U} - E_0 \dot{u} \sin\theta_f. \quad \text{(S65)}$$

Hence, for absolutely continuous u,

$$\mathcal{H}_0(t) \le \overline{E} := \mathcal{H}_0(0) + E_0\big(|u(0)| + \|u\|_{\infty,[0,T]} + \mathrm{TV}_{[0,T]}(u)\big), \quad 0 \le t \le T. \quad \text{(S66)}$$

This follows by integration of (S65), discarding the nonpositive dissipation term and bounding each sine by one. The recorded inputs have the form

$$u(t) = R(t) \sum_{j=1}^{2} a_j \sin(\varpi_j t + \phi_j),$$
$$R(t) = 1 - \exp[-(t/t_r)^2],$$

where $a_j = A_j/B_0$, $t_r > 0$, $R(0) = 0$, $0 \le R \le 1$ and $\int_0^T R'\,dt \le 1$. Thus

$$\|u\|_\infty \le \sum_j |a_j|, \qquad \mathrm{TV}(u) \le \sum_j |a_j|(1 + |\varpi_j| T). \quad \text{(S67)}$$

Positive definiteness now bounds both U and $\dot{U}$ for the whole interval. In particular, these bounds also preclude finite-time escape of the smooth native ODE on $[0, T]$.

For reproducible positive-definiteness checks, let T_A be an invertible triangular matrix of exact binary64 entries. If interval arithmetic verifies

$$\|T_A A T_A^{\mathsf{T}} - I\|_\infty \le \tau_A < 1, \quad \text{(S68)}$$

then symmetry implies $T_A A T_A^{\mathsf{T}} \succeq (1 - \tau_A) I$. For any row l and vector z,

$$|lz| \le \frac{\|l T_A^{\mathsf{T}}\|_2}{\sqrt{1 - \tau_A}} \sqrt{z^{\mathsf{T}} A z}. \quad \text{(S69)}$$

Indeed, substitute $z = T_A^{\mathsf{T}} y$ and apply Cauchy–Schwarz. A numerical Cholesky factor supplies only a candidate T_A; the proof uses (S68), not exactness of that factor. Applying (S69) with $A = K_c$ to the rows of B_Q and to e^{T} bounds $B_Q U$ and θ_f. If $|\theta_f| \le \Theta$ and $|u| \le U_b$, then

$$|g(\theta_f, u)| \le \min\{\Theta^3/6 + U_b,\ \Theta + 1 + U_b\}.$$

The cubic bound follows by integrating $0 \le 1 - \cos s \le s^2/2$. Together with (S63), this bounds every component of r_{mat}. The same dual estimate with $A = M_c$ bounds p and therefore r_q.

The saved $t = 0$ displacement and velocity vectors define eight exact initial-value problems, one for each training or validation input. Arb evaluation at 160 bits, independently repeated at 256 bits, verifies

$$\begin{aligned} \tau_{M_c} &\le 3.100544\,10^{-15}, \\ \tau_{K_c} &\le 2.705468\,10^{-11}, \\ \sup_{0\le t\le 1} \|r_{\mathrm{mat}}(t)\|_2 &\le 9.504461\,10^{-9}, \\ \sup_{0\le t\le 1} \|r_q(t)\|_2 &\le 2.756043\,10^{-16}. \end{aligned} \quad \text{(S70)}$$

The maxima cover all eight inputs. The proof uses the analytic input variation and energy bounds, without a time-grid coverage assumption or a new trajectory integration.

For these exact IVPs and the observed full-angle port, $\eta_{\mathrm{chart}} = 0$, $\eta_C = 0$ and the bound for r_{mat} supplies η_{phys} in the conservative-gradient separation. The nonzero r_q describes the literal coordinate dynamics and is reported separately. Errors in numerical states, full-angle observations, matrix-product evaluation and time integration remain observation terms in the weak-loss analysis. The certificate does not bound those terms, the finite-element discretization error relative to a continuous beam, or the discrepancy from a configuration-dependent mass model.

S3.4 Weak-form joint identification and conservative error bounds

We derive deterministic conservative-gradient bounds for joint Hamiltonian and damping identification from finite-element trajectory observations. The analysis relates a five-node weak momentum objective and a nodewise kinematic penalty to pointwise and domain-wide errors. It accounts explicitly for damping identification, the velocity chart, physical reduction, quadrature and observation uncertainty.

S3.4.1 Weak observations and the joint objective Write $x = (q, p) = (X, Y) \in \mathbb{R}^4$, use dimensionless time $\tau = \omega_0 t$,

and set $N = \mathrm{diag}(\nu_1, \nu_2)$. The model and the positive damping matrix are

$$\begin{aligned} H_\theta &= H_2 + h_\theta, \\ H_2 &= \tfrac{1}{2}(q^{\mathsf{T}} N q + p^{\mathsf{T}} N p), \\ D_\theta(q) &= \mathrm{diag}(\gamma_{\theta,1}(q)/\nu_1, \gamma_{\theta,2}(q)/\nu_2). \end{aligned} \tag{S71}$$

The Hamiltonian branch has two trainable complex $\tanh$ layers; the positive q-dependent damping branch is trained jointly. The state extracted from the finite-element observations uses the fixed modal velocity scaling. Its ideal kinematic relation is $q' = Np$; the literal saved coefficients give $q' = Np + r_q$, with the explicit rounding defect bounded in (S70). Agreement with a physical target requires the target Hamiltonian and this observation chart to satisfy the stated normalization or to retain the corresponding coordinate defect.

For a window $I_i = [a_i, a_i + h_i]$, define $s_\ell = \ell/4$, $b = (7, 32, 12, 32, 7)/90$, and

$$\begin{aligned} Q_i f &= \sum_{\ell=0}^{4} b_\ell f(a_i + h_i s_\ell), \\ A_i f &= \frac{1}{h_i} \int_{I_i} f(\tau)\, d\tau. \end{aligned} \tag{S72}$$

The prediction field used in the momentum observations is

$$\begin{aligned} F_{\theta,p} &= -H_{\theta,q} - D_\theta H_{\theta,p} + C_{\mathrm{obs}}, \\ k_\theta &= H_{\theta,p} - Np = \partial_p h_\theta. \end{aligned} \tag{S73}$$

Here C_{obs} includes measured forcing and the explicitly retained discarded-mode magnetic correction. It need not be a function of the four retained state coordinates alone.

For numerical conditioning, let γ_0 denote a damping initializer estimated from the training trajectories, with componentwise products understood. The recentered target and prediction are

$$\begin{aligned} T_i^{\mathrm{rec}} &= \frac{p(a_i + h_i) - p(a_i)}{h_i} - Q_i(-Nq + C_{\mathrm{obs}}) + \gamma_0 Q_i p, \\ P_i^{\mathrm{rec}} &= Q_i\{-h_{\theta,q} - D_\theta h_{\theta,p} - (\gamma_\theta - \gamma_0)p\}. \end{aligned} \tag{S74}$$

Since $D_\theta N p = \gamma_\theta p$, direct subtraction gives

$$P_i^{\mathrm{rec}} - T_i^{\mathrm{rec}} = w_{p,i} := Q_i F_{\theta,p} - \frac{p(a_i + h_i) - p(a_i)}{h_i}. \tag{S75}$$

Thus the recentered objective fits the physical residual in (S75); its conditioning requires only the training-based initializer and no true damping labels.

Put hats on quantities computed from discrete observations. For M windows and fixed positive component scales s_j^{loss}, the joint objective is

$$\begin{aligned} L_{\mathrm{mom}} &= \frac{1}{2M} \sum_{i=1}^{M} \sum_{j=1}^{2} \left(\frac{\widehat{w}_{p,ij}}{s_j^{\mathrm{loss}}} \right)^2, \\ L_{\mathrm{kin}} &= \frac{1}{10M} \sum_{i=1}^{M} \sum_{\ell=0}^{4} \sum_{j=1}^{2} \left(\frac{\widehat{k}_{i\ell j}}{s_j^{\mathrm{loss}}} \right)^2, \\ L_{\mathrm{tot}} &= L_{\mathrm{mom}} + 0.1 L_{\mathrm{kin}}. \end{aligned} \tag{S76}$$

The kinematic term L_{kin} penalizes $\partial_p(H_\theta - H_2)$ at observation nodes. A weak position residual additionally contains $Q_i(Np) - [q(a_i + h_i) - q(a_i)]/h_i$.

The full-beam observations and the low-mode energy/Rayleigh decoder retain different magnetic contributions. Let d be the retained modal slope vector, $\theta_\ell = d^{\mathsf{T}} q$, θ_f the full finite-element tip rotation, and u the normalized magnetic input. The observed contribution is

$$\begin{aligned} C_{\mathrm{obs}} = d\,[&\theta_f - \sin\theta_f - \theta_\ell + \sin\theta_\ell \\ &+ u\cos\theta_f], \end{aligned} \tag{S77}$$

and the closed low-mode decoder uses $C_{\mathrm{port}} = u\cos\theta_\ell\, d$. Therefore

$$\begin{aligned} &C_{\mathrm{obs}} - C_{\mathrm{port}} \\ &\quad = d\,\big[\theta_f - \sin\theta_f - \theta_\ell + \sin\theta_\ell \\ &\qquad + u(\cos\theta_f - \cos\theta_\ell)\big]. \end{aligned} \tag{S78}$$

Replacing the observed field by the closed decoder therefore introduces (S78) into the port/reduction error. This term is independent of the decoder's exact power-balance identity.

S3.4.2 Finite sample bounds and a weak-to-pointwise lemma All vector norms in this section are Euclidean, and matrix norms are the associated operator norms.

Proposition S7. Finite-sample residual bounds. *With $s_{\max} = \max_j s_j^{\mathrm{loss}}$, the objective (S76) gives*

$$\begin{aligned} \max_i \|\widehat{w}_{p,i}\| &\le s_{\max}\sqrt{2ML_{\mathrm{mom}}}, \\ \max_{i,\ell} \|\widehat{k}_{i\ell}\| &\le s_{\max}\sqrt{10ML_{\mathrm{kin}}}. \end{aligned} \tag{S79}$$

These inequalities require no independence of windows, but apply only to the finite set appearing in the sums.

Proof. For any fixed window, its sum of two nonnegative scaled squared components is no larger than the sum over all windows. Multiplication by $s_{\max}^2$ and taking square roots gives the first inequality. The same argument for one of the $5M$ node occurrences gives the second. Repeated nodes are counted with the same multiplicity as in the loss.

Proposition S8. Boole averages and window-midpoint residuals. *Let p be continuously differentiable on $I = [a, a+h]$ and let $F_{\theta,p}(\tau)$ be evaluated along the corresponding exact trajectory and its specified input/retained correction. Suppose that $F_{\theta,p}$ and $e_p = F_{\theta,p} - p'$ are Lipschitz in time on the whole window, with constants L_F and L_e. Suppose the five computed field values each have error at most η_F, the two computed endpoint momenta each have error at most η_p, and the times are exact. Then, at $m = a + h/2$,*

$$\|e_p(m)\| \le \|\widehat{w}_p\| + \eta_F + \frac{2\eta_p}{h} + \frac{239}{3240}hL_F + \frac{1}{4}hL_e. \tag{S80}$$

For a uniform bound throughout I, replace the last coefficient $1/4$ by $1/2$. If a Lipschitz bound L_v for p' is available, an alternative midpoint bound replaces the last two terms by $(239/3240)hL_v + (23/90)hL_e$.

Proof. First consider a vector-valued L-Lipschitz function f on $[0,1]$. Let $\mu = \sum_{\ell=0}^{4} b_\ell \delta_{s_\ell}$ and let λ be uniform probability measure. On the four open node intervals, the distribution function of μ has the successive values

$$c_0 = 7/90, \quad c_1 = 39/90, \quad c_2 = 51/90, \quad c_3 = 83/90.$$

For a unit vector v, integration by parts for the Lipschitz scalar function $v^\mathsf{T} f$ gives

$$\left| \int v^\mathsf{T} f \, d(\mu - \lambda) \right| \le L \int_0^1 |F_\mu(s) - s| \, ds.$$

The boundary term vanishes because the signed measure has zero total mass. Each c_ℓ lies within its node interval, so the integral is the explicitly checkable rational sum

$$\sum_{\ell=0}^{3} \frac{1}{2} \left[(c_\ell - \ell/4)^2 + ((\ell+1)/4 - c_\ell)^2 \right] = \frac{239}{3240} =: c_B. \tag{S81}$$

Taking the supremum over unit v and rescaling to length h yields $\|Qf - Af\| \le c_B h L$. Positive weights also give

$$\|Qf - f(m)\| \le hL \sum_{\ell=0}^{4} b_\ell |s_\ell - 1/2| = \frac{23}{90} hL.$$

The corresponding continuous-average bound is $\|Af - f(m)\| \le hL \int_0^1 |s - 1/2| \, ds = hL/4$. For an arbitrary point of the window, the latter integral is at most $1/2$. The weighted discrete absolute distance is also at most $1/2$: it is convex in the chosen point and equals $1/2$ at either endpoint.

The fundamental theorem of calculus gives $Ap' = \Delta p/h$ and hence $w_p - Ae_p = QF_{\theta,p} - AF_{\theta,p}$. The preceding bounds imply $\|e_p(m)\| \le \|w_p\| + c_B h L_F + hL_e/4$. The positive weights sum to one, so the stated observation errors give $\|\widehat{w}_p - w_p\| \le \eta_F + 2\eta_p/h$. Combining the two inequalities proves (S80) and its uniform version. Alternatively, use $w_p = Qe_p + (Qp' - Ap')$ and the discrete midpoint bound to obtain the stated expression involving L_v.

The midpoint is an observation node. If evaluating k_θ at the exact state instead of its observed approximation incurs error at most η_k, Proposition S7 gives directly

$$\|k_\theta(x(m_i))\| \le K_i := s_{\max}\sqrt{10 M L_{\mathrm{kin}}} + \eta_k. \tag{S82}$$

If a uniform temporal Lipschitz constant L_k for $k_\theta(x(\tau))$ is independently known, the whole-window version adds $hL_k/8$, since every point is within $h/8$ of one of the five nodes. Interpretation as physical velocity error additionally requires agreement of the observation and target velocity charts.

The regularity terms in Proposition S8 cannot be omitted. For any scalar A_0, take

$$e(a + hs) = A_0[(s - 1/2)^2 - 1/12],$$
$$F = 0, \qquad p' = -e.$$

Its continuous mean and Boole mean both vanish, so $w = 0$, while $|e(a + h/2)| = |A_0|/12$ is arbitrarily large. Hence analytic cancellation alone prevents a pointwise estimate from the weak residual. Finite-difference slopes at observation nodes provide no uniform continuous Lipschitz bound without additional regularity information. The term η_F includes state, field-evaluation and input errors; uncertain time stamps require additional error terms.

S3.4.3 Separating the conservative and dissipative errors

Proposition S9. Conservative gradient bound with model and observation errors. *Suppose the physical target is expressed in the same state coordinates as (S71), with differentiable reference Hamiltonian H_*, coordinate defect r_q, and equations*

$$q' = H_{*,p} + r_q, \qquad p' = -H_{*,q} - D_* H_{*,p} + C_* + r_{\mathrm{phys}}. \tag{S83}$$

At a point of the trajectory, put $\Delta H = H_\theta - H_*$, $\Delta D = D_\theta - D_*$, and $\Delta C = C_{\mathrm{obs}} - C_*$. Then the following

identities are exact:

$$\begin{aligned}\Delta H_p &= k_\theta + (Np - H_{*,p}),\\ \Delta H_q &= -e_p - D_\theta \Delta H_p - \Delta D H_{*,p} + \Delta C - r_{\text{phys}}.\end{aligned} \tag{S84}$$

Assume bounds $\|k_\theta\| \le K$, $\|e_p\| \le E_p$, $\|Np - H_{*,p}\| \le \eta_{\text{chart}}$, $\|D_\theta\| \le D_{\max}$, $\|\Delta D\| \le \eta_D$, $\|H_{*,p}\| \le V$, $\|\Delta C\| \le \eta_C$, *and* $\|r_{\text{phys}}\| \le \eta_{\text{phys}}$. *Set* $E_v = K + \eta_{\text{chart}}$. *Then*

$$\begin{aligned}E_q &= E_p + D_{\max} E_v + \eta_D V + \eta_C + \eta_{\text{phys}},\\ \|\nabla(H_\theta - H_*)\| &\le (E_v^2 + E_q^2)^{1/2}.\end{aligned} \tag{S85}$$

The K *and* E_p *terms can be supplied by* (S82) *and* (S80) *at recorded midpoints, provided all their observation and regularity hypotheses have been verified.*

Proof. Subtract $H_{*,p}$ from $H_{\theta,p} = Np + k_\theta$. Subtracting the physical momentum equation from (S73), and writing $D_\theta H_{\theta,p} - D_* H_{*,p} = D_\theta \Delta H_p + \Delta D H_{*,p}$, gives (S84). The triangle inequality and the operator-norm bound give $\|\Delta H_p\| \le E_v$ and $\|\Delta H_q\| \le E_p + D_{\max} E_v + \eta_D V + \eta_C + \eta_{\text{phys}}$. The squared Euclidean norm of the full gradient is the sum of the two block norms squared.

The representation in (S83) specifies the physical applicability of the bound. The proof only uses the momentum equation, so an explicit r_q does not change the conservative-gradient identity. It must still be included when comparing full physical and learned vector fields. For a fixed-mass nominal model, $H_{*,p} = Np$ follows when its modal normalization has been validated. For a configuration-dependent mass model with physical canonical momentum, the kinetic derivative is $M(q)^{-1}p$ before any further canonical transformation. Application to such a model requires a justified canonical transformation and its associated chart-error bound. Fixed velocity scaling and a small momentum residual alone provide neither condition.

The damping-discrepancy term requires information beyond positivity of D_θ. For the specified constant-drag nominal model, a separately verified coefficient α_{mass} gives $D_* = \text{diag}(\alpha_{\text{mass}}/(\omega_0 \nu_j))$. Post-training comparison with that coefficient bounds η_D without using true damping labels in the joint objective. For an unknown physical damping law, an independent measurement or a proved joint identifiability argument is required.

For the reported complex model with optimizer seed 43, interval evaluation bounds the damping discrepancy on $[-0.091, 0.091]^4$. Interpreting the specified physical constants as their exact binary64 values, rational endpoint subtraction gives

$$\sup_{[-0.091,0.091]^4} \|D_\theta - D_*\|_2 \le 2.940580 \times 10^{-7}. \tag{S86}$$

For a real diagonal matrix the Euclidean operator norm is the largest absolute diagonal entry. Subtracting each exact nominal diagonal coefficient from its certified interval, taking the larger absolute endpoint, and rounding upward therefore proves the displayed bound. The accompanying artifact index provides the exact rational endpoints, model identities, domain specification and interval records. The bound applies to the specified nominal damping law on this domain. Configuration-dependent damping and larger domains require their own bounds. Application to trajectory windows additionally requires verification that the relevant states belong to the certified domain.

A zero-loss counterexample demonstrates the need for damping identifiability. In one degree of freedom, on $0 \le \tau \le 1$, take

$$\begin{aligned} q(\tau) &= -0.2 + 0.1\tau, & p(\tau) &= 0.1,\\ H_*(q,p) &= \tfrac{1}{2}p^2 - 0.02q, & D_* &= 0.2,\\ \widetilde{H}(q,p) &= \tfrac{1}{2}p^2 - 0.01q, & \widetilde{D} &= 0.1. \end{aligned}$$

With $N = 1$ and no input, both pairs produce $q' = 0.1$, $p' = 0$ on this trajectory. Thus $k = 0$, the pointwise momentum residual is zero, and every weak momentum residual is zero, but $\partial_q(\widetilde{H} - H_*) = 0.01$. Both damping coefficients are positive. The losses therefore do not identify the conservative component within this analytic function class. An identifiability result restricted to the finite-width constrained network requires an additional argument; the counterexample does not assume exact network representation of these polynomials.

S3.4.4 Domain extension and applicability

Proposition S10. A deterministic state-coverage extension. *Let* $K \subset \mathbb{R}^4$ *be compact,* $g = \nabla(H_\theta - H_*)$, *and assume that* g *is* L_g*-Lipschitz on* K. *Suppose corrected pointwise bounds* $\|g(x_i)\| \le \varepsilon_i$ *have been established at centers* $x_i \in K$, *and that every* $x \in K$ *is within distance* r *of a center. Then*

$$\sup_{x \in K} \|g(x)\| \le \max_i \varepsilon_i + L_g r. \tag{S87}$$

If additionally K *is convex,* $x_0 \in K$, *and* $|H_\theta(x_0) - H_*(x_0)| \le \eta_0$, *then*

$$\sup_K |H_\theta - H_*| \le \eta_0 + \sup_{x \in K} \|x - x_0\| (\max_i \varepsilon_i + L_g r). \tag{S88}$$

Proof. Choose a center within r and apply the Lipschitz inequality and the triangle inequality. For the energy bound, integrate g along the segment from x_0 to x, which lies in K by convexity.

This proposition uses conservative-gradient bounds that account for the error terms in Proposition S9. The conservative gradient depends on the canonical state alone. Applying an analogous coverage argument directly to the raw forced residual instead would require coverage and regularity in an extended domain containing time, input and the discarded-mode variables. Covering only (q, p) does not control those quantities. The four-dimensional fill distance r requires an independent coverage bound; trajectory count and training loss do not determine it.

For the reported complex model with optimizer seed 43, six training trajectories give $M = 1494$, $s_{\max} = 2.76020557256 \times 10^{-6}$, $L_{\mathrm{mom}} = 1.57502043026 \times 10^{-5}$ and $L_{\mathrm{kin}} = 1.42356958715 \times 10^{-6}$. Substitution into (S79) gives the finite-set bounds

$$\max_i \|\widehat{w}_{p,i}\| \le 5.98790388583 \times 10^{-7},$$
$$\max_{i,\ell} \|\widehat{k}_{i\ell}\| \le 4.02537252211 \times 10^{-7}.$$

up to the reported floating-point rounding; these decimal evaluations are not outward-rounded certificates. The window length is $h = 0.004\omega_0 \simeq 0.0456691$. These finite-sample estimates, together with (S86), quantify the observation-node residuals and the nominal damping discrepancy. A domain-wide conservative-gradient bound inferred from these losses additionally requires continuous observation-error and regularity bounds, port and physical-model discrepancies on the same domain, and the coverage constants L_g, r. Such bounds are not established by the reported finite-sample estimates. Extension to configuration-dependent mass also requires a validated canonical chart. For the specified fixed-mass nominal target, Section S3.5 instead verifies the frozen-function error directly; this bypasses inference from sampled losses without asserting the missing observation or statistical hypotheses.

The deterministic results apply to correlated windows under the stated regularity and model-error hypotheses. Statistical generalization requires an appropriate sampling argument, while application of KAM theory requires its specified complex-domain or higher-derivative norms. Analyticity alone supplies no quantitative conversion from a real first-derivative estimate: $f_n(q) = n^{-2}\sin(nq)$ has real derivative bounded by $1/n$, while $|f_n(\mathrm{i}\rho)| = \sinh(n\rho)/n^2$ grows for fixed $\rho > 0$. An a posteriori KAM certificate establishes an invariant torus of the specified learned Hamiltonian. Transfer of that conclusion to a physical target requires the identification and reduction bounds above.

S3.5 Deterministic identification error of the frozen deep energy

The nominal conservative energy in (S60) is known independently of the training objective. This permits a deterministic comparison with the exact exported complex seed-43 Hamiltonian. Set

$$\begin{aligned} e_H &= H_\theta - H_* = h_\theta - g(d^{\mathsf{T}} q), \\ g(s) &= 1 - \cos s - s^2/2. \end{aligned} \tag{S89}$$

The identical quadratic priors cancel before interval evaluation. The energy gauge is $H_\theta(0) = H_*(0) = 0$. All exported coefficients, including the input and energy scales, retain their exact binary64 values. The reference coefficients d, N use the same convention. These are bounds for a specified function and target, not a statistical consequence of the loss.

Proposition S11. Cellwise enclosure. *Let $X = c + \prod_j [-r_j, r_j]$ be a real cell on which e_H is twice continuously differentiable. Suppose interval evaluation gives $|e_H(c)| \le a_0$, $|\partial_i e_H(c)| \le a_i$ and $|\partial_{ij} e_H(x)| \le B_{ij}$ for all $x \in X$. Then*

$$\begin{aligned} \sup_X |\partial_i e_H| &\le a_i + \sum_j B_{ij} r_j, \\ \sup_X |e_H| &\le a_0 + \sum_i a_i r_i + \tfrac{1}{2} \sum_{i,j} B_{ij} r_i r_j. \end{aligned} \tag{S90}$$

The maxima of these bounds over a finite closed-cell cover are valid domain bounds.

Proof. Apply the fundamental theorem of calculus to $\partial_i e_H$ along the segment from c to x. Taylor's formula with integral remainder gives the second inequality, because the segment remains in the convex cell. Taking maxima over a cover retains every point, including cell boundaries.

For $D_r = [-r, r]^4$, use the norms

$$E_0 = \sup_{D_r} |e_H|, \qquad E_1 = \sup_{D_r} \|\nabla e_H\|_\infty,$$
$$E_2 = \sup_{D_r} \|D^2 e_H\|_\infty.$$

The Hessian norm is the induced maximum-row-sum norm. Since the canonical J is a signed permutation, E_1 is also the maximum-component conservative-vector-field error. On $D_{0.091}$, divide each axis into 16 equal intervals and apply

Table S2. Certified errors on $[-0.091, 0.091]^4$. The middle column contains upper bounds, not estimates of the exact suprema. The last column reports the exact quadratic-model norm rounded for display.

Norm	Deep upper bound	Quadratic error
E_0	$2.137046\,10^{-8}$	$2.566601\,10^{-7}$
E_1	$3.937778\,10^{-6}$	$6.631067\,10^{-6}$
E_2	$6.873080\,10^{-4}$	$2.185885\,10^{-4}$

Proposition S11 on all 65 536 cells. Arb arithmetic at 128 bits yields the outward-rounded bounds in Table S2. The network Hessian is computed by the exact chain rule, with the quadratic prior omitted. The reference derivatives are $g'(s) = \sin s - s$ and $g''(s) = -2\sin^2(s/2)$. The centre evaluations are point intervals, so subtracting $\sin s - s$ there does not introduce a finite cell-width loss.

The comparison has an analytic baseline. Let $S = r\sum_j |d_j|$ and $d_{\max} = \max_j |d_j|$. For $S \le \pi$, the exact errors of H_2 relative to H_* are

$$\begin{aligned} E_{0,2} &= S^2/2 + \cos S - 1, \\ E_{1,2} &= d_{\max}(S - \sin S), \\ E_{2,2} &= d_{\max}\Big(\sum_j |d_j|\Big)(1 - \cos S). \end{aligned} \tag{S91}$$

These follow from monotonicity on $[0, S]$ and are attained at same-sign position corners. The certified deep-model upper bounds are strictly below the exact baseline norms for E_0 and E_1. Thus the network improves both nominal energy and conservative-field approximation uniformly on this local box. The E_2 upper bound does not establish an improvement of the linearized field.

S3.5.1 Limits of extending the real domain Local accuracy does not imply accuracy on the entire pole-free architecture domain. At $q = (-0.2, -0.2)$, $p = (0, 0)$, point interval evaluation proves

$$\|\nabla e_H(q, p)\|_\infty \ge 6.813327\,10^{-5}. \tag{S92}$$

This is more than 0.9683 times the exact quadratic-model gradient error on $D_{0.2}$. At the declared binary64 half-width $r_{\mathrm{dec}} \simeq 0.6$, the point $q = (-r_{\mathrm{dec}}, -r_{\mathrm{dec}})$, $p = (r_{\mathrm{dec}}, -r_{\mathrm{dec}})$ gives the lower bound $1.890704\,10^{-3}$, more than 0.9999999 times its quadratic baseline. These are lower bounds on the domain error, verified at 128 and 256 bits; they are not inferred from sparse samples to be upper bounds. They rule out extending the small local error claim to those boxes. They do not state that every observed trajectory visits these witness states.

S3.5.2 A direct analytic error bound

Proposition S12. Vertical extension of a verified gradient bound. *Suppose e_H is holomorphic on a neighbourhood of $\mathcal{T} = D + i\prod_j[-\eta_j, \eta_j]$, and for $x \in D$ and $z \in \mathcal{T}$ one has $|\partial_i e_H(x)| \le G_i$ and $|\partial_{ij} e_H(z)| \le B^{\mathbb{C}}_{ij}$. Then*

$$\sup_{\mathcal{T}} |\partial_i e_H| \le G_i + \sum_j B^{\mathbb{C}}_{ij}\eta_j. \tag{S93}$$

Proof. For $z = x + iy$, integrate $\frac{d}{dt}\partial_i e_H(x + ity) = i\sum_j \partial_{ij} e_H(x + ity)y_j$ from 0 to 1. The whole segment is contained in $\mathcal{T}$.

For this frozen function choose $D = [-0.0405, 0.0405]^4$ and $\eta_j = 0.00631$. The inherited Fourier-envelope certificate places the same KAM candidate in this tube even on the angle strip $1/40$. Using 4^4 real subdivisions of the tube, both conjugate-coefficient branches of the holomorphic energy extension are evaluated before averaging; taking the real part at a complex state would be invalid. Both layers' preactivation imaginary parts are certified strictly below $\pi/2$. The complex Hessian bounds and the real gradient bounds above give

$$\sup_{z\in\mathcal{T}} \|\nabla(H_\theta - H_*)(z)\|_\infty \le 4.014626\,10^{-5}. \tag{S94}$$

The tube calculation is repeated at 128 and 256 bits, with the first precision providing the reported bound.

For any candidate in this tube, its residuals for H_θ and H_* differ by at most (S94). This establishes a numerical real-to-analytic comparison for the actual frozen network without an iid or optimizer-convergence assumption. It does not transfer the learned-model KAM certificate to H_*: the bound is much larger than the existing learned-model residual budget, and applying the theorem to another Hamiltonian also requires its own domain, derivative and nondegeneracy bounds. The independent torus proof for H_θ remains a statement about that learned conservative function.

S3.5.3 A positive discrepancy on the unchanged candidate The limitation of the fixed-candidate comparison is not solely an overestimate of the error. Write K_θ for the validated Fourier candidate of the frozen network, before its a-posteriori correction, and set $c = (1/25, 1/25, 0, 0)$. Its Fourier envelope supplies $\|K_\theta(0) - c\|_\infty \le r_K$. On the convex real box $D_{0.091}$, the previously certified Hessian bound B for e_H therefore gives, for every component i,

$$|\partial_i e_H(K_\theta(0))| \ge |\partial_i e_H(c)| - Br_K.$$

Evaluation of the right-hand side using the exact exported coefficients and outward arithmetic at both 192 and 256 bits proves

$$\|\nabla e_H(K_\theta(0))\|_\infty > 9.65618\,10^{-9}. \tag{S95}$$

The candidate residual for H_θ is below $2.497\,10^{-98}$. By the reverse triangle inequality, the residual for H_* on this same embedding consequently exceeds $9.65617\,10^{-9}$, already on real angles. This exceeds the original learned-model allowance $5.107\,10^{-82}$. Increasing a validation sample cannot remove this deterministic discrepancy between two fixed functions. This comparison rules out retaining that embedding with the old allowance; it does not supply the theorem constants for H_* or rule out a separately constructed nominal torus. Such a torus is certified in Section S3.9.

S3.6 Deep complex learning: analytic domain, dissipation and KAM transfer

This section concerns the trained two-hidden-layer complex $\tanh$ model. The finite-feature COMSOL identification and interval KAM records in Sections S4.5 and S4.3 retain their original fitted Hamiltonian. The following propositions characterize the analytic domain and certificate-transfer conditions for the deep Hamiltonian. A certificate for the finite-feature Hamiltonian alone does not establish these conditions for a different function.

S3.6.1 A computable pole-free neighbourhood Write $x = (X, Y) \in \mathbb{R}^m$, $m = 4$, and $a_1 = C_1 x + b_1$. The complex layer on z and its conjugate has this affine form on the real canonical variables. For its analytic continuation, x is complexified and the two affine input combinations are continued independently: no complex conjugation of the continued x is performed. Let D be the closed real box on which the row constraints in the main text hold. Assume the certified imaginary-preactivation bounds are $\rho_1, \rho_2 < \pi/4$, and put

$$\begin{aligned}
\widehat{\rho}_\ell &= (\rho_\ell + \pi/4)/2, \\
L_1 &= \|C_1\|_\infty, \\
L_2 &= \|W_2\|_\infty \sec^2(\widehat{\rho}_1) L_1, \\
\eta &= \tfrac{1}{2} \min\left\{ \frac{\widehat{\rho}_1 - \rho_1}{L_1}, \frac{\widehat{\rho}_2 - \rho_2}{L_2} \right\}.
\end{aligned} \tag{S96}$$

Matrix norms are induced by the complex componentwise maximum norm. A zero denominator gives no restriction; if both vanish, any fixed finite radius can be used.

Proposition S13. Holomorphic extension of the frozen network. *For the exact frozen coefficients, every preactivation and output in the deep network is holomorphic on $D_\eta = \{x + e : x \in D,\ e \in \mathbb{C}^m,\ |e|_\infty < \eta\}$. If $F(x) = w_o^{\mathsf{T}} h_2^{\mathrm{net}}(x)$, its real readout has the holomorphic continuation*

$$r^{\mathbb{C}}(x) = \tfrac{1}{2}\big(F(x) + F^\sharp(x)\big), \qquad F^\sharp(x) = \overline{F(\bar{x})}. \tag{S97}$$

The expression $F^\sharp$ is evaluated by conjugating the coefficients of the analytic expression for F. Thus it is holomorphic, and the continuation equals $\operatorname{Re} F$ on real states.

Proof. Fix $x \in D$ and $|e|_\infty < \eta$. The change in the first preactivation along $x + te$, $0 \le t \le 1$, is at most $L_1\eta < \widehat{\rho}_1 - \rho_1$. Consequently its imaginary part stays inside $(-\widehat{\rho}_1, \widehat{\rho}_1)$. The identity $|\cosh(a + \mathrm{i}b)|^2 = \sinh^2 a + \cos^2 b$ gives $|\tanh'(a + \mathrm{i}b)| \le \sec^2 \widehat{\rho}_1$ there. Integrating the derivative along that segment bounds the change in h_1 by $\sec^2(\widehat{\rho}_1) L_1 \eta$ and the change in a_2 by $L_2\eta < \widehat{\rho}_2 - \rho_2$. Neither layer reaches a zero of $\cosh$, so composition is holomorphic. Conjugated coefficients satisfy the same bounds on the conjugation-invariant domain D_η; this proves the readout claim. The fixed prior and gauge subtraction preserve holomorphicity.

The row constraints are imposed by smooth parameterization. Replace each absolute value in a raw row budget by $a_\epsilon(t) = \sqrt{t^2 + \epsilon^2} \ge |t|$, with $\epsilon = 10^{-12}$, sum the appropriately weighted terms to obtain $U \ge 0$, and multiply that row's constrained coefficients by $\rho/(\rho + U)$. Its resulting budget is at most $\rho U/(\rho + U) < \rho$. For the first layer the constrained coefficients are its imaginary affine coefficients and imaginary bias; for the second they are both weight components and the imaginary bias. Real biases remain free. This construction avoids a state-dependent modification of the activation. Before these constraints, complex Gaussian initialization follows the variance criterion in Trabelsi et al. (2018, Section 3.6); the effective constrained weights no longer follow that unconstrained initialization law.

This is a sufficient radius for the particular frozen network, not a uniform radius for unconstrained weights. Floating-point evaluations of (S96) are diagnostics. A computer-assisted certificate must instead round the row sums and L_1, L_2 upward and the margins and η downward, including the coefficient export and parameterization operations. Real samples that avoid poles alone do not establish these complex-domain bounds.

S3.6.2 Joint energy–dissipation training and power accounting The joint branch uses separate energy and position-only damping networks with a common optimizer and a common vector-field objective. It does not share the full position–momentum hidden state with the damping head. For $R_\theta(X) \succeq 0$, define

$$\mathcal{D}_\theta(X, v) = \tfrac{1}{2} v^\mathsf{T} R_\theta(X) v, \qquad g(X) = d\cos(d^\mathsf{T} X). \tag{S98}$$

Holding X fixed while differentiating the independent velocity gives $\partial_v \mathcal{D}_\theta = R_\theta(X)v$. Hence

$$f_\theta(x, u) = \begin{pmatrix} H_Y \\ -H_X - R_\theta(X) H_Y + g(X)u \end{pmatrix}, \tag{S99}$$

and direct multiplication by (H_X, H_Y) yields

$$\nabla H_\theta^\mathsf{T} f_\theta = -H_Y^\mathsf{T} R_\theta H_Y + H_Y^\mathsf{T} g(X) u. \tag{S100}$$

The cross terms cancel. This cancellation holds for the learned Hamiltonian itself, without requiring a kinetic–potential split. The static magnetic bias is already included in the stored energy; the transverse field enters once through $u = B_y/B_0$. Multiplication by $E_0\omega_0$ restores dimensional power. If the Hamiltonian is explicitly time dependent, the right side gains $\partial_\tau H$; if a time-dependent field is also an energy argument, its chain-rule term must be included.

For comparison, a decoder with an arbitrary state-dependent PSD matrix $R(x)$ also obeys (S100); derivatives of R are not required when the force is defined directly as $-R(x)H_Y$. However, interpreting that force as $-\partial_v[\frac{1}{2} v^\mathsf{T} R(X, v) v]$ would introduce additional velocity derivatives of R. The position-only head avoids this ambiguity. PSD structure establishes nonpositive unforced power, not accuracy of the inferred damping or positive definiteness of the learned energy.

S3.6.3 Analytic error and the fourth-order Taylor jet Let H_F denote one explicitly frozen finite-feature Hamiltonian and H_D a frozen deep Hamiltonian, expressed in the same canonical chart and energy gauge. Choose a common complex polydisc P_r inside their domains of holomorphicity. Write

$$\begin{aligned} e(x) &= H_D(x) - H_F(x) \\ &= \sum_{|\alpha| \le 4} c_\alpha x^\alpha + R_5(x), \\ c_\alpha &= \frac{\partial^\alpha e(0)}{\alpha!}. \end{aligned} \tag{S101}$$

If interval bounds $|c_\alpha| \le b_\alpha$ and $\|R_5\|_{P_r} \le b_5$ are available, then

$$\|e\|_{P_r} \le \epsilon_0 := \sum_{|\alpha| \le 4} b_\alpha r^{|\alpha|} + b_5. \tag{S102}$$

The tail bound must come from the full analytic network, for example by interval Taylor arithmetic with a bounded analytic remainder. For $0 < r' < r$, Cauchy's formula gives

$$\|\partial^\alpha e\|_{P_{r'}} \le \frac{\alpha!\, \epsilon_0}{(r - r')^{|\alpha|}}, \qquad |\alpha| \le 4. \tag{S103}$$

These inequalities follow respectively from the triangle inequality and iterated one-variable Cauchy estimates. When the domains are tubes rather than concentric polydiscs, the same estimates use a verified coordinate-polydisc margin around each point of the inner domain. Tensor-to-operator norm factors must be included in the norms used by the target KAM theorem.

An automatic-differentiation jet evaluated in ordinary floating point can reveal an equilibrium shift, quadratic error or cubic terms, but does not bound b_5 or the complex-domain supremum. The deep residual in this study fixes only its value at zero. Consequently the direct quartic-to-twist map of the finite-feature model is inapplicable unless its additional hypotheses are proved for the deep function. A local normal-form calculation must first enclose its equilibrium, perform a canonical linear normalization of its actual Hessian, check the required nonresonances, and eliminate its cubic terms before computing the fourth-order Birkhoff coefficients. In particular, the cubic generating function contributes to fourth order.

S3.6.4 A conditional transfer of an a posteriori certificate There is also a route that does not normalize about equilibrium. Reuse a specified Fourier candidate $K : \mathbb{T}^n_\rho \to \mathbb{C}^{2n}$ and exact frequency ω from a finite-feature certificate as a candidate for the new function. It remains necessary to verify the new Hamiltonian on a complex neighbourhood of that candidate.

Proposition S14. Reusing a torus candidate for a new learned function. *Let K, ω satisfy the frequency, embedding and domain hypotheses of Haro and Luque (2019, Theorem 3.1) with its specified Appendix B constants. Let H_F, H_D be holomorphic on a common neighbourhood containing the whole complex image of K and the required correction tube. Suppose a certified residual bound for H_F is $\|J\nabla H_F(K) - DK\omega\|_\rho \le \epsilon_F$, and a certified derivative bound is $\|\nabla(H_D - H_F)\| \le \delta_1$ in the vector maximum norm*

on that neighbourhood. Then

$$\|J\nabla H_D(K) - DK\omega\|_\rho \le \epsilon_F + \delta_1. \tag{S104}$$

Enlarge the old Hamiltonian derivative enclosures by certified error bounds through order four, and reevaluate every Hamiltonian-dependent torsion, inverse and smallness test in the cited theorem for H_D. If these tests and the domain margins pass with $\epsilon_F + \delta_1$ as residual bound, the theorem supplies an invariant Lagrangian torus of the actual frozen deep Hamiltonian.

Proof. Subtract the two invariance equations for the same candidate and frequency. Since J is a signed permutation in canonical coordinates, its induced maximum norm equals one; the triangle inequality proves (S104). Bounds such as (S103) enclose the new derivatives by adding their error intervals to the old enclosures. Evaluating the theorem's finite algebraic expressions with these intervals encloses the new torsion and theorem constants, including every required inverse. For an inverse one may, for example, verify $\|A_F^{-1}\Delta A\| < 1$ and use the Neumann-series bound $\|(A_F + \Delta A)^{-1}\| \le \|A_F^{-1}\|/(1 - \|A_F^{-1}\Delta A\|)$. The stipulated domain, nondegeneracy and smallness tests are exactly the additional hypotheses required to apply the cited a posteriori theorem to H_D. This proves existence for H_D, rather than for the comparison Hamiltonian H_F.

This test can fail even when field or trajectory errors are small. The very small residual threshold of the existing finite-feature certificate makes direct reuse particularly demanding. Refining a new Fourier candidate for H_D can lower its residual, but still requires new interval domain and torsion calculations. The transfer criterion is sufficient only when all of these function-specific bounds satisfy the a posteriori theorem. Direct verification for the COMSOL-trained complex Hamiltonian uses its own refined candidate and interval bounds, identified in the reproducibility index. The damping and magnetic input are switched off for this conservative test. Transferring a verified low-mode result to the controlled full beam would additionally require evaluating the magnetic realization and its bounds for that same newly certified energy.

S3.7 From an energy generalization bound to an analytic KAM residual

We relate the energy generalization estimate of Chen et al. (2022) to the analytic invariance residual required by Haro and Luque (2019, Theorem 3.1). The argument combines an interior gradient estimate with interpolation on a complex torus strip. Theorem S3 instead uses a uniform higher Sobolev bound to obtain a real C^ℓ estimate for finite-smoothness KAM theory. Both statistical results require hypotheses on the sampling distribution and the entire hypothesis class. A deterministic a-posteriori certificate for an individual learned Hamiltonian has a distinct scope: it establishes the existence of an invariant torus of that Hamiltonian without a population-risk or generalization statement.

S3.7.1 The conditional statistical input and its energy gauge Work in one fixed canonical coordinate and time normalization, with phase dimension $m = 2n$ and $n \ge 2$. Let $D \subset \mathbb{R}^m$ be a bounded, connected Lipschitz domain. The conservative target is $H_\star$, and $\mathcal{H} = \{H_\vartheta : \vartheta \in \Theta\}$ is a hypothesis class fixed before the data are drawn. All energies are real valued on D, even when the network parameters are complex. Fix a finite exponent $p > m$ and a confidence parameter $0 < \alpha < 1$. Assume $H_\star, H_\vartheta \in W^{1,p}(D)$, using their continuous representatives, and define the pointwise conservative loss

$$\ell_\vartheta(x) = \|\nabla H_\vartheta(x) - \nabla H_\star(x)\|_p^p. \tag{S105}$$

Assume that $X_1, \dots, X_N$ are independent and identically distributed with law P supported on D, whose density relative to Lebesgue measure satisfies $f_P \ge m_P > 0$. Assume $0 \le \ell_\vartheta \le L_{\max}$ uniformly on the entire class and domain. Let $\overline{\mathfrak{R}}_N$ be a valid upper bound for the empirical Rademacher complexity of this loss class, with the normalization in Lemma 2 of Chen et al. (2022). Define

$$\widehat{L}_{p,N}(\vartheta) = \frac{1}{N}\sum_{i=1}^{N} \ell_\vartheta(X_i),$$
$$A_N(\vartheta, \alpha) = \widehat{L}_{p,N}(\vartheta) + 2\overline{\mathfrak{R}}_N + 3L_{\max}\sqrt{\frac{2\log(4/\alpha)}{N}}. \tag{S106}$$

Under these assumptions the cited generalization bound supplies an event $\mathcal{G}_\alpha$, with $\mathbb{P}(\mathcal{G}_\alpha) \ge 1 - \alpha$, on which $\mathbb{E}_P \ell_\vartheta \le A_N(\vartheta, \alpha)$ simultaneously for every $\vartheta \in \Theta$. This uniformity is essential when the same data select the fitted parameters.

Define the gauge-fixed error

$$e_\vartheta = H_\vartheta - H_\star - c_\vartheta, \qquad c_\vartheta = \frac{1}{|D|}\int_D (H_\vartheta - H_\star)\, dx. \tag{S107}$$

Let C_P satisfy $\|e\|_{L^p(D)} \le C_P\|\nabla e\|_{L^p(D)}$ for mean-zero functions, and let C_S be a valid constant in $W^{1,p}(D) \hookrightarrow C^0(\overline{D})$, using $\|e\|_{W^{1,p}}^p = \|e\|_{L^p}^p + \|\nabla e\|_{L^p}^p$.

Proposition S15. Conditional energy bound. *On $\mathcal{G}_\alpha$, simultaneously for every member of the class,*

$$\begin{aligned} \|e_\vartheta\|_{C^0(\overline{D})} &\le \varepsilon_\vartheta, \\ \varepsilon_\vartheta &:= C\left(\frac{A_N(\vartheta,\alpha)}{m_P}\right)^{1/p}, \\ C &= C_S(1+C_P^p)^{1/p}. \end{aligned} \tag{S108}$$

Proof. The density lower bound and the uniform risk event give

$$\int_D \|\nabla e_\vartheta\|_p^p \, dx \le m_P^{-1}\mathbb{E}_P \ell_\vartheta \le m_P^{-1} A_N(\vartheta,\alpha).$$

Poincare's inequality applies because (S107) has zero mean. Consequently $\|e_\vartheta\|_{W^{1,p}} \le (1+C_P^p)^{1/p}(A_N/m_P)^{1/p}$. The Sobolev embedding, valid because $p > m$, proves the assertion.

The exponent $1/p$ in (S108) follows from the pth-power definition of the loss (S105); a linear dependence on that loss would require different definitions. For the four-dimensional phase space considered here, this first-order embedding requires $p > 4$. Using weak-increment observations or a combined Hamiltonian–damping residual to bound $\widehat{L}_{p,N}$ additionally requires conservative-gradient label-error estimates, including errors in dissipation and input subtraction. The trajectory sampling scheme, with consecutive and overlapping windows, does not meet the iid state-sampling hypothesis of Proposition S15.

S3.7.2 A real gradient bound with explicit domain loss

Proposition S16. Interior gradient estimate from energy values. *Let $e \in C^2(D)$ satisfy $\sup_D |e| \le \varepsilon$ and $\max_j \sup_D |\partial_{jj} e| \le B$, with $B \ge 0$. Let $S \subset D$ and $r > 0$ satisfy $x + [-r, r]\mathbf{e}_j \subset D$ for every $x \in S$ and coordinate unit vector $\mathbf{e}_j$. Then*

$$\sup_{x\in S} \|\nabla e(x)\|_\infty \le b_r(\varepsilon, B) := \inf_{0<h\le r}\left(\frac{\varepsilon}{h} + \frac{Bh}{2}\right). \tag{S109}$$

For $B > 0$ this is

$$b_r(\varepsilon,B) = \begin{cases} \sqrt{2B\varepsilon}, & 0 \le \varepsilon \le Br^2/2, \\ \varepsilon/r + Br/2, & \varepsilon > Br^2/2. \end{cases} \tag{S110}$$

For $B = 0$, take $b_r(\varepsilon, 0) = \varepsilon/r$.

Proof. For $0 < h \le r$, the fundamental theorem of calculus gives

$$\begin{aligned} &\left|\partial_j e(x) - \frac{e(x+h\mathbf{e}_j) - e(x-h\mathbf{e}_j)}{2h}\right| \\ &\quad \le \frac{1}{2h}\int_{-h}^{h} |\partial_j e(x+t\mathbf{e}_j) - \partial_j e(x)|\, dt \le \frac{Bh}{2}. \end{aligned}$$

The difference quotient has magnitude at most ε/h. Taking the infimum proves (S109). For $B\varepsilon > 0$, the unconstrained minimizer is $h = \sqrt{2\varepsilon/B}$; otherwise use $h = r$ or take the appropriate zero limit. This proves (S110).

The estimate only needs bounds on the pure second derivatives of the *error*; a full Hessian operator bound is a stronger sufficient condition. A certified sum of target and network Hessian bounds also suffices. An additive energy constant has no effect on this step.

Let $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$, and fix a reference embedding $K_0 : \mathbb{T}^n \to D$ and frequency $\omega \in \mathbb{R}^n$. Suppose its real image has the coordinate-segment margin in Proposition S16, and that an independently justified target residual obeys

$$\|J\nabla H_\star(K_0) - DK_0\omega\|_0 \le \delta_\star. \tag{S111}$$

The subscript 0 denotes the supremum over real angles, with the maximum norm for vector components. Since $\|J\|_{\infty\to\infty} = 1$, the learned invariance residual satisfies

$$\begin{aligned} E_\vartheta &= J\nabla H_\vartheta(K_0) - DK_0\omega, \\ \|E_\vartheta\|_0 &\le \delta_\star + b_r(\varepsilon_\vartheta, B) =: \delta_\vartheta \quad \text{on } \mathcal{G}_\alpha. \end{aligned} \tag{S112}$$

In particular, a certificate for an invariant torus of H_ϑ cannot be substituted for the target residual in (S111). If K_0 is an exact torus of $H_\star$, then $\delta_\star = 0$; otherwise its defect must be included.

The weak-increment objective also admits a deterministic application. If Proposition S9 and the coverage conditions of Proposition S10 supply a uniform conservative gradient bound G_{det} on $K_0(\mathbb{T}^n)$, then $\|E_\vartheta\|_0 \le \delta_\star + G_{\mathrm{det}}$ directly. The Euclidean bound used in those propositions also bounds the maximum norm here. Substitution of G_{det} for $b_r(\varepsilon_\vartheta, B)$ in the analytic step below without a statistical assumption gives a deterministic residual bound. Its applicability requires the stated nuisance-term and domain-coverage estimates, which are independent of finite-sample weak-residual bounds. No iid probability interpretation is attached to correlated windows.

S3.7.3 Analytic continuation of the residual with a certified outer bound

Write $\mathbb{T}^n_R = \{\theta \in \mathbb{C}^n/\mathbb{Z}^n : |\operatorname{Im}\theta|_\infty <$

$R\}$. For a vector function, $\|F\|_R = \max_j \sup_{\mathbb{T}^n_R} |F_j|$. Assume that K_0 extends holomorphically to this strip and that $K_0(\mathbb{T}^n_R)$ lies in a pole-free complex domain of H_ϑ. Assume a *certified upper bound*

$$\|E_\vartheta\|_R \le M. \tag{S113}$$

For example, if verified constants satisfy $\sup_{K_0(\mathbb{T}^n_R)} \|J\nabla H_\vartheta\|_\infty \le C_X$ and $\|DK_0\|_R \le C_K$, then one may *choose* $M = C_X + C_K\|\omega\|_\infty$. Here the matrix norm is $\|A\|_R = \max_i \sum_j \sup_{\mathbb{T}^n_R} |A_{ij}|$, as in the analytic norm convention of Haro and Luque (2019). Point samples of a complex extension do not certify these suprema.

Lemma S17. Residual interpolation on a torus strip. *Let F be a bounded holomorphic periodic vector function on $\mathbb{T}^n_R$ with $\|F\|_0 \le \delta$ and $\|F\|_R \le M$. For $0 < \rho < R$, set $\bar\delta = \min\{\delta, M\}$. Then*

$$\|F\|_\rho \le \bar\delta^{\,1-\rho/R} M^{\rho/R}. \tag{S114}$$

If either bound is zero, the right-hand side is interpreted by continuity.

Proof. First suppose $M \ge \bar\delta > 0$ and that the extension is continuous on the closed strip. Given $x + iy$ with $t = |y|_\infty > 0$, put $v = y/t$. For each component define the one-variable function $g(w) = F_j(x + wv)$ on $0 \le \operatorname{Im} w \le R$. The real parts $x + (\operatorname{Re} w)v$ range over real angles, so the lower boundary has bound $\bar\delta$ and the upper boundary has bound M. The function is bounded throughout this strip. Set

$$\beta = R^{-1}\log(M/\bar\delta), \qquad G(w) = \bar\delta^{-1} g(w) e^{i\beta w}.$$

Both horizontal boundaries satisfy $|G| \le 1$. To justify the maximum principle on this unbounded domain, use $G_\eta(w) = G(w)\exp(-\eta w^2 - \eta R^2)$, with $\eta > 0$. Its modulus is at most one on the horizontal boundaries and tends to zero on the vertical boundaries of expanding rectangles. The rectangle maximum principle, followed by expansion and then $\eta \downarrow 0$, gives $|G(w)| \le 1$. In particular,

$$|F_j(x + iy)| = |g(it)| \le \bar\delta \exp(\beta t) = \bar\delta^{1-t/R} M^{t/R}.$$

Since $\bar\delta \le M$, this expression is nondecreasing in t. Taking $t \le \rho$ and the componentwise supremum proves the claim. For an open outer strip, apply the argument with $\rho < R' < R$ and let $R' \uparrow R$. Zero bounds follow by replacing the bounds with positive ones and taking a limit. This also covers real angles $t = 0$.

Combining the energy and gradient estimates with the strip lemma gives, on $\mathcal{G}_\alpha$, the analytic residual bound

$$\begin{aligned} &\|E_\vartheta\|_\rho \le \mathcal{E}_\vartheta, \\ &\mathcal{E}_\vartheta := \bar\delta_\vartheta^{\,1-\rho/R} M^{\rho/R}, \\ &\bar\delta_\vartheta := \min\{M, \delta_\vartheta\}, \end{aligned} \tag{S115}$$

where δ_ϑ is defined by (S112). There is no dimension factor for this maximum-component norm.

For the real energy readout of the complex network, analytic continuation means complexification of the real canonical variables, not application of a real-part operation at complex arguments. If $H(x) = \operatorname{Re} F(x)$ on real x and F has a holomorphic extension, its extension is $[F(z) + F^\#(z)]/2$, where $F^\#(z) = \overline{F(\bar z)}$ is a holomorphic function obtained by conjugating coefficients. Features written as $(z, \bar z)$ on real phase space become the independent linear expressions $(Q + iP)/\sqrt{2}$ and $(Q - iP)/\sqrt{2}$ after complexification. Intermediate complex $\tanh$ layers must remain separated from their poles on the whole declared complex domain. Bounds for one trained network do not establish corresponding uniform bounds for a parameter class.

S3.7.4 The precise a-posteriori KAM endpoint We use the ordinary theorem of Haro and Luque (2019, Theorem 3.1, equations (3.9)–(3.11), arXiv v1), specialized to $d = n \ge 2$ and canonical geometry. This specialization has no additional first integrals; their constants are zero as in the cited Remark 3.2. In this paragraph $\mathcal{B}$ denotes the complex phase domain and is distinct from the scalar Hessian bound B. The following are the corresponding H1–H5 requirements, imposed independently of the empirical training loss:

H1. The exact symplectic and metric structures and H_ϑ extend analytically to $\mathcal{B}$, with the specified analytic bounds. In canonical coordinates the geometric structures are constant, as in the cited Remark 3.3. The required Hamiltonian bounds concern $\|DH_\vartheta\|_\mathcal{B}$, $\|X_{H_\vartheta}\|_\mathcal{B}$, $\|DX_{H_\vartheta}\|_\mathcal{B}$, $\|D^2X_{H_\vartheta}\|_\mathcal{B}$ and $\|D(X^{\mathsf{T}}_{H_\vartheta})\|_\mathcal{B}$ in the cited tensor-norm convention. As emphasized in its Section 3.1, this last derivative of a row-vector function and the transpose of the Jacobian have different tensor-norm definitions and are not interchangeable. The optional first integral c of that theorem may be chosen constant, with zero derivative bound.

H2. The embedding K_0 and DK_0 extend continuously to the closed strip of width ρ, are holomorphic inside

it, and satisfy $\operatorname{dist}(K_0(\mathbb{T}^n_\rho), \partial\mathcal{B}) > 0$, $\|DK_0\|_\rho < \sigma_K$ and $\|(DK_0)^{\mathsf{T}}\|_\rho < \sigma_{K^{\mathsf{T}}}$.

H3. The tangent frame $L = DK_0$ has maximal rank throughout the closed complex strip, and its metric Gram inverse $\mathsf{B} = (L^{\mathsf{T}}L)^{-1}$ satisfies $\|\mathsf{B}\|_\rho < \sigma_B$. The transpose is bilinear, not a Hermitian adjoint. The normal frame is defined by equations (3.2)–(3.5) of the cited theorem, with the same symplectic sign convention.

H4. For the resulting torsion matrix T of equations (3.6)–(3.7), computed using X_{H_ϑ}, the real-angle average has an inverse satisfying $|\langle T\rangle^{-1}| < \sigma_T$. Small gradient error by itself does not establish this condition.

H5. For fixed $\gamma > 0$ and $\tau \geq n-1$, the frequency obeys $|k\cdot\omega| \geq \gamma|k|_1^{-\tau}$ for every $k \in \mathbb{Z}^n \setminus \{0\}$. A finite search for small divisors does not prove this all-integer assertion.

For $0 < \rho_\infty < \rho$, these data determine the explicit theorem constants C_1, C_2, C_3 of the cited Appendix B. Its smallness condition is exactly

$$\frac{C_1\|E_\vartheta\|_\rho}{\gamma^4\rho^{4\tau}} < 1. \tag{S116}$$

Accordingly, checking

$$\frac{C_1\mathcal{E}_\vartheta}{\gamma^4\rho^{4\tau}} < 1 \tag{S117}$$

and H1–H5 supplies, on $\mathcal{G}_\alpha$, an invariant torus K_∞ for the learned conservative Hamiltonian with the prescribed frequency, analytic on the strip of width ρ_∞. The cited theorem also gives

$$\|K_\infty - K_0\|_{\rho_\infty} \leq \frac{C_2\|E_\vartheta\|_\rho}{\gamma^2\rho^{2\tau}} \leq \frac{C_2\mathcal{E}_\vartheta}{\gamma^2\rho^{2\tau}}. \tag{S118}$$

The zero-residual case is already invariant and admits $K_\infty = K_0$; the weak inequality also covers this case. The constants and geometric bounds must correspond to this Hamiltonian, domain and pair of strip widths. The conclusion concerns a conservative torus, not attraction, dissipative control, all trajectories, or a positive-measure family without an additional family argument.

Proposition S18. Deterministic certification of an arbitrary training output. *Let* A *be any possibly randomized training and checkpoint-selection procedure, and let* $\widehat{\vartheta} = \mathsf{A}(\mathcal{Z}, \xi)$ *be its finite output parameters, including data-dependent preprocessing constants. For each realized dataset* $\mathcal{Z}$ *and random seed* ξ*, consider the exact exported Hamiltonian* $H_{\widehat{\vartheta}}$ *and any candidate* K_0*, frequency* ω *and verification domain. Suppose these objects satisfy H1–H5 above in the same coordinate, time and norm conventions, and a valid deterministic bound* $\overline{E} \geq \|J\nabla H_{\widehat{\vartheta}}(K_0) - DK_0\omega\|_\rho$ *satisfies*

$$\frac{C_1\overline{E}}{\gamma^4\rho^{4\tau}} < 1.$$

Then $H_{\widehat{\vartheta}}$ *has the invariant torus and correction bound stated above, with* $\overline{E}$ *in place of* $\mathcal{E}_\vartheta$*. No sampling-independence, descent, stationarity or optimality condition on the training output is required.*

Proof. Fix an arbitrary realization $(\mathcal{Z}, \xi)$ and its exact output. The verified hypotheses and residual bound satisfy Haro and Luque (2019, Theorem 3.1), which gives the asserted torus and correction estimate. This implication holds for every realization; no property of the procedure producing the candidate is used.

This result concerns acceptance of a particular frozen function. It requires verified bounds for that function, rather than class-wide statistical constants, and does not guarantee that the training procedure produces an acceptable candidate. The finite-sample weak-loss bounds and conservative-error identities likewise apply to arbitrary selected parameters under their stated observation and model hypotheses.

S3.7.5 An explicit risk threshold and the probability statement For $B > 0$, $M > 0$, set

$$s = \rho/R, \qquad \kappa = \frac{\gamma^4\rho^{4\tau}}{C_1}, \qquad T_0 = \left(\frac{\kappa}{M^s}\right)^{1/(1-s)}. \tag{S119}$$

If $T_0 > \delta_\star$, define the positive admissible energy error

$$\varepsilon_{\text{allow}} = \min\left\{\frac{Br^2}{2}, \frac{(T_0-\delta_\star)^2}{2B}\right\}. \tag{S120}$$

A sufficient empirical-risk and complexity condition is

$$A_N(\vartheta,\alpha) < m_P\left(\frac{\varepsilon_{\text{allow}}}{C}\right)^p. \tag{S121}$$

Indeed, (S108) then yields $\varepsilon_\vartheta < \varepsilon_{\text{allow}} \leq Br^2/2$ and hence $\delta_\star + \sqrt{2B\varepsilon_\vartheta} < T_0$. Raising to $1-s$ and multiplying by M^s proves (S117). The general criterion (S117) remains available when $B = 0$ or the optimal difference step is limited by r; these cases are covered directly by (S117).

Corollary S19. Uniform event and post-fit acceptance. *Suppose the statistical assumptions preceding Proposition S15 hold. Let* $\widehat{\vartheta}$ *be any data-dependent selection from the fixed class. Define acceptance to require valid deterministic bounds for all analytic, reference-residual and geometric assumptions above, together with* (S117) *evaluated using*

(S108). *Let $\mathcal{A}$ be this acceptance event and $\mathcal{K}$ the event that the asserted learned invariant torus exists. Then*

$$\mathbb{P}(\mathcal{A} \cap \mathcal{K}^c) \le \alpha. \tag{S122}$$

Proof. On the uniform event $\mathcal{G}_\alpha$, the energy estimate applies to every parameter, including $\widehat{\vartheta}$. All subsequent implications are deterministic whenever the acceptance conditions hold. Thus erroneous acceptance is contained in $\mathcal{G}_\alpha^c$, whose probability is at most α.

Equation (S122) is not the claim $\mathbb{P}(\mathcal{K} \mid \mathcal{A}) \ge 1 - \alpha$. If the acceptance probability is positive, the immediate conditional bound is instead at most $\alpha/\mathbb{P}(\mathcal{A})$ for failure. An unconditional success probability of at least $1 - \alpha$ requires the remaining conditions to hold for the selected model; a procedure that can reject that model gives only the joint-event guarantee above. Application of this statistical result requires uniform parameter-class bounds, a valid complexity estimate for the realified complex architecture, iid state samples with a positive density lower bound, and conservative pointwise label-error estimates. The correlated weak-window sampling scheme does not satisfy the iid hypothesis. Moreover, bounds for the fitted Hamiltonian are model-specific and do not provide the class-wide complexity or distributional constants required in (S106). The numerical trajectory-based results are therefore interpreted through their deterministic estimates, without a population-confidence claim for the fitted model.

Remark S1. Necessity of quantitative regularity bounds. Real energy closeness alone does not control a vector field: on a bounded real interval, $e_k(q) = k^{-1}\sin(k^2 q)$ has sup norm tending to zero and derivatives of unbounded amplitude. Even a real Hessian bound does not supply (S113): the entire, 2π-periodic functions $e_k(q) = k^{-2}\sin(kq)$ have $\|e_k\|_0 \to 0$, $\|e_k'\|_0 \to 0$ and $\|e_k''\|_0 \le 1$, while $|e_k'(i\rho)| = k^{-1}\cosh(k\rho)$ diverges at any fixed $\rho > 0$. A pole-free extension together with a quantitative outer bound, not analyticity alone, is what makes Lemma S17 useful. These examples concern scalar functions and do not assert unboundedness within a separately constrained finite network class.

S3.8 Validated fourth-order normal form of the frozen deep energy

The finite-feature normal form cannot be substituted for that of the end-to-end energy: the latter has a shifted equilibrium and a nonzero cubic jet. We therefore evaluate its complete fourth-order normal form directly from the frozen complex network. Throughout this subsection, the nominal coefficients ν, d are the same exact binary64 numbers as in the training model, interpreted as rational numbers. Define

$$H_\lambda = (1-\lambda)H_\theta + \lambda H_*, \qquad 0 \le \lambda \le 1, \tag{S123}$$

where H_* retains the full cosine. This interpolation compares conservative energies; it is not a training path or an actuated motion.

Proposition S20. Certified elliptic nondegeneracy. *For the frozen complex seed-43 energy, every Hamiltonian in (S123) has exactly one critical point in $\|x\|_\infty < 10^{-8}$. This point is elliptic, its frequencies are nonresonant through order four, and its complete quartic Birkhoff matrix A_λ is nonsingular. Outward interval bounds valid for every $\lambda \in [0,1]$ are*

$$\begin{aligned}
\lambda_{\min}(D^2 H_\lambda(x_\lambda)) &> 0.9999993,\\
\min_{0<|k|_1\le 4} |k\cdot\omega_\lambda| &> 0.9999998,\\
-1.418772\,10^{-6} < \det A_\lambda &< -7.254483\,10^{-7},\\
\sigma_{\min}(A_\lambda) &> 3.75252\,10^{-4}.
\end{aligned} \tag{S124}$$

At the learned endpoint, narrower enclosures give

$$\begin{aligned}
\omega_{\theta,1} &\in [0.999999898957840, 0.999999898957843],\\
\omega_{\theta,2} &\in [6.15211711018629, 6.15211711018630],\\
\det A_\theta &\in [-8.182369, -8.182367]10^{-7},\\
\sigma_{\min}(A_\theta) &> 4.39754\,10^{-4}.
\end{aligned} \tag{S125}$$

These are bounds on the actual network, including the cubic elimination term, rather than on a fitted polynomial surrogate.

Proof. The reproducibility record contains exact endpoints for the calculations below, performed with Arb/Acb outward ball arithmetic at 160 and 256 bits. The 32 closed intervals $[j/32, (j+1)/32]$ cover the entire parameter interval; parameter uncertainty is propagated through each calculation. The common quadratic prior is cancelled algebraically before evaluating the difference of the two Hamiltonians.

For a point centre c, a point rational approximate inverse Y of the Hessian, and a box $B = c + [-r, r]^4$, let

$$\beta = \sup_\lambda \|Y\nabla H_\lambda(c)\|_\infty, \quad t = \sup_{x\in B,\lambda} \|I - YD^2 H_\lambda(x)\|_\infty.$$

The verifier establishes $t < 1$ and $\beta + tr < r$. Thus $x \mapsto x - Y\nabla H_\lambda(x)$ is a contraction from B into itself for every parameter value in the interval. All resulting root enclosures lie in $(-10^{-8}, 10^{-8})^4$. A separate strict diagonal-dominance

bound for the symmetric Hessian on this entire common box is greater than 0.9999993, for all λ. Strict convexity therefore makes these local roots the unique critical points in the common box. It also identifies the roots on overlapping parameter intervals. The implicit function theorem makes x_λ analytic locally in λ.

Write $K = D^2H_\lambda(x_\lambda)$, $L = JK$. Set

$$\begin{aligned} a &= -\tfrac{1}{2}\operatorname{tr}(L^2), \qquad b = \det K, \\ \ell_{1,2} &= \tfrac{1}{2}(a \mp \sqrt{a^2 - 4b}), \qquad \omega_j = \sqrt{\ell_j}. \end{aligned} \tag{S126}$$

The interval calculation proves $K > 0$, $a^2 - 4b > 0$ and $\ell_j > 0$. Hence L has distinct pairs $\pm i\omega_j$. Its real spectral projectors are

$$\Pi_j = \frac{L^2 + \ell_{3-j} I}{\ell_{3-j} - \ell_j}, \qquad v_j = \Pi_j e_{q_j}, \qquad w_j = -Lv_j/\omega_j.$$

The positive quantities $r_j = v_j^{\mathsf{T}} J w_j$ are also enclosed away from zero. Indeed, $r_j = v_j^{\mathsf{T}} K v_j/\omega_j$. The columns $v_j/\sqrt{r_j}$ and $w_j/\sqrt{r_j}$ define a matrix S satisfying

$$S^{\mathsf{T}} J S = J, \qquad S^{\mathsf{T}} K S = \operatorname{diag}(\omega_1, \omega_2, \omega_1, \omega_2). \tag{S127}$$

These are exact identities: L^2 is self-adjoint for the symplectic pairing, so its distinct spectral planes are symplectically orthogonal; $Lw_j = \omega_j v_j$ gives the within-plane identities. Computed interval residuals enclosing zero serve only as implementation checks.

In $x = x_\lambda + Sy$, write $H_\lambda = H_\lambda(x_\lambda) + H_2 + H_3 + H_4 + O(|y|^5)$. Multivariate Taylor arithmetic through total degree four propagates the actual affine layers and complex hyperbolic tangents. Taking real parts of the Taylor coefficients on real inputs gives the derivatives of the real energy and hence its holomorphic extension. Both conjugate-coefficient branches are pole-free on $|\Re x_j|, |\Im x_j| \le 10^{-6}$: their first- and second-layer preactivation imaginary parts are respectively below 0.132 and 0.291, strictly less than $\pi/2$. The full cosine is entire. The resulting jets and analytic remainders therefore belong to the same functions as the equilibrium calculation.

With $z_j = (Q_j + iP_j)/\sqrt{2}$, $I_j = z_j \bar{z}_j$, and $\{z_j, \bar{z}_k\} = -i\delta_{jk}$, put $\Delta_{\alpha\beta} = (\alpha - \beta)\cdot\omega$. All nontrivial integer combinations of order at most four are enclosed away from zero. The degree-three homological equation and the resonant quartic part are consequently

$$\begin{aligned} (\chi_3)_{\alpha\beta} &= i(H_3)_{\alpha\beta}/\Delta_{\alpha\beta}, \\ G_4 &= H_4 + \tfrac{1}{2}\{H_3, \chi_3\}, \\ Z_4 &= \Pi_{\mathrm{res}} G_4 = \tfrac{1}{2} I^{\mathsf{T}} A_\lambda I. \end{aligned} \tag{S128}$$

Here $\{H_2, \chi_3\} = -H_3$; the second homological equation removes the nonresonant quartic terms. Finite-time flows of these polynomial generators are analytic symplectic maps on sufficiently small neighbourhoods. They leave an analytic $O(|y|^5)$ remainder. The diagonal entries of A are twice the I_j^2 coefficients; the off-diagonal entries are the $I_1 I_2$ coefficient. Interval determinants give the strict negative bounds in (S124). Finally, $\sigma_{\min}(A) = |\det A|/\sigma_{\max}(A) \ge |\det A|/\|A\|_F$ yields the stated lower bounds.

At $\lambda = 1$, an independent closed-form check recovers $A_{*,jj} = -d_j^4/8$, $A_{*,12} = -d_1^2 d_2^2/4$ and $\det A_* = -3d_1^4 d_2^4/64$.

Corollary S21. Local invariant families along the conservative interpolation. *For every $\lambda \in [0,1]$, arbitrarily small neighbourhoods of x_λ contain a positive-measure family of analytic invariant Lagrangian two-tori of H_λ.*

Proof. Fix λ and use its exact normal form $\omega \cdot I + \frac{1}{2} I^{\mathsf{T}} A I + R_5$. On positive action boxes $I \in \varepsilon^2 D$, with D a fixed box compactly contained in the positive quadrant, the action–angle map is analytic on a fixed angle strip and an action neighbourhood of width proportional to ε^2. Its remainder is $O(\varepsilon^5)$. Since A is invertible, the integrable frequency map $I \mapsto \omega + AI$ is a local analytic diffeomorphism, with a frequency domain of diameter proportional to ε^2.

Use these frequencies as parameters in Pöschel (2001, Theorems A–B). Take the local action radius $r = \varepsilon^{5/2}$, a frequency complex width $h = c\varepsilon^2$, a fixed sufficiently small angle width $s > 0$, and Diophantine constant $\alpha = \varepsilon^{2+\delta}$ with $0 < \delta < 1/4$. The perturbation after linearizing the integrable part at each action centre is $O(r^2) + O(\varepsilon^5) = O(\varepsilon^5)$. Consequently

$$\frac{\|P\|}{\alpha r s^{\tau+1}} = O(\varepsilon^{1/2-\delta}) \longrightarrow 0, \qquad \alpha s^{\tau+1} \le h$$

for sufficiently small ε, with any fixed $\tau > 1$. All smallness conditions of the cited parameter theorem hold. Its parameter displacement is $O(\|P\|/r) = O(\varepsilon^{5/2})$, which is $o(\alpha)$ and hence smaller than the boundary buffer in the Diophantine parameter domain.

The Diophantine set is nonempty with positive relative measure. Indeed, in a two-dimensional frequency box of diameter $O(\varepsilon^2)$, the strip $|k \cdot \varpi| < \alpha |k|^{-\tau}$ occupies at most $C\varepsilon^2 \alpha |k|^{-\tau-1}$ area. Summing over nonzero $k \in \mathbb{Z}^2$ converges for $\tau > 1$, and its relative excluded measure is $O(\alpha/\varepsilon^2) = O(\varepsilon^\delta)$. The boundary layer of thickness α has area $O(\varepsilon^2 \alpha)$ and obeys the same relative estimate.

To obtain positive phase-space measure, write the torus map from Theorem A as $(U(\vartheta,\varpi),\vartheta+V(\vartheta,\varpi))$ and its parameter correction as φ. In the original action coordinates the combined embedding is

$$\Psi(\vartheta,\varpi)=\begin{pmatrix}A^{-1}(\varphi(\varpi)-\omega)+U(\vartheta,\varpi)\\ \vartheta+V(\vartheta,\varpi)\end{pmatrix}.$$

Choose a frequency box $\varpi_c+\varepsilon^2\widehat{\Omega}$ with fixed $\widehat{\Omega}$, and rescale the frequency increment and the action increment from $A^{-1}(\varpi_c-\omega)$ by ε^2. The weighted Lipschitz estimates of Theorem A, together with Cauchy estimates on a smaller fixed angle strip, give

$$\mathrm{Lip}(\widehat{\Psi}-\widehat{\Psi}_0)=O(\varepsilon^{1/2-2\delta})\longrightarrow 0,$$
$$\widehat{\Psi}_0(\vartheta,\eta)=(A^{-1}\eta,\vartheta).$$

For example, the angle correction has parameter Lipschitz bound $O(\|P\|/(\alpha^2 r))$ before scaling, yielding the displayed dominant order after multiplication by ε^2. Since A is invertible and $\delta<1/4$, the combined map is bi-Lipschitz onto its image for sufficiently small ε. It maps the positive-measure product of the angle torus and the retained parameter set to a set of positive four-dimensional measure. The inverse symplectic normal-form map preserves this positivity. The resulting tori lie arbitrarily close to x_λ as $\varepsilon\to 0$.

The corollary certifies local invariant families for the actual learned energy and for every intervening conservative energy. It does not identify these families at a common frequency, give a certified finite amplitude threshold, or continue a specified torus along λ. The equilibrium frequencies of the two endpoints differ, so equality of their local frequency domains cannot be presumed. A prescribed frequency comparison requires separate a posteriori verification.

S3.9 A nominal physical torus at the certified learned frequency

The analytic identification bound alone does not fit the residual budget of the previously verified learned embedding. A separate nominal embedding can nevertheless be constructed and certified at exactly the same frequency. This calculation uses the full-cosine fixed-mass target H_*, without a fitted quartic correction.

Proposition S22. Same-frequency conservative embeddings. *Put* $\phi=(1+\sqrt{5})/2$ *and define*

$$w_1=\frac{499999368111}{500000000000},\qquad a=\frac{7198999\phi+2261342}{1170165\phi+367571},$$
$$\varpi=\frac{1}{2\pi}(w_1,w_1 a).\tag{S129}$$

Angles are measured in cycles and time is the common dimensionless time τ. *There exist analytic embedded Lagrangian tori* K_θ *and* K_* *satisfying*

$$J\nabla H_\theta(K_\theta)=DK_\theta\varpi,\qquad J\nabla H_*(K_*)=DK_*\varpi.\tag{S130}$$

The learned embedding is analytic on $|\Im\vartheta|_\infty<1/320$, *and the nominal embedding on* $|\Im\vartheta|_\infty<1/80$. *The latter lies within* $3.170\,10^{-66}$, *in the maximum-component analytic norm, of its supplied rational Fourier candidate.*

Proof. The first embedding is the independently verified frozen deep-model embedding reported in the main article. The exact frequency in (S129) is reconstructed from its continued-fraction definition, rather than from rounded decimal frequencies. Its continued-fraction partial quotients are bounded by eight, giving the conservative all-integer bound

$$|k\cdot\varpi|\ge\frac{9}{700}|k|_1^{-1}\ge\frac{9}{700}|k|_1^{-3},\qquad k\neq 0.$$

The already verified frequency certificate also supplies a stronger constant. The weaker displayed constant is used consistently in the nominal verification.

For H_*, a Fourier Newton calculation starts from a harmonic embedding with unequal amplitudes. The resulting finite candidate $\widehat{K}_*$ has a 128×128 coefficient array, with every coefficient stored as an exact rational number and with exact conjugate symmetry. Its model coefficients and exact frequency are checked against the same ν,d and (S129). Newton convergence is used only to construct this candidate.

Arb evaluation on a 384×384 grid encloses the residual Fourier interpolant. For $\rho=1/40$, $\widehat{\rho}=1/4$, and $N=384$, the analytic interpolation tail is bounded by

$$\frac{8M\exp[-\pi(\widehat{\rho}-\rho)N](1+q)}{(1-q)^2},\qquad q=\exp[-2\pi(\widehat{\rho}-\rho)],\tag{S131}$$

where M bounds the residual on the outer strip. The full cosine is evaluated there, and the bound is added to the

interpolant norm. For the frame, the reference embedding is

$$K_a = \begin{pmatrix} a_1 \cos 2\pi\vartheta_1 \\ a_2 \cos 2\pi\vartheta_2 \\ -a_1 \sin 2\pi\vartheta_1 \\ -a_2 \sin 2\pi\vartheta_2 \end{pmatrix},$$

with $a_1 = 45041/10^6$ and $a_2 = 75319/(2\,10^6)$. Its Gram matrix is exactly $\operatorname{diag}((2\pi a_1)^2, (2\pi a_2)^2)$ on complex strips as well. Neumann bounds for the candidate perturbation certify the frame inverse on the required strip. This uses the smaller reference amplitude in the Gram lower bound and the larger one in derivative upper bounds.

The average torsion is computed on the 128×128 grid. On the wider strip of width 0.07, the normal frame and its transpose have norms below 64. Cauchy estimates on width 0.06 and the global derivative bounds give a torsion norm below 10^7. With $z = \exp(-2\pi\, 0.06\, 128)$, each entry of the torsion average therefore receives the tail allowance $10^7((1+z)^2/(1-z)^2 - 1)$. The inverse is taken only after this allowance has been included. The resulting bounds are

Verified quantity	Outward upper bound
$\|\widehat{K}_*\|_\rho$	0.052703
$\|D\widehat{K}_*\|_\rho$	0.331140
$\|(D\widehat{K}_*)^{\mathsf{T}}\|_\rho$	0.662280
Gram inverse norm	17.862
Average torsion inverse norm	80516.636
Analytic residual norm	$2.585\,10^{-109}$
A posteriori smallness quotient	$7.713\,10^{-20}$
Embedding correction norm	$3.170\,10^{-66}$

The derivative bounds for the canonical vector field on the complex polydisc of radius 0.3 are $(3, 8, 32, 1)$ for the vector field, its Jacobian, the derivative of its row-vector transpose, and its second derivative, respectively. In particular, the third bound uses the sum of all Jacobian entries required by that convention. The frame bound is enlarged to $\sigma_B = 64$ in the rational constants of Haro and Luque (2019, Theorem 3.1); the corresponding residual threshold exceeds $3.351\,10^{-90}$. The smallness quotient is strictly below one and all frame, domain and torsion margins are preserved. The theorem gives the second invariant embedding and the displayed correction estimate. An injective phase projection relative to K_a, with Lipschitz deviation below $5.148\,10^{-5}$ including the correction, establishes embeddedness. The computations pass at both 512 and 640 bits.

Corollary S23. Phase-paired comparison. *With the phases fixed by the supplied Fourier candidates,*

$$\sup_{\vartheta\in\mathbb{T}^2} |K_*(\vartheta) - K_\theta(\vartheta)|_\infty < 0.005041552,$$
$$\|K_* - K_\theta\|_{1/320} < 0.005042524. \tag{S132}$$

These bounds use the dimensionless canonical coordinates. The embedding map $F = K_* \circ K_\theta^{-1}$ conjugates the two conservative flows restricted to these tori. For matched initial phases, the real bound holds at every positive and negative time. At phase $(0,0)$, however, the first configuration components satisfy

$$(K_*)_{q_1}(0,0) - (K_\theta)_{q_1}(0,0) > 0.005040448. \tag{S133}$$

Proof. The deep candidate is compared to the same cosine/negative-sine reference with amplitudes $(1/25, 1/25)$. Both the phase convention and the complete candidate identity are checked against its geometric certificate. Let e_*, e_θ denote the certified candidate-to-reference errors, and c_*, c_θ the respective a posteriori corrections. For $\rho_c = 1/320$, the triangle inequality gives

$$\|K_* - K_\theta\|_{\rho_c} \le e_* + e_\theta + c_* + c_\theta + 0.005041\cosh(2\pi\rho_c).$$

Here the available reference-error bounds hold on wider strips, so they also bound this common strip and the real torus. Setting the hyperbolic cosine factor to one gives the real estimate. At phase zero the reference q_1 difference is exactly 0.005041; subtracting the same four error bounds gives (S133). The exact interval arithmetic for these comparisons is included in the reproduction record. Finally, both equations in (S130) give the same phase evolution $\vartheta(\tau) = \vartheta_0 + \varpi\tau$. Embeddedness makes F well defined and invertible and proves the flow conjugacy and uniform-in-time bound.

The correction estimate measures the distance from the nominal Fourier candidate to its exact nominal torus. It is not the error between the learned and physical models. The two embeddings have the same frequency but different amplitudes and need not coincide. This result does not establish persistence under damping or magnetic forcing, nor does it identify the configuration-dependent-mass rod with the fixed-mass nominal target.

S3.10 Configuration-dependent inertia and its fourth-order invariant

Configuration-dependent inertia changes both the canonical momentum and the coefficients entering the Birkhoff invariant. The following results give the corresponding reduction and fourth-order calculation for an analytic mechanical model. They do not identify its coefficients with those of the fixed-mass two-mode target used in the numerical certificate.

Proposition S24. Canonical reduction of a kinetic metric. *Let $q \mapsto (r(s,q), D(s,q))$ be an autonomous analytic immersion into admissible rod configurations, with fixed essential boundary conditions. Suppose the line density ρA and body inertia $\mathcal{J}_b$ are positive, and define*

$$r_i = \partial_{q_i} r, \qquad a_i = \operatorname{vee}(D^{\mathsf{T}} \partial_{q_i} D),$$
$$M_{ij}(q) = \int_0^L \left(\rho A r_i \cdot r_j + a_i^{\mathsf{T}} \mathcal{J}_b a_j\right) ds. \tag{S134}$$

Assume the integral and its local derivatives exist, so that M is analytic, and let V be an analytic scalar potential. The variationally restricted system has canonical variables $p = M(q)\dot{q}$ and Hamiltonian

$$H(q,p) = \tfrac{1}{2} p^{\mathsf{T}} M(q)^{-1} p + V(q). \tag{S135}$$

For the pulled-back nonconservative force Q, its equations are

$$\dot{q} = v = M^{-1} p,$$
$$\dot{p}_i = \tfrac{1}{2} v^{\mathsf{T}} (\partial_i M) v - \partial_i V + Q_i, \tag{S136}$$

and $\dot{H} = v^{\mathsf{T}} Q$. Equivalently,

$$M\dot{v} + c(q,v) + \nabla V = Q,$$
$$c_i(q,v) = \sum_{j,k} (\partial_k M_{ij}) v_j v_k - \tfrac{1}{2} v^{\mathsf{T}} (\partial_i M) v. \tag{S137}$$

Proof. For any $b \neq 0$, immersion implies that the corresponding translational and rotational tangent fields cannot both vanish identically. Positivity of their kinetic weights gives $b^{\mathsf{T}} M(q) b > 0$. The inverse exists locally and is analytic. If π_r and ℓ_b are the linear and body angular momentum densities, the restricted virtual-work one-form is

$$\int_0^L (\pi_r \cdot \delta r + \ell_b \cdot a_i \delta q_i) ds = \sum_i p_i \delta q_i,$$
$$p_i = \int_0^L (r_i \cdot \pi_r + a_i \cdot \ell_b) ds.$$

Repeated indices in this one-form are summed. For compatible velocities this is exactly $p = M\dot{q}$. The Legendre transform of $\dot{q}^{\mathsf{T}} M \dot{q}/2 - V$ therefore gives (S135) in the canonical cotangent coordinates. Differentiating $MM^{-1} = I$ yields $\partial_i M^{-1} = -M^{-1}(\partial_i M) M^{-1}$, proving (S136). Differentiating $p = Mv$ gives (S137). Substitution into $\dot{H}$ cancels the Hamiltonian terms and leaves $v^{\mathsf{T}} Q$.

For example, an admissible unshearable local chart is

$$D(s,q) = D_0(s) \exp\big(\widehat{\psi(s,q)}\big),$$
$$r(s,q) = r_b + \int_0^s [1 + e(\sigma,q)] D(\sigma,q) e_3 \, d\sigma, \tag{S138}$$

with analytic ψ, e, $\psi(0,q) = 0$, and $e > -1$; $e = 0$ imposes inextensibility as well. The hat denotes the cross-product matrix. The identity $D^{\mathsf{T}} r_s = (1+e)e_3$ is exact. The rotational tangent is $a_i = J_r(\psi)\partial_i \psi$, where $J_r(\psi) = \int_0^1 \exp(-t\widehat{\psi})\, dt$ is analytic at zero. Local immersion remains a separate condition on the chosen basis. In finite-dimensional notation a chart $U = F(q)$, with $B = DF(q)$, pulls back a full momentum covector by

$$p = B^{\mathsf{T}} P_{\text{full}}, \qquad \dot{p} = B^{\mathsf{T}} \dot{P}_{\text{full}} + (DB[\dot{q}])^{\mathsf{T}} P_{\text{full}}. \tag{S139}$$

Thus a varying chart requires the second transport term. A dimensional restriction is not an invertible symplectic change of the full phase space: agreement with unrestricted dynamics requires either an invariant restriction or a bound on the omitted residual.

Theorem S25. Complete kinetic contribution to the fourth-order normal form. *Suppose* (S135) *is expanded at a conservative equilibrium $(q,p) = (0,0)$, with $DV(0) = 0$ and $M_0, K_0 \succ 0$. Write the homogeneous Taylor terms as*

$$M(q) = M_0 + M_1(q) + M_2(q) + O(|q|^3),$$
$$M_1(q) = \sum_i (\partial_i M)(0) q_i,$$
$$M_2(q) = \tfrac{1}{2} \sum_{i,j} (\partial_{ij} M)(0) q_i q_j,$$
$$V(q) = V_0 + \tfrac{1}{2} q^{\mathsf{T}} K_0 q + V_3(q) + V_4(q) + O(|q|^5). \tag{S140}$$

Put $G_0 = M_0^{-1}$. In total phase-space degree the Hamiltonian is $H = V_0 + H_2 + H_3 + H_4 + O(|(q,p)|^5)$, where

$$H_2 = \tfrac{1}{2} p^{\mathsf{T}} G_0 p + \tfrac{1}{2} q^{\mathsf{T}} K_0 q,$$
$$H_3 = V_3 - \tfrac{1}{2} p^{\mathsf{T}} G_0 M_1 G_0 p,$$
$$H_4 = V_4 + \tfrac{1}{2} p^{\mathsf{T}} (G_0 M_1 G_0 M_1 G_0 - G_0 M_2 G_0) p. \tag{S141}$$

Let an orthogonal U diagonalize $M_0^{-1/2}K_0M_0^{-1/2}$ as $U^{\mathsf{T}}M_0^{-1/2}K_0M_0^{-1/2}U=\Omega^2$, where $\Omega=\operatorname{diag}(\omega_1,\dots,\omega_n)\succ 0$. Set $L=M_0^{-1/2}U\Omega^{-1/2}$ and use the canonical substitution $q=LQ$, $p=L^{-\mathsf{T}}P$ in all three polynomials. In the variables $z=(Q+iP)/\sqrt{2}$, assume

$$k\cdot\omega\neq 0 \quad \textit{for every } k\in\mathbb{Z}^n\setminus\{0\},\quad |k|_1\le 4. \tag{S142}$$

For a coefficient h_{ab} of $z^a\bar z^b$ in the transformed cubic polynomial, define

$$(\chi_3)_{ab}=\frac{ih_{ab}}{(a-b)\cdot\omega}. \tag{S143}$$

The fourth-order Birkhoff invariant is

$$Z_4=\Pi_{\mathrm{res}}\left(H_4+\tfrac12\{H_3,\chi_3\}\right)=\tfrac12 I^{\mathsf{T}}AI,\qquad I_j=z_j\bar z_j, \tag{S144}$$

where Π_{res} retains exactly $a=b$ and $\{f,g\}=-i\sum_j(f_{z_j}g_{\bar z_j}-f_{\bar z_j}g_{z_j})$. In particular, $A_{ii}=2[I_i^2]Z_4$ and $A_{ij}=[I_iI_j]Z_4$ for $i\neq j$.

Proof. Keeping matrix products in their stated order, multiplication by M verifies

$$M^{-1}=G_0-G_0M_1G_0+G_0M_1G_0M_1G_0 - G_0M_2G_0+O(|q|^3).$$

Multiplying by $p^{\mathsf{T}}/2$ and p raises each configuration degree by two, yielding (S141). The substitution preserves $p^{\mathsf{T}}dq=P^{\mathsf{T}}dQ$ and gives $H_2=\sum_j\omega_jz_j\bar z_j$. Since $\{H_2,z^a\bar z^b\}=i((a-b)\cdot\omega)z^a\bar z^b$, (S143) solves $\{H_2,\chi_3\}=-H_3$. There are no cubic resonant monomials under (S142). The time-one Hamiltonian flow of χ_3 transforms the degree-four term into

$$H_4+\{H_3,\chi_3\}+\tfrac12\{\{H_2,\chi_3\},\chi_3\}=H_4+\tfrac12\{H_3,\chi_3\}.$$

Another homogeneous generating polynomial of degree four removes its nonresonant terms. The remaining terms have $a=b$ by (S142), and therefore form the quadratic polynomial in I displayed in (S144). These polynomial flows are analytic on a sufficiently small neighbourhood. Reality of the original polynomials ensures reality of the transformations and of A. Terms omitted in this argument have total degree at least five.

Both the first and second derivatives of M affect A. The first derivatives enter H_3, the quadratic product in H_4, and the cubic elimination bracket. Consequently the magnetic quartic potential alone does not determine nondegeneracy of a model with variable inertia. The theorem computes its invariant; it does not assert $\det A\neq 0$ without values or bounds for the specified coefficients.

Corollary S26. One-dimensional invariant. *For $M(q)=1+\alpha q+\beta q^2+O(q^3)$ and $V(q)=\omega^2q^2/2+\lambda q^3+\mu q^4+O(q^5)$, $\omega>0$, the normal form $H=\omega I+AI^2/2+O(|(Q,P)|^5)$ has*

$$A=\frac{\alpha^2}{8}-\frac{\beta}{2}+\frac{3\mu}{\omega^2}+\frac{3\alpha\lambda}{2\omega^2}-\frac{15\lambda^2}{2\omega^4}. \tag{S145}$$

Proof. An independent canonical flattening is $y=\int_0^q\sqrt{M(s)}\,ds$ and $p_y=p/\sqrt{M(q)}$. It preserves $p\,dq=p_y\,dy$ and changes the kinetic energy exactly to $p_y^2/2$. Inverting its Taylor series gives $q=y-\alpha y^2/4+(\alpha^2-\beta)y^3/6+O(y^4)$. The new cubic and quartic potential coefficients are

$$\widetilde\lambda=\lambda-\omega^2\alpha/4,$$
$$\widetilde\mu=\mu-3\alpha\lambda/4+\omega^2(19\alpha^2/96-\beta/6).$$

For unit mass, direct cubic elimination gives $A=3\widetilde\mu/\omega^2-15\widetilde\lambda^2/(2\omega^4)$. Substitution yields (S145), agreeing with (S141)–(S144).

Proposition S27. Axis probes do not identify the kinetic two-jet. *Let mass observations be available at the origin and at $q=\pm h_\ell e_j$, with $\ell=1,2$ and $j=1,\dots,n$, where $h_1,h_2>0$. Even exact values of all kinetic energies at these configurations, together with analyticity and local positive definiteness, do not determine $D^2M(0)$ or the fourth-order invariant of* (S135).

Proof. Fix a coordinate j and a symmetric matrix C. The analytic perturbation

$$\Delta M(q)=\frac{q_j^2(q_j^2-h_1^2)(q_j^2-h_2^2)}{h_1^2h_2^2}\,C \tag{S146}$$

vanishes at every observed configuration, as do its changes to both central first differences. It preserves $M(0)$ and $DM(0)$, whereas $\partial_{jj}\Delta M(0)=2C$. For every fixed C, continuity and $M(0)\succ 0$ preserve positivity in some neighbourhood of zero. No common prescribed radius is claimed. Thus the two-jet is not identified. In the one-dimensional subcase $C=c$, this changes β by c, so (S145) changes A by $-c/2$ without changing any observed energy. Embedding this example in a diagonal multi-dimensional oscillator gives the same failure for the general class. Extra derivative bounds or an independently specified mass function are therefore necessary to infer the invariant.

The archived eight-coordinate kinetic diagnostic uses precisely this $1+4n=33$ configuration design, with steps 10^{-7} and 5×10^{-8}. Its centre is a recorded transient state rather than a certified conservative equilibrium. The saved energy values permit reconstruction of a kinetic matrix at each probe; they provide neither a certified second derivative at an equilibrium nor a compatible potential four-jet. A finite-difference agreement test does not remove the obstruction in Proposition S27. Moreover, no verified nonlinear cotangent map identifies this historical eight-coordinate chart with the two-mode chart of the present trained Hamiltonian. These are coefficient and model-identification conditions, separate from the proved algebra in Theorem S25. A separate four-coordinate archive contains mixed finite-difference mass derivatives and fitted potential derivatives at a recentered state. These provide numerical regression targets, but no certified derivative remainder or identification with the present two-mode chart.

Proposition S28. Fixed-mass comparison on a common canonical chart. *Let* $H_{\mathrm{var}}=p^{\mathsf T}M(q)^{-1}p/2+V(q)$ *and* $H_{\mathrm{fix}}=p^{\mathsf T}G_0p/2+V_{\mathrm{fix}}(q)$ *be expressed on the same chart and in the same units. Suppose, on the domain in question,*

$$\begin{aligned}\|M_0^{-1/2}(M-M_0)M_0^{-1/2}\|_2\le\eta<1,\quad |p|_2\le P,\\ \max_i\|\partial_iM\|_2\le m_1,\quad |V-V_{\mathrm{fix}}|\le v_0,\\ \max_i|\partial_i(V-V_{\mathrm{fix}})|\le v_1.\end{aligned}$$

Writing $e=H_{\mathrm{var}}-H_{\mathrm{fix}}$ *gives*

$$\begin{aligned}|e|&\le\tfrac12P^2\|G_0\|_2\frac{\eta}{1-\eta}+v_0,\\ \|\partial_pe\|_2&\le P\|G_0\|_2\frac{\eta}{1-\eta},\\ \max_i|\partial_{q_i}e|&\le\tfrac12P^2\left(\frac{\|G_0\|_2}{1-\eta}\right)^2m_1+v_1.\end{aligned}\tag{S147}$$

Proof. Writing $M=M_0^{1/2}(I+E)M_0^{1/2}$, the Neumann bound gives $\|M^{-1}\|_2\le\|G_0\|_2/(1-\eta)$ and $\|M^{-1}-G_0\|_2\le\|G_0\|_2\eta/(1-\eta)$. Apply these estimates to the energy and momentum derivatives. The configuration derivative follows from $\partial_iM^{-1}=-M^{-1}(\partial_iM)M^{-1}$.

These estimates can be added to a learned-to-fixed error bound only after the common-chart hypotheses have been verified. The stored eight-coordinate probes alone do not supply the uniform constants or the chart identification required for that substitution. Accordingly, the reduction and normal-form theorem alone does not certify that historical native model. Section S3.11 instead defines a complete analytic planar restriction and verifies its coefficients directly. The earlier numerical torus certificates retain their specified fixed-mass target and frozen learned energy.

S3.11 An explicit planar Kirchhoff restriction with variable inertia

Consider a planar variational restriction of the rod equations with configuration-dependent inertia. It uses the declared material constants and the two frozen Hermite basis functions of the small-slope example. This is a separate physical specialization; the preceding COMSOL training and control results retain their specified models. Let $f_i=S_i\phi_i$ include the recorded displacement scale, and put $\beta_i=f_i'$. The fixed-end conditions give $f_i(0)=\beta_i(0)=0$. Define the director and centreline by

$$\begin{aligned}\theta(s,X)&=\beta(s)^{\mathsf T}X,\qquad D(s,X)=\exp(\theta(s,X)\widehat{e}_2),\\ r(s,X)&=\int_0^s(\sin\theta(\sigma,X),0,\cos\theta(\sigma,X))^{\mathsf T}d\sigma.\end{aligned}\tag{S148}$$

Thus $D^{\mathsf T}r_s=e_3$ holds exactly: the restriction is unshearable and inextensible, with fixed base and planar bending. The free-end magnetic moment enters through the potential. Rotational inertia is retained. The displacement $\sum_i f_iX_i$ is the transverse tangent at $X=0$; the finite-amplitude centreline is the integral in (S148). Write $\Delta\theta=\theta(\sigma,X)-\theta(\zeta,X)$. The full kinetic metric is

$$\begin{aligned}M_{ij}(X)=\rho A\int_0^L\int_0^s\int_0^s\beta_i(\sigma)\beta_j(\zeta)\cos(\Delta\theta)\,d\sigma\,d\zeta\,ds\\ +\rho I\int_0^L\beta_i(s)\beta_j(s)\,ds,\end{aligned}\tag{S149}$$

and the conservative potential is

$$\begin{aligned}V(X)=\frac{EI}{2}\int_0^L(\beta'(s)^{\mathsf T}X)^2ds\\ +E_0[1-\cos(d^{\mathsf T}X)].\end{aligned}\tag{S150}$$

Here $d=\beta(L)$, $E_0=\mu B_0>0$, and the field is static and uniform. This finite-dimensional restriction excludes extension, torsion, contact, and nonconservative forcing; it does not assert that the two-dimensional configuration manifold is invariant under the unrestricted rod dynamics.

For $\tau=\omega_0t$, define

$$\begin{aligned}G(X)&=\frac{\omega_0^2}{E_0}M(X),\qquad P=G(X)X',\\ h_K(X,P)&=\tfrac12P^{\mathsf T}G(X)^{-1}P+V(X)/E_0.\end{aligned}\tag{S151}$$

The dimensional momentum conjugate to X is $(E_0/\omega_0)P$. Consequently (X,P) are canonical for h_K in time τ.

In particular, the earlier fixed-mass velocity-to-momentum labels cannot be reused by renaming their momentum coordinate.

Proposition S29. Analytic kinetic domain. *The metric G is entire and even in X. For real X, $\lambda_{\min}(G(X)) > 2.2257\,10^{-5}$. It is invertible throughout the closed complex polydisc $|X_i| \le 0.1$. Hence h_K is holomorphic on a neighbourhood of that polydisc, for arbitrary finite complex momenta, and is real analytic on the entire real phase space.*

Proof. The integrands are entire functions on finite intervals, with locally uniform bounds. Their integrals are therefore entire. The evenness follows from the cosine kernel. For real X the translational part is a Gram matrix and is nonnegative. Put $R = (\omega_0^2/E_0)\rho I \int_0^L \beta\beta^{\mathsf T} ds$. Exact polynomial integration proves $R \succ 0$ and $\det R/\operatorname{tr} R > 2.2257\,10^{-5}$, a lower bound for $\lambda_{\min}(R)$ and thus for $\lambda_{\min}(G(X))$.

For a quantitative complex-domain bound let $b_i = \max_s |\beta_i(s)|$, $B = b_1 + b_2$, $\kappa = \omega_0^2/E_0$, and $C_{ij} = \kappa\rho A L^3 b_i b_j/3$. The b_i are computed exactly by checking the endpoints and stationary point of each quadratic element polynomial. If $|X_i| \le r$, then $|\theta(\sigma, X) - \theta(\zeta, X)| \le 2rB$. Writing $G_2(X) = D^2G(0)[X, X]/2$ gives the entrywise bound

$$|G_{ij}(X) - G_{ij}(0) - (G_2)_{ij}(X)| \le C_{ij}\frac{(2rB)^4\cosh(2rB)}{24}. \quad \text{(S152)}$$

This follows by summing the absolute cosine series beyond degree two and using $\int_0^L s^2 ds = L^3/3$. Add the absolute coefficient bound for G_2 to the right-hand side of (S152) to obtain the nonnegative matrix $E(r)$. Entrywise absolute values then imply

$$\|G_0^{-1}(G - G_0)\|_\infty \le \big\||G_0^{-1}|E(r)\big\|_\infty =: \eta(r).$$

Outward interval evaluation at both 160 and 256 bits yields $\eta(0.1) < 0.002654 < 1$ and $\|G^{-1}\|_\infty < 6.165944$. The Neumann series proves invertibility; its strict margin and compactness also give an open complex neighbourhood. Positive real mass gives the corresponding local inverse around every real X.

Proposition S30. Exact kinetic two-jet and nondegeneracy. *For the frozen basis and constants specified above, the origin is an elliptic equilibrium of* (S151). *Its kinetic metric is positive definite for every real X. The first kinetic derivatives vanish. With*

$$B_{ik}(s) = \int_0^s \beta_i\beta_k\, d\sigma, \qquad C_{ikl}(s) = \int_0^s \beta_i\beta_k\beta_l\, d\sigma,$$

the second derivatives are

$$\partial_{kl}M_{ij}(0) = \rho A\int_0^L (B_{ik}B_{jl} + B_{il}B_{jk})\, ds - \rho A\int_0^L (f_iC_{jkl} + f_jC_{ikl})\, ds. \quad \text{(S153)}$$

The fourth-order Birkhoff invariant is nondegenerate. In the canonical normalization $h_2 = \omega_K \cdot I$, validated enclosures give

$$\begin{aligned} \omega_{K,1} &\in [0.99996796645, 0.99996796646],\\ \omega_{K,2} &\in [6.15072521693, 6.15072521695],\\ \det A_K &\in [-8.580864, -8.580863]\,10^{-4},\\ \sigma_{\min}(A_K) &> 0.00227500005. \end{aligned} \quad \text{(S154)}$$

The frequencies are nonresonant through degree four.

Proof. Differentiating (S148) shows that the translational kinetic term is a Gram matrix. The rotary Gram matrix $\rho I \int \beta\beta^{\mathsf T} ds$ is strictly positive definite, as verified by exact rational principal minors for the frozen basis. It is independent of X, so it proves the stated real-domain positivity. Both M and V are analytic even functions. The origin is therefore an exact equilibrium, $DM(0) = 0$, and the potential Hessian is positive definite by the same exact principal-minor check.

The translational tangent has components

$$r_i = \begin{pmatrix} f_i - \frac{1}{2}\sum_{k,l} C_{ikl}X_kX_l \\ 0 \\ -\sum_k B_{ik}X_k \end{pmatrix} + O(|X|^3),$$

which gives (S153). On each of the 32 elements f_i is cubic, β_i quadratic, and the displayed integrals are polynomial integrals with exact rational coefficients. Their antiderivatives retain all accumulated constants across elements. The stored binary64 basis and material constants are interpreted literally; products such as $E_0 = \mu B_0$ are performed exactly, and differences from previously rounded derived constants are retained.

There is no cubic Hamiltonian term. The quartic term is exactly

$$h_4 = -\frac{1}{4}\sum_{i,j,k,l}(G_0^{-1}(\partial_{kl}G)_0G_0^{-1})_{ij}X_kX_lP_iP_j - \frac{(d^{\mathsf T}X)^4}{24}. \quad \text{(S155)}$$

The spectral-projector normalization of (S127) is applied to the exact block Hessian $\operatorname{diag}(D^2(V/E_0)(0), G_0^{-1})$. Arb enclosures at 160 and 256 bits verify all nonzero divisors through degree four. Transforming (S155) and retaining $z^a\bar{z}^b$ with $a = b$ gives $I^{\mathsf{T}} A_K I/2$. Its homological identity and real structure are checked, and its determinant enclosure excludes zero. Finally $\sigma_{\min}(A_K) \geq |\det A_K|/\|A_K\|_F$ gives (S154).

The rounded matrix is

$$A_K \simeq \begin{pmatrix} 0.0007410266167 & -0.02407752390 \\ -0.02407752390 & -0.3756399220 \end{pmatrix}.$$

Corollary S31. Local invariant families of the planar rod. *Arbitrarily small neighbourhoods of the origin contain a family of analytic invariant Lagrangian two-tori of h_K whose union has positive four-dimensional measure.*

Proof. Proposition S29 supplies a holomorphic Hamiltonian near the exact elliptic equilibrium. The verified nonresonance and invertible A_K give the analytic normal form $\omega_K \cdot I + I^{\mathsf{T}} A_K I/2 + O(|(Q,P)|^5)$. The action-box, Diophantine-measure and joint-embedding argument in the proof of Corollary S21 applies with $A = A_K$, using Pöschel (2001, Theorems A–B). In particular, $r = \varepsilon^{5/2}$ and $\alpha = \varepsilon^{2+\delta}$, $0 < \delta < 1/4$, make its smallness quotient tend to zero and its joint torus embedding bi-Lipschitz on a positive-measure parameter set. The inverse normal-form map preserves positive measure.

The complex radius 0.1 proves analyticity, not a finite-amplitude KAM threshold. This corollary supplies no numerical lower bound for the admissible torus amplitudes.

Comparison with the frozen learned energy. The old and new Hamiltonians can be compared as functions of the same canonical coordinates (X, P), while retaining their different physical velocity-to-momentum maps. On the real box $|X_i|, |P_i| \leq 0.04$, the kinetic Neumann bounds and the exact quadratic and magnetic coefficient differences give

$$\begin{aligned} \sup |h_K - H_*| &< 8.622807\,10^{-6}, \\ \sup \|\nabla(h_K - H_*)\|_\infty &< 2.155701\,10^{-4}. \end{aligned} \tag{S156}$$

For completeness, differentiate the cosine kernel and subtract its linear sine term: the derivative remainder is bounded entrywise by $2b_k C_{ij}(2rB)^3 \cosh(2rB)/6$. Thus the coefficient bound for $\partial_k G_2$ plus this remainder bounds $\partial_k G$. Apply $\partial_k G^{-1} = -G^{-1}(\partial_k G)G^{-1}$ to bound the configuration gradient. The exact difference between G_0^{-1} and $\operatorname{diag}\nu$ is retained in the momentum and energy bounds. For the potential, retain the exact Hessian difference and bound the change in $F(u) = 1 - \cos u - u^2/2$ using $|F'(u)| \leq |u|^3 \cosh|u|/6$. These steps, evaluated with outward rounding, yield (S156). Combining this result with the certified frozen-to-nominal bounds of Section S3.5 on the containing box gives the deterministic comparison

$$\begin{aligned} \sup_{[-0.04,0.04]^4} |H_\theta - h_K| &< 8.644178\,10^{-6}, \\ \sup_{[-0.04,0.04]^4} \|\nabla(H_\theta - h_K)\|_\infty &< 2.195079\,10^{-4}. \end{aligned} \tag{S157}$$

The discrepancy is demonstrably nonzero. At the exact point $(X_1, X_2, P_1, P_2) = (0, 0, 0, 1/25)$, $\partial_P h_K = G_0^{-1}(0, 1/25)^{\mathsf{T}}$ and $\partial_X h_K = 0$. Direct interval evaluation of the same frozen deep network gives

$$|\partial_{P_2} H_\theta - \partial_{P_2} h_K| > 1.1136795\,10^{-4}. \tag{S158}$$

This point lies inside the previously certified fixed-mass box. Consequently its $3.937778\,10^{-6}$ conservative-field error bound cannot also hold for the new rod on that box. The witness concerns one velocity component in dimensionless canonical coordinates; it is not a trajectory or tip-position error. Equation (S157) gives a valid, larger uniform error bound for the frozen function. A native-data learning claim for this rod still requires matching physical trajectories and momentum labels; this deterministic comparison does not supply them.

Theorem S32. Conservative interpolation from the trained energy to the rod. *Let $F_\lambda = (1-\lambda)H_\theta + \lambda h_K$ for $0 \leq \lambda \leq 1$, using the exact frozen network and (S151) on the same canonical phase space. Each F_λ has a unique elliptic equilibrium in $[-10^{-8}, 10^{-8}]^4$, and its frequencies are nonresonant through order four. Its complete fourth-order invariant satisfies*

$$\det A_\lambda < -7.4388\,10^{-9}, \qquad \sigma_{\min}(A_\lambda) > 6.6235\,10^{-7}. \tag{S159}$$

For every λ, arbitrarily small neighbourhoods of that equilibrium contain positive-measure families of analytic invariant Lagrangian two-tori of F_λ.

Proof. It is necessary to control the complete energy at the shifted equilibria, rather than interpolate only the endpoint quartic matrices. On the four-dimensional complex polydisc of radius $R = 0.1$, Proposition S29 gives

$$\begin{aligned} |h_K| \leq R^2 \|G^{-1}\|_\infty + \frac{R^2}{2} \sum_{i,j} |K_{b,ij}| \\ + \cosh(R\|d\|_1) - 1 < 1, \end{aligned}$$

where $K_b = D^2(V/E_0)(0) - dd^{\mathsf{T}}$. Write $h_K = h_2 + h_4 + \mathcal{R}_6$ at the origin. Even parity and the exact four-jet imply that $\mathcal{R}_6$ starts at total degree six. Cauchy's coefficient estimate bounds each original degree-k coefficient by R^{-k}. For $|x_i| \le r$, $t = r/R$, and a multi-index $|\beta| = m \le 4$, provided $10t/(7-m) < 1$, it follows that

$$|[u^\beta]\mathcal{R}_6(x+u)| \le R^{-m}\binom{9}{m+3}\frac{t^{6-m}}{1-10t/(7-m)}. \tag{S160}$$

Indeed, the sum of the binomial factors at original degree k is $\sum_{|a|=k}\prod_j \binom{a_j}{\beta_j} = \binom{k+3}{m+3}$. The successive contributions have ratio at most $10t/(7-m)$ for $k \ge 6$. This ratio is strictly below one on every verified root box. The estimate bounds Taylor coefficients: a pure Hessian entry therefore carries the additional factor two.

Interval evaluation combines these remainders with the exact quadratic and quartic rod terms and the actual network derivatives. On 89 closed parameter intervals with exact endpoints covering $[0,1]$, a rational preconditioner gives a strictly self-mapping contraction for the equilibrium equation. A common real box has Hessian lower bound 0.99968, so the roots on overlapping parameter cells belong to the same unique branch. A common complex box with real and imaginary half-width 10^{-6} satisfies both the network pole margins and the rod's kinetic inverse-domain bound.

At each root enclosure, the spectral symplectic normalization and complete cubic elimination of Section S3.8 are applied to the full local jets. The resulting enclosures verify (S159) at both 160 and 256 bits. Independent endpoint evaluations reproduce the original deep and rod invariants. Finally the proof of Corollary S21 applies for each λ.

The theorem links the actual trained function to the specified variable-inertia energy through a nondegenerate conservative family. It does not continue a selected finite-amplitude torus at fixed frequency; that assertion would require separate a posteriori certification along the family.

S3.12 *Mechanical complex learning and a matched dissipative scheme*

The variable-inertia restriction admits a structured learning extension that keeps the physical momentum and the trained energy compatible. Following the mechanical decomposition of Zhong et al. (2020, Section 3.1), retain the specified kinetic metric and magnetic energy and learn a potential correction c_θ:

$$h_\theta^M(X,P) = \tfrac{1}{2}P^{\mathsf{T}}G(X)^{-1}P + v_K(X) + c_\theta(X), \tag{S161}$$

where $v_K = V/E_0$ is defined in (S150). This extension has its own parameters. It does not alter the previously fitted H_θ. The mass is known in this specialization; learning it again would introduce an additional inverse problem.

Proposition S33. Velocity observations in the mechanical chart. *Let G be differentiable and positive definite, and put $L_\theta(X,v) = v^{\mathsf{T}}G(X)v/2 - v_K(X) - c_\theta(X)$. For independent velocity observations $v = X'$ the momentum is $P = G(X)v$. With nonconservative force Q, the mechanical equations are equivalent to*

$$\begin{aligned} P' &= \partial_X L_\theta(X,v) + Q, \\ (\partial_X L_\theta)_i &= \tfrac{1}{2}v^{\mathsf{T}}(\partial_i G)v - \partial_i(v_K + c_\theta). \end{aligned} \tag{S162}$$

Here the partial derivative holds v fixed. On each time window $[a,b]$ these equations imply the exact residual identities

$$\begin{aligned} \mathcal{R}_P &= [G(X)v]_a^b - \int_a^b (\partial_X L_\theta + Q)\,d\tau = 0, \\ \mathcal{R}_X &= X(b) - X(a) - \int_a^b v\,d\tau = 0. \end{aligned} \tag{S163}$$

Proof. Euler–Lagrange differentiation gives $d(\partial_v L_\theta)/d\tau = \partial_X L_\theta + Q$ and $\partial_v L_\theta = Gv$. The Legendre transform is (S161), so these are precisely its forced canonical equations. Integrating gives (S163). The derivative of $G(X)v$ automatically includes its configuration transport term.

The velocity-coordinate construction uses the Euler–Lagrange automatic-differentiation component of Cranmer et al. (2020, Eqs. (4)–(6)). Equation (S163) permits training from position and velocity windows without acceleration labels. It uses the actual metric rather than a fixed modal momentum surrogate. Its quadrature, observation and projection errors remain additional quantities; a small computed window residual alone does not bound their size. This interface requires observations on the stated configuration manifold and the stated dimensionless time scale.

Proposition S34. A complex correction preserving the certified four-jet. *Let $\mathcal{A}_6 = \{\alpha \in \mathbb{N}^2 : |\alpha| = 6\}$ and define*

$$c_\theta(X) = \sum_{\alpha\in\mathcal{A}_6} X^\alpha \frac{N_{\theta,\alpha}(X) + N_{\theta,\alpha}(-X)}{2}. \tag{S164}$$

Each $N_{\theta,\alpha}$ is a real readout of two complex fully connected tanh *layers with the pole constraints of Section S3.6, evaluated at $(X,0)$. For every fixed finite admissible parameter vector, h_θ^M has the same equilibrium, quadratic normalization and fourth-order Birkhoff invariant as h_K.*

It therefore has positive-measure local families of analytic invariant Lagrangian two-tori.

Proof. There are seven monomials in $\mathcal{A}_6$. The coefficient networks are real analytic near zero, with holomorphic extensions, and their symmetrizations are even. Thus c_θ is even and vanishes through total degree five. Its addition changes none of the derivatives used in Propositions S29 and S30 at the origin. In particular $\det A_K \neq 0$ remains the same exact invariant. Applying the local KAM argument of Corollary S31 proves the assertion for each fixed network.

This parameterization fixes known low-order physics. It cannot identify an unknown quadratic or quartic correction, and its degree-six prefactors restrict its hypothesis class. A need to learn those lower orders would require a different parameterization and new coefficient verification. The local KAM neighbourhood may depend on the trained weights; no uniform finite-amplitude radius or accuracy guarantee is inferred from jet preservation. The supplied implementation verifies complex-layer training gradients. The corresponding fitted planar model and its post-fit certificates are reported in Sections S3.15, S3.16 and S3.17.

For the mass-proportional damping specialization choose a constant $\gamma \geq 0$, a prescribed open-loop magnetic input $u(\tau)$, and

$$\begin{aligned} \mathcal{D}(X, v) &= \tfrac{\gamma}{2} v^{\mathsf{T}} G(X) v, \\ Q &= -\partial_v \mathcal{D} + u\, g(X) = -\gamma P + u\, g(X), \\ g(X) &= d \cos(d^{\mathsf{T}} X). \end{aligned} \tag{S165}$$

This is the positive-semidefinite dissipation construction of Zhong et al. (2020, Eq. (3)) with $R(X) = \gamma G(X)$. It gives a power-consistent interpretation of a packaged pair $h_\theta^M + i\mathcal{D}$: the real energy and Rayleigh function are decoded separately using (S165). For the autonomous stored energy,

$$(h_\theta^M)' = -\gamma P^{\mathsf{T}} G^{-1} P + u\, g^{\mathsf{T}} G^{-1} P. \tag{S166}$$

The input is work, and the first term is nonpositive on the real domain.

Theorem S35. A Birkhoff representation with variable inertia. *Let $h(X, P, \tau)$ be a real C^2 Hamiltonian and consider*

$$X' = h_P, \qquad P' = -h_X - \gamma P. \tag{S167}$$

Write $z = (X, P)$ and

$$F = e^{\gamma\tau}(P, 0), \quad B = e^{\gamma\tau} h, \quad K_{ij} = \partial_i F_j - \partial_j F_i. \tag{S168}$$

Then $K = -e^{\gamma\tau} J$ is nondegenerate and $Kz' = \nabla_z B + \partial_\tau F$ reproduces (S167). Its flow obeys

$$D\Phi_{\tau_1,\tau_0}^{\mathsf{T}} K(\tau_1) D\Phi_{\tau_1,\tau_0} = K(\tau_0). \tag{S169}$$

For (S161) and (S165), take $h = h_\theta^M - u(\tau)\sin(d^{\mathsf{T}} X)$. No constant-mass or separable-kinetic-energy assumption is required.

Proof. The two block rows of the Birkhoff equation are $-e^{\gamma\tau} P' = e^{\gamma\tau}(h_X + \gamma P)$ and $e^{\gamma\tau} X' = e^{\gamma\tau} h_P$. For $A = D_z(z')$, symmetry of $D_z^2 h$ yields $A^{\mathsf{T}} J + JA = -\gamma J$. Differentiating $D\Phi^{\mathsf{T}} e^{\gamma\tau} J D\Phi$ proves (S169). Equivalently, $\Pi = e^{\gamma\tau} P$ gives the canonical Hamiltonian $\widetilde{h}(X, \Pi, \tau) = e^{\gamma\tau} h(X, e^{-\gamma\tau}\Pi, \tau)$.

The construction specializes the time-dependent Birkhoff structures of Su and Qin (2003, Eqs. (28), (30)). Both endpoint times in (S169) are essential. In the original phase space the flow is conformally symplectic. This result establishes a dissipative geometric structure; conservative KAM theorems do not therefore become theorems for that dissipative flow.

Proposition S36. Matched second-order discrete structure. *For a step h_s, let $D_a(X, P) = (X, e^{-\gamma a} P)$ and let M_{h_s} be implicit midpoint for the Hamiltonian field of $h(\cdot, \tau + h_s/2)$, with the input value prescribed independently of the state. On a locally unique implicit solution branch, define*

$$\Psi_{h_s} = D_{h_s/2} \circ M_{h_s} \circ D_{h_s/2}. \tag{S170}$$

Then $(D\Psi_{h_s})^{\mathsf{T}} J D\Psi_{h_s} = e^{-\gamma h_s} J$. The method is second order for smooth h and input on a regular branch.

Proof. At the implicit midpoint let $A = JD^2 h$ and $a = h_s/2$. Its derivative is $(I - aA)^{-1}(I + aA)$. Since $A^{\mathsf{T}} J + JA = 0$, this derivative is symplectic. Each $D_{h_s/2}$ scales J by $e^{-\gamma h_s/2}$, giving the stated product. Midpoint has second-order local consistency, and the symmetric splitting with the exact damping flows retains order two. The same argument applies to the nonautonomous system after adjoining the clock variable. Local invertibility follows for sufficiently small steps from the implicit-function theorem.

These conditions distinguish the matched specialization from the earlier general joint decoder. For mechanical h, a momentum force $-R(X)h_P$ has conformal factor γ in

this fixed canonical structure only if $R(X)G(X)^{-1} = \gamma I$, hence $R = \gamma G$. This follows from the mixed block of $A^{\mathsf{T}}J + JA$. Position-dependent damping outside this class can still satisfy the power identity, but requires a different structure analysis. Likewise an arbitrary feedback law cannot be substituted for the prescribed open-loop midpoint input in the proposition without checking its state derivatives. The existing frozen joint network and feedback experiments retain their earlier verified scope.

The implementation supplies a differentiable full-geometry kinetic evaluator, the complex potential component, the velocity-window residual and the matched implicit step. A positive scalar $\gamma_\eta = \log(1 + e^\eta)$ can be optimized jointly with the complex potential using those window residuals. For each frozen parameter vector it is constant in space and time, as required by Theorem S35. Both parameter groups have nonzero implementation-test gradients; no optimizer-convergence or fitting accuracy is inferred from that check. Floating-point Gauss quadrature and automatic differentiation are checked against the exact rod jets and by quadrature refinement. These are implementation checks; the analytic-rod existence results use the exact polynomial integration and outward interval bounds of the preceding section.

S3.13 A training rule matched to the mechanical model

The smooth-descent argument associated with Zhang et al. (2022, Theorem 3) can be realized by a specified optimizer for (S161). This requires control of derivatives with respect to the trainable parameters, rather than only the state derivative bounds used for KAM. The following construction supplies that control on the relevant optimization sets without imposing a global parameter-Hessian estimate.

Let $\xi \in \mathbb{R}^d$ collect every trainable real and imaginary weight, bias, and the scalar η, with $\gamma_\eta = \log(1 + e^\eta)$. Keep the mechanical metric, the normalization, and a finite collection of observed windows fixed. All positions lie in the declared real box of the coefficient networks, and each window has a strictly positive duration. Let $\widehat{\mathcal{R}}_{P,j}(\xi)$ and $\widehat{\mathcal{R}}_{X,j}$ be the fixed-quadrature versions of (S163). Choose fixed positive weights w_j with $\sum_j w_j = 1$, fixed positive component scales, and let S_j denote the corresponding diagonal residual-scaling matrix, including the inverse window duration. Define

$$F_\lambda(\xi) = \sum_j w_j \left\| S_j \begin{pmatrix} \widehat{\mathcal{R}}_{P,j}(\xi) \\ \widehat{\mathcal{R}}_{X,j} \end{pmatrix} \right\|_2^2 + \frac{\lambda}{2}\|\xi\|_2^2, \qquad \lambda > 0. \tag{S171}$$

The penalty includes all raw trainable coordinates, including η. It changes the fitted objective and introduces regularization bias. The velocity residual is independent of the learned potential when the metric and observations are fixed; it remains a diagnostic of the supplied chart and sampling, not an additional learned quantity.

Theorem S37. Full-batch descent for the mechanical complex network. *Use the smooth imaginary-row constraints of the implemented complex layers, with $\sqrt{a^2 + \epsilon^2}$ and fixed $\epsilon > 0$, and the degree-six potential of (S164). Assume the fixed metric matrices used to evaluate the windows are positive definite. Choose $0 < c < 1$, $0 < \beta < 1$, and $a_{\max} > 0$. At every iteration evaluate the same full objective and its real gradient $g_k = \nabla_\xi F_\lambda(\xi_k)$. If $g_k \neq 0$, choose the first $a_k = a_{\max}\beta^j$, $j = 0, 1, \ldots$, satisfying*

$$F_\lambda(\xi_k - a_k g_k) \le F_\lambda(\xi_k) - c a_k \|g_k\|_2^2, \qquad \xi_{k+1} = \xi_k - a_k g_k. \tag{S172}$$

In exact arithmetic every search terminates. The iterates are bounded, the step sizes have a positive uniform lower bound, and $\|\nabla_\xi F_\lambda(\xi_k)\|_2 \to 0$. Every accumulation point is stationary for (S171). At every iterate and every accumulation point, the conservative mechanical energy retains the local KAM conclusion of Proposition S34.

Proof. For each fixed observed position the constrained affine layers keep the imaginary preactivations strictly below their prescribed bounds, uniformly separated from the poles of complex $\tanh$. The smoothed row denominators are positive. Consequently the coefficient networks, their position derivatives, and γ_η are smooth functions of all finite raw parameters. Fixed finite quadrature and fixed positive metric matrices therefore make F_λ a nonnegative C^2 function on $\mathbb{R}^d$.

Put $F_0 = F_\lambda(\xi_0)$. The level set $\mathcal{L} = \{\xi : F_\lambda(\xi) \le F_0\}$ is compact, since $\|\xi\| \le R_0 = (2F_0/\lambda)^{1/2}$ there. Let G_0 bound $\|\nabla F_\lambda\|$ on this level set. Every trial segment from an accepted iterate lies in the closed ball of radius $R_0 + a_{\max}G_0$. Continuity of the parameter Hessian on that ball gives a finite bound $L_0 > 0$. Taylor's formula along a trial segment then yields

$$F_\lambda(\xi - ag) - F_\lambda(\xi) \le -a(1 - L_0 a/2)\|g\|^2.$$

Thus every $a \le 2(1-c)/L_0$ satisfies (S172), and the accepted step obeys

$$a_k \ge a_{\min} := \min\{a_{\max}, 2\beta(1-c)/L_0\} > 0.$$

Induction keeps all accepted iterates in $\mathcal{L}$. Summing their decreases gives, for every $K \ge 1$,

$$ca_{\min} \sum_{k=0}^{K-1} \|g_k\|^2 \le F_0, \qquad \min_{k<K} \|g_k\|^2 \le \frac{F_0}{ca_{\min}K}. \tag{S173}$$

The summability proves gradient convergence; compactness and continuity prove stationarity of every accumulation point. If an iterate has zero gradient, keep it fixed and the conclusions remain valid. Each finite parameter vector still gives an analytic degree-six correction with unchanged four-jet. The same is true at any finite accumulation point, so Proposition S34 applies to their conservative energies.

The real-coordinate gradient is equivalent to the complex convention $g_w = \nabla_a F + i\nabla_b F = 2\partial_{\bar{w}} F$ for $w = a + ib$. The theorem uses a real positive step and the full objective; it does not apply retroactively to the earlier Adam/L-BFGS fits. It proves stationarity, not convergence to a global optimum, vanishing data residual, or identification of unknown low-order physical terms. The KAM statement remains local for each frozen conservative energy; no numerical finite-amplitude radius, common torus, or dissipative KAM conclusion is inferred from the optimizer.

Corollary S38. Uniform local existence on the optimization level set. *For the fixed initial loss and regularization in Theorem S37, there exists $\varepsilon_0 > 0$ such that every conservative energy whose raw parameters belong to $\mathcal{L}$ has positive-measure invariant two-torus families at each amplitude scale $0 < \varepsilon < \varepsilon_0$, on the positive action boxes used in Corollary S21. This common existence scale applies to all accepted iterates and their accumulation points. It is not a shared embedding or a prescribed common frequency.*

Proof. The set $\mathcal{L}$ is compact. At real X in a fixed smaller closed box about zero, the first and second complex preactivations have a uniform positive distance from their poles. Their dependence on the real parameters and on complexified X is continuous. Compactness therefore supplies a common smaller complex ball on which every coefficient network and its real-energy complexification are holomorphic and uniformly bounded. Intersect it with the verified complex domain of the physical kinetic energy. Cauchy estimates now give a uniform bound for the degree-six-and-higher remainder.

The equilibrium, linear frequencies, nonresonant divisors through order four, and invertible twist matrix are the same throughout $\mathcal{L}$. The fixed finite-order symplectic normalization and the action–angle scaling in Corollary S21 consequently have uniform constants. In that proof the remainder and quadratic frequency expansion obey the common $O(\varepsilon^5)$ upper bound, and both $O(\varepsilon^{1/2-\delta})$ smallness and $O(\varepsilon^{1/2-2\delta})$ joint embedding estimates tend to zero uniformly for $0 < \delta < 1/4$. The excluded relative frequency measure is uniformly $O(\varepsilon^{\delta})$. Choosing one sufficiently small ε_0 proves the result.

This corollary uses compactness in addition to jet preservation. It supplies a uniform qualitative scale, but does not compute its value or certify a requested physical amplitude. Its tori belong to the conservative members of the model class, irrespective of their fit to the observed system.

The theorem and corollary also hold with a fixed finite nonnegative damping coefficient, omitting η from ξ, provided every remaining trainable raw coordinate is penalized. They remain valid when the fixed base potential is replaced by a fixed analytic even potential differing from v_K by $O(|X|^6)$, with the same kinetic energy, and when the degree-six correction is multiplied by a fixed finite factor. In particular,

$$v_4(X) = \tfrac{1}{2}X^{\mathsf{T}}K_bX + \tfrac{1}{2}(d^{\mathsf{T}}X)^2 - \tfrac{1}{24}(d^{\mathsf{T}}X)^4$$

satisfies $v_K - v_4 = (d^{\mathsf{T}}X)^6/720 + O(|X|^8)$. The unchanged four-jet and compact parameter sets give the same local conservative conclusion; stationarity refers to the remaining trainable coordinates. Fixed finite quadrature, including the composite Simpson rule, defines the same smooth finite-objective setting; quadrature accuracy remains a separate requirement.

The KAM assertions concern the continuous analytic energy (S161). Fixed quadrature defines the computed training objective and selects its parameters; transferring finite-precision prediction errors to that continuous energy still requires the corresponding quadrature error bounds.

The supplied implementation uses `logaddexp` for the smooth positive damping map and restores the preceding parameters if a finite numerical line search fails. It records the measured objective, gradient norm and Armijo margin at every accepted step. These floating-point records check the implementation, while the convergence claim concerns the exact algorithm above. A finite accepted run is not an infinite-iteration convergence certificate. Tests include a quadratic problem with a known minimizer and the actual

mechanical window objective; they do not constitute a new physical training dataset or a new fitted COMSOL model.

Finally, compactness of an optimization level set is not by itself the statistical input of Section S3.7. That section requires a hypothesis class fixed independently of its iid samples, a justified complexity bound, and conservative-gradient label errors. A radius inferred from the realized training loss is data dependent. The fixed correlated windows used here retain their deterministic interpretation. Thus optimization convergence, conservative KAM structure, and physical identification remain distinct claims with their stated dependencies.

S3.14 Local nondegeneracy for the implemented quadrature metrics

The metric used in the mechanical implementation is a nested eight-point Gauss approximation of (S149). Its local conservative nondegeneracy can be checked for the implemented coefficients, without identifying that function with the continuous integral. First consider ideal Gauss nodes and weights in exact arithmetic. Each element function β_i is a polynomial of degree at most two. The coefficient of a state monomial of degree n in a centreline tangent integral therefore has element-polynomial degree at most $2n+3$. Accumulated integrals from preceding elements contribute constants. A product of total state degree $2k$ has element-polynomial degree at most $4k+6$. Consequently the outer integrands determining $G(0)$ and $D^2G(0)$ have degrees at most six and ten, respectively; the inner integrands needed for the latter have degree at most six. An eight-point Gauss rule integrates polynomials through degree 15 exactly. The rotational Gram integrand has degree four. Thus the ideal rule preserves the kinetic two-jet in (S153) and the Hamiltonian four-jet in (S155).

The stored numerical constants require a separate check. We consider three specified analytic functions: the real-arithmetic interpretation of the binary64 buffers in the training evaluator; the exported COMSOL expression graph with its decimal literals interpreted exactly; and the same graph after each literal is rounded to binary64. The exporter precomputes some products before printing the literals, so these functions need not be identical. In each interpretation the constants are converted to exact rational numbers. Trigonometric expressions are then expanded symbolically through degree two in X, giving exact $G(0)$ and $D^2G(0)$. The stored potential coefficients give the corresponding exact potential four-jet. This procedure defines an analytic function for each set of frozen constants; it does not regard floating-point evaluations of that function as exact.

For all three functions, exact rational tests give $G(0) \succ 0$ and a positive-definite potential Hessian. Parity gives $DG(0) = 0$ and $h_3 = 0$. Applying the interval symplectic normalization and quartic homological calculation of Proposition S30, independently at 160 and 256 bits, gives the common enclosures

$$\begin{aligned} \omega_1 &\in [0.99996796645, 0.99996796646], \\ \omega_2 &\in [6.15072521693, 6.15072521695], \\ \min_{0<|k|_1\le 4} |k\cdot\omega| &> 0.9999679, \\ \det A &\in [-8.580864, -8.580863]\,10^{-4}. \end{aligned} \tag{S174}$$

The maximum entrywise differences from the continuous-model exact jets are less than $6.4\,10^{-16}$ for $G(0)$ and $2.9\,10^{-17}$ for $D^2G(0)$. These differences are not set to zero: the nondegeneracy tests use each function's own jets. The exact coefficients, outward enclosures, and source hashes are recorded in the accompanying reproducibility material.

Each frozen metric is analytic and is invertible on some complex neighbourhood of the origin, since its value there is positive definite. The analytic elliptic KAM argument of Corollary S31 therefore applies separately to each associated conservative Hamiltonian. Replacing its full-cosine potential by its quartic Taylor polynomial and adding an analytic even $O(|X|^6)$ network correction leaves the certified four-jet unchanged. The resulting frozen conservative energies consequently retain positive-measure families of invariant two-tori at sufficiently small amplitudes. On a compact raw-parameter set, the uniform local argument of Corollary S38 applies with the chosen quadrature metric held fixed.

This establishes local conservative nondegeneracy for the actual coefficient conventions. It does not transfer the continuous model's quantitative complex-domain bounds to these functions, compute a common numerical amplitude, or bound the quadrature error throughout the observation domain. Solver, rounding, and velocity-to-momentum transport errors remain separate requirements for interpreting COMSOL observations and fitted predictions. The certificate also does not establish equivalence to a three-dimensional beam model or a KAM result for the damped trajectories.

S3.15 A matched planar COMSOL identification prototype

This experiment uses the planar inextensible Kirchhoff restriction of Section S3.11, with the implemented Gauss-eight metric examined in Section S3.14. COMSOL Global

Equations integrate its four canonical equations with fixed mass-proportional damping $\gamma = 0.035$ and zero input. The learning model retains that metric and damping but replaces the full potential by its prescribed quartic part v_4 and a trainable correction. Thus the deliberately hidden target is

$$R(s) = 1 - \cos s - \tfrac{1}{2}s^2 + \tfrac{1}{24}s^4, \qquad s = d^{\mathsf{T}} q.$$

This is a reduced-model identification benchmark; it does not use the archived three-dimensional finite-element beam as its data source.

Seven even coefficient networks multiply the degree-six monomials $q_1^a q_2^{6-a}$, $a = 0, \dots, 6$, with fixed amplitude $a_0 = 10^{-5}$. Each network has two width-two complex fully connected hidden layers, the constrained complex $\tanh$ activation, and a real energy readout. All 252 real raw coordinates are trained. The analytic target is excluded from the supervision and initialization.

Proposition S39. Approximation capacity of the frozen architecture. *In exact real arithmetic this potential class contains* $R_6(s) = s^6/720$. *For* $0 < |s| \le A < 1$,

$$\frac{\|\nabla_q (R - R_6)\|}{\|\nabla_q R\|} \le \frac{A^2}{42 - A^2}, \qquad \frac{|R - R_6|}{|R|} \le \frac{A^2}{56 - A^2}.$$

The force estimate holds in every vector norm; both ratios extend continuously by zero on $s = 0$.

Proof. For coefficient a, choose $C_a = \binom{6}{a} d_1^a d_2^{6-a}/(720 a_0)$. Set the first hidden layer and second-layer weights to zero. A real second-layer bias $(\operatorname{atanh}(1/2), 0)$ and real readout $(2C_a, 0)$ give the constant raw output C_a. The implemented constraints preserve these zero weights and real bias; even symmetrization preserves the constant. The binomial theorem gives $a_0 \sum_a C_a q_1^a q_2^{6-a} = R_6(s)$. Alternating-series estimates give $|R - R_6| \le |s|^8/8!$ and $R \ge |s|^6(1 - s^2/56)/6!$, together with $|R' - R_6'| \le |s|^7/7!$ and $|R'| \ge |s|^5(1 - s^2/42)/5!$. Division yields the bounds, since $\nabla_q R = dR'$. The absolute estimates also establish the continuous extensions.

Exact rational comparison of the stored constants on $|q_i| \le 0.1$ permits $A = 0.054747$. The relative force and potential bounds are then respectively below $7.136781\,10^{-5}$ and $5.352491\,10^{-5}$. This representation witness is used only to assess capacity; it is not supplied to training, proved to lie in the realized optimization level set, or asserted to be found by gradient descent.

Native data and frozen optimization. Fourteen training and four held-out initial conditions are fixed before batch generation. Each trajectory contains 801 uniform nodes on $0 \le \tau \le 0.2$, where $\tau = \omega_0 t$ and $\omega_0 = 11.417287282852458$. COMSOL uses manual Dormand–Prince fifth-order steps of $\Delta\tau = 0.00025$, disabled consistent initialization and the explicitly evaluated initial ODE derivative. Native solution and derivative vectors are mapped by their exact degree-of-freedom names and checked against independent endpoint expressions. The observed velocity is the saved derivative $\mathrm{d}q/\mathrm{d}\tau$, whereas the separate algebraic field $G^{-1}P$ is reserved for interface diagnostics. The loss reconstructs $P = G(q)\dot{q}$ in that same chart.

The selected training-only pilot passes the original one-percent signal gate. Its refined full-physics momentum and energy balance defects are respectively below $2.387\,10^{-15}$ and $4.522\,10^{-17}$; the hidden impulse has maximum component $7.229\,10^{-11}$. Historical initialization and BDF precision failures are retained. Before fitting, every training window is tested against the frozen absolute gate $\max\{\text{original floor}, \text{pilot tolerance } Delta\tau/0.2\}$; the pointwise power gate retains its full-pilot rate bound. Local signal-to-numerical-sensitivity ratios are reported separately, so a small absolute defect does not imply one-percent relative accuracy for each low-signal window.

The full interval and its four quarters provide five fixed windows per trajectory, with weights $1/2, 1/8, 1/8, 1/8, 1/8$ and equal trajectory weights. The resulting 70-window objective uses fixed composite Simpson integration, momentum and kinematic rate scales 10^{-9} and 10^{-8}, and raw-coordinate regularization $\lambda = 10^{-10}$. Starting from seed 20261016, the full real gradient is used at every Armijo step with $a_{\max} = 1$, $c = 10^{-4}$, $\beta = 1/2$, at most 40 backtracks and a fixed budget of 400 accepted steps. Held-out numerical values are first read after this optimization has stopped. Analytic remainder errors are evaluated in that post-fit pass.

Native training and held-out remainder errors. All 1989 training-data checks pass. Across the 70 windows the largest full-physics momentum, position and energy defects are $2.651\,10^{-15}$, $1.534\,10^{-15}$ and $3.059\,10^{-16}$, respectively. The seven-column integrated monomial-gradient matrix has rank seven and column-scaled condition number 4.534; this does not identify all raw network coordinates uniquely. Eight training windows have hidden impulses smaller than their diagnostic noise scales, with the smallest signal-to-noise ratio 0.01385.

The fixed run accepts all 400 steps, each at step size one with no backtracks. The total objective decreases from 0.03467314043 to $4.89603438\,10^{-5}$. Its unregularized part decreases from 0.03467313740 to $4.89570966\,10^{-5}$; the fixed kinematic term is $2.66968823\,10^{-13}$ at both endpoints.

Table S3. Hidden-gradient errors after the frozen native fit. H0–H3 denote the four fixed held-out initial conditions; Pool combines all their nodes and components. Relative RMS uses each row's own target RMS. Absolute values use the dimensionless canonical normalization.

Set	Absolute RMS	Absolute max	Rel. RMS (%)
Pool	$2.438\,10^{-12}$	$7.960\,10^{-12}$	3.406
H0	$1.980\,10^{-12}$	$3.075\,10^{-12}$	10.83
H1	$3.549\,10^{-12}$	$7.960\,10^{-12}$	2.502
H2	$1.368\,10^{-12}$	$2.385\,10^{-12}$	23.25
H3	$2.320\,10^{-12}$	$3.204\,10^{-12}$	1074

The final parameter penalty is $3.24722792\,10^{-9}$. The run stops at its prescribed budget, with a nonzero last recorded pre-step squared gradient norm $2.82336\,10^{-7}$, rather than a stationarity certificate. An independent per-window replay of the initial and final checkpoints, source identities and scalar descent records passes 4249 checks.

The post-fit held-out quality audit passes 568 checks. Table S3 reports componentwise RMS and maximum errors of the hidden gradient. Pooled relative RMS decreases from 99.42% to 3.406%, and the final maximum error is 2.981% of the pooled target maximum. The predeclared pooled thresholds, 10% and 20% respectively, are met. This conclusion is restricted to the pooled criterion: three individual trajectories exceed 10% relative RMS, and H3 remains poorly identified relative to its very small target RMS of $2.1603\,10^{-13}$. Four held-out windows have signal-to-noise ratios below one. Their failures of relative accuracy are reported without changing the protocol or refitting.

The pooled hidden-potential RMS error is $3.324\,10^{-14}$, or 2.048% of its target RMS. The reported full-field relative RMS, $8.870\,10^{-12}$, compares two analytic Python fields at the same reconstructed canonical states and with the same metric, damping and port. Their position-field difference is algebraically zero and their momentum-field difference is minus the hidden-gradient error. This small full-field ratio therefore does not replace the remainder errors in Table S3, nor measure native derivative prediction or time-rollout accuracy.

Interpretation of the matched prototype. The actual fitted function belongs to the class of Theorem S37. Its unchanged four-jet and the exact coefficient checks of Section S3.14 retain the qualitative local conservative KAM conclusion. The finite optimization record does not establish asymptotic convergence or a numerical KAM radius. The deterministic trajectories and overlapping windows do not instantiate the iid assumptions of Section S3.7. The small approximation witness also shows that this benchmark cannot establish an advantage of depth or complex arithmetic over a sixth-order polynomial. Because the native input is zero and the damping is prescribed, the experiment provides neither joint damping identification nor input-port or feedback-control validation. Those extensions and the archived three-dimensional beam correspondence remain separate requirements.

S3.16 Deterministic identification and uniform errors of the joint model

The joint fit of Section S3.17 uses the same seven complex coefficient networks as Section S3.15, while identifying the scalar drag from the weak momentum equations. This section separates a conditional inference from those equations from a verified error bound for the resulting frozen function. Neither argument treats correlated trajectory windows as independent samples.

The effective coefficients in the weak equations. Write $m_a(q) = q_1^a q_2^{6-a}$ and

$$c_\theta(q) = a_0 \sum_{a=0}^{6} m_a(q) C_{\theta,a}(q), \qquad C_{\theta,a}(-q) = C_{\theta,a}(q).$$

The leading effective parameter vector and its target are

$$\beta_\theta = \big(\gamma_\theta, a_0 C_{\theta,0}(0), \dots, a_0 C_{\theta,6}(0)\big)^{\mathsf{T}},$$
$$\beta_\star = \left(\gamma_\star, \left(\binom{6}{a} d_1^a d_2^{6-a}/720\right)_{a=0}^{6}\right)^{\mathsf{T}}. \tag{S175}$$

These eight effective coefficients do not identify the 252 raw neural coordinates. Let $\mathcal{Q}_j$ denote the fixed positive quadrature on window j, and let S_j contain its fixed component and trajectory weights. For the cached reconstructed momentum P, define the two-row block

$$A_j = S_j\big[\mathcal{Q}_j P, \mathcal{Q}_j \nabla m_0, \dots, \mathcal{Q}_j \nabla m_6\big].$$

Stack the blocks into $A \in \mathbb{R}^{140\times 8}$. Let b be the corresponding cached contribution from the momentum increments, known kinetic energy, quartic potential and input. The learned discrete residual has the exact decomposition

$$r_\theta = b + A\beta_\theta + t_\theta,$$
$$t_{\theta,j} = S_j \mathcal{Q}_j \nabla \left[a_0 \sum_a m_a(q)\big(C_{\theta,a}(q) - C_{\theta,a}(0)\big)\right]. \tag{S176}$$

The target analog is $r_\star = b + A\beta_\star + t_\star$, where $t_{\star,j} = S_j \mathcal{Q}_j \nabla(R - R_6)$ and $R_6(s) = s^6/720$. The same cached equations and arithmetic conventions are essential in both expressions. In particular, $r_\star$ need not vanish for numerical COMSOL observations.

Proposition S40. A deterministic weak-data error interface. *Let D be a positive diagonal column scaling. Suppose a verified $\sigma_0 > 0$ satisfies $\sigma_{\min}(AD^{-1}) \geq \sigma_0$. If $\|r_\theta\|_2 \leq \varepsilon_{\mathrm{fit}}$, $\|r_\star\|_2 \leq \nu$, and $\|t_\theta - t_\star\|_2 \leq T$, then*

$$\|D(\beta_\theta - \beta_\star)\|_2 \leq \frac{\varepsilon_{\mathrm{fit}} + \nu + T}{\sigma_0}. \tag{S177}$$

Proof. Subtract the two decompositions in (S176). Then $A(\beta_\theta - \beta_\star) = r_\theta - r_\star - (t_\theta - t_\star)$. Apply the lower singular-value bound to $AD^{-1}D(\beta_\theta - \beta_\star)$ and the triangle inequality.

For the frozen discrete design, Arb arithmetic gives

$$\sigma_{\min}(AD^{-1}) > 0.2915834456765604. \tag{S178}$$

To verify this bound, form the interval enclosure of $G = (AD^{-1})^{\mathsf{T}}(AD^{-1})$ and an exact binary64 approximate inverse R_G. The verified estimates

$$\begin{aligned} \|I - R_G G\|_\infty &< 4.444\,10^{-15} < 1, \\ \sigma_{\min}(AD^{-1}) &\geq \sqrt{\frac{1 - \|I - R_G G\|_\infty}{\|R_G\|_\infty}} \end{aligned}$$

follow from the Neumann inverse bound and the symmetry of G^{-1}. The seven monomial columns are enclosed from the stored position samples and Simpson weights, and the cached drag integrals and row scales are interpreted as exact binary64 constants. Thus this is a certificate for the stated discrete design. The numerical source, velocity observation, momentum reconstruction, force evaluation and quadrature defects require bounds when inferring an ideal continuous mechanical equation from this system. Diagnostic acceptance tests alone do not supply those bounds. For the present known benchmark, however, $r_\star$ can be enclosed directly in the same cached discrete equations, as follows.

At all 11207 distinct stored training positions, Arb evaluates the frozen raw network and the analytic remainder gradient, retaining the original Simpson weights. The 70 windows give certified bounds

$$\begin{aligned} \varepsilon_{\mathrm{fit}} &< 1.292084\,10^{-5}, \\ \nu_{\mathrm{discrete}} := \|r_\star\|_2 &< 1.053117\,10^{-5}, \\ \|t_\theta - t_\star\|_2 &< 6.564356\,10^{-5}, \\ \|D(\beta_\theta - \beta_\star)\|_2 &< 3.055577\,10^{-4}. \end{aligned} \tag{S179}$$

The last bound follows from (S177), using the certified singular value rather than a floating-point rank test. Here all cached base increments, drag integrals, positions, weights and row scales denote their exact stored binary64 values. Thus ν_{discrete} is a rigorous consistency bound for this finite system. It is not a COMSOL solution-error bound or a bound relating its cached base terms to an ideal continuous trajectory.

Uniform certification after fitting. For this controlled benchmark the hidden target R is known for post-fit verification, although its values and coefficients are excluded from supervision and initialization. Let $e(q) = c_{\widehat{\theta}}(q) - R(d^{\mathsf{T}}q)$ for the frozen joint checkpoint. The following construction certifies suprema on an entire box rather than estimating them from validation nodes.

Let r bound the modulus of each position coordinate, and suppose $B_{a,jk}$ bounds $|\partial_j \partial_k C_{\widehat{\theta},a}|$ throughout a symmetric star-shaped box. Evenness gives $\nabla C_{\widehat{\theta},a}(0) = 0$, so Taylor's formula gives

$$\begin{aligned} |C_{\widehat{\theta},a}(q) - C_{\widehat{\theta},a}(0)| &\leq \frac{r^2}{2} \sum_{j,k} B_{a,jk} =: U_a, \\ |\partial_j C_{\widehat{\theta},a}(q)| &\leq r \sum_k B_{a,jk} =: V_{a,j}. \end{aligned} \tag{S180}$$

Writing $\Delta_a = a_0 C_{\widehat{\theta},a}(0) - \binom{6}{a} d_1^a d_2^{6-a}/720$ and $\alpha_a = (a, 6-a)$, this yields

$$\begin{aligned} |e(q)| &\leq r^6 \sum_a (|\Delta_a| + a_0 U_a) + \mathcal{T}_0, \\ |\partial_j e(q)| &\leq \sum_a \left[\alpha_{a,j} r^5 (|\Delta_a| + a_0 U_a) + a_0 r^6 V_{a,j}\right] + \mathcal{T}_j. \end{aligned} \tag{S181}$$

With $S = r(|d_1| + |d_2|)$, valid real or complex tail bounds are $\mathcal{T}_0 = S^8 \cosh(S)/8!$ and $\mathcal{T}_j = |d_j| S^7 \cosh(S)/7!$. The analogous Hessian bound follows by differentiating the monomial products twice and bounding the target tail by $|d_j d_k| S^6 \cosh(S)/6!$.

The conditional data bound can be inserted directly into (S181). Write $D = \mathrm{diag}(D_\gamma, D_1, \ldots, D_7)$, so that D_{a+1} is the column scale of m_a, and put

$$\begin{aligned} B_5(r) &= r^5 \max_j \left(\sum_{a=0}^{6} \frac{\alpha_{a,j}^2}{D_{a+1}^2} \right)^{1/2}, \\ U_5(r) &= \max_j \left\{ a_0 \sum_a (\alpha_{a,j} r^5 U_a + r^6 V_{a,j}) + \mathcal{T}_j \right\}. \end{aligned}$$

Cauchy–Schwarz and Proposition S40 give

$$\sup \|\nabla e\|_\infty \leq B_5(r) \frac{\varepsilon_{\mathrm{fit}} + \nu + T}{\sigma_0} + U_5(r). \tag{S182}$$

The same statement holds on a complex rectangle with r bounding its coordinate moduli and with the corresponding

complex coefficient Hessian bounds. Since $\varepsilon_{\mathrm{fit}}$ is at most the square root of the unregularized squared momentum loss, this displays the deterministic training-error contribution separately from true-data defects and nonlinear tails. Without a valid bound for ν it remains conditional. The direct certification below uses the frozen coefficients instead and supplies an unconditional function comparison for the stated known benchmark.

With the discrete bounds in (S179), this route gives the whole-box conservative-force estimate

$$\sup_{|q_i|\le 0.1} \|\nabla e(q)\|_\infty < 1.526835\,10^{-11}. \tag{S183}$$

This explicitly instantiates the discrete residual–coefficient–uniform error chain for the frozen model. The direct post-fit comparison below is slightly sharper. Both are statements for the declared known benchmark; neither removes the separate continuous-data and physical-model discrepancy terms.

All these quantities are evaluated by outward-rounded Arb/Acb arithmetic. The stored raw binary64 parameters define exact rational coordinates; the smooth row constraints are reevaluated by interval arithmetic with the implementation's stored scalar constants. The certified function is the mathematical network defined by these coordinates. The certification does not include floating-point forward evaluation error. On the real box $|q_i| \le 0.1$, the frozen checkpoint satisfies

$$\begin{aligned} \sup|e| &< 2.772\,10^{-13},\\ \sup\|\nabla e\|_\infty &< 1.269\,10^{-11},\\ \sup\|D^2 e\|_{\infty\to\infty} &< 8.789\,10^{-10},\\ |\gamma_{\widehat\theta} - 7/200| &< 1.168\,10^{-13}. \end{aligned} \tag{S184}$$

These are absolute errors of the deliberately hidden reduced-model potential and scalar drag. They do not bound a three-dimensional beam discrepancy or the Gauss-eight quadrature error relative to a continuous rod.

The two fitted channels also give an error bound for the complete forced vector field. Let $F_\star(x,u)$ and $F_{\widehat\theta}(x,u)$ use the same finite inertia graph and magnetic port, with target drag $7/200$ and learned drag $\gamma_{\widehat\theta}$, respectively. Then

$$F_{\widehat\theta} - F_\star = \begin{pmatrix} 0 \\ -\nabla e(q) - (\gamma_{\widehat\theta} - 7/200)P \end{pmatrix}.$$

Consequently, for $|q_i| \le 0.1$, $|P_i| \le P_{\max}$ and the same arbitrary input u,

$$\|F_{\widehat\theta} - F_\star\|_\infty < 1.269\,10^{-11} + 1.168\,10^{-13} P_{\max}. \tag{S185}$$

The port terms cancel exactly. This bound includes the identified proportional damping, but retains the same reduced-model and exact-parameter scope as (S184).

The quantitative learning-to-KAM interface. For the complex rectangle $|\operatorname{Re} q_i| \le 0.1$, $|\operatorname{Im} q_i| \le 0.001$, the same computation certifies

$$\begin{aligned} \sup\|\nabla e\|_\infty &< 1.670\,10^{-11},\\ \sup\|D^2 e\|_{\infty\to\infty} &< 1.160\,10^{-9}. \end{aligned} \tag{S186}$$

For complex arguments the real energy readout is extended by conjugating coefficients, as specified in Section S3.7; one never takes a real part after inserting complex coordinates. Interval lower bounds for the intermediate $|\cosh|$ denominators certify the absence of activation poles on this rectangle.

The kinetic evaluator has a compatible numerical complex domain. Each finite tangent in the Gauss rule can be written $T_i(q) = \sum_k a_{ik}(\cos(\beta_k^{\mathsf T} q), -\sin(\beta_k^{\mathsf T} q))$ with its own frozen real constants. The bilinear Gram identity gives

$$T_i(q)^{\mathsf T} T_j(q) = \sum_{k,l} a_{ik} a_{jl} \cos((\beta_k - \beta_l)^{\mathsf T} q).$$

Applying $|\cos z - 1| \le |z|^2 \cosh(|z|)/2$ to the finite graph, with exact rational coefficient moments, proves on the same whole rectangle

$$\begin{aligned} \|G_Q(0)^{-1}(G_Q(q) - G_Q(0))\|_\infty &< 0.004987788 < 1,\\ \|G_Q(q)^{-1}\|_\infty &< 6.180410. \end{aligned} \tag{S187}$$

The inverse bound follows from the Neumann series. The constant rotary term cancels from the difference; the signed tangent coefficients and positive outer weights are retained in the enclosure. Thus the same complete conservative Hamiltonian is holomorphic here for every finite momentum. This certificate concerns the finite Gauss graph, rather than the continuous quadrature error, and supplies a domain condition for subsequent KAM estimates.

Suppose a target candidate K_0 has its position component in this rectangle on the angle strip $\mathbb{T}^2_\rho$, and the conservative target defect is bounded by $\delta_{\star,\rho}$. Because the mass metric and base potential coincide, the learned conservative defect is

$$\begin{aligned} E_{\widehat\theta} &= E_\star + \begin{pmatrix} 0 \\ -\nabla e \end{pmatrix} \circ K_0,\\ \|E_{\widehat\theta}\|_\rho &< \delta_{\star,\rho} + 1.670\,10^{-11}. \end{aligned}$$

This is a quantitative, deterministic learning-error-to-residual interface for this particular checkpoint. Applying

a finite-amplitude KAM theorem still requires the same candidate's strip inclusion, Diophantine constants, torsion bounds, Hamiltonian derivative bounds, domain margins and smallness inequality. The direct construction below verifies these conditions for the learned conservative Hamiltonian itself. Preserving the degree-four jet also provides the local conservative KAM statement of Section S3.13. The dissipative system with $\gamma_{\widehat{\theta}} > 0$ is treated by the energy inequality in Section S3.17, rather than by claiming that its unforced conservative KAM tori persist unchanged.

A finite-amplitude candidate for the joint Hamiltonian. The a-posteriori construction can be applied directly to the learned conservative Hamiltonian, without adding the learning-error bound to the defect of a different model. Let $\vartheta \in \mathbb{R}^2/\mathbb{Z}^2$, and write $\nu = \omega/(2\pi)$ for the frequency in period-one angles. For the frozen joint checkpoint, a Newton construction with the full nonseparable Hessian gives a finite Fourier polynomial K_0 in the physical canonical coordinates (q, P). Its initial reference has equal amplitudes $a \simeq 0.04$ in the linear normal coordinates; the saved value is exactly $5764607523034235/144115188075855872$. The frequency is fixed before the Newton corrections by

$$\omega_1 = \frac{499974648628101}{500000000000000},$$
$$\omega_2/\omega_1 = [6; 6, 1, 1, 1, 2, 1, 5, 14, 17, 4, \overline{1}],$$

where $\overline{1}$ denotes an infinite tail of ones. Thus $\omega_2 \simeq 6.150405443$. All partial quotients are at most 17. The bounded-type continued-fraction estimate, including the case of a zero second integer component, gives the all-integer bound

$$|k \cdot \nu| \geq \frac{0.0083761512}{|k|_1} \geq \frac{0.0083761512}{|k|_1^3}, \qquad \text{(S188)}$$

This holds for every $0 \neq k \in \mathbb{Z}^2$ and is not inferred from a finite search over integers. The computations below use the certified constant $\gamma = 4828522443384859/576460752303423488$; the displayed decimal in (S188) is rounded downward.

The exponent $\tau = 1$ is admissible in the two-frequency theorem of Haro and Luque (2019). A uniform Rüssmann constant can be obtained without a finite small-divisor cutoff. For $R > 0$, the positive values $k \cdot \nu$ with $0 < |k|_1 \leq R$ have separation at least $\gamma/(2R)$, by (S188). Thus

$$\sum_{0<|k|_1\leq R} \frac{1}{(2\pi k \cdot \nu)^2} \leq \frac{R^2}{3\gamma^2}.$$

Writing each exponential as a positive tail integral gives, for every $\delta > 0$,

$$\sum_{k\neq 0} \frac{e^{-4\pi\delta|k|_1}}{(2\pi k \cdot \nu)^2} \leq \frac{1}{24\pi^2\gamma^2\delta^2}.$$

For bounded analytic v on $\mathbb{T}^2_\rho$, Parseval's identity on the four imaginary corner shifts, followed by a limit from smaller strips, yields $\sum_k |\widehat{v}_k|^2 e^{4\pi\rho|k|_1} \leq 4\|v\|_\rho^2$. Cauchy–Schwarz therefore bounds the zero-average cohomological inverse by

$$\|\mathcal{R}_\nu v\|_{\rho-\delta} \leq \frac{\|v\|_\rho}{\sqrt{6}\,\pi\gamma\delta} < \frac{\|v\|_\rho}{7\gamma\delta}. \qquad \text{(S189)}$$

The same componentwise bound applies to the analytic row-sum matrix norm. The choice $c_R = 1/7$ is consequently uniform in all iteration strip losses; it is not an assumed numerical value from the reference.

The retained polynomial has modes $|k_j| \leq 31$ and exact rational coefficients with conjugate symmetry, obtained by high-precision Newton correction of a finite Fourier seed. Using all those coefficients, outward-rounded calculations at 192 and 256 bits verify the following bounds on $\max_j |\operatorname{Im}\vartheta_j| \leq 1/40$:

$$\begin{aligned} \max_i |\operatorname{Re}(K_0)_i| &< 0.04050724, \\ \max_i |\operatorname{Im}(K_0)_i| &< 0.00631145, \\ \|DK_0\|_\rho &< 0.294197, \\ \|DK_0^{\mathsf{T}}\|_\rho &< 0.588324, \\ \|(DK_0^{\mathsf{T}} DK_0)^{-1}\|_\rho &< 15.903630. \end{aligned} \qquad \text{(S190)}$$

The matrix norms are maximum row sums of componentwise analytic suprema. For a conjugate pair of Fourier modes, the real and imaginary image bounds use $2|c_k|\cosh(2\pi\rho|k|_1)$ and $2|c_k|\sinh(2\pi\rho|k|_1)$, respectively. The Gram enclosure retains the nonidentity matrix $S^{\mathsf{T}}S$ of the linear reference; S is not assumed orthogonal. In the inverse reference coordinates, the phase-lift deviation has Lipschitz constant below $0.003142 < 1$, which also verifies that the real polynomial parametrizes an embedded two-torus. These are properties of the candidate, rather than a claim that the torus is invariant.

A full-Hamiltonian analytic neighbourhood. The same raw checkpoint is holomorphic on the four-dimensional rectangle $|\operatorname{Re} x_i| \leq 0.08$, $|\operatorname{Im} x_i| \leq 0.03$, $x = (q, P)$. The finite Gauss Neumann argument and all seven networks' pole exclusions are repeated on this larger domain. They give $|H_{\widehat{\theta}}| < 0.072222575$. For example, with $R =$

$\sqrt{0.08^2 + 0.03^2}$,

$$|\tfrac{1}{2} P^{\mathsf{T}} G_Q^{-1} P| \le \|G_Q^{-1}\|_\infty R^2,$$
$$|1 - \cos(d^{\mathsf{T}} q)| \le \tfrac{1}{2}(\|d\|_1 R)^2 \cosh(\|d\|_1 R).$$

and the certified learned-potential error completes the energy bound. On the inner rectangle $|\operatorname{Re} x_i| \le 0.06$, $|\operatorname{Im} x_i| \le 0.01$, a complex polydisc of radius $\delta = 0.01$ remains strictly inside the outer rectangle. Cauchy's estimates yield bounds for the complete field $X = J\nabla H_{\widehat{\theta}}$:

$$\begin{aligned} \|X\| &< 7.222258, & \|DX\| &< 3611.128750, \\ \|D(X^{\mathsf{T}})\| &< 14444.515000, & \|D^2 X\| &< 2166677.250. \end{aligned} \tag{S191}$$

Here the norm of $D(X^{\mathsf{T}})$ is the total sum of its componentwise derivative suprema, as in Haro and Luque (2019); it is not the induced matrix norm of $(DX)^{\mathsf{T}}$. In four dimensions, the Cauchy factors for these four bounds are M/δ, $5M/\delta^2$, $20M/\delta^2$ and $30M/\delta^3$, respectively. The candidate in (S190), together with its complex maximum-coordinate ball of radius 0.003, lies inside the inner rectangle. The physical inertia $G_Q(q)$ enters H; the symplectic matrix and the auxiliary phase-space metric remain canonical constants.

Sharper bounds follow by differentiating the finite mechanical graph before applying the norm estimates. Let $A(q) = G_Q(q)^{-1}$ and let g_j, a_j bound the matrix row norm with the sum over all ordered j configuration-derivative indices of G_Q, A, respectively. The cosine-difference second moments used above imply entrywise derivative majorants proportional to $R\cosh(RD)$, $\cosh(RD)$ and $RD^2\cosh(RD)$ for orders one, two and three; here D bounds each pair's phase-vector diameter. These majorants are summed over the complete finite graph. Repeated differentiation of $AG_Q = I$ gives

$$\begin{aligned} a_1 &\le a_0^2 g_1, \\ a_2 &\le 2a_0^3 g_1^2 + a_0^2 g_2, \\ a_3 &\le 6a_0^4 g_1^3 + 6a_0^3 g_1 g_2 + a_0^2 g_3. \end{aligned}$$

The zeroth-order bound uses the entrywise Neumann majorant $\sup|A| \le \sum_{m\ge 0}(|G_Q(0)^{-1}|B)^m |G_Q(0)^{-1}|$, where B bounds $|G_Q - G_Q(0)|$ entrywise. Thus the supremum of each entry is taken before the row sum, consistently with the theorem.

For the potential, write $V_{\widehat{\theta}} = V_\star + e$ exactly. The certified first and second derivatives of e are used directly; its third derivative is bounded by differentiating each analytic Hessian entry with Cauchy radius $1/20$ inside the larger configuration rectangle $|\operatorname{Re} q_i|, |\operatorname{Im} q_i| \le 3/20$. The derivatives of $V_\star$ are explicit. If v_j bound the potential derivative row norms and $R_P = \sqrt{0.06^2 + 0.01^2}$, the kinetic contractions give

$$\begin{aligned} c_{X,0} &\le \max\{R_P a_0, R_P^2 a_1 + v_1\}, \\ c_{X,1} &\le \max\{a_0 + R_P a_1, R_P^2 a_2 + 2R_P a_1 + v_2\}, \\ c_{X,2} &\le \max\{R_P a_2 + 2a_1, R_P^2 a_3 + 4R_P a_2 + 2a_1 + v_3\}. \end{aligned}$$

All momentum contractions and both ordered mixed derivative blocks are included. The total Hessian-entry sum instead gives $c_{X^\top,1} \le 2a_0 + 4R_P a_1 + R_P^2 a_2 + \sum_i \sum_j \|\partial_i \partial_j V_{\widehat{\theta}}\|$. Outward-rounded evaluation on the same inner rectangle proves

$$\begin{aligned} c_{X,0} &< 0.384443, & c_{X,1} &< 6.365862, \\ c_{X^\top,1} &< 19.878534, & c_{X,2} &< 2.266875. \end{aligned} \tag{S192}$$

These estimates retain the complete variable-inertia kinetic energy and every learned coefficient network.

Continuous mean torsion and whole-strip defect. Put $L = DK_0$, $B = (L^{\mathsf{T}}L)^{-1}$ and $N_s = JLB$, with $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$. The saved evaluator computes

$$T_s = N_s^{\mathsf{T}} J\big(DX(K_0)N_s - D_\nu N_s\big), \qquad D_\nu = \nu \cdot \partial_\vartheta.$$

The canonical convention in Haro and Luque (2019) uses $\Omega_0 = -J$, so its normal and torsion are $N = -N_s$ and $T = -T_s$. This sign change preserves the norm of the inverse mean torsion. The derivative of N_s is evaluated analytically, including $D_\nu B = -B((D_\nu L)^{\mathsf{T}} L + L^{\mathsf{T}} D_\nu L)B$. A 256-bit, 32×32 trapezoidal evaluation of the full-Hamiltonian torsion retains all candidate modes by exact frequency folding. On the additional angle strip of width $1/4$, separate Gram and Hamiltonian-domain bounds give $\|T_s\|_{1/4} \le M_T$. The infinite quadrature alias in each mean entry is bounded by $4M_T r/(1-r)^2$, $r = e^{-2\pi(1/4)32}$, which is less than $1.644\,10^{-15}$. Interval inversion of the resulting continuous mean enclosure proves

$$\|\langle T\rangle^{-1}\|_\infty < 18390.633082. \tag{S193}$$

For the invariance defect $E = X(K_0) - DK_0\nu$, let I_m denote the two-dimensional Fourier interpolant on an even $m \times m$ grid, with frequency representatives $[-m/2, m/2)^2 \cap \mathbb{Z}^2$. If $G = X \circ K_0$ is analytic on a wider strip $\widehat{\rho}$ and $\|G\|_{\widehat{\rho}} \le M_G$, define $r = e^{-2\pi(\widehat{\rho}-\rho)}$. Fourier coefficient decay and the exact discrete alias identity imply

$$\|G - I_m G\|_\rho \le 2M_G \left(\frac{1+r}{1-r}\right)^2 \left(2r^{m/2} - r^m\right). \tag{S194}$$

Indeed, both the omitted modes and their aliases are bounded by the sum of $r^{|k|_1}$ outside the representative square; that sum equals $((1+r)/(1-r))^2(2r^{m/2}-r^m)$. The factor two includes both contributions. The finite transport polynomial $DK_0\nu$ lies strictly inside the retained block, so $E - I_m E = G - I_m G$.

For this exact candidate, separate pole exclusion and metric inversion on the physical rectangle with real and imaginary half-widths 0.45 certify $\widehat{\rho} = 19/40$ and $M_G < 407.825$. At $m = 80$, a 384-bit evaluation of the original parameter graph, followed by an outward-rounded discrete Fourier transform, gives

$$\begin{aligned} \|I_{80}E\|_{1/40} &< 5.108\,10^{-46}, \\ \|G - I_{80}G\|_{1/40} &< 1.578\,10^{-46}, \\ \|E\|_{1/40} &< 6.685\,10^{-46}. \end{aligned}$$

The interpolant bound is the maximum, over vector components, of the Fourier coefficient-ball moduli weighted by $e^{2\pi\rho|k|_1}$. Thus arithmetic uncertainty, unsampled points and the infinite analytic tail are included; a real-grid residual alone is not used as evidence.

Theorem S41. Finite-amplitude torus of the joint learned Hamiltonian. *The frozen learned conservative Hamiltonian $H_{\widehat{\theta}}$ of Section S3.17 has a real analytic invariant embedded two-torus with the exact frequency ν specified above. On $\mathbb{T}^2_{1/400}$ its parametrization K_∞ satisfies*

$$X(K_\infty) = DK_\infty\nu, \qquad \|K_\infty - K_0\|_{1/400} < 8.062\,10^{-28}.$$

Proof. Apply Theorem 3.1 of Haro and Luque (2019) in its canonical case $n = d = 2$, with auxiliary metric I_4, $\tau = 1$, $c_R = 1/7$, $\rho = 1/40$ and domain margin $3/1000$. Use the bounds in (S190), (S192) and (S193), retaining their unrounded certified endpoints. Each norm input is rounded upward to a rational with denominator 10^{12}. The four strict iteration bounds for DK_0, DK_0^{T}, B and $\langle T\rangle^{-1}$ are $21/20$ times these upward bounds. Choose $a_1 = 2$, $\rho_\infty = 1/400$ and iteration parameter $\nu_{\mathrm{it}} = 1/1000$, distinct from the frequency vector. The iteration parameters in Section 5.3 of that reference are then $a_2 = 10$, $a_3 = 20/3$ and $\delta_0 = 3/800$. Exact rational evaluation of its Tables 1–3 gives

$$\frac{\gamma^4\rho^4}{\mathfrak{C}_1} > 6.739\,10^{-38}, \qquad \frac{\mathfrak{C}_1\|E\|_\rho}{\gamma^4\rho^4} < 9.918\,10^{-9} < 1.$$

All eight expanded iteration inequalities are strictly satisfied: contraction, initial residual size, approximate symplectic inversion, preservation of both tangent norms, Gram and mean-torsion inversion, and distance to the domain boundary. The theorem's correction estimate gives the stated bound. Cauchy's estimate on the final strip bounds the real derivative correction by $2\|K_\infty - K_0\|_{\rho_\infty}/\rho_\infty$. Combining it with the reference-coordinate phase estimate above gives a phase-lift deviation less than $0.003291 < 1/2$, preserving the real embedding. All inputs are bound to the same exact rational candidate and raw trained checkpoint.

This theorem concerns the learned finite planar Gauss model's conservative dynamics. The deterministic learning-error bounds above compare it to the specified target function. They do not certify continuous COMSOL observation or time-solver errors, the continuous-rod quadrature and reduction errors, or statistical generalization. Positive damping and magnetic input remain governed by the separate power and safety results of Section S3.17.

S3.17 Joint drag identification and magnetic safety in the same chart

We now remove the prescribed-damping assumption of Section S3.15. The metric, coordinates, quartic prior and seven complex coefficient networks are unchanged. Only the scalar mass-proportional drag specialization is considered; an arbitrary configuration-dependent dissipative tensor is not identified by this construction.

The profiled training problem. Stack the weighted, scaled momentum-window residuals into $b(\theta) + \gamma a$, where a is the fixed stack of integrated observed momenta and b contains the prescribed quartic prior and the trainable potential. The kinematic loss is a fixed constant $K_X \geq 0$. With $a \neq 0$, solve

$$\gamma_+(\theta) = \max\left\{0, -\frac{a^{\mathsf{T}}b(\theta)}{a^{\mathsf{T}}a}\right\}, \tag{S195}$$

$$F_+(\theta) = \|b(\theta) + \gamma_+(\theta)a\|_2^2 + K_X + \frac{\lambda}{2}\|\theta\|_2^2, \qquad \lambda > 0. \tag{S196}$$

This is exact elimination of a fitted nonnegative scalar in the joint weak-loss problem. It does not supply a damping label. Unlike (S171), it penalizes only the 252 real potential coordinates, rather than a softplus drag coordinate.

Theorem S42. Descent for jointly profiled damping. *Under the fixed-window and smooth-network hypotheses of Theorem S37, assume $a \neq 0$ and apply full-batch Armijo descent to (S196), with unlimited backtracking in exact arithmetic. Every line search terminates, the potential iterates and their fitted drag coefficients are bounded, and $\|\nabla F_+(\theta_k)\| \to 0$. Every accumulation point, paired with (S195), is first-order stationary for the nonnegatively constrained joint objective.*

Proof. Write $t = a^{\mathsf{T}}b$ and $A = a^{\mathsf{T}}a > 0$. Eliminating γ gives

$$F_+ = \|b\|^2 - \frac{\min(t,0)^2}{A} + K_X + \frac{\lambda}{2}\|\theta\|^2.$$

The scalar function $\min(t,0)^2$ is continuously differentiable with a Lipschitz derivative, including at zero. Since b is smooth, F_+ has a locally Lipschitz gradient; twice differentiability at the active-set boundary is unnecessary. Its gradient is

$$\nabla F_+ = 2(Db)^{\mathsf{T}}(b + \gamma_+ a) + \lambda\theta.$$

For $\gamma_+ > 0$ the omitted drag derivative vanishes by the scalar normal equation; at $\gamma_+ = 0$ the same formula follows by continuity and satisfies the drag Karush–Kuhn–Tucker condition.

Nonnegativity of the residual gives compact potential sublevel sets, with $\|\theta\| \le \sqrt{2F_+(\theta_0)/\lambda}$. Continuity of b and (S195) then bounds the fitted drag. On a compact ball containing all trial segments the gradient has a finite Lipschitz constant. The descent lemma and the Armijo summation in (S173) prove a uniform positive lower bound for accepted steps and square summability of the gradients. Continuity gives stationarity at each accumulation point. The scalar normal equation or its boundary inequality supplies the remaining constrained first-order condition.

This theorem proves neither a global minimum nor unique mechanical identification. The profile can attain zero drag; such a boundary value must be stored as a nonnegative scalar, since no finite softplus coordinate represents zero exactly. A run remaining strictly positive can instead freeze $\eta = \log(\exp(\gamma_+) - 1)$ in the existing mechanical decoder. Its conservative energy still has the same four-jet. As before, that local KAM property is structural and does not certify successful learning. Section S3.16 treats accuracy separately for the frozen fitted function.

Power and dissipative structure. For the frozen learned model set

$$H_\theta = \tfrac{1}{2}P^{\mathsf{T}}G_Q(q)^{-1}P + V_\theta(q),$$
$$\mathcal{D}_\theta = \tfrac{\widehat{\gamma}}{2}P^{\mathsf{T}}G_Q(q)^{-1}P.$$

Here G_Q is the specified positive-weight Gauss metric used in training, not the unevaluated continuous integral. Decode $H_\theta + i\mathcal{D}_\theta$ as in (S165). With $v = G_Q^{-1}P$ and $g(q) = d\cos(d^{\mathsf{T}}q)$, the same learned field obeys

$$H'_\theta = -\widehat{\gamma}P^{\mathsf{T}}G_Q^{-1}P + uy, \qquad y = g(q)^{\mathsf{T}}v. \tag{S197}$$

Indeed $\nabla H_\theta^{\mathsf{T}} J \nabla H_\theta = 0$ and $\partial_P H_\theta = v$. The first term is nonpositive. For constant frozen $\widehat{\gamma}$ and prescribed open-loop input, Theorem S35 and Proposition S36 supply the time-dependent Birkhoff form and conformal discrete structure for this same metric. They do not imply that arbitrary state feedback is conformally symplectic, or that its numerical step satisfies an exact work balance.

True energy and geometric barriers. For control, let the actual reduced plant have the same G_Q, chart and magnetic port, but true potential

$$V_*(q) = \tfrac{1}{2}q^{\mathsf{T}}K_b^Q q + 1 - \cos(d^{\mathsf{T}}q), \qquad K_b^Q \succ 0,$$

and nonnegative true drag $\gamma_* \ge 0$. Its energy is H_*. The assumption on true drag is physical; positivity of the fitted coefficient alone does not establish it. The planar geometry uses $\theta(s,q) = \beta(s)^{\mathsf{T}}q$, $\theta(0,q) = 0$ and $r_x(s,q) = \int_0^s \sin\theta(\sigma,q)\,d\sigma$. Let $K_b^{\mathrm{geom}} = (EI/E_0)\int_0^L \beta_s\beta_s^{\mathsf{T}}\,ds$ be the exact dimensionless bending matrix for this geometry and choose a verified $0 < \chi_b \le 1$ such that $K_b^Q \succeq \chi_b K_b^{\mathrm{geom}}$. This comparison includes the rounding of stored stiffness buffers. Then

$$|r_x(s,q)|^2 \le \frac{2E_0 s^3}{3EI\chi_b}H_*(q,P), \qquad E_c^Q = \chi_b\frac{3EIc^2}{2E_0L^3}. \tag{S198}$$

To see this, integrate θ_s from the clamped base and use

$$|r_x(s)| \le \int_0^s |\theta(\sigma)|\,d\sigma \le \left(\frac{s^3}{3}\int_0^s \theta_s^2\,d\sigma\right)^{1/2}.$$

The last integral is bounded by $2E_0H_*/(EI\chi_b)$. This argument retains the sine geometry; it uses no small-angle replacement of r_x.

For the network box $D = \{|q_i| < r_i\}$ a separate configuration barrier is

$$E_D^Q = \min_i \frac{r_i^2}{2((K_b^Q)^{-1})_{ii}}. \tag{S199}$$

The inequality $q_i^2 \le (q^{\mathsf{T}}K_b^Q q)((K_b^Q)^{-1})_{ii}$ proves that $H_* < E_D^Q$ prevents exit through any face of D.

Theorem S43. A work filter for the jointly identified planar model. *Assume the true reduced plant satisfies the power relation $H'_* = -\gamma_* P^{\mathsf{T}}G_Q^{-1}P + uy$ and the preceding geometric assumptions. Suppose an initial potential certificate gives $|V_\theta(q_0) - V_*(q_0)| \le \varepsilon_0$. Choose*

$$W(0) = H_\theta(q_0,P_0) + \varepsilon_0 < E_{\max} < \min\{E_c^Q, E_D^Q\}.$$

Measure the actual port output y, integrate $W' = uy$, and take the scalar Euclidean projection of a locally Lipschitz request $u_{\mathrm{req}}(q, P, W, \tau)$ onto

$$\begin{aligned} &|u| \le U_{\max}, \quad U_{\max} > 0, \\ &uy \le \kappa(E_{\max} - W), \quad \kappa > 0. \end{aligned} \tag{S200}$$

Under exact sensing and actuation the filter is feasible and $H_* \le W < E_{\max}$ for all forward time. The trajectory remains inside D, and $|r_x(s)| < c$ for the whole centreline.

Proof. The shared kinetic metric and initial certificate give $H_*(0) \le W(0)$. Integrating true power and using nonnegative drag gives $H_* \le W$ up to a possible first exit. Zero input belongs to the filter while $W < E_{\max}$, and the constraint yields

$$E_{\max} - W(\tau) \ge (E_{\max} - W(0))e^{-\kappa\tau} > 0.$$

Equations (S198) and (S199) exclude a finite first exit through the clearance or configuration boundary. On any finite time interval the strict energy margin also keeps q in a compact subset of D. Continuity and positive definiteness of G_Q bound the momentum there by the kinetic-energy bound. The projected command is locally Lipschitz while $W < E_{\max}$: near $y = 0$ the power constraint is inactive, and away from zero its endpoint is the smooth ratio $\kappa(E_{\max} - W)/y$. Bounded actuation and local regularity permit continuation, proving the all-time claim.

This theorem uses the fitted energy to initialize a certified budget and permits its force and drag estimates to generate the requested command. It proves safety, not exact tracking, a complex-network advantage, or KAM persistence of the damped closed loop. The current port has rank one in two mechanical coordinates; it cannot realize an arbitrary two-component compensating force. The earlier two-port full-beam realization is therefore not invoked here. The theorem also requires actual power: sampled measurements, nonideal actuators and numerical work integration need additional error budgets before a physical implementation can inherit the guarantee.

Frozen joint identification. The joint experiment reuses the fourteen accepted training and four held-out COMSOL trajectories, with the same 70 windows and component scales. Only q, observed q', τ and u are opened during fitting. Stored damping metadata and analytic remainder labels are excluded until the fit has stopped. The metric remains prescribed; the experiment identifies one scalar drag and a potential correction.

Table S4. Post-fit hidden-force errors for the joint planar fit. Each row uses its own held-out target RMS; Pool combines all nodes and components. These are analytic benchmark errors at reconstructed states.

Set	Absolute RMS	Relative RMS (%)
Pool	$6.503\,10^{-15}$	0.009085
H0	$1.659\,10^{-15}$	0.009075
H1	$9.947\,10^{-15}$	0.007011
H2	$2.856\,10^{-15}$	0.04853
H3	$7.702\,10^{-15}$	3.566

An eight-column weighted weak design contains the integrated momentum and seven degree-six monomial gradients. Column-scaled floating-point singular values give rank eight and condition number 5.046. This concerns the leading surrogate coefficients, not unique identification of the nonlinear raw parameters. A training-only linear fit supplies those coefficients. Each random coefficient network's final readout is minimally adjusted to match its fitted constant at zero; all 252 real potential coordinates then participate in the profiled Armijo fit. No exact cosine-remainder coefficients are supplied to either initialization or optimization.

With $\lambda = 10^{-10}$ and the previous Armijo settings, the fixed 400-step run accepts every step at size one. The total objective decreases from $1.65027750\,10^{-8}$ to $1.18505807\,10^{-8}$; its momentum part decreases from $4.81914905\,10^{-9}$ to $1.66948170\,10^{-10}$. The fixed kinematic term is $2.66968823\,10^{-13}$. The profiled drag stays positive and freezes at 0.03500000000011677. Its difference from the stored reference, opened only after fitting, is $1.168\,10^{-13}$. A high-precision mathematical softplus evaluation is enclosed separately in Section S3.16. This numerical agreement is not a bound on unknown physical drag or COMSOL solver error.

Table S4 retains the low-signal individual results. The pooled maximum hidden-force error is $1.440\,10^{-14}$, and pooled potential RMS error is $1.971\,10^{-16}$. The initialized network already has pooled relative force RMS 0.03143% before Armijo descent. Consequently the improvement cannot be assigned solely to deep nonlinear optimization, much less to a complex-versus-real advantage. The sixth-order approximation witness in Proposition S39 remains relevant.

An independent checkpoint replay compares the profiled and direct joint losses, with an absolute difference below $2.474\,10^{-20}$. At 198 observed states and prescribed nonzero test inputs, the learned power identity has maximum binary64 defect $4.467\,10^{-17}$ and nonnegative evaluated dissipation. These checks complement the exact algebraic identity; they do not certify time integration. Source hashes,

raw parameters, the calibration, all descent steps and retained observations bind this experiment to the same checkpoint as the uniform error certificate.

The implementation centers the large damping contribution using training data before adding the small potential residual. On the retained Windows platform, NumPy `longdouble` has the same 52 explicit mantissa bits as binary64; this is an operation-order improvement, not an extended-precision computation. An independent exact-rational replay of the uncentered cached residual bounds the centering displacement by $6.701\,10^{-11}$ in the scaled residual rows. Initial and final momentum-loss shifts are respectively below $4.104\,10^{-16}$ and $9.110\,10^{-17}$. The actual parameter gradient is also checked against directional finite differences, and positive-drag, zero-drag and active-set-boundary examples are checked independently against the constrained joint problem.

A nonempty same-model safety budget. Exact rational comparisons give $E_D^Q > 0.00474100399328$ and $E_c^Q > 0.03976078202199$ for $r_i = 0.1$ and $c = 4$ mm. The stiffness-buffer comparison permits $\chi_b > 1 - 1.508\,10^{-16}$. The smaller barrier here is the configuration proof box. A strictly feasible choice is $E_{\max} = 0.0023705019966407895$, approximately 1.4223 nJ in dimensional energy. For $q_0 = (1/80, 1/80)$ and $P_0 = 0$, $1 - \cos s \le s^2/2$ gives true initial energy below 0.000558759179723. The learned-energy initialization with the certified potential-error allowance is therefore feasible as well. These constants are recomputed for this planar model; no earlier full-beam control trajectory or KAM radius is transferred.

Same-checkpoint closed-loop diagnostics. For the scalar output $s = d^{\mathsf{T}} q$, a nominal feedback request uses the learned potential, learned drag, and the exact configuration transport of G_Q. Its input coefficient is $d^{\mathsf{T}} G_Q^{-1} g(q) > 0$ on the declared box. Set $s_{\mathrm{ref}} = 0.08 \sin(0.5\tau)$, position and velocity gains 4 and 3, $U_{\max} = 0.1$ and $\kappa = 1$. These choices, the preceding initial condition and $0 \le \tau \le 20$ are fixed before simulation. Initializing W with the learned energy plus $2.772\,10^{-13} + 10^{-16}$ includes the potential certificate and a declared numerical-evaluation reserve. The latter is a simulation allowance, not a rigorous floating-point evaluation bound.

Both amplitude-only and power-filtered requests are integrated in the full-cosine reduced plant with the same Gauss metric. Two DOP853 settings, $(\mathrm{rtol}, \mathrm{atol}, h_{\max}) = (10^{-9}, 10^{-12}, 0.02)$ and $(2\,10^{-11}, 2\,10^{-14}, 0.01)$, provide a numerical refinement check. A separately coded NumPy metric, its derivatives, and the frozen complex potential are compared with automatic differentiation at twelve fixed random states before integration.

In the refined filtered run, the sampled maxima are $H_* = 0.001740966181$, $W = 0.001858865533$ and $\max_i |q_i| = 0.05773832$. The energy-derived centreline bound is 0.83701 mm. The continuous-time theorem with this energy cap gives the slightly larger all-time bound 0.976681 mm. The maximum difference between the two filtered numerical state solutions is $1.545\,10^{-9}$, and the refined accumulated energy-balance defect is below $3.904\,10^{-18}$. These are ODE diagnostics, not validated time integration or a native COMSOL feedback replay.

The input amplitude limit is active for approximately 93.95% of sampled nominal requests; the power filter changes the amplitude-limited request at 6.30% of samples. Output RMS tracking error remains 0.04831, against reference amplitude 0.08. The amplitude-only comparison also stays inside the box and clearance, although it violates the instantaneous work constraint by up to 0.000699924. This test therefore demonstrates implementation of the budget and a feasible safe response, without establishing collision avoidance relative to that comparison or successful reference tracking.

S3.18 A complex factor network for dissipation

Retain the frozen conservative Hamiltonian and the same canonical chart,

$$H_{\widehat{\theta}}(q,P) = \tfrac{1}{2} P^{\mathsf{T}} G_Q(q)^{-1} P + V_{\widehat{\theta}}(q), \qquad v = \partial_P H_{\widehat{\theta}} = G_Q(q)^{-1} P. \tag{S201}$$

Overdots in this subsection denote derivatives with respect to the dimensionless time τ used in the mechanical decoder. Here G_Q is smooth and positive definite on the declared real configuration box $\mathcal{Q}$. A separate network maps q to three complex entries of a lower-triangular matrix $L_\phi(q)$. Its two complex fully connected hidden layers use literal complex $\tanh$, followed by a complex affine readout. Thus the affine arithmetic has the real block form described by Trabelsi et al. (2018); the activation and pole constraints are specified below. Let $G_Q = S_G S_G^{\mathsf{T}}$ be its real Cholesky factorization and set

$$M_\phi(q) = \mathrm{Re}\big(L_\phi(q) L_\phi(q)^*\big), \qquad R_\phi(q) = S_G(q) M_\phi(q) S_G(q)^{\mathsf{T}}. \tag{S202}$$

The star denotes conjugate transpose. This factor construction adapts the positive-semidefinite dissipation decoder of Zhong et al. (2020) to a complex network. The trainable coordinates ϕ are all raw real and imaginary affine parameters.

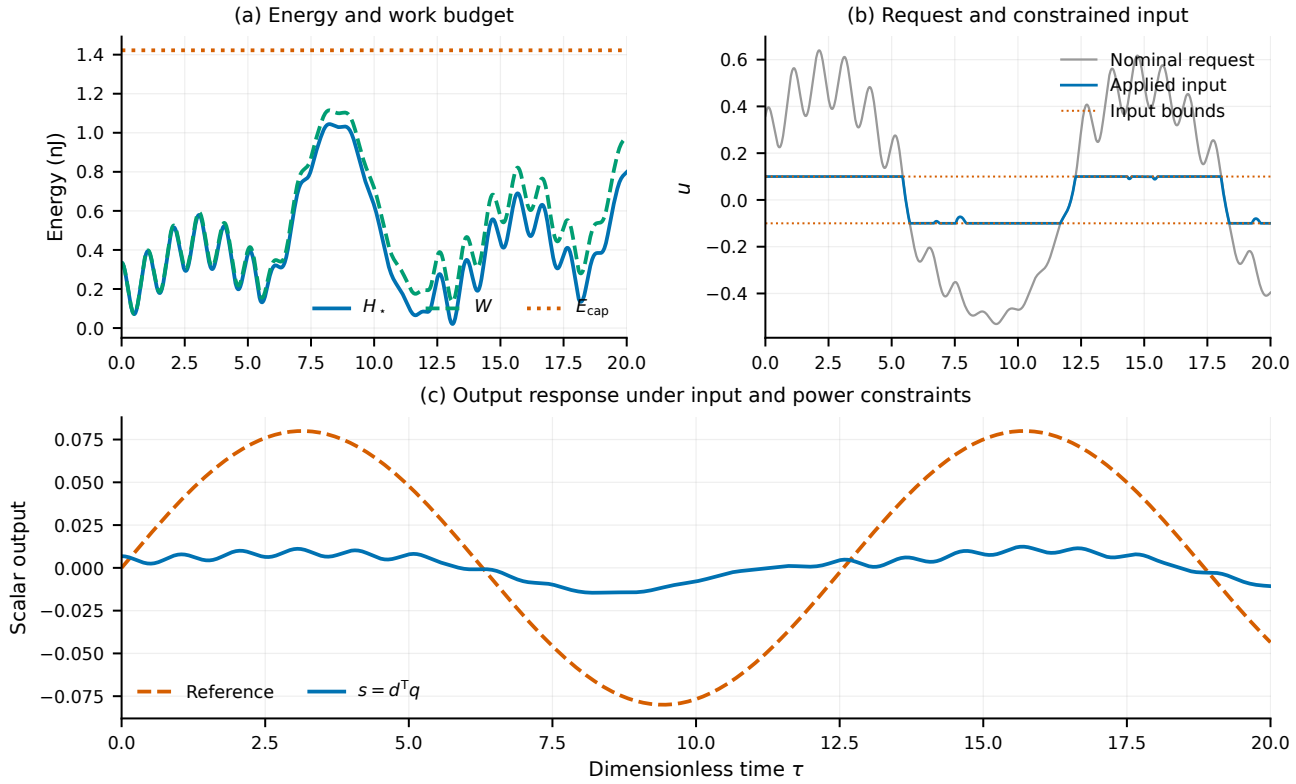


Figure S1. Same-checkpoint planar magnetic control diagnostics. The refined filtered run shows true energy and accumulated input work (top left), the nominal and constrained scalar commands (top right), and the substantial output tracking error (bottom). Only the energy axis is dimensional; input and output follow the declared canonical normalization.

Proposition S44. Rayleigh decoding and exact port power. *For every finite ϕ, $R_\phi(q)$ is real symmetric positive semidefinite. Define*

$$\mathcal{D}_\phi(q,v) = \tfrac{1}{2} v^\mathsf{T} R_\phi(q) v,$$
$$D_\phi(q,P) = \mathcal{D}_\phi(q, G_Q(q)^{-1} P).$$

With $g(q) = d\cos(d^\mathsf{T} q)$, the mechanical decoder

$$\dot{q} = \partial_P H_{\widehat{\theta}}, \qquad \dot{P} = -\partial_q H_{\widehat{\theta}} - R_\phi(q) v + g(q) u \tag{S203}$$

has dissipative force

$$F_\mathrm{d} = -\partial_v \mathcal{D}_\phi = -G_Q \partial_P D_\phi = -R_\phi v \tag{S204}$$

and satisfies

$$\begin{aligned} \frac{d}{d\tau} H_{\widehat{\theta}} &= -v^\mathsf{T} R_\phi(q) v + uy \\ &= -2\mathcal{D}_\phi(q,v) + uy \le uy, \\ y &= g(q)^\mathsf{T} v. \end{aligned} \tag{S205}$$

These identities hold for open-loop input and state feedback wherever the corresponding solution exists.

Proof. Write $L_\phi = A_\phi + iB_\phi$ with real matrices. Then $M_\phi = A_\phi A_\phi^\mathsf{T} + B_\phi B_\phi^\mathsf{T}$ and, for every real vector a,

$$a^\mathsf{T} R_\phi a = \|A_\phi^\mathsf{T} S_G^\mathsf{T} a\|_2^2 + \|B_\phi^\mathsf{T} S_G^\mathsf{T} a\|_2^2 \ge 0.$$

Since R_ϕ depends only on q, differentiation at fixed q gives $\partial_v \mathcal{D}_\phi = R_\phi v$ and $\partial_P D_\phi = G_Q^{-1} R_\phi v$. Finally, multiplication of (S203) by the energy gradient cancels the two canonical Hamiltonian terms and yields (S205).

Thus $H_{\widehat{\theta}} + iD_\phi$ can record the two learned functions, with dynamics defined by (S203)–(S204). In canonical momentum coordinates the mass factor in $-G_Q \partial_P D_\phi$ is essential. Configuration dependence of G_Q is already included in $\partial_q H_{\widehat{\theta}}$ at fixed P. Magnetic actuation retains the separate supply uy, whose sign is unrestricted. The specialization $L_\phi \equiv \sqrt{\gamma_0} I$ gives $R_\phi = \gamma_0 G_Q$ and $F_\mathrm{d} = -\gamma_0 P$ for any $\gamma_0 \ge 0$; $L_\phi = 0$ represents zero dissipation exactly.

Uniform bounds from the complex readout. Normalize q to $\widetilde{q}$ with $|\widetilde{q}_i| \le 1$. For the first layer, multiply the raw imaginary weights and bias of row j by $s_j = \rho_1/(\rho_1 + U_j)$, where

$$U_j = \sum_i \sqrt{(B_{ji}^{(1)})^2 + \epsilon^2} + \sqrt{(b_{j,\mathrm{im}}^{(1)})^2 + \epsilon^2}, \qquad \epsilon > 0.$$

Its preactivation has imaginary part strictly below ρ_1 in modulus. For the second layer, scale both raw weight parts and the imaginary bias using $\rho_2/(\rho_2 + U_j^{(2)})$, with

$$U_j^{(2)} = \sum_i \Big(\sqrt{(A_{ji}^{(2)})^2 + \epsilon^2} + \sqrt{(B_{ji}^{(2)})^2 + \epsilon^2} \Big) + \sqrt{(b_{j,\mathrm{im}}^{(2)})^2 + \epsilon^2}.$$

Take $0 < \rho_1, \rho_2 < \pi/4$. The identity

$$|\tanh(x+iy)|^2 = \frac{\cosh(2x) - \cos(2y)}{\cosh(2x) + \cos(2y)} \le 1, \qquad |y| < \pi/4,$$

first bounds every first-layer feature in modulus by one. In particular, both its real and imaginary parts have modulus at most one, so the stated second-layer budget also bounds its imaginary preactivation strictly by ρ_2. Every final hidden feature $h_j(q)$ therefore satisfies $|h_j(q)| \le 1$ throughout $\mathcal{Q}$, for every finite raw parameter vector.

Proposition S45. An explicit whole-box dissipation error bound. *Write the final complex readout as $\ell_i(q) = b_i + \sum_j W_{ij} h_j(q)$, where $(\ell_1, \ell_2, \ell_3) = (L_{11}, L_{21}, L_{22})$. For a fixed reference $\gamma_0 \ge 0$, put $c_0 = \sqrt{\gamma_0}$, $\iota = (1, 0, 1)$ and*

$$\begin{aligned} e_i &= |b_i - c_0 \iota_i| + \sum_j |W_{ij}|, \\ \Delta_L &= \left(\sum_{i=1}^{3} e_i^2 \right)^{1/2}, \\ \eta_\phi &= 2c_0 \Delta_L + \Delta_L^2. \end{aligned} \tag{S206}$$

Then, uniformly on $\mathcal{Q}$,

$$\begin{aligned} \|L_\phi - c_0 I\|_F &\le \Delta_L, \\ \|M_\phi - \gamma_0 I\|_2 &\le \eta_\phi, \\ \|R_\phi - \gamma_0 G_Q\|_2 &\le \|G_Q\|_2 \eta_\phi. \end{aligned} \tag{S207}$$

All constants can be enclosed directly from the stored readout parameters using outward-rounded arithmetic. Alternatively, define

$$\begin{aligned} e_{i,\mathrm{r}} &= |\operatorname{Re} b_i - c_0 \iota_i| + \sum_j |W_{ij}|, \\ e_{i,\mathrm{i}} &= |\operatorname{Im} b_i| + \sum_j |W_{ij}|, \\ \Delta_{\mathrm{r}}^2 &= \sum_i e_{i,\mathrm{r}}^2, \qquad \Delta_{\mathrm{i}}^2 = \sum_i e_{i,\mathrm{i}}^2, \\ \eta_\phi^{\mathrm{split}} &= 2c_0 \Delta_{\mathrm{r}} + \Delta_{\mathrm{r}}^2 + \Delta_{\mathrm{i}}^2. \end{aligned} \tag{S208}$$

Then η_ϕ in the last two bounds of (S207) may be replaced by $\eta_\phi^{\mathrm{split}}$, or by the smaller of the two. This removes the first-order contribution of a purely imaginary readout perturbation.

Proof. The hidden-feature bound and the triangle inequality give the first estimate. Put $E = L_\phi - c_0 I$. Expansion gives

$$L_\phi L_\phi^* - \gamma_0 I = c_0 (E + E^*) + E E^*,$$

whose spectral norm is at most $2c_0 \|E\|_2 + \|E\|_2^2 \le \eta_\phi$. For real unit a, $a^{\mathsf{T}}(M_\phi - \gamma_0 I)a = a^*(L_\phi L_\phi^* - \gamma_0 I)a$; the variational characterization for real symmetric matrices gives the second estimate. Congruence by S_G gives the third, since $\|S_G\|_2^2 = \|G_Q\|_2$. For the alternative bound write $L_\phi = c_0 I + E_{\mathrm{r}} + i E_{\mathrm{i}}$ with real matrices. The feature bound gives $\|E_{\mathrm{r}}\|_F \le \Delta_{\mathrm{r}}$ and $\|E_{\mathrm{i}}\|_F \le \Delta_{\mathrm{i}}$. Taking the real part before estimating yields the exact identity

$$M_\phi - \gamma_0 I = c_0 (E_{\mathrm{r}} + E_{\mathrm{r}}^{\mathsf{T}}) + E_{\mathrm{r}} E_{\mathrm{r}}^{\mathsf{T}} + E_{\mathrm{i}} E_{\mathrm{i}}^{\mathsf{T}}.$$

The spectral norm triangle inequality gives $\|M_\phi - \gamma_0 I\|_2 \le \eta_\phi^{\mathrm{split}}$; the same metric congruence completes the proof.

For a target with the same metric and magnetic port, potential V_* and drag $R_* \succeq 0$, suppose

$$\begin{aligned} \|\nabla(V_{\widehat{\theta}} - V_*)\|_\infty &\le \varepsilon_V, \\ \sup_{\mathcal{Q}} \|G_Q\|_2 &\le G_{\max}, \\ \sup_{\mathcal{Q}} \|R_* - \gamma_0 G_Q\|_2 &\le \varepsilon_*. \end{aligned}$$

On any declared phase-space set with $\|v\|_2 \le V_{\max}$, the complete learned and target fields at the same state and input obey

$$\|f_{\widehat{\theta},\phi} - f_*\|_\infty \le \varepsilon_V + (G_{\max} \eta_\phi + \varepsilon_*) V_{\max}. \tag{S209}$$

The configuration components agree exactly, and the momentum difference is $-\nabla(V_{\widehat{\theta}} - V_*) - (R_\phi - R_*)v$. For the mass-proportional target $R_* = \gamma_* G_Q$, one may take $\varepsilon_* = G_{\max}|\gamma_* - \gamma_0|$; choosing $\gamma_0 = \gamma_*$ sets this term to zero. A previously fitted γ_0 is also a valid reference when this additional discrepancy is retained. The bound is a computable sufficient estimate and need not be sharp. More explicitly, if $|P_i| \le P_{\max}$ and $\sup_{\mathcal{Q}} \|G_Q^{-1}\|_\infty \le A_{\max}$, symmetry implies $\|G_Q^{-1}\|_2 \le A_{\max}$. Put $\varepsilon_R = G_{\max}\eta_\phi + \varepsilon_*$. Then

$$\begin{aligned} \|f_{\widehat{\theta},\phi} - f_*\|_\infty &\le \varepsilon_V + \sqrt{2}\, \varepsilon_R A_{\max} P_{\max}, \\ |v^{\mathsf{T}}(R_\phi - R_*)v| &\le 2\varepsilon_R A_{\max}^2 P_{\max}^2. \end{aligned} \tag{S210}$$

These follow from $\|P\|_2 \le \sqrt{2}P_{\max}$ and $\|v\|_2 \le A_{\max}\|P\|_2$. The same conclusions hold with $\eta_\phi^{\mathrm{split}}$ in place of η_ϕ.

Training with fixed observed weak momentum equations. All observations, windows, positive quadrature weights, component scales and the conservative Hamiltonian are fixed before this optimization. At each observed node reconstruct $P_j = G_Q(q_j)v_j$. Let $\mathcal{L}_H(q,v) = \frac{1}{2}v^{\mathsf{T}}G_Q(q)v - V_{\widehat{\theta}}(q)$, and let a_{wj} be the fixed integration weights in window w. Its physical unscaled weak momentum residual is

$$\begin{aligned} r_w^{\mathrm{phys}}(\phi) = P_{w,b} - P_{w,a} \\ - \sum_j a_{wj}\big[\partial_q \mathcal{L}_H(q_j, v_j) \\ + g(q_j)u_j - R_\phi(q_j)v_j\big]. \end{aligned} \tag{S211}$$

The derivative $\partial_q \mathcal{L}_H$ holds v fixed; it equals $-\partial_q H_{\widehat{\theta}}$ evaluated at $P = G_Q v$, where the latter derivative holds P fixed. The implemented objective treats the stored observation and weak-equation caches as fixed constants. Denote the stored node arrays by $\bar{q}_j, \bar{v}_j, \bar{G}_j, \bar{P}_j$, the stored quadrature weights by $\bar{a}_{wj}$, the stored baseline row vector by b_{c}, and the diagonal positive row-scale matrix by A_{c}. For each positive-definite cached metric set

$$R_{\phi,\mathrm{c}}(\bar{q}_j) = \mathrm{chol}(\bar{G}_j) M_\phi(\bar{q}_j)\, \mathrm{chol}(\bar{G}_j)^{\mathsf{T}}.$$

Fix a previously fitted centering coefficient γ_{ref}. The exact real-arithmetic objective defined by the stored constants is

$$\begin{aligned} r_{\mathrm{c}}(\phi) &= b_{\mathrm{c}} + A_{\mathrm{c}}\, \mathrm{col}_w\left(\sum_j \bar{a}_{wj}[R_{\phi,\mathrm{c}}(\bar{q}_j)\bar{v}_j - \gamma_{\mathrm{ref}}\bar{P}_j]\right), \\ F(\phi) &= \|r_{\mathrm{c}}(\phi)\|_2^2 + K_X + \frac{\lambda}{2}\|\phi\|_2^2, \qquad K_X \ge 0, \quad \lambda > 0. \end{aligned} \tag{S212}$$

The stacked baseline b_{c} already includes the same window and component scaling. Here K_X is the fixed kinematic consistency term. The penalty includes every raw real and imaginary head coordinate. In exact arithmetic before cache rounding, the physical centering identity is

$$r_w^{\mathrm{phys}}(\phi) = r_w^{\mathrm{phys}}(\gamma_{\mathrm{ref}}G_Q) + \sum_j a_{wj}[R_\phi(q_j)v_j - \gamma_{\mathrm{ref}}P_j].$$

It explains the cached construction without introducing a damping label or variable projection for the matrix head. Independently rounded cached momenta need not equal $\bar{G}_j\bar{v}_j$ exactly when all cached entries are subsequently interpreted as exact real numbers. Thus (S212) is the precise optimization objective; its agreement with (S211) requires observation, quadrature and cache-rounding error bounds. These discrepancies are distinct from finite-precision evaluation of F itself. No acceleration or independent power labels enter this fit.

Theorem S46. Full-batch descent for the fixed-energy drag objective. *Assume every observed q_j belongs to $\mathcal{Q}$ and its fixed metric is positive definite. Use the smooth pole-budget parameterization above and hold all observations, scalings, caches and conservative parameters fixed. For constants $0 < c_A < 1$, $0 < \beta < 1$ and $\overline{\alpha} > 0$, apply exact full-batch gradient descent to (S212). At each nonstationary iterate choose the first $\alpha_k = \overline{\alpha}\beta^m$ satisfying*

$$F(\phi_k - \alpha_k \nabla F(\phi_k)) \le F(\phi_k) - c_A\alpha_k\|\nabla F(\phi_k)\|_2^2,$$

with unrestricted backtracking. Every line search terminates, the iterates are bounded, and $\sum_k \|\nabla F(\phi_k)\|_2^2 < \infty$. Consequently $\|\nabla F(\phi_k)\|_2 \to 0$, and every accumulation point is stationary for the same objective.

Proof. The smooth absolute-value majorant, positive budget denominators and strict avoidance of complex $\tanh$ poles make every sampled network output smooth in all raw real parameters. The PSD decoder is polynomial in the real and imaginary factor entries, with fixed metric factors. Hence F is smooth and nonnegative, and

$$\|\phi\|_2^2 \le 2F(\phi_0)/\lambda \quad \text{whenever } F(\phi) \le F(\phi_0).$$

Accepted iterates lie in this compact ball. The gradient is bounded there, so all trial segments with length at most $\overline{\alpha}\|\nabla F(\phi_k)\|_2$ lie in one larger compact ball. Smoothness gives a finite common gradient Lipschitz constant $L_F > 0$ on that ball. The descent lemma guarantees acceptance whenever $\alpha \le 2(1-c_A)/L_F$. Therefore

$$\begin{aligned} \alpha_k \ge \alpha_{\min} := \min\{\overline{\alpha}, 2\beta(1-c_A)/L_F\} > 0, \\ c_A\alpha_{\min}\sum_{k=0}^{N-1}\|\nabla F(\phi_k)\|_2^2 \le F(\phi_0). \end{aligned}$$

Taking $N \to \infty$ proves square summability and gradient convergence. Continuity gives stationarity at every accumulation point. If a zero gradient is reached, the iteration may stop.

Corollary S47. Uniformly conditioned descent directions. *Under the fixed-cache hypotheses of Theorem S46, let the proposed directions d_k satisfy, for constants $m_d, M_d > 0$,*

$$\begin{aligned} -\nabla F(\phi_k)^{\mathsf{T}} d_k \ge m_d\|\nabla F(\phi_k)\|_2^2, \\ \|d_k\|_2 \le M_d\|\nabla F(\phi_k)\|_2. \end{aligned}$$

Choose $\alpha_k = \overline{\alpha}\beta^m$ by unrestricted Armijo backtracking with

$$F(\phi_k + \alpha_k d_k) \le F(\phi_k) + c_A \alpha_k \nabla F(\phi_k)^{\mathsf{T}} d_k.$$

The boundedness, finite line-search termination, gradient square-summability and stationary-accumulation conclusions remain valid. In particular, the direction conditions hold for $d_k = -B_k \nabla F(\phi_k)$ when the symmetric positive-definite raw-coordinate operators satisfy $m_d I \preceq B_k \preceq M_d I$.

Proof. The same compactness argument gives a common L_F on all trial segments. The descent lemma guarantees acceptance whenever

$$\alpha \le \frac{2(1 - c_A) m_d}{L_F M_d^2}.$$

Thus accepted steps have a uniform positive lower bound, and each decrease is at least $c_A \alpha_{\min} m_d \|\nabla F(\phi_k)\|_2^2$. Summation and continuity finish the proof.

Corollary S48. Same-objective scaled Gauss–Newton directions. *Let $r = r_c(\phi)$, $J = Dr_c(\phi)$ and $g = \nabla F(\phi) = 2J^{\mathsf{T}} r + \lambda\phi$. Fix a positive diagonal matrix $D = \mathrm{diag}(s_i)$, a raw-coordinate direction damping $\eta > 0$ and a scaled direction damping $\mu \ge 0$. Solve exactly*

$$\min_y \left\| \begin{bmatrix} \sqrt{2}JD \\ \sqrt{\lambda}D \\ \sqrt{\eta}D \\ \sqrt{\mu}I \end{bmatrix} y - \begin{bmatrix} -\sqrt{2}r \\ -\sqrt{\lambda}\phi \\ 0 \\ 0 \end{bmatrix} \right\|_2^2, \qquad d = Dy. \quad \text{(S213)}$$

Under the preceding fixed-cache hypotheses, these directions satisfy the uniform conditioning assumptions of Corollary S47 on the initial sublevel set. The same conclusion holds if any proposed direction is replaced by the fixed scaled steepest-descent direction $-D^2 g$. With exact arithmetic, exact direction solves and unrestricted Armijo backtracking, the gradient therefore tends to zero and every accumulation point is stationary for the same F.

Proof. The augmented matrix has full column rank because $\eta > 0$ and D is invertible. Multiplying its normal equation by D^{-1} gives

$$A(\phi) d = -g, \qquad A(\phi) = 2J^{\mathsf{T}} J + (\lambda + \eta) I + \mu D^{-2}. \quad \text{(S214)}$$

In particular, η damps the direction and does not alter the objective F. The pole budgets and finite fixed cache make J continuous in all finite raw coordinates. Since $F \ge \lambda\|\phi\|_2^2/2$, the closed initial sublevel set $K = \{\phi : F(\phi) \le F(\phi_0)\}$ is compact. Hence $C_J = \sup_{\phi \in K} \|J(\phi)\|_2 < \infty$. Put $s_{\min} = \min_i s_i > 0$ and $s_{\max} = \max_i s_i < \infty$. For every $\phi \in K$,

$$a_{\min} I \preceq A(\phi) \preceq a_{\max} I,$$
$$a_{\min} = \lambda + \eta + \mu / s_{\max}^2 > 0,$$
$$a_{\max} = 2C_J^2 + \lambda + \eta + \mu / s_{\min}^2 < \infty.$$

Thus $a_{\max}^{-1} I \preceq A(\phi)^{-1} \preceq a_{\min}^{-1} I$, independently of the rank of J. For the optional fallback, $s_{\min}^2 I \preceq D^2 \preceq s_{\max}^2 I$. Both choices are therefore covered with

$$m_d = \min\{a_{\max}^{-1}, s_{\min}^2\} > 0,$$
$$M_d = \max\{a_{\min}^{-1}, s_{\max}^2\} < \infty.$$

The previous corollary applies to their unrestricted Armijo iteration, beginning at ϕ_0; every accepted point stays in K, so the same constants apply at the next step.

For example, choose the scales once from the initial residual Jacobian as

$$s_i = 1 / \max\{\|J(\phi_0)_{:,i}\|_2, \sqrt{\lambda}\}.$$

They are finite and strictly positive, and remain fixed throughout optimization. The compactness proof supplies existence of C_J and the uniform conditioning constants; it is not a numerical interval enclosure of their values.

This is a first-order optimization result for the regularized observed weak loss. It asserts neither global optimality nor unique function or factor identification. A finite-precision implementation with a finite backtracking budget can instead terminate at its last accepted iterate; that finite record alone does not establish asymptotic convergence.

Conservative persistence and true-power safety. Training only ϕ leaves the raw Hamiltonian, mechanical buffers and canonical vector field $J\nabla H_{\widehat{\theta}}$ unchanged. Thus the finite-amplitude conservative torus of Theorem S41 remains the same certified object when dissipation and input are switched off. The enlarged dissipative field in (S203) has the separate power identity (S205); positive dissipation does not inherit conservative invariant tori. A general $R_\phi(q)$ also requires a separate argument before any conformal symplectic or Birkhoff conclusion. The PSD identity alone does not impose entropy-degeneracy conditions of a full metriplectic formulation.

Corollary S49. Work filtering with general nonnegative true dissipation. *Retain the geometry, true-energy lower bounds defining E_c^Q and E_D^Q, initial energy certificate, common kinetic metric, magnetic port, exact sensing and actuation, and local regularity hypotheses of Theorem S43. Replace its*

mass-proportional true power relation by

$$\dot{H}_* = -\delta_*(q,P) + uy, \qquad \delta_*(q,P) \geq 0.$$

The same work filter and the same strict initial energy budget still imply its configuration-domain and whole-centreline clearance conclusions. In particular this holds for $\delta_* = v^{\mathsf{T}} R_*(q) v$ *with any locally regular* $R_*(q) \succeq 0$.

Proof. With the actual measured port output and $\dot{W} = uy$, $\frac{d}{d\tau}(W - H_*) = \delta_* \geq 0$. The certified initial inequality therefore gives $H_* \leq W$ until a possible first exit. The same input constraint implies $E_{\max} - W(\tau) \geq (E_{\max} - W(0))e^{-\kappa\tau} > 0$. The unchanged true-energy barriers exclude that exit, and the same local regularity and bounded-energy argument gives continuation for all forward time.

The nonnegative sign in this corollary is a property of the true plant; algebraic positivity of a fitted head does not establish that physical premise. The head may inform the requested command without being used to subtract an uncertified dissipation estimate from the work budget.

Frozen-energy complex-head identification. The conservative checkpoint in Section S3.16 is held fixed while a separate width-two, two-hidden-layer complex factor network is optimized. Its complex affine readout produces the three lower-triangular entries of L_ϕ; all 42 raw real and imaginary coordinates, including both hidden layers, participate in the weak objective (S212). The imaginary budgets are $\rho_1 = \rho_2 = 0.5$, and the real configuration domain is $|q_i| \leq 0.1$. The previously fitted scalar drag supplies only an initial factor and numerical centering reference. No target drag, acceleration or independent power labels enter this optimization.

The fourteen accepted training trajectories supply 70 windows and 140 scalar momentum rows. With $\lambda = 10^{-10}$, fixed initial Jacobian column scaling, $\eta = 10^{-2}$ and $\mu = 10^{-12}$, the augmented direction in (S213) is accepted by Armijo backtracking in all 12 prescribed steps. The pure momentum weak loss decreases from 42.8534 to $1.27751\,10^{-10}$; a nested constant factor initialized from the previously fitted scalar gives $1.66948\,10^{-10}$ under the same cache and scaling. Section S3.19 distinguishes its stored-factor coefficient from the original checkpoint scalar and compares both continuous equations. Every parameter tensor changes. An independent implementation replays the accepted iterates, final raw parameters and objective components exactly in binary64. This is a finite optimization record, not a certificate of stationarity or global optimality.

After the head and training choices are fixed, four trajectories excluded from this head optimization supply 20 evaluation windows. Both models use the same frozen conservative energy and the same observed position–velocity quadrature. Table S5 reports all four results. The equally weighted pooled normalized weak loss is $9.32505\,10^{-11}$ for the factor head and $1.10234\,10^{-10}$ for the scalar reference. One trajectory becomes worse. The corresponding unscaled weighted momentum-rate component RMS values are $6.82827\,10^{-15}$ and $7.42409\,10^{-15}$, respectively. These values characterize the fixed numerical weak equations; such small residual differences do not establish physical identification accuracy, statistical generalization or a benefit from complex arithmetic. All four trajectories have zero applied input and span $0 \leq \tau \leq 0.2$. In particular, they do not identify an arbitrary state-dependent dissipative law from driven motion.

Table S5. Normalized weak momentum loss on every trajectory excluded from the complex-head optimization. The scalar reference and the new factor head share the same frozen conservative checkpoint.

Case	Scalar	Complex PSD	Ratio
Direction 0	$2.314\,10^{-11}$	$1.971\,10^{-11}$	0.852
Direction 1	$3.373\,10^{-10}$	$2.390\,10^{-10}$	0.709
Direction 2	$4.314\,10^{-11}$	$1.049\,10^{-10}$	2.432
Direction 3	$3.739\,10^{-11}$	$9.359\,10^{-12}$	0.250

Whole-box errors and preservation of the conservative certificate. For the frozen trained head, outward-rounded 256-bit evaluation of the separate real and imaginary readout bounds gives

$$\Delta_r < 9.797\,10^{-10}, \qquad \Delta_i < 1.819\,10^{-6}.$$

The split PSD estimate in Proposition S45, combined with the same certified Gauss metric and conservative force error, yields

$$\begin{aligned} \|f_{\widehat{\theta},\phi} - f_*\|_\infty &< 1.269\,10^{-11} + 3.236\,10^{-9} P_{\max}, \\ |v^{\mathsf{T}}(R_\phi - R_*)v| &< 2.829\,10^{-8} P_{\max}^2. \end{aligned} \tag{S215}$$

Here $|q_i| \leq 0.1$, $|P_i| \leq P_{\max}$, and the benchmark has the same metric, magnetic port and input, full cosine potential and $R_* = (7/200)G_Q$. This benchmark coefficient is opened for post-fit certification, not supplied as a training label. Independent 512-bit arithmetic encloses all 20 scalar quantities of the coarse and split certificates inside their original 256-bit enclosures. These are entire-box function bounds, not extrapolations from trajectory nodes or error bounds for a continuous rod or a native time solver.

The raw conservative checkpoint is unchanged, so the finite-amplitude torus of Theorem S41 retains its previous frequency, candidate and error certificate for the conservative subsystem.

The trained head in the single-port work filter. The same frozen head replaces the scalar drag in the nominal acceleration request of the magnetic controller; the full cosine benchmark plant retains its specified true damping. Initial state, gains, reference, input bound and safety budget are those of Section S3.17, with no retuning. The filter continues to constrain the actual supplied power and does not credit predicted dissipation. Both DOP853 tolerance settings complete $0 \le \tau \le 20$. For the refined filtered run, the energy-based centreline deviation bound is below 0.838 mm against a 4 mm clearance, and $\max_i |q_i| < 0.05774$. The maximum budget is 0.001859, below the cap 0.002371. The two filtered numerical trajectories differ by less than $2.375\,10^{-9}$ in the saved augmented state. All saved controller requests, port powers, work constraints and reported statistics are recomputed from the saved trajectories.

Tracking remains limited: the dimensionless output RMS error is 0.04830, and the requested input exceeds the amplitude bound at approximately 93.95% of the output samples. The filter changes the amplitude-limited request at approximately 6.30% of samples. The unfiltered amplitude-limited comparison also stays within the clearance budget in this test. Accordingly this experiment checks the implementation of the conditional work-filter theorem; it does not demonstrate a collision avoided by the filter, a tracking guarantee, native COMSOL feedback or validated time integration.

S3.19 From native observations to a continuous weak equation

Fix the conservative energy and complex dissipation head of Section S3.18. This subsection validates their final output against the same finite-Gauss reference equation. Its stored observations, raw network parameters and mechanical buffers denote exact binary64 rational values. The target damping constant is separately defined as the exact rational $\gamma_* = 7/200$. Put $A = G_Q^{-1}$ and

$$V_*(q) = \tfrac{1}{2} q^{\mathsf{T}} B q + 1 - \cos(d^{\mathsf{T}} q).$$

For the prescribed zero input, the target canonical field is

$$\begin{aligned} f_{*,q} &= AP, \\ f_{*,P_i} &= -\partial_i V_* - \tfrac{1}{2} P^{\mathsf{T}} A_i P - \gamma_* P_i, \end{aligned} \tag{S216}$$

where $A_i = \partial_i A$. The native generation source fixes the input identically to zero throughout each solve; zero values at the observation nodes alone would not prove this premise.

A fixed reconstruction and its initial-value problem. Write $x_j = (q_j, P_j)$ for the native canonical records and $a_j = (v_j^{\mathrm{obs}}, \dot{P}_j^{\mathrm{obs}})$ for the stored native derivatives, normalized to the dimensionless time τ. A piecewise quintic Hermite curve $\bar{x}$ interpolates these values and slopes, together with one fixed binary64 second-derivative design at every shared node. The latter only selects the curve; it is not assumed to equal an exact target derivative. The design and all endpoint data are identified by hashes before the curve is reused. Consequently $\bar{x}$ is globally C^2 and is an explicit polynomial on every original time interval.

Set $h_j = \tau_{j+1} - \tau_j$ using the exact stored times, and $z = 2(\tau - \tau_j)/h_j - 1$. Neither nominal equal spacing nor a floating-point solver's internal dense output is substituted for this reconstruction. Its full four-dimensional defect is

$$r(\tau) = \bar{x}'(\tau) - f_*(\bar{x}(\tau)). \tag{S217}$$

The target solution below starts at $x_*(\tau_0) = x_0$. In all eighteen cases this first native node agrees bitwise with the initial configuration and momentum parsed from the generation protocol. This does not identify a binary64 initial value with an unrounded physical measurement or exact decimal input.

An energy norm retaining the linear oscillation. Let $G_0 = G_Q(0)$, $K_0 = B + dd^{\mathsf{T}}$ and

$$\begin{aligned} A_0 &= \begin{pmatrix} 0 & G_0^{-1} \\ -K_0 & -\gamma_* I \end{pmatrix}, \\ W &= \begin{pmatrix} K_0 & \gamma_* I/2 \\ \gamma_* I/2 & G_0^{-1} \end{pmatrix}. \end{aligned} \tag{S218}$$

The condition $K_0 - \gamma_*^2 G_0/4 \succ 0$ gives $W \succ 0$ by the Schur complement. Direct multiplication gives

$$A_0^{\mathsf{T}} W + W A_0 = -\gamma_* W. \tag{S219}$$

For the frozen buffers, exact rational LDL factorization verifies positive definiteness and the identity without a tolerance.

Proposition S50. A posteriori trajectory enclosure. *Let $\mathcal{B}$ be a compact convex box in a smooth domain of f_*. Suppose*

$\epsilon, d_j \geq 0$ *satisfy*

$$\sup_{x\in\mathcal{B}} \|W^{1/2}(Df_*(x) - A_0)W^{-1/2}\|_2 \leq \epsilon,$$
$$\sup_{\tau\in[\tau_j,\tau_{j+1}]} \|r(\tau)\|_W \leq d_j.$$

Let $R(\tau_0) \geq \|x_*(\tau_0) - \bar{x}(\tau_0)\|_W$ *and propagate* $R' = \ell R + d_j$ *on each interval, with* $\ell = -\gamma_*/2 + \epsilon$. *If the entire resulting tube lies strictly inside* $\mathcal{B}$, *the target IVP has a unique solution throughout the partition and*

$$\|x_*(\tau) - \bar{x}(\tau)\|_W \leq R(\tau).$$

Proof. Until a possible first exit, put $e = x_* - \bar{x}$. Convexity and the integral mean-value formula give

$$e' = A_0 e + \int_0^1 [Df_*(\bar{x} + se) - A_0]e\, ds - r.$$

Equation (S219) implies $D^+\|e\|_W \leq \ell\|e\|_W + d_j$. Scalar comparison proves the bound until that exit. Strict tube inclusion excludes it, and compactness inside the smooth domain continues the solution.

The exact scalar endpoint map and its outward-rounded use are

$$\begin{aligned} \widetilde{R}_{j+1} &= e^{\ell h_j} R_j + d_j \frac{\operatorname{expm1}(\ell h_j)}{\ell}, \\ R_{j+1} &= \operatorname{up}(\widetilde{R}_{j+1}) \geq \widetilde{R}_{j+1}, \end{aligned} \tag{S220}$$

where up returns a certified upper bound. The exact endpoint map has continuous interpretation $R_j + d_j h_j$ at $\ell = 0$. Radii propagate from the initial node and are never reset at window boundaries. For the present positive ℓ, the endpoint radius also bounds its whole interval. Coordinate inclusion uses $|e_i| \leq R\sqrt{(W^{-1})_{ii}}$.

The Jacobian enclosure retains the oscillatory linear part. Entrywise inverse-derivative bounds follow from

$$\begin{aligned} A_i &= -AG_iA, \\ A_{ij} &= AG_jAG_iA + AG_iAG_jA - AG_{ij}A. \end{aligned}$$

If $B_E \geq |Df_* - A_0|$ entrywise, then

$$\|W^{1/2}(Df_* - A_0)W^{-1/2}\|_2^2 \leq \operatorname{tr}(|W|B_E|W^{-1}|B_E^{\mathsf{T}}) \tag{S221}$$

is a sufficient upper bound, by the Frobenius norm and entrywise absolute-value expansion. The certificate evaluates its right side first and takes an outward upper square root. On $|q_i|, |P_i| \leq 1/10$ it yields

$$\epsilon < 0.051162, \qquad \ell < 0.033662.$$

Thus the large linear mechanical frequencies do not enter a crude Gronwall exponent.

Whole-interval finite-graph evaluation. The metric polynomial is an enclosure of the exact finite-Gauss graph, rather than a fit. Each Gram contribution has the form $w\sum_{a,b} c_{ia}c_{kb}\cos((\beta_a - \beta_b)^{\mathsf{T}}q)$, with $w \geq 0$. Let $S_i = \sum_a |c_{ia}|$ and let D bound the ℓ^1 diameter of its phases. For an even total Taylor degree N, on $|q_i| \leq r_q$, each derivative of order $m = 0, 1, 2$ has remainder at most

$$wS_iS_kD^m \frac{(Dr_q)^{N+2-m}}{(N+2-m)!}. \tag{S222}$$

Indeed, differentiate the cosine-difference series before applying the real Taylor remainder. Its first omitted degrees are $N+2, N+1, N$, respectively, and every real sine or cosine derivative has modulus at most one. Sum this bound over the Gram terms. The implementation uses $N = 10$ and $r_q = 1/10$; it never differentiates an unstructured remainder.

On each time interval, a Taylor model means $p(z) + [-e, e]$ for every real $|z| \leq 1$, with interval polynomial coefficients. Polynomial products place every truncated coefficient in the remainder; inversion uses a contractive geometric series. These operations, independent metric derivative tails and real sine tails enclose the defect on the whole interval. All calculations use outward-rounded Arb arithmetic. The final certificates cover all 14 400 intervals of the fourteen training and four held-out trajectories. In the norm of (S218),

$$\max_\tau \|x_*(\tau) - \bar{x}(\tau)\|_W < 2.114\,10^{-13}. \tag{S223}$$

Both configuration and momentum component errors are below $2.114\,10^{-13}$; the full curves and their tubes remain strictly inside the stated box. The final stored time is the exact binary64 value represented by 0.2, and every step uses the actual stored endpoint difference.

Native momentum and the training observables. The native canonical momentum is used only in this post-training validation. The frozen objective still uses only $q, v^{\mathrm{obs}}, \tau, u$, reconstructing $\Pi_j = G_Q(q_j)v_j^{\mathrm{obs}}$. A separate certificate at all 14 418 original nodes gives

$$\max_j \|\Pi_j - P_j\|_\infty < 3.185\,10^{-16}. \tag{S224}$$

This differs from the cached floating-point momentum versus exact $G_Q v^{\mathrm{obs}}$ error, whose upper bound is $1.178\,10^{-16}$. The latter belongs to the implementation cache and cannot replace (S224).

The continuous residual of the same learned output. Let $\widehat{V} = V_4 + C_{\widehat{\theta}}$ and $\widehat{R} = S_G M S_G^{\mathsf{T}}$ be the frozen learned functions. Because their kinetic metric and input port equal those of the target, the momentum-field difference is exactly

$$\begin{aligned} h(q,P) &= f_{*,P} - \widehat{f}_P \\ &= \nabla(\widehat{V} - V_*) + S_G(M - \gamma_* I)S_G^{-1}P. \end{aligned} \tag{S225}$$

For an original window $w = [a,b]$, its learned continuous weak residual on the exact target solution is therefore

$$\begin{aligned} \rho_w^* &= P_*(b) - P_*(a) - \int_a^b \widehat{f}_P(x_*(\tau))\,d\tau \\ &= \int_a^b h(x_*(\tau))\,d\tau. \end{aligned} \tag{S226}$$

Proposition S51. Continuous weak residual from a validated tube. *For each original interval j, add its certified coordinate tube bounds to the unstructured remainders of the polynomial reconstruction. Suppose rigorous composition with h_i gives*

$$h_i(x_*(\tau)) \in \sum_{k=0}^{n} b_{jk} z^k + [-e_{j,i}, e_{j,i}].$$

Then, for a window whose endpoints are partition nodes,

$$\rho_{w,i}^* \in \sum_{j \subset w} \left[h_j \sum_{\substack{k \le n \\ k \text{ even}}} \frac{b_{jk}}{k+1} + [-h_j e_{j,i}, h_j e_{j,i}] \right]. \tag{S227}$$

The unknown trajectory error need not be analytic, differentiable or polynomial. If $I_{w,i}^c$ encloses the exact analytic cache row and $s_{w,i}$ is its stored row scale, the total discrepancy obeys

$$r_{w,i}^c - s_{w,i}\rho_{w,i}^* \in I_{w,i}^c - s_{w,i} I_{w,i}^{\rho},$$

where I^{ρ} is the enclosure in (S227). The recorded binary64 row can be substituted with its separately certified rounding discrepancy.

Proof. The tube gives a pointwise real coordinate enclosure, so the composition enclosure holds for every time. Integrate its center polynomial exactly and bound the integral of its arbitrary remainder by $h_j e_{j,i}$. Summation proves (S227). Interval subtraction then encloses the two exact residuals' difference, with each row scale applied once. No derivative of the unknown error is used.

For this composition, complex $\tanh$ is expanded about the exact dyadic midpoint of its constant coefficient. A radius-$1/8$ complex circle is verified strictly inside $|\operatorname{Im} z| < \pi/4$, where $|\tanh z| \le 1$. For an argument increment of modulus at most $r < 1/8$, the degree-n Cauchy tail is at most $(8r)^{n+1}/(1-8r)$. Real metric square roots use a contractive binomial expansion. For $X = \alpha(1+\eta)$ with $\alpha > 0$ and $|\eta| \le r < 1$, its degree-n tail is at most $\sqrt{\alpha}\, r^{n+1}/(1-r)$, since $|\binom{1/2}{k}| \le 1$. The time-model norm certifies this bound including the unstructured tube remainder. The decoder retains $L = cI + E_r + iE_i$, with $c = \sqrt{\gamma_*}$, so that

$$M - \gamma_* I = c(E_r + E_r^{\mathsf{T}}) + E_r E_r^{\mathsf{T}} + E_i E_i^{\mathsf{T}}.$$

Likewise, writing $\theta = d^{\mathsf{T}} q$, the potential calculation uses

$$\begin{aligned} \theta - \theta^3/6 - \sin\theta &= \sum_{k=2}^{7} \frac{(-1)^{k+1}\theta^{2k+1}}{(2k+1)!} + \mathcal{R}_{17}, \\ |\mathcal{R}_{17}| &\le \frac{|\theta|^{17}}{17!}. \end{aligned}$$

The time-model norm supplies the uniform bound on $|\theta|$. The exact cancellation of the linear and cubic terms holds under arbitrary tube perturbations. These cancellations are algebraic identities of the frozen output, not fitted corrections.

Every original full window and each of its four quarter windows keeps its actual nodes, frozen weights, row scales and grouping. The integration is performed separately on the original intervals; it does not invoke a globally C^4 Simpson error formula for a curve that is only C^2 across its joints. Direct interval subtraction certifies the total cache-to-continuous discrepancy; it does not assign its quadrature, observation or cache contributions the value zero.

This result concerns one frozen output and the declared finite-Gauss reference IVP. It does not make the error bound uniform over training iterates, certify a continuum reduction or hidden solver steps, or establish statistical generalization and unique dissipation identification. A small weak integral can also contain cancellation. The entire-box conservative error estimate and finite-amplitude KAM certificate remain their separate, unchanged statements for the same conservative checkpoint.

Continuous residuals under the original normalization. For each trajectory the five windows are its full interval and its four successive quarter intervals. All original row scales are kept: the training group has 140 scalar rows and each excluded trajectory has ten. No additional average is applied to these scaled rows. Let r_F denote the recorded floating-point residual vector, r_C the exact analytic cache vector, r_D the exact mechanical vector with the stored quadrature weights, and r_I the continuous vector from (S226), with the same stored scales. Table S6 encloses $\|r_I\|_2^2$ and the

direct difference $\|r_F - r_I\|_2$. The bounds include the entire reference trajectories, rather than only their observation nodes.

For the training group the separately certified layer differences satisfy

$$\begin{aligned} \|r_F - r_C\|_2 &< 1.097\,10^{-10}, \\ \|r_C - r_D\|_2 &< 6.578\,10^{-7}, \\ \|r_D - r_I\|_2 &< 1.054\,10^{-5}. \end{aligned} \tag{S228}$$

Their triangle bound is below $1.120\,10^{-5}$, while direct interval subtraction yields the tighter bound in the table. The $r_D - r_I$ comparison includes the combined effect of the observations and quadrature; it is not a certificate assigning either effect zero separately. Exact endpoint Simpson weights and exact row scales are also evaluated as auxiliary variants, without replacing the objective used in training.

The continuous training loss is approximately $2.13219\,10^{-10}$, whereas the recorded discrete loss is $1.27751\,10^{-10}$. Thus a decrease of the implemented discrete objective cannot be relabelled as a decrease of the continuous objective. The maximum unscaled window-average momentum-field discrepancy for the continuous training equations is below $4.329\,10^{-14}$; its small absolute size reflects this benchmark's scale and does not supply a statistical error or an identifiability result. The original discrete descent statement and the entire-box function bounds retain their stated meanings.

A scalar comparison on the same continuous IVP. The original scalar checkpoint defines $\widehat{\gamma} = \log(1 + \exp(\widehat{a}))$, with the stored binary64 raw coordinate $\widehat{a}$ evaluated as an exact real parameter. Its conservative energy is identical to that of the complex-head model. Hence its continuous momentum-field error is

$$\begin{aligned} h_{\mathrm{scalar}}(q,P) = h(q,P) - S_G(M - \gamma_* I)S_G^{-1}P \\ + (\widehat{\gamma} - \gamma_*)P. \end{aligned} \tag{S229}$$

Applying the same tube and integration bounds to this correction gives the scalar column in Table S6, without retraining or changing the reference IVP. For the training group, the continuous loss ratio is enclosed by $[0.8597977,\ 0.8597981]$, a decrease of approximately 14.0202%. The two independent loss intervals already give a strict comparison; retaining the shared correction in $\|r_I\|_2^2 - \|r_{I,\mathrm{scalar}}\|_2^2$ also gives the tighter enclosure

$$[-3.4768280,\ -3.4768260]\,10^{-11}.$$

Directions 0 and 1 have larger continuous losses with the complex head, while directions 2 and 3 have smaller losses. Thus the continuous comparison retains a training improvement but changes the per-trajectory ordering of the discrete comparison. It does not establish uniform held-out improvement or an advantage over a matched real neural network.

The discrete training reference in Section S3.18 uses a nested constant factor initialized with the stored binary64 square root of the fitted floating-point scalar. Its exact mathematical coefficient is that stored bias squared, whereas the checkpoint scalar in (S229) is the exact softplus above. The continuous calculation also evaluates this nested variant. Its training loss lies in $[2.4798582,\ 2.4798587]\,10^{-10}$, and all five comparison signs remain the same. This distinction preserves the identities of both original comparators; it is not a change to the training objective or the frozen functions.

S3.20 *Finite-horizon safety with sampled port data and execution errors*

Theorem S43 has a separate implementation question: a port sample is held between decisions, and the applied input may differ from its command. We address that question on the same planar chart, with the same conservative Hamiltonian and frozen complex PSD head generating the requested command. The safety argument uses the true energy and a deterministic error reserve. It requires no new network fit and does not replace the preceding all-time theorem.

True power and a guard region. Let $A = G_Q^{-1}$, $v = AP$, and

$$\begin{aligned} H_*(q,P) &= \tfrac{1}{2}P^{\mathsf{T}}AP + V_*(q), \\ V_*(q) &= \tfrac{1}{2}q^{\mathsf{T}}K_b^Q q + 1 - \cos(d^{\mathsf{T}}q), \\ g(q) &= d\cos(d^{\mathsf{T}}q), \qquad y = g^{\mathsf{T}}v. \end{aligned} \tag{S230}$$

Assume the actual power relation is

$$\dot{H}_* = -\delta_*(q,P) + (u + e_a)y + p_{\mathrm{ext}}, \qquad \delta_* \geq 0. \tag{S231}$$

Dots denote derivatives in the dimensionless time τ. The true dissipation need not equal the learned head. Choose

$$\begin{aligned} E_{\max} < E_g < \min\{E_c^Q, E_D^Q\}, \\ Q_g = \{q \in D : V_*(q) \leq E_g\}. \end{aligned} \tag{S232}$$

where E_c^Q and E_D^Q are the same clearance and configuration barriers as in the exact theorem. Since $K_b^Q \succ 0$ and $E_g < E_D^Q$, the configuration sublevel is compact inside D; the continuous positive definite metric and the kinetic-energy

Table S6. Certified continuous pure weak losses for the frozen complex factor head and the original checkpoint scalar. Loss endpoints are displayed outwards in units of 10^{-10}. The last column bounds the complex head's discrete-to-continuous discrepancy, rather than an estimated standard error. Every original case is included.

Group	Complex PSD	Checkpoint scalar	$\|r_F - r_I\|_2$ upper bound
Training	[2.1321856, 2.1321861]	[2.4798683, 2.4798688]	$1.054\,10^{-5}$
Direction 0	[0.19935060, 0.19935063]	[0.086614827, 0.086614840]	$3.144\,10^{-6}$
Direction 1	[1.1256197, 1.1256201]	[1.0020625, 1.0020629]	$1.034\,10^{-5}$
Direction 2	[0.14613189, 0.14613192]	[0.44942806, 0.44942811]	$1.055\,10^{-5}$
Direction 3	[0.76717998, 0.76718005]	[1.5276436, 1.5276438]	$6.739\,10^{-6}$

bound then bound the momenta. Assume local uniqueness and continuation of the admitted true dynamics on compact subsets of this guard. Constants certified on $H_* \le E_g$, rather than only on $H_* \le E_{\max}$, can then be used in a first-exit proof without presupposing safety.

The complete holding interval. There are finitely many actual command switches $0 = \tau_0 < \cdots < \tau_N = T$. On each interval the command is u_k, the stored port sample is y_k, and a known bound $0 < h_k := \tau_{k+1} - \tau_k \le \Delta_k$ covers its duration. Until a possible first guard exit, assume

$$\begin{aligned} |y(\tau) - y_k| \le \varepsilon_y, \quad |e_a(\tau)| \le \xi_a, \\ |y(\tau)| \le Y_g, \quad p_{\mathrm{ext}}(\tau) \le \overline{p}, \end{aligned} \tag{S233}$$

with nonnegative bounds. If the measurement error at its timestamp is at most ε_m and the maximum sample age is Δ, a uniform bound $|\dot{y}| \le L_y$ gives

$$\varepsilon_y = \varepsilon_m + L_y \Delta. \tag{S234}$$

The age must include the hold and any admitted latency; a bound only at sample instants is insufficient. The numerical instance below has no external canonical force and no latency. More general force classes require new derivative bounds.

Let $U_{\lim}$ be the actual actuator limit. Choose a command limit $U_c > 0$ satisfying $U_c + \xi_a \le U_{\lim}$. In particular, for an actuator $a = (1 + e_r)u + e_b$, where $|e_r| \le \rho$, $|e_b| \le \varepsilon_b < U_{\lim}$ and $\rho \ge 0$, it is sufficient to take

$$U_c \le \frac{U_{\lim} - \varepsilon_b}{1 + \rho}, \qquad \xi_a = \rho U_c + \varepsilon_b. \tag{S235}$$

Here $e_a = e_r u + e_b$. Downward rounding of the command limit preserves this inequality. Merely clipping u to $U_{\lim}$ would not bound the applied input.

Proposition S52. Two sufficient power reserves. *Under* (S231) *and* (S233)*, every command* $|u_k| \le U_c$ *satisfies*

$$\dot{H}_* \le u_k y_k + \eta_{\mathrm{dir}}, \qquad \eta_{\mathrm{dir}} = U_c \varepsilon_y + \xi_a Y_g + \overline{p}. \tag{S236}$$

If the additional true-plant bound

$$\delta_* \ge \gamma_0 v^{\mathsf{T}} G_Q v, \quad \gamma_0 > 0, \qquad D_g \ge \sup_{q \in Q_g} g^{\mathsf{T}} A g \tag{S237}$$

is available, another sufficient rate is

$$\eta_{\mathrm{diss}} = U_c \varepsilon_y + \frac{\xi_a^2 D_g}{4\gamma_0} + \overline{p}. \tag{S238}$$

The first rate assumes only nonnegative true dissipation. The second requires true coercivity; a PSD learned head alone does not supply it.

Proof. The first bound follows from $u_k y \le u_k y_k + |u_k| \varepsilon_y$ and $e_a y \le \xi_a Y_g$, after dropping $-\delta_*$. For the second, complete the square in the true kinetic metric:

$$\begin{aligned} e_a g^{\mathsf{T}} v - \delta_* &\le -\gamma_0 \left\| v - \frac{e_a}{2\gamma_0} A g \right\|_{G_Q}^2 + \frac{e_a^2}{4\gamma_0} g^{\mathsf{T}} A g \\ &\le \frac{\xi_a^2 D_g}{4\gamma_0}. \end{aligned} \tag{S239}$$

Combining this with the same sensing bound proves the claim.

Initialization and exact work accumulation. Let $H_*(x_0) \le \overline{H}_0$ be a certified initial energy upper bound. For a specified uncertain full-state set X_0, a sufficient bound is $\overline{H}_0 \ge \sup_{x \in X_0} \widehat{H}(x) + \varepsilon_V$ when the shared kinetic metric and uniform potential certificate give $|\widehat{V} - V_*| \le \varepsilon_V$ on its projection. Evaluation only at the measured center does not cover unknown momentum. The numerical instance below instead fixes the original binary64 initial state exactly and certifies its true energy; it does not certify an initial full-state measurement tolerance.

Define the nominal held-data ledger by

$$\begin{aligned} Z_0 = \overline{H}_0, \qquad Z_{k+1} = Z_k + h_k u_k y_k, \\ Z(\tau_k + s) = Z_k + s u_k y_k, \qquad 0 \le s \le h_k. \end{aligned} \tag{S240}$$

Stored binary64 samples, commands and timestamps are interpreted as exact dyadic numbers for this calculation. Rational products, sums and acceptance checks give exact

work updates: a constant $u_k y_k$ requires no quadrature. This addresses numerical work integration without identifying a digital sample with an error-free physical measurement. Sensing and execution errors remain in (S233). A physical clock error needs its own verified budget; the theorem uses the actual holding times and the stated duration bounds.

Theorem S53. Finite-horizon sampled single-port filter. *Retain the guard, regularity and error-envelope hypotheses above. Choose either sufficient nonnegative rate η from Proposition S52, and require*

$$C = E_{\max} - \eta T, \qquad \overline{H}_0 < C. \tag{S241}$$

Fix $\kappa > 0$. At every switch choose the actual stored command to satisfy

$$|u_k| \le U_c, \qquad (1 + \kappa\Delta_k) u_k y_k \le \kappa(C - Z_k). \tag{S242}$$

Then the filter is recursively feasible, zero command is always feasible, and for $0 \le \tau \le T$,

$$H_*(\tau) \le Z(\tau) + \eta\tau < E_{\max}, \qquad |u_k + e_a(\tau)| \le U_{\lim}. \tag{S243}$$

The configuration remains in D and the entire centerline retains strict clearance. This is a finite-horizon statement.

Proof. Set $m_k = C - Z_k$ and $p_k = u_k y_k$. Initially $m_0 > 0$. If $p_k \ge 0$, (S242) gives, for every $0 \le s \le h_k \le \Delta_k$,

$$\begin{aligned} p_k &\le \kappa(m_k - s p_k), \\ m_k - h_k p_k &\ge \frac{m_k}{1 + \kappa\Delta_k} > 0. \end{aligned} \tag{S244}$$

For $p_k < 0$ the gap increases and the same inequalities hold. Induction proves $Z < C$ on every whole hold, including its endpoint. In particular $u_k = 0$ is feasible, even at $y_k = 0$. This endpoint tightening preserves the continuous nominal-work barrier inequality during the hold. The left-end condition $p_k \le \kappa m_k$ alone would not do so.

Up to a possible first guard exit, the chosen reserve gives $\dot{H}_* \le u_k y_k + \eta$. Integration, using the certified initial value and (S240), yields $H_* \le Z + \eta\tau < C + \eta\tau \le E_{\max}$. Since $E_{\max} < E_g$, this excludes an energy exit from the guard. The strict inequalities to E_D^Q and E_c^Q exclude configuration and clearance exits. The kinetic-energy bound, positive definite metric and compact configuration sublevel bound P; the assumed local regularity continues the solution through the finite partition. Finally $|u_k + e_a| \le U_c + \xi_a \le U_{\lim}$ proves the input bound.

For implementation, first clip any request to $a_k \in [-U_c, U_c]$. If it violates (S242), its nominal power is positive and scalar projection gives

$$u_k = \frac{\kappa(C - Z_k)}{(1 + \kappa\Delta_k) a_k y_k}\, a_k.$$

After rounding towards zero, check the stored command by exact rational arithmetic; fall back to zero if that check fails. There is no division at $y_k = 0$. A floating-point ODE state used as a work accumulator does not implement this exact ledger. Such an accumulator would require a separately certified numerical reserve, including its future accumulation, before the proof applies.

Uniform same-plant port constants. The numerical certificate bounds the stronger quantity $\sup_{q \in Q_g} d^{\mathsf{T}} A(q) d$ by D_g, which also bounds $g^{\mathsf{T}} A g$ because $|\cos s| \le 1$. The kinetic bound gives both

$$|d^{\mathsf{T}} v| \le Y_g, \quad |y| \le Y_g, \qquad Y_g^2 \ge 2E_g \sup_{q \in Q_g} d^{\mathsf{T}} A(q) d. \tag{S245}$$

The bound on $d^{\mathsf{T}} v$ is also needed when differentiating $y = \cos(s) w$, where $s = d^{\mathsf{T}} q$ and $w = d^{\mathsf{T}} v$:

$$\dot{y} = -\sin(s) w^2 + \cos(s) d^{\mathsf{T}} \dot{v}. \tag{S246}$$

For the specified true drag $\gamma_* = 7/200$ and canonical force f, differentiation of $P = G_Q v$ gives

$$\begin{aligned} G_Q \dot{v} &= -\nabla V_* - \gamma_* G_Q v + (u + e_a) g + f + c(v), \\ c_i(v) &= \tfrac{1}{2} v^{\mathsf{T}} (\partial_i G_Q) v - \sum_j [(\partial_j G_Q) v]_i v_j. \end{aligned} \tag{S247}$$

This cancellation retains the configuration-dependent kinetic term. The damping contribution to $|\dot{y}|$ is at most $\gamma_* Y_g$ and the actual input contribution is at most $U_{\lim} D_g$. Bounding the restoring and $c(v)$ terms on a guard box, and using $|\sin s| \le \min\{1, |s|\}$, supplies L_y.

The certificate uses positive finite-Gauss Gram phase moments, not a sampled maximum. Weighted second moments bound each real second derivative of G_Q. Since $DG_Q(0) = 0$, these give entrywise bounds $B_G \ge |G_Q - G_Q(0)|$ and first-derivative bounds throughout a box containing Q_g. With $A_0 = G_Q(0)^{-1}$, the verified condition $\||A_0| B_G\|_\infty < 1$ implies

$$|A(q)| \le (I - |A_0| B_G)^{-1} |A_0| \tag{S248}$$

entrywise. Exact rational sums and outward square roots then bound the kinetic components, the connection term and

Table S7. Deterministic budget for the same single-port reference plant, with $f = 0$ and no latency. These are conditional bounds, not measured hardware specifications.

Quantity	Value or certified bound
Horizon T	20
Maximum hold age Δ	$100000000001/10^{14}$
Sensor error ε_m	10^{-5}
Relative / additive error	$\rho = 10^{-2}, \varepsilon_b = 10^{-4}$
Actual input limit	$U_{\mathrm{lim}} = 1/10$
Command limit	$\mathrm{down}_{64}((1/10 - 10^{-4})/1.01)$
Port constants	$D_g < 0.688048, Y_g < 0.057142803$
Output derivative	$L_y < 0.43902962$
Neumann row bound	< 0.000747612
Initial energy gap	$> 1.8117429083019\,10^{-3}$
Direct reserve $\eta_{\mathrm{dir}}T$	$> 2.1329730906931\,10^{-3}$
Damping reserve $\eta_{\mathrm{diss}}T$	$< 1.0048689056874\,10^{-3}$
Remaining initial ledger gap	$> 8.0687400261464\,10^{-4}$

(S245). No differentiation of a numerical time series enters this calculation.

The derivative certificate used here sets $f \equiv 0$, not merely $p_{\mathrm{ext}} = 0$. A force can preserve energy and still rotate momentum arbitrarily rapidly. Thus a scalar external-power bound does not by itself supply L_y. To admit additional canonical forces one must bound that force class and recompute L_y; otherwise a conservative range-only sensing envelope is needed. The actual input bound already includes execution error, so no input-slew assumption is used to bound $\dot{y}$ almost everywhere.

A nonzero reference-plant budget. All buffers and initial records in this calculation denote their exact stored binary64 values. The initial state is

$$q_{0,1} = q_{0,2} = \frac{3602879701896397}{2882303761517117 44}, \qquad P_0 = 0,$$

which is not substituted by the exact decimal $1/80$. Directed rounding gives

$$\begin{aligned} 5.5875908833881 79\,10^{-4} &\le H_*(x_0) \\ &\le 5.5875908833881 82\,10^{-4}. \end{aligned} \tag{S249}$$

The underlying Arb upper endpoint, before display rounding, initializes Z_0. The energy cap is the unchanged stored value $E_{\max} \simeq 2.3705019966407895\,10^{-3}$ and $E_g = (1001/1000)E_{\max}$, with both below the configuration and clearance barriers. Table S7 reports sufficient outward bounds; the ledger uses the underlying exact rational constants.

The direct budget fails for this retained tolerance group; its reserve exceeds the available initial energy gap. The smaller completed-square budget succeeds using the true reference damping, without altering the selected sensor, execution or sampling errors. It gives $C > 1.36563309095 34\,10^{-3} > \overline{H}_0$. This distinction records which physical assumption makes the nonzero budget possible. No coercivity claim is inferred from the learned PSD head.

The normalization is $\tau = \omega_0 t$, $\omega_0 \simeq 11.417287282852458\,\mathrm{s}^{-1}$, $u = B_y/B_0$ and $H = E/E_0$, with stored $B_0 \simeq 0.03$ T and $E_0 \simeq 6\,10^{-7}$ J. Thus T corresponds to approximately 1.75173 s, the hold bound to less than $87.586479629\,\mu$s, and the sampling scale to approximately 11.4173 kHz. The actual and additive field bounds are 3 mT and $3\,\mu$T, respectively. Physical power is $E_0\omega_0 uy$; equivalently its magnetic port coefficient is $\mu\omega_0 y$ when the dipole is μ. These conversions specify the numerical operating assumptions; they do not certify a device's bandwidth, delay or sensing accuracy.

The reserve is finite and is not renewed automatically after T. The proof supplies clearance under its declared envelopes, rather than tracking accuracy, robustness of a conservative invariant torus, a complex-network advantage or validated numerical ODE integration. The preceding exact-sensing theorem and conservative KAM certificate retain their distinct hypotheses and conclusions.

Fixed sampled-control diagnostics. The same frozen complex Hamiltonian and PSD head generate the nominal request in the original sine-output task $d^{\mathsf{T}}q \approx 0.08\sin(\tau/2)$, with gains $k_p = 4$, $k_d = 3$ and the original initial state. No new fitting or post-rollout tuning of the task, gains or error waveforms is used. There are 20 000 actual holding intervals over $T = 20$. The port-sample error is $\varepsilon_m\{0.75\sin(3\tau_k) + 0.20\cos(7\tau_k)\}$, and the applied input is

$$a(\tau) = \{1 + 0.9\rho\sin(2.3\tau)\}u_k + 0.9\varepsilon_b\cos(3.7\tau).$$

These waveforms stay within the declared error classes. Only the scalar power-port sample is perturbed; the nominal request uses the simulated full state. Additional full-state sensor errors, transport delay and clock uncertainty are not modeled by this diagnostic.

Each hold is solved separately by DOP853. The two settings use $(\mathrm{rtol}, \mathrm{atol}) = (10^{-9}, 10^{-12})$ and $(2\,10^{-11}, 2\,10^{-14})$, with maximum internal steps equal to one hold and one half-hold, respectively. Table S8 reports the finer run. Plant diagnostics are evaluated at the midpoint and endpoint of every hold. Their maxima describe these numerical probes; they are not interval enclosures of the ODE.

Independent exact reconstruction checks every saved command, measurement and timestamp difference in both runs: all 40 000 work updates and the endpoint-tightened

Table S8. Sampled diagnostics with fixed nonzero sensing and execution errors. The exact ledger invariant is checked separately from these numerical plant values.

Quantity	Finer-run value
Maximum sampled true energy	$< 1.188930\,10^{-3}$
Maximum held-data ledger	$< 1.274947\,10^{-3}$
Maximum sampled applied input	< 0.099891
Sampled clearance energy bound	< 0.692 mm
Output RMS error	0.0486557
Power-filter-active holds	25.265%
Amplitude-saturated requests	92.500%

inequalities agree with the recorded ledger and inward-rounded scalar projection. The two settings' endpoint states, including their separately integrated dissipation and actual work, differ by less than 10^{-12}. The finer run's sampled power-identity and energy-balance defects are below 10^{-17} and 10^{-15}, respectively. These small numerical defects do not constitute validated time integration. The analytic guarantee follows instead from Theorem S53 and the uniform guard-domain bounds.

For any exact reference-plant solution meeting those hypotheses, the energy cap and the existing geometric comparison imply the uniform whole-centerline bound $|r_x| < 0.977$ mm over the declared horizon, below the required 4 mm. The smaller sampled value in Table S8 is a diagnostic of this particular computed trajectory. Substantial saturation and tracking error remain. No tracking improvement, complex-network advantage or measured hardware collision avoidance is inferred.

S3.21 Continuous separation of conservative and dissipative errors

The scalar-drag design in Section S3.16 does not cover the final configuration-dependent complex factor. Here the continuous reference-IVP enclosures of Section S3.19 give a deterministic identification interface for that same final output. Its effective coordinates comprise seven potential coefficients and three entries of a symmetric normalized damping matrix. No independent-sample interpretation of the trajectory windows is used.

The common mechanical chart. Write $G_Q = SS^{\mathsf{T}}$ for the specified positive Cholesky factor, and let $m_a(q) = q_1^a q_2^{6-a}$. The frozen model is

$$\begin{aligned} \widehat{H} &= \tfrac{1}{2} P^{\mathsf{T}} G_Q^{-1} P + V_4 + \widehat{c}, \\ \widehat{c} &= a_0 \sum_{a=0}^{6} m_a C_a(q), \qquad C_a(-q) = C_a(q), \\ \widehat{F}_d &= -S\widehat{M}(q) S^{-1} P, \qquad \widehat{M} = \mathrm{Re}(L_\phi L_\phi^*). \end{aligned} \tag{S250}$$

The damping expression follows from the same variable-mass Rayleigh decoder. Replacing it by $-\widehat{M}P$ would change the model. Assume the target has the same metric, known potential V_4 and magnetic port, and write its remaining functions as

$$\begin{aligned} c_* &= \sum_{a=0}^{6} b_{*,a} m_a + \tau_*, \\ M_*(q) &= M_{*,0} + \Delta M_*(q), \qquad M_{*,0}^{\mathsf{T}} = M_{*,0}. \end{aligned} \tag{S251}$$

Bounds on $\nabla\tau_*$ and ΔM_* are required; an arbitrary unknown dissipative law is not covered. In the specified full-cosine, constant-damping reference, $\Delta M_* = 0$ and the potential tail begins at degree eight. The values of its leading coefficients and damping scalar need not be supplied as labels in the inference below.

Let $b_a = a_0 C_a(0)$ and $m = (\widehat{M}_{11}(0), \widehat{M}_{12}(0), \widehat{M}_{22}(0))$. Use the symmetric bases E_{11}, $E_{12} + E_{21}$ and E_{22}, denoted E_l, $1 \le l \le 3$. These are ten coordinates of the output functions, not a unique representation of the raw potential or complex-factor parameters.

The continuous design and its remainder. For every original window $w = [a, b]$ and component i, retain its stored positive row scale s_{wi}. All integrals below follow the same exact reference IVP $x_*(\tau)$ enclosed by the native trajectory certificate. Define

$$\begin{aligned} (A_V)_{wi,a} &= s_{wi} \int_a^b \partial_i m_a(q_*)\,\mathrm{d}\tau, \\ (A_D)_{wi,l} &= s_{wi} \int_a^b [S(q_*) E_l S(q_*)^{-1} P_*]_i\,\mathrm{d}\tau. \end{aligned} \tag{S252}$$

Stacking fourteen training trajectories and their original five windows gives $A = [A_V, A_D] \in \mathbb{R}^{140\times 10}$. Let $\widehat{\tau}_V = \widehat{c} - \sum_a b_a m_a$ and $\Delta\widehat{M} = \widehat{M} - \widehat{M}(0)$. The continuous learned residual and tail are

$$\begin{aligned} \widehat{r}_{wi} &= s_{wi} \left\{ P_{*,i}(b) - P_{*,i}(a) - \int_a^b \widehat{f}_{P,i}(x_*, u)\,\mathrm{d}\tau \right\}, \\ t_{wi} &= s_{wi} \int_a^b [\nabla(\widehat{\tau}_V - \tau_*)]_i\,\mathrm{d}\tau \\ &\quad + s_{wi} \int_a^b [S(\Delta\widehat{M} - \Delta M_*) S^{-1} P_*]_i\,\mathrm{d}\tau. \end{aligned} \tag{S253}$$

The shared kinetic, known-potential and input terms cancel, giving the exact identity

$$\widehat{r} = A_V(b - b_*) + A_D(m - m_*) + t. \tag{S254}$$

The target weak residual is zero here because x_* solves the declared reference IVP. The original cached COMSOL

equations do not thereby have zero residual. The residual in (S253) is the previously enclosed continuous quantity with the original stored scales, rather than the discrete training objective.

Proposition S54. Continuous effective-coordinate error. *Let $D = \mathrm{diag}(D_V, D_D)$ have positive fixed entries. Suppose*

$$\sigma_{\min}(AD^{-1}) \geq \sigma_0 > 0, \qquad \|\widehat{r}\|_2 \leq \epsilon_c, \qquad \|t\|_2 \leq T_c.$$

For $\beta = (b, m)$, the same output then satisfies

$$\|D(\widehat{\beta} - \beta_*)\|_2 \leq \frac{\epsilon_c + T_c}{\sigma_0} =: E_\beta. \tag{S255}$$

In particular, the conservative subvector obeys this bound.

Proof. Equation (S254) gives $AD^{-1}D(\widehat{\beta} - \beta_*) = \widehat{r} - t$. Apply the verified singular-value bound and the triangle inequality. No uniqueness of the raw factor coordinates is required.

The constant damping coefficients can alternatively be eliminated by an orthogonal projection. Set $B_V = A_V D_V^{-1}$, $B_D = A_D D_D^{-1}$ and, when B_D has full column rank,

$$Q = I - B_D(B_D^{\mathsf{T}} B_D)^{-1} B_D^{\mathsf{T}}.$$

Then $\|Q\|_2 = 1$ and $QB_D = 0$. A verified lower bound $\sigma_{\min}(QB_V) \geq \sigma_V > 0$ replaces σ_0 in the conservative-only conclusion. This accounts for the three damping components as nuisance coefficients; dropping their columns without projection would not separate conservative error.

Tail bounds without the target leading coefficients. The already certified Hessian envelopes for C_a, together with evenness, give on a box with coordinate moduli at most r

$$|C_a - C_a(0)| \leq U_a = \tfrac{1}{2} r^2 \sum_{jk} B_{a,jk},$$
$$|\partial_j C_a| \leq V_{a,j} = r \sum_k B_{a,jk}.$$

Consequently, for $\alpha_a = (a, 6-a)$, the learned potential tail has component bounds

$$|\partial_j \widehat{\tau}_V| \leq a_0 \sum_a \{\alpha_{a,j} r^5 U_a + r^6 V_{a,j}\}. \tag{S256}$$

For the specified full-cosine reference, writing $s_r = r\|d\|_1$, its degree-eight-and-higher tail satisfies

$$|\partial_j \tau_*| \leq |d_j| s_r^7 \cosh(s_r)/7!.$$

These use the target class's higher-order tail, not the actual leading coefficient differences.

The damping head is centered at its own $L_0 = L_\phi(0)$. Writing $L_0 = A_0 + iB_0$ and $L_\phi(q) - L_0 = E_r + iE_i$, bounds $\|E_r\|_F \leq e_r$, $\|E_i\|_F \leq e_i$ imply

$$\|\widehat{M}(q) - \widehat{M}(0)\|_2 \leq 2\|A_0\|_2 e_r + e_r^2 + 2\|B_0\|_2 e_i + e_i^2. \tag{S257}$$

This follows by expanding the real part of $(L_0 + E)(L_0 + E)^* - L_0 L_0^*$ and applying matrix norm inequalities. A direct interval enclosure of the centered entries is another valid bound. Neither method substitutes the previously computed discrepancy from $\sqrt{\gamma_*} I$ for the variation about $L_\phi(0)$.

If $\|\Delta\widehat{M} - \Delta M_*\|_2 \leq \eta_M$, the damping-tail component in a window is bounded by

$$s_{wi}\eta_M \int_a^b \sqrt{(G_Q)_{ii}} \sqrt{P_*^{\mathsf{T}} G_Q^{-1} P_*}\, \mathrm{d}\tau. \tag{S258}$$

Indeed $\|e_i^{\mathsf{T}} S\|_2 = \sqrt{(G_Q)_{ii}}$ and $\|S^{-1}P\|_2^2 = P^{\mathsf{T}} G_Q^{-1} P$. Combining the potential and damping bounds in every row gives T_c by a Euclidean sum. Time subdivision may tighten these bounds without altering the fitted objective or the windows.

From the coefficient estimate to conservative accuracy. For this reference, on a real box or a complex rectangle with the established analytic extensions and Hessian envelopes, define

$$\begin{aligned} B_5(r) &= r^5 \max_j \left(\sum_{a=0}^{6} \frac{\alpha_{a,j}^2}{D_{V,a}^2} \right)^{1/2}, \\ U_5(r) &= \max_j \Big\{ a_0 \sum_a (\alpha_{a,j} r^5 U_a + r^6 V_{a,j}) \\ &\qquad + |d_j| s_r^7 \cosh(s_r)/7! \Big\}. \end{aligned} \tag{S259}$$

Cauchy–Schwarz applied to the seven leading monomials gives

$$\sup \|\nabla(\widehat{c} - c_*)\|_\infty \leq B_5(r) \frac{\epsilon_c + T_c}{\sigma_0} + U_5(r). \tag{S260}$$

For a complex rectangle, r bounds the coordinate moduli, not only the real half-width. The real-readout extension uses the conjugate-coefficient analytic construction already specified for the frozen network; taking real parts after substituting complex variables would not be that extension.

What this supplies to a KAM argument. Equation (S260) is a deterministic learning-error interface for the final dual-function output. For a candidate K_0 whose complex image is contained in the certified box, the two conservative

fields differ by at most this gradient bound in the same pointwise supremum norm and canonical coordinates. A changed chart requires its transformation constants. A Fourier-weighted residual requires its own norm conversion, with the required strip loss, before this bound can enter its a posteriori theorem. Domain inclusion, torsion, Diophantine and smallness conditions must still be checked for that same candidate. The separately verified frozen-Hamiltonian torus does not turn this estimate into a passed target-to-learned finite-amplitude transfer test. Likewise, the sampling and density conditions in the statistical learning route of Chen et al. (2022) are not supplied by the correlated finite windows.

Certified design and error estimates. The preceding bounds are evaluated for the same frozen conservative checkpoint and final complex damping factor used in Sections S3.18 and S3.19. All fourteen training reference IVPs, their 11 200 original intervals and seventy original windows are retained. The ten column scales are the exact dyadic values of the column norms frozen before the continuous calculation. Each stored row scale is applied once. The seven potential columns integrate the complete degree-at-most-25 polynomial products of the quintic reconstruction, with a monomial-Hessian bound for the pointwise trajectory tube. The three damping columns use the same finite-Gauss metric, positive Cholesky factor and validated time Taylor arithmetic. Neither the trajectories nor the networks are refitted for this calculation.

For $B = AD^{-1}$, interval evaluation of the 10×10 Gram matrix $\mathcal{G} = B^{\mathsf{T}}B$ and a fixed dyadic approximate inverse R gives

$$\|I - R\mathcal{G}\|_\infty < 3.409\,10^{-10},$$
$$\sigma_{\min}(B) > 0.2880896913927310.$$

Indeed the Neumann estimate bounds $\|\mathcal{G}^{-1}\|_\infty$ by $\|R\|_\infty/(1 - \|I - R\mathcal{G}\|_\infty)$. The exact inverse is symmetric, so its spectral norm is at most its infinity norm. This proves the stated singular-value bound without treating a floating-point singular-value decomposition as a certificate. The largest interval width in a normalized design entry is below $4.633\,10^{-12}$.

The continuous residual enclosure from Section S3.19, with precisely the same windows and row scales, and the separately bounded tails give

$$\|\widehat{r}\|_2 < 1.460200701226\,10^{-5},$$
$$\|t\|_2 < 1.650063983\,10^{-3}, \qquad \text{(S261)}$$
$$\|D(\widehat{\beta} - \beta_*)\|_2 < 0.005778290718.$$

Table S9. Continuous weak-data bounds for the final conservative output. Each entry is a strict upper bound.

Quantity	Real box	Complex rectangle
B_5	$9.941982977\,10^{-9}$	$9.944468659\,10^{-9}$
U_5	$1.223049693\,10^{-11}$	$1.624553850\,10^{-11}$
$\sup \|\nabla(\widehat{c} - c_*)\|_\infty$	$6.967816488\,10^{-11}$	$7.370756945\,10^{-11}$

Real box: $|q_i| \le 1/10$. Complex rectangle: $|\operatorname{Re} q_i| \le 1/10$, $|\operatorname{Im} q_i| \le 1/1000$.

The potential-tail and learned-damping-variation contributions have Euclidean bounds below $1.610972935\,10^{-3}$ and $7.310467623\,10^{-5}$, respectively. Their rowwise combination, followed by the Euclidean norm, gives the displayed total; the two separate Euclidean bounds need not add to that total. The damping variation is centered at the learned factor's own $L_\phi(0)$ and $\widehat{M}(0)$.

For the specified constant-damping target, the resulting uniform conservative-gradient estimates are given in Table S9. All displayed upper bounds are rounded outward from exact certificate endpoints. The complex rectangle uses the already certified analytic extension and Hessian envelopes; its coordinate-modulus radius is $r = \sqrt{(1/10)^2 + (1/1000)^2}$.

The inferred real-box estimate is more conservative than the direct frozen-function comparison in Section S3.16. Its role is different: the continuous weak residual and a verified design now imply a conservative error bound while retaining the final nonconstant learned damping factor. The actual leading target-coefficient errors and the discrepancy from the true constant damping factor are not supplied as inference inputs. The target's specified high-order potential-tail class and constant-damping premise remain necessary. The upstream trajectory-tube and continuous-residual certificates use the known reference vector field. Thus this is a deterministic validation against that specified benchmark, with an explicit data-to-reference bridge; it does not infer an arbitrary unknown plant from observations alone. If instead $\sup \|\Delta M_*\|_2 \le \eta_*$, the same rowwise calculation replaces $T_c(0)$ by a bound $T_c(0) + \eta_* S_D$, where S_D is the Euclidean norm of the nonnegative row coefficients in (S258). It cannot set that additional uncertainty to zero for an unknown plant.

The ten-coordinate singular-value bound is certified here; a separate projected seven-column singular-value improvement is not claimed. These deterministic estimates concern the final output and the declared reference IVPs. They do not identify its raw network parameters, supply a statistical sampling hypothesis or establish a target-to-learned finite-amplitude KAM transfer. The latter still

requires the same-candidate domain, norm, nondegeneracy and smallness checks stated above.

A domain-compatible interface at the certified candidate. The complex rectangle in Table S9 does not contain the candidate K_0 of Theorem S41 on its initial strip $\rho = 1/40$. This is verified at the interior complex point $\theta = (1/4 + i/80, 0)$, where $|\operatorname{Im}(K_0)_{q_1}| > 0.003142936259 > 0.001$. Consequently, that rectangle's error bound cannot be composed with this candidate on the initial strip.

The same conservative checkpoint also has the coefficient-Hessian enclosures on the larger-imaginary-width domain

$$|\operatorname{Re} q_i| \le 3/50, \qquad |\operatorname{Im} q_i| \le 1/100.$$

The certified candidate image, including the specified $3/1000$ coordinate neighbourhood, lies in this domain. Reusing only those Hessian envelopes in (S259), with the same E_β from the continuous data, gives

$$\sup \|\nabla(\widehat{c} - c_*)\|_\infty < 5.133240866\,10^{-12}.$$

Using the candidate's certified coordinate-modulus bounds in the monomial factors further gives the same-strip composed bound

$$\|X_{\widehat{H}} \circ K_0 - X_{H_*} \circ K_0\|_{1/40} < 6.857321746\,10^{-13}. \tag{S262}$$

Here the norm is the maximum over components of the pointwise supremum on the period-one complex strip. This is the strip norm used by the cited a posteriori theorem. The weighted Fourier coefficient sum used to certify the earlier residual dominates that supremum; it does not require an additional Fourier-norm conversion for (S262). The canonical symplectic matrix has infinity operator norm one, and the two conservative fields have the same kinetic term.

For $E_* = X_{H_*} \circ K_0 - DK_0\nu$ and $\widehat{E} = X_{\widehat{H}} \circ K_0 - DK_0\nu$, this proves the valid same-domain estimate

$$\|E_*\|_{1/40} \le \|\widehat{E}\|_{1/40} + \|X_{H_*} \circ K_0 - X_{\widehat{H}} \circ K_0\|_{1/40}.$$

The resulting upper bound does not establish the residual budget of the saved learned-system certificate, whose scale is $6.740\,10^{-38}$. This is an insufficiency of the error budget, not a lower bound on the actual residual or a nonexistence result. Moreover, this K_0 was refined for the learned Hamiltonian: the calculation bounds the target residual at that candidate, rather than supplying an independent target torus. The target's derivative and torsion bounds require separate verification. Section S3.22 supplies these bounds uniformly along the learned–reference homotopy and re-evaluates its scalar test; the available residual upper bound still does not establish the required smallness.

This direct estimate concerns the fixed candidate. Section S3.23 supplies a separate residual verification for an explicit parameter-response candidate and the same declared conservative pair.

S3.22 Uniform nondegeneracy along the learned–reference homotopy

The continuous-window error also controls the derivative and nondegeneracy hypotheses for the reference Hamiltonian. Throughout this subsection, K_0 is the unchanged exact Fourier candidate of Theorem S41, with its saved frequency ν and initial strip $\rho = 1/40$. Define

$$\Delta V(q) = H_*(q,P) - \widehat{H}(q,P),$$
$$H_\lambda = \widehat{H} + \lambda \Delta V, \quad 0 \le \lambda \le 1.$$

The difference is independent of P because the two Hamiltonians have the same finite-Gauss kinetic graph. The reference class and known low-order mechanical terms are those used in the preceding continuous-identification estimate. No additional unknown metric error is included. On the wide rectangle, let ϵ denote the exact certified upper bound, so that

$$\sup \max_{j=1,2} |\partial_j \Delta V| \le \epsilon < 5.133240866\,10^{-12}. \tag{S263}$$

Proposition S55. Uniform derivative and average-torsion bounds. *For every $\lambda \in [0,1]$, the average torsion at this fixed K_0 is invertible and satisfies*

$$\|\langle T_\lambda \rangle^{-1}\|_\infty < 18409.653468. \tag{S264}$$

The four vector-field bounds in Table S10 hold uniformly on the open complex domain

$$\mathcal{B} = K_0(\mathbb{T}^2_{1/40}) + \{z : \|z\|_\infty < 61/20000\}.$$

In particular, the distance of the candidate image from $\mathcal{B}^c$ is at least $61/20000 > 3/1000$.

Proof. The exact conjugate symmetry of the Fourier coefficients makes K_0 real on $\mathbb{T}^2$. Its certified componentwise image bounds place complex coordinate polydiscs of radius $r_0 = 1/125$ about this real image strictly inside the wide rectangle. About every point of $\mathcal{B}$, polydiscs of radius $r_1 = 1/2000$ also remain inside that rectangle: the smallest

certified real and imaginary margins, after both displacements, exceed $159/10000$ and $1/8000$, respectively. These comparisons use the exact image enclosures, including all four canonical coordinates.

Apply Cauchy's estimate to the two holomorphic components of $\nabla_q \Delta V$. At either radius $r = r_0, r_1$, the resulting field-difference bounds are

$$\begin{aligned} \|\Delta X\|_{\max} &\le \epsilon, & \|D\Delta X\|_{\infty} &\le 2\epsilon/r, \\ \sum_{a,b} \sup |\partial_b \Delta X_a| &\le 4\epsilon/r, & \|D^2 \Delta X\|_{\infty,\mathrm{bil}} &\le 6\epsilon/r^2. \end{aligned} \tag{S265}$$

Only the two q inputs and two momentum outputs can contribute. For the last bound, the two repeated second derivatives cost 2! each and the two ordered mixed derivatives cost one each. The bilinear norm is majorized by the sum over ordered input indices for each output. The total-entry sum in the third bound is the conservative row-vector differential majorant used for $c_{X^\mathsf{T},1}$; it is not an assertion that ordinary transpose matrix norms require the same factor four. Adding these bounds at r_1 to the certified learned-field bounds proves Table S10, since $|\lambda| \le 1$.

For the average torsion, put $L = DK_0$, $B = (L^\mathsf{T} L)^{-1}$ and $N = JLB$. With the saved canonical convention,

$$T_H = N^\mathsf{T} J\big(JD^2 H(K_0)N - D_\nu N\big).$$

All geometric factors are fixed along this homotopy. Thus $J^2 = -I$ gives the exact identity

$$T_\lambda - T_{\widehat{H}} = -\lambda N^\mathsf{T} D^2 \Delta V(K_0) N. \tag{S266}$$

Use the separately certified induced infinity bounds for N and N^T, denoted by n_0 and n_0^T, and the continuous-mean inverse bound b_0 for the learned Hamiltonian. The period-one average has volume one, so

$$\|\langle T_\lambda - T_{\widehat{H}} \rangle\|_\infty \le n_0^\mathsf{T}(2\epsilon/r_0) n_0 < 5.617939218\,10^{-8}.$$

Exact rational evaluation gives

$$\eta := b_0 n_0^\mathsf{T}(2\epsilon/r_0) n_0 < 0.001033175 < 1.$$

The Neumann estimate $b_0/(1-\eta)$ proves (S264) for every real $\lambda \in [0,1]$. The overall sign change under the Haro torsion convention leaves this inverse norm unchanged.

The direct smallness test at the fixed candidate. The geometric, arithmetic and domain data are unchanged. Combining them with Proposition S55 supplies uniform inputs to the same Haro–Luque scalar test. Re-evaluating its tables gives a strict residual threshold of approximately $6.73939\,10^{-38}$. On the other hand, the available continuous-data estimate gives only the common upper bound

$$\begin{aligned} &\|X_{H_\lambda} \circ K_0 - DK_0 \nu\|_{1/40} \\ &\quad \le \|\widehat{E}\|_{1/40} + \|X_{H_*} \circ K_0 - X_{\widehat{H}} \circ K_0\|_{1/40} \\ &\quad < 6.857321746\,10^{-13}. \end{aligned}$$

Inserting that available upper bound into the uniform test gives a smallness quotient of approximately $1.01750\,10^{25}$. Consequently the derivative and average-torsion hypotheses are now verified, while this residual estimate does not establish the sufficient smallness condition. The quotient is not a lower bound on the actual defect. The proposition concerns nondegeneracy at a fixed approximate candidate; it does not assert the existence of invariant tori throughout the homotopy.

This fixed-candidate calculation motivates the explicit response polynomial in Section S3.23. Its geometry and actual invariance defect are verified separately at the moving candidates. The fixed-candidate upper bound above is not used as their residual certificate.

Table S10. Uniform field bounds on $\mathcal{B}$ for all $\lambda \in [0,1]$. Displayed upper bounds are rounded outward; the scalar test uses the stored exact rational bounds.

Quantity	Upper bound
$c_{X,0}$	0.384442890794
$c_{X,1}$	6.365861808
$c_{X^\mathsf{T},1}$ (total-entry majorant)	19.878533236
$c_{X,2}$ (bilinear majorant)	2.266997899

S3.23 Explicit parameter response and same-frequency conservative transfer

We fix one learned Hamiltonian $\widehat{H}$ and its declared full-cosine reference H_* in the same canonical coordinates. Both use the same stored finite-Gauss kinetic metric; their difference is a configuration potential $\Delta V = H_* - \widehat{H}$. Define

$$H_\lambda = \widehat{H} + \lambda \Delta V, \qquad P_m(\theta, \lambda) = K_0(\theta) + \sum_{a=1}^{m} \lambda^a U_a(\theta). \tag{S267}$$

Here K_0 and all U_a are explicitly supplied finite Fourier polynomials with exact rational complex coefficients and exact Fourier reality. The coefficients in (S267) are ordinary power coefficients, with no factorial convention. This construction supplies an approximate embedding for each parameter; it does not assume a family of exact invariant tori.

The frequency remains fixed throughout. For the candidate used in this response computation, its exact period-one definition is

$$\begin{aligned}\phi &= \frac{1+\sqrt{5}}{2}, \qquad w = \frac{499974648628101}{500000000000000},\\ \alpha &= \frac{2524851\phi + 622102}{410497\phi + 101143},\\ \nu &= \frac{1}{2\pi}(w, w\alpha).\end{aligned} \tag{S268}$$

Rounded frequencies are not used in differentiation or small-divisor estimates. Prescribing ν does not impose additional independent constraints fixing all actions.

For $s > 0$, write $\mathbb{T}^2_s = \{\theta \in (\mathbb{C}/\mathbb{Z})^2 : |\operatorname{Im}\theta|_\infty < s\}$ and $\|f\|_s = \max_v \sup_{\theta\in\mathbb{T}^2_s} |f_v(\theta)|$. For $s = 0$ the supremum is taken on the real torus. For matrix-valued functions we use the row-sum majorant required by the cited theorem,

$$\|M\|_s = \max_i \sum_j \sup_{\theta\in\mathbb{T}^2_s} |M_{ij}(\theta)|,$$

with transposes bounded separately. This bounds the supremum of the pointwise induced infinity norm. Set

$$\begin{aligned}G(\theta,\lambda) &= X_{H_\lambda}(P_m(\theta,\lambda)),\\ T(\theta,\lambda) &= D_\theta P_m(\theta,\lambda)\nu,\\ F(\theta,\lambda) &= G(\theta,\lambda) - T(\theta,\lambda) = \sum_{j\ge 0} e_j(\theta)\lambda^j.\end{aligned} \tag{S269}$$

The complete coefficients e_j include the base residual, the errors of the response equations, and every projection or rational-midpoint error. None of these terms is set to zero because a construction equation has been solved numerically.

For completeness, the fixed-base response operator is $\mathcal{L}_0 U = DX_{\widehat{H}}(K_0)U - D_\theta U\nu$. If $P^{(j-1)} = K_0 + \sum_{a<j}\lambda^a U_a$ and

$$f_j = [\lambda^j]\big(X_{H_\lambda}(P^{(j-1)}) - D_\theta P^{(j-1)}\nu\big),$$

then the coefficient of order j after inserting U_j is $e_j = \mathcal{L}_0 U_j + f_j$. Inserting higher-order responses leaves this and every earlier coefficient unchanged. The retained rational U_j are approximate solutions of these construction equations; the residual lemma below uses the actual e_j, not the formal zero suggested by an exactly solved equation.

Lemma S56. Residual bound from two complex product domains. *Let $0 \le \rho < \widehat{\rho}$, $R > 1$, $r > 0$, and $L \ge m$. Assume the following bounds for the same explicit map (S267) and the same Hamiltonian family.*

1. *On $\mathbb{T}^2_\rho \times \{|\lambda| < R\}$, the map P_m takes values in a phase-space domain on which the composition defining F is holomorphic, and $\sup_{|\lambda|<R}\|F(\cdot,\lambda)\|_\rho \le M_R$.*
2. *On $\mathbb{T}^2_{\widehat{\rho}} \times \{|\lambda| < r\}$, the same explicit map has a separately verified holomorphic composition defining G, and $\sup_{|\lambda|<r}\|G(\cdot,\lambda)\|_{\widehat{\rho}} \le M_X$.*
3. *The even integer $N \ge 2$ strictly contains every Fourier mode of $K_0, U_1, \dots, U_m$: each such mode k satisfies $|k_a| < N/2$ for $a = 1, 2$. For every $0 \le j \le L$, a rigorous normalized $N \times N$ DFT of the actual coefficient e_j gives a bound $b_{j,N,\rho}$ for its Fourier interpolant in the component-maximum weighted Fourier ℓ^1 norm, including all sample and DFT uncertainty.*

Define

$$\begin{aligned}q &= \exp[-2\pi(\widehat{\rho} - \rho)],\\ \mathcal{A}_N(\widehat{\rho},\rho) &= 2\left(\frac{1+q}{1-q}\right)^2 (2q^{N/2} - q^N),\\ \mathcal{E}_{m,L,N} &= \sum_{j=0}^{L} b_{j,N,\rho} + M_X\mathcal{A}_N(\widehat{\rho},\rho)\sum_{j=0}^{L} r^{-j}\\ &\quad + \frac{M_R}{R^L(R-1)}.\end{aligned} \tag{S270}$$

Then

$$\sup_{|\lambda|\le 1}\|F(\cdot,\lambda)\|_\rho \le \mathcal{E}_{m,L,N}. \tag{S271}$$

The two parameter radii R and r need not agree; in particular, $r = 1$ is admissible.

Proof. Write $G = \sum_{j\ge0} g_j\lambda^j$ and $T = \sum_{j=0}^{m} t_j\lambda^j$, where $t_j = D_\theta P_j\nu$, $P_0 = K_0$ and $P_j = U_j$ for $j \ge 1$. Set $t_j = 0$ for $j > m$. Cauchy's estimate in the second parameter disk gives

$$\|g_j\|_{\widehat{\rho}} \le M_X r^{-j}.$$

If a bound is stated on an open disk or strip, apply Cauchy's formula on smaller contours and pass to the limit. The same finite upper bound must hold throughout the stated open domain.

Let $C_N = \{-N/2, \dots, N/2-1\}^2$. The normalized interpolant is

$$\begin{aligned}I_N f &= \sum_{k\in C_N} \widetilde{f}_{k,N} e^{2\pi i k\cdot\theta},\\ \widetilde{f}_{k,N} &= \frac{1}{N^2}\sum_{a,b=0}^{N-1} f(\theta_{ab}) e^{-2\pi i k\cdot\theta_{ab}},\\ \theta_{ab} &= (a/N, b/N).\end{aligned}$$

Differentiation does not enlarge finite Fourier support. The strict support condition therefore gives $I_N t_j = t_j$ and

$$e_j - I_N e_j = g_j - I_N g_j.$$

Thus the analytic interpolation error only requires a bound for g_j, rather than a bound for the transport on the wider strip.

For a scalar function bounded by M on $\mathbb{T}^2_{\widehat{\rho}}$, Fourier decay bounds its omitted weighted coefficient sum on $\mathbb{T}^2_{\rho}$ by

$$M \sum_{k \notin C_N} q^{|k|_1} = M \left(\frac{1+q}{1-q} \right)^2 (2q^{N/2} - q^N).$$

Every omitted Fourier mode aliases to a representative in C_N with no larger ℓ^1 frequency. Adding the omitted terms and their aliases therefore bounds $\|f - I_N f\|_\rho$ by $M\mathcal{A}_N(\widehat{\rho}, \rho)$. This estimate includes the Nyquist aliases for the stated signed cube. Applying it component by component, and then taking their maximum, yields

$$\|e_j\|_\rho \le b_{j,N,\rho} + M_X r^{-j} \mathcal{A}_N(\widehat{\rho}, \rho), \qquad 0 \le j \le L.$$

There is no factor equal to the number of state components. The weighted Fourier bounds include the numerical enclosures and dominate the pointwise strip norm. A real-grid maximum alone would not supply the required $b_{j,N,\rho}$.

Independently, Cauchy's estimate on the first product domain gives $\|e_j\|_\rho \le M_R R^{-j}$. Hence for $|\lambda| \le 1$,

$$\left\| \sum_{j>L} e_j \lambda^j \right\|_\rho \le M_R \sum_{j>L} R^{-j} = \frac{M_R}{R^L(R-1)}.$$

Adding the finite coefficient bounds proves (S271).

Theorem S57. Pointwise same-frequency transfer along the real homotopy. *Fix $0 < \rho_\infty < \rho$ and a distance $\mathfrak{d} > 0$. Suppose Lemma S56 applies. For every real $\lambda \in [0,1]$, assume the canonical hypotheses of Haro and Luque (2019, Theorem 3.1, Case III, $n = d = 2$) hold at $P_m(\cdot, \lambda)$ with common bounds, in particular:*

1. *H_λ is real analytic on a common complex domain containing the $\mathfrak{d}$-neighbourhood of every candidate image. The symplectic form is the exact canonical form, the auxiliary metric is I_4, and the vector-field and derivative bounds use the operator and transpose conventions of that theorem.*
2. *Each candidate is a real analytic embedding. With $L_\lambda = D_\theta P_m$, its Gram inverse $B_\lambda = (L_\lambda^\mathsf{T} L_\lambda)^{-1}$ is analytic on $\mathbb{T}^2_\rho$ with a uniform bound. Define*

$$\mathcal{T}_\lambda = -N_\lambda^\mathsf{T} D^2 H_\lambda(P_m) N_\lambda - N_\lambda^\mathsf{T} J D_\nu N_\lambda,$$

where $N_\lambda = J L_\lambda B_\lambda$. Its averages are invertible with a common inverse bound, and $D_\nu = D_\theta(\cdot)\nu$. The common iteration bounds $\sigma_K, \sigma_{K^\tau}, \sigma_B, \sigma_T$ exceed their respective candidate bounds strictly.
3. *The same exact frequency obeys the all-integer estimate $|k \cdot \nu| \ge \gamma |k|_1^{-\tau}$ for every $0 \ne k \in \mathbb{Z}^2$, and the required Rüssmann estimate is verified with the common arithmetic constants.*

Let C_1, C_2 be the constants in the cited theorem, computed from these uniform bounds and the chosen iteration parameters. Assume all of its strict scalar conditions hold with $E = \mathcal{E}_{m,L,N}$, including

$$\frac{C_1 E}{\gamma^4 \rho^{4\tau}} < 1, \qquad \delta_K := \frac{C_2 E}{\gamma^2 \rho^{2\tau}} < \mathfrak{d}. \tag{S272}$$

Assume also that the verified embedding criterion remains strict after adding the correction bounds supplied by that theorem. Then, for every real $\lambda \in [0,1]$, there exists a real analytic embedded Lagrangian torus K_λ^{inv} on $\mathbb{T}^2_{\rho_\infty}$ such that

$$\begin{aligned} X_{H_\lambda}(K_\lambda^{\mathrm{inv}}) &= D_\theta K_\lambda^{\mathrm{inv}} \nu, \\ \|K_\lambda^{\mathrm{inv}} - P_m(\cdot, \lambda)\|_{\rho_\infty} &\le \delta_K. \end{aligned} \tag{S273}$$

In particular the two endpoint Hamiltonians possess such tori at the same exact prescribed frequency. The quantifier is $\forall \lambda \in [0,1]\ \exists K_\lambda^{\mathrm{inv}}$; no joint holomorphic dependence on λ is asserted.

Proof. Fix a real $\lambda \in [0,1]$. The explicit polynomial is an admissible candidate by the geometric and analytic assumptions. Lemma S56 supplies its actual whole-strip defect bound $\|F(\cdot, \lambda)\|_\rho \le E$. The common constants and strict conditions therefore permit a direct application of the cited canonical a posteriori theorem. It gives an invariant torus with the same frequency and the correction estimate in (S273). The additional strict embedding check preserves embeddedness. Every hypothesis and correction bound is uniform in the chosen real parameter, so the argument applies to each such parameter. It neither differentiates an exact invariant embedding with respect to λ nor assumes that an invariant family already exists.

From identification error to KAM. The deterministic continuous-data argument supplies a bound for the discrepancy of this fixed learned/reference pair on a compatible complex configuration domain. Because the kinetic metric

is shared, ΔV is a potential discrepancy. Cauchy estimates then control the required derivatives, and the fixed-candidate torsion identity in Section S3.22 supplies the first uniform nondegeneracy bound. Finite Fourier bounds for the explicit displacement $P_m - K_0$, followed by Gram and average-torsion perturbation estimates on the common domain, transfer these geometric bounds to the actual candidates P_m. The response construction and Lemma S56 address the remaining invariance defect. The argument proceeds from continuous identification error to uniform derivatives and nondegeneracy, then to explicit-response residual verification and a posteriori KAM. The final implication requires the strict tests above.

The large complex parameter disk is used only for the residual tail. Real-parameter geometric estimates are not extended to that disk. Conversely, the wide angular strip is used with its own parameter radius r, so its image proof does not require the candidate to remain admissible on the product with radius R. The verified response calculation uses $m = 3$, $L = 8$, $\rho = 1/40$, $\widehat{\rho} = 19/40$, $R = 10^5$, and $r = 1$; its actual low-coefficient enclosures, final scalar assembly, and independent acceptance are supplied in the accompanying numerical certificate for Theorem S57.

This statement concerns one declared conservative finite-Gauss learned/full-cosine pair. It supplies neither iid statistical generalization nor a uniform result over a mixed seven-dimensional coefficient-error ball. It also supplies no joint analytic parameterization of the exact tori, no continuum-model identification, and no persistence claim for damping or magnetic forcing. In particular, it does not complete the separate forced-dissipative Birkhoff programme.

Numerical verification for the fixed response polynomial. We use the actual rational Fourier map $P_3 = K_0 + \lambda U_1 + \lambda^2 U_2 + \lambda^3 U_3$, with the same Hamiltonians, canonical coordinates and exact frequency as above. The bounds below are rounded outwards from exact rational endpoints in the actual coefficient and scalar assembly. In particular, the displayed residual bounds are rounded upwards and the displayed sufficient threshold is rounded downwards; none is obtained from a real-grid maximum or a formal response cancellation.

The two product domains use

$$m = 3, \quad L = 8, \quad N = 80,$$
$$\rho = \frac{1}{40}, \quad \widehat{\rho} = \frac{19}{40}, \quad R = 10^5, \quad r = 1.$$

Their composition bounds can be displayed as $M_R < 0.67239765$ and $M_X < 666.91867$. The full original support is retained, its largest coordinate mode is 31, and it lies strictly inside the signed 80×80 cube. The 256-bit coefficient calculation retains all actual coefficients $e_0, \dots, e_8$, with normalized DFT coefficient balls. Table S11 gives their component-maximum weighted Fourier bounds. The dominant interpolant is e_4; the lower-order construction errors remain in the sum.

Table S11. Outward upper bounds for the actual low-order residual interpolants on the inner angular strip.

j	$b_{j,80,\rho}$	j	$b_{j,80,\rho}$
0	$5.1072283\,10^{-46}$	5	$7.9879444\,10^{-51}$
1	$1.8886333\,10^{-44}$	6	$7.7234366\,10^{-60}$
2	$1.0489745\,10^{-53}$	7	$7.1910422\,10^{-69}$
3	$1.5292573\,10^{-62}$	8	$4.3780938\,10^{-78}$
4	$1.2036573\,10^{-41}$		

Table S12. Residual assembly and fresh a posteriori test for the same explicit P_3. Every row is uniform in the stated parameter set.

Quantity	Outward display bound
Sum of actual interpolant bounds $\sum_{j=0}^{8} b_{j,80,\rho}$	$< 1.2055970\,10^{-41}$
Sum of angular truncation and alias tails	$< 2.3212724\,10^{-45}$
Parameter tail $M_R/[R^8(R-1)]$	$< 6.7240438\,10^{-46}$
Complete defect $\sup_{\lvert\lambda\rvert\le 1} \lVert F(\cdot,\lambda)\rVert_{1/40}$	$< 1.2058963\,10^{-41}$
Sufficient residual threshold	$> 5.8247458\,10^{-38}$
Smallness quotient $C_1 E/(\gamma^4\rho^4)$	$< 2.0702986\,10^{-4}$
Uniform correction on $\mathbb{T}^2_{1/400}$	$< 1.5643707\,10^{-23}$

The exact assembly first adds the rational interpolant bounds and then the two analytic tail allowances in Eq. (S270). This yields Table S12. The sum in the first row is rounded from its exact value, rather than recomputed from the individually rounded entries in Table S11.

The moving-candidate geometry uses the common domain radius $61/20000$ around the base image and the theorem distance $\mathfrak{d} = 3/1000$. Its uniform response displacement is below $2.0339487\,10^{-11}$, and its average-torsion inverse bound is 19781.921092. The strict iteration margins are re-evaluated for these moving candidates. In the canonical scalar tables we retain

$$\gamma = \frac{4828522443384859}{576460752303423488}, \quad \tau = 1,$$
$$c_R = \frac{1}{7}, \quad \rho_\infty = \frac{1}{400}, \quad a_1 = 2, \quad \eta_{\mathrm{it}} = \frac{1}{1000},$$

where η_{it} denotes the auxiliary iteration parameter, distinct from the frequency ν. The separate expanded scalar gates

are bounded by

quadratic contraction	$2.0702986\,10^{-4}$
residual / iteration scale	$3.2157235\,10^{-36}$
approximate symplectic inverse	$2.0155646\,10^{-32}$
preservation of DK	$8.5079068\,10^{-19}$
preservation of $(DK)^{\mathsf{T}}$	$8.5089052\,10^{-19}$
preservation of the Gram inverse	$1.0844892\,10^{-17}$
mean-torsion inverse	$4.7682962\,10^{-9}$
domain distance	$5.2145688\,10^{-21}$.

All are strictly below one. The correction is strictly below $\mathfrak{d}$, and the normalized phase-projection criterion, including that correction, is below $0.0032905506 < 1/2$. This last quantity is an embedding criterion, separate from the correction norm in Table S12.

Application to the declared conservative pair. An independent acceptance calculation reproduces all 36 component interpolant bounds exactly at the original 256-bit precision, and separately assembles a valid 384-bit residual envelope with fresh strict scalar gates. All bounds retain the same frozen inputs and use outward interval arithmetic or exact rational arithmetic. The preceding bounds verify the numerical hypotheses of Theorem S57 for this fixed P_3. Consequently, for every real $\lambda \in [0,1]$ there exists an analytic embedded Lagrangian torus K_λ^{inv} on $\mathbb{T}^2_{1/400}$ with the exact frequency in Eq. (S268), satisfying

$$X_{H_\lambda}(K_\lambda^{\mathrm{inv}}) = D_\theta K_\lambda^{\mathrm{inv}} \nu,$$
$$\|K_\lambda^{\mathrm{inv}} - P_3(\cdot,\lambda)\|_{1/400} < 1.5643707\,10^{-23}.$$

In particular, the learned and full-cosine endpoint systems each possess such a same-frequency torus. This statement quantifies over real parameters separately. It does not assert joint analytic parameter dependence, iid generalization, a mixed coefficient-error ball, or persistence under damping or forcing.

S3.24 Continuous identification of configuration-dependent dissipation

We extend the effective-coordinate argument of Section S3.21 to the quadratic configuration dependence of the same frozen dissipative output. All quantities use its canonical coordinates, dimensionless time, known potential and port, and the same finite-Gauss metric $G_Q(q) = S(q)S(q)^{\mathsf{T}}$ with positive Cholesky factor. In particular, $\widehat{F}_d = -S\widehat{M}S^{-1}P$, where $\widehat{M}(q) = \mathrm{Re}(L_\phi(q)L_\phi(q)^*)$; this decoder is retained throughout.

The function class and continuous design. Put $\ell = 1/10$, $m_a(q) = q_1^a q_2^{6-a}$, and

$$\psi(q) = \big(1, q_1/\ell, q_2/\ell, q_1^2/\ell^2, q_1q_2/\ell^2, q_2^2/\ell^2\big)\,.$$

Let $E_1 = E_{11}$, $E_2 = E_{12} + E_{21}$ and $E_3 = E_{22}$. Write θ for the frozen learned output, so that $c_\theta = \widehat{c}$ and $M_\theta = \widehat{M}$. For $\nu \in \{\theta, *\}$, the required output class has

$$c_\nu(q) = \sum_{a=0}^{6} b_{\nu,a} m_a(q) + \tau_\nu^V(q),$$
$$M_\nu(q) = \sum_{k=0}^{5} C_{\nu,k}\psi_k(q) + \mathcal{R}_\nu(q), \quad C_{\nu,k} = \sum_{l=1}^{3} c_{\nu,kl}E_l. \tag{S274}$$

Each $C_{\nu,k}$ is real symmetric. The potential coefficients include the fixed correction amplitude, as in the preceding ten-coordinate argument. Bounds on $\nabla\tau_\nu^V$ and $\mathcal{R}_\nu$ are premises; an arbitrary unknown function M_* is not covered by a finite design.

Retain every original window I_w and its positive component scale s_{wi}. Along the same exact reference IVPs define

$$(A_V)_{wi,a} = s_{wi}\int_{I_w} \partial_i m_a(q_*)\,\mathrm{d}\tau,$$
$$(A_D)_{wi,kl} = s_{wi}\int_{I_w} \psi_k(q_*)[S(q_*)E_lS(q_*)^{-1}P_*]_i\,\mathrm{d}\tau. \tag{S275}$$

Thus $A = [A_V \;\; A_D] \in \mathbb{R}^{140\times 25}$ for fourteen trajectories, five windows each and two momentum components. The damping columns are ordered first by k, then by l. These are coefficients of the real symmetric output, rather than unique parameters of its complex factor. The whole window is the sum of four quarter windows before row scaling; their resulting linear dependencies are retained.

To enclose the new integrals, on a normalized time interval $u \in [-1,1]$ write $q_* = \bar{q} + \delta q$, where $\bar{q}$ is the existing quintic and $|\delta q_j| \le \epsilon_j$. The complete polynomial $p_k = \psi_k(\bar{q})$ has degree at most ten. With $X \ge \sup|\bar{q}_1/\ell|$, $Y \ge \sup|\bar{q}_2/\ell|$ and $(e_x, e_y) = (\epsilon_1, \epsilon_2)/\ell$, its pointwise error is bounded by

$$(\eta_k)_{k=0}^5 = (0, e_x, e_y, 2Xe_x + e_x^2, \\ Xe_y + Ye_x + e_xe_y, 2Ye_y + e_y^2).$$

The positive Cholesky graph on the same full (q,P) tube gives $[SE_lS^{-1}P_*]_i = d_{il}(u) + \rho_{il}(u)$, with $\deg d_{il} \le 9$ and $|\rho_{il}| \le r_{il}$. Hence the complete product polynomial has degree at most nineteen, and its remainder is at most

$$r_{il}\|p_k\|_\infty + \eta_k\big(\|d_{il}\|_\infty + r_{il}\big). \tag{S276}$$

For an interval of length h, every even coefficient of degree j is integrated with factor $h/(j+1)$, and the displayed remainder with factor h. No product coefficient is discarded and no unstructured tube error is differentiated. The conservative columns retain the complete degree-twenty-five center polynomial and Hessian-times-tube estimate.

Let D be the positive diagonal matrix of the twenty-five fixed preprojection column scales, interpreted as exact rational constants, and put $B = AD^{-1} = [B_0 \;\; B_N]$. Here B_0 consists of the seven conservative and three constant-damping columns, and B_N of the fifteen nonconstant columns. The scales need not be exact continuous column norms.

Proposition S58. Continuous damping coordinates. *Suppose $\sigma_{\min}(B) \ge s > 0$. With $P_0 = B_0(B_0^{\mathsf T}B_0)^{-1}B_0^{\mathsf T}$, one has*

$$\sigma_{\min}((I-P_0)B_N) \ge s, \qquad B_N^{\mathsf T}(I-P_0)B_N \succeq s^2 I_{15}. \tag{S277}$$

If the same continuous weak residual obeys $\widehat r = A(\widehat\beta - \beta_*) + t$, $\|\widehat r\|_2 \le \epsilon_c$ and $\|t\|_2 \le T$, then

$$\begin{aligned} \|D(\widehat\beta - \beta_*)\|_2 &\le E_{25} := \frac{\epsilon_c + T}{s}, \\ \|D_N(\widehat c_N - c_{*,N})\|_2 &\le E_{25}. \end{aligned} \tag{S278}$$

Proof. Full column rank makes $B_0^{\mathsf T}B_0$ invertible. For every v,

$$\begin{aligned} \|(I-P_0)B_N v\|_2 &= \inf_u \|B_0 u + B_N v\|_2 \\ &\ge \inf_u s\sqrt{\|u\|_2^2 + \|v\|_2^2} \ge s\|v\|_2. \end{aligned}$$

The Schur bound follows by squaring. Applying the full singular-value bound to $BD(\widehat\beta - \beta_*) = \widehat r - t$ proves the error bound. No projected column is renormalized.

The validated continuous design gives

$$\begin{aligned} s &> 0.01326602753076006, \\ \lambda_{\min}\big(B_N^{\mathsf T}(I-P_0)B_N\big) &> 0.0001759874864468840. \end{aligned} \tag{S279}$$

These bounds follow from the full Gram matrix $\Gamma = B^{\mathsf T}B$: an exact rational approximate inverse R satisfies $\|I - R\Gamma\|_\infty \le \delta < 1$, whence $\sigma_{\min}(B) \ge \sqrt{(1-\delta)/\|R\|_\infty}$. All continuous entries, products and normalizations are enclosed before this inequality is applied.

The actual head and its real cubic remainder. For the frozen head, $\widehat C_k$ in (S274) are its ordinary quadratic Taylor coefficients. Entrywise they are

$$\begin{pmatrix} M(0), \ell\partial_1 M(0), \ell\partial_2 M(0), \\ \tfrac12\ell^2\partial_{11}M(0), \ell^2\partial_{12}M(0), \tfrac12\ell^2\partial_{22}M(0) \end{pmatrix}.$$

The coefficient of E_2 is M_{12}, without an additional factor two. In particular $\widehat C_0 = \widehat M(0)$ is the same origin value used in the earlier decomposition $\widehat M - \widehat M(0)$.

The strict imaginary-affine budgets of both complex $\tanh$ layers hold on the entire real box $|q_i| \le \ell$. There $|\tanh z| \le 1$ and its first three derivatives have bounds $2, 4, 16$. Propagating the full third-order chain rule gives Frobenius bounds $\mathcal L_j$ on $\mathrm d^j L_\phi(t\ell u)/\mathrm dt^j$, $j = 0,1,2,3$, uniformly for $\|u\|_\infty \le 1$, $0 \le t \le 1$. Differentiating $\mathrm{Re}(LL^*)$ then yields

$$\left\| \frac{\mathrm d^3 \widehat M(t\ell u)}{\mathrm dt^3} \right\|_2 \le 2\mathcal L_0\mathcal L_3 + 6\mathcal L_1\mathcal L_2.$$

Real Taylor's theorem along the segment from zero to q proves

$$\|\mathcal R_\theta(q)\|_2 \le \eta_3 \|q/\ell\|_\infty^3, \qquad \eta_3 < 5.292679\,10^{-9}. \tag{S280}$$

The nonconstant coefficients satisfy $\max_{k>0,l}|c_{\theta,kl}| < 7.610\,10^{-13}$. This is a real-box statement. The quadratic polynomial is used in analysis; the deployed positive-semidefinite decoder is unchanged.

Two complementary tail bounds and the certified error. Let $P_N \ge \sum_{k=1}^5 \|\widehat C_k\|_2$ and let η_0 bound $\|\widehat M(q) - \widehat M(0)\|_2$ on the box. Since $|\psi_k| \le 1$ there, the uniform learned remainder also satisfies $\|\mathcal R_\theta\|_2 \le \widehat\eta_2 := \min(\eta_3, \eta_0 + P_N)$. For each row, retain the earlier potential-tail bound p_{wi} and the earlier local learned-variation bound h_{wi} for the absolute integral of $S(\widehat M - \widehat M(0))S^{-1}P_*$ with its row scale. Let

$$\xi_{wi} \ge s_{wi} \int_{I_w} \sqrt{(G_Q)_{ii}(q_*)}\sqrt{P_*^{\mathsf T}G_Q^{-1}P_*}\,\mathrm d\tau.$$

For the specified constant-damping reference, $\mathcal R_* = 0$, and a valid quadratic-tail budget is

$$\begin{aligned} T_{wi} &= p_{wi} + \min\{\xi_{wi}\eta_3, h_{wi} + \xi_{wi}P_N\}, \\ T &= \left(\sum_{w,i} T_{wi}^2\right)^{1/2}. \end{aligned} \tag{S281}$$

The second route follows by subtracting the nonconstant quadratic polynomial from the same origin-based variation. For a different reference class with $\|\mathcal R_*\|_2 \le \eta_{*,2}$, $\xi_{wi}\eta_{*,2}$ must additionally be included in each row.

For the present frozen output, continuous residual and reference class, the enclosed quantities satisfy

$$\begin{aligned} \epsilon_c &< 1.460201\,10^{-5}, \qquad T < 0.001801, \\ E_{25} &< 0.136861, \\ P_N &< 2.37003\,10^{-12}, \qquad \widehat{\eta}_2 < 3.775274\,10^{-10}. \end{aligned} \tag{S282}$$

Writing D_{kl} for the damping entries of D, the Frobenius conversion, including its off-diagonal multiplicity, is

$$\begin{aligned} C_M^2 &= \sum_{k=0}^{5} \left(D_{k1}^{-2} + 2D_{k2}^{-2} + D_{k3}^{-2} \right), \\ \sup_{|q_i| \le \ell} \|\widehat{M}(q) - M_*(q)\|_2 &\le C_M E_{25} + \widehat{\eta}_2 + \eta_{*,2}. \end{aligned} \tag{S283}$$

Thus the same specified reference, for which $\eta_{*,2} = 0$, obeys

$$\sup_{|q_i| \le 1/10} \|\widehat{M}(q) - M_*(q)\|_2 < 1.020023\,10^{-7}.$$

The nonconstant learned coefficients are small, but this does not detect a nonconstant reference law. In fact the complete fifteen-dimensional zero block is not excluded by the certified error ball:

$$\|D_N \widehat{c}_N\|_2^2 < 3.717665\,10^{-10} < E_{25}^2.$$

Here the specified reference has $M_*(q) = \gamma_* I$ and $R_*(q) = \gamma_* G_Q(q)$; the metric-induced variation of R_* is not an unknown nonconstant normalized law. This is an uncertainty-ball statement, not proof of physical realizability of every point of that ball. The reference trajectories remain those of the specified reduced finite-Gauss model, with constant normalized damping; they are not physical measurements. No arbitrary off-trajectory function or redundant neural parameters are uniquely identified. The sharper earlier ten-coordinate conservative bound remains the input to the conservative KAM transfer of Section S3.23; the present damping certificate does not replace it or extend that conclusion to damped dynamics.

From output accuracy to true dissipation and sampled safety. The output certificate also supplies the true coercivity premise of Section S3.20; positive semidefiniteness of the learned decoder alone is insufficient for that premise.

Proposition S59. Identified true coercivity. *On $|q_i| \le \ell$, let μ_0 be a certified Gershgorin lower bound for $\widehat{M}(0)$, let η_0 bound $\|\widehat{M}(q) - \widehat{M}(0)\|_2$, and let ε_M be the uniform output-error bound in (S283). If $\gamma_{\mathrm{cert}} := \mu_0 - \eta_0 - \varepsilon_M > 0$, then*

$$\begin{aligned} M_*(q) &\succeq \gamma_{\mathrm{cert}} I, \\ R_*(q) := S(q) M_*(q) S(q)^{\mathsf{T}} &\succeq \gamma_{\mathrm{cert}} G_Q(q), \\ \delta_*(q,P) := v^{\mathsf{T}} R_* v &\ge \gamma_{\mathrm{cert}} P^{\mathsf{T}} G_Q^{-1} P, \\ v &= G_Q^{-1} P. \end{aligned} \tag{S284}$$

For the specified reference class and actual frozen output, $\gamma_{\mathrm{cert}} > 0.034999897$ on the entire box, which contains the existing control guard.

Proof. Gershgorin and the two uniform spectral bounds give $\widehat{M}(q) \succeq (\mu_0 - \eta_0) I$ and then $M_*(q) \succeq \gamma_{\mathrm{cert}} I$. Congruence by the real invertible S gives the bound for R_*, and its quadratic form gives the power inequality.

An upper output bound also closes the damping contribution to the port-derivative estimate. If $\overline{\mu}_0$ is a certified spectral upper bound for $\widehat{M}(0)$, then

$$\|M_*(q)\|_2 \le \overline{m} := \overline{\mu}_0 + \eta_0 + \varepsilon_M. \tag{S285}$$

With $A = G_Q^{-1}$ and the same D_g, Y_g as in Section S3.20, use its stronger bounds $D_g \ge \sup_{Q_g} d^{\mathsf{T}} A d$ and $Y_g^2 \ge 2E_g D_g$. Cauchy–Schwarz yields

$$|d^{\mathsf{T}} A R_* v| \le \overline{m} \sqrt{d^{\mathsf{T}} A d} \sqrt{v^{\mathsf{T}} G_Q v} \le \overline{m} Y_g \quad \text{on } H_* \le E_g.$$

Thus one can replace the earlier damping contribution to L_y by $\overline{m} Y_g$, retaining the independently certified restoring, metric-connection, trigonometric and actual-input terms. No derivative of M_* is required. Put

$$\begin{aligned} L_y^{\mathrm{cert}} &= L_{\mathrm{rest}} + \overline{m} Y_g, \qquad \varepsilon_y^{\mathrm{cert}} = \varepsilon_m + L_y^{\mathrm{cert}} \Delta, \\ \eta_{\mathrm{cert}} &= U_c \varepsilon_y^{\mathrm{cert}} + \frac{\xi_a^2 D_g}{4\gamma_{\mathrm{cert}}}, \quad C_{\mathrm{cert}} = E_{\max} - \eta_{\mathrm{cert}} \mathcal{T}, \end{aligned} \tag{S286}$$

where $\mathcal{T}$ denotes the control horizon. Here L_{rest} is the sum of those four retained derivative bounds. The same sensor, execution and hold-age tolerances give

$$\begin{aligned} \overline{m} &< 0.035000103, \qquad \mathcal{T} = 20, \\ \eta_{\mathrm{cert}} \mathcal{T} &< 0.001004870, \qquad C_{\mathrm{cert}} - \overline{H}_0 > 0.000806873. \end{aligned} \tag{S287}$$

The filter was evaluated with the new cap C_{cert} under the same sensor, execution and hold-age tolerances. Both predeclared precision settings completed 20 000 holds over $\mathcal{T} = 20$. An independent exact-arithmetic replay verifies every stored command against the endpoint-tightened inequality of Theorem S53; the minimum work-ledger

margin exceeds 9.06865×10^{-5} in each run. The numerical state probes give $H_* < 0.001188930$ and $|u_{\mathrm{act}}| < 0.099891$; the maximum endpoint-state difference between the two precisions is below 4.03×10^{-16}. The exact command and ledger checks are separate from the numerical state integration. Continuous-time safety follows under the theorem's declared plant, sensing, execution and timing assumptions; the DOP853 curves are not validated trajectory enclosures.

The reference remains the declared finite-Gauss model with constant normalized true damping; no nonconstant physical law is inferred. The derivative calculation still uses its specified true potential, metric and force class $f \equiv 0$, and does not identify an arbitrary unknown plant from data.

S3.25 Geometry-induced linear viscous dissipation

A distributed Rayleigh law $f_d(s) = -c_{\mathrm{line}}\dot{r}(s)$ contributes the generalized force $-c_{\mathrm{line}}D(q)\dot{q}$, where $D = \int J(s,q)^{\mathsf{T}}J(s,q)\,\mathrm{d}s$. An endpoint law adds $-c_{\mathrm{tip}}T(q)\dot{q}$. These laws retain the conservative scalar energy and the magnetic port; their dissipated power is the corresponding nonnegative quadratic form. The following result concerns the declared finite-quadrature constitutive model.

Proposition S60. Distributed and mixed viscosity in the frozen Gauss geometry. *Use the original chart, positive stored Gauss8 weights and unchanged metric $G = SS^{\mathsf{T}} = fD + K_r$, where D is the translational geometric Gram matrix, $f > 0$ is the stored translation factor and $K_r \succ 0$ is the stored rotary Gram matrix. Let $T = J_{\mathrm{tip}}^{\mathsf{T}}J_{\mathrm{tip}}$ use the same endpoint graph. On $|q_i| \le 1/10$, the exact frozen constants imply*

$$D(q) \succeq \mu_D I, \qquad \mu_D > 5.95075\,10^{-7}\ \mathrm{m}^3.$$

If $r_+ I \succeq K_r$, put $\nu_D = \mu_D/(f\mu_D + r_+) > 0$. For fixed SI coefficients $c_{\mathrm{line}} > 0$ and $c_{\mathrm{tip}} \ge 0$, define

$$R = \frac{\omega_0}{E_0}(c_{\mathrm{line}}D + c_{\mathrm{tip}}T), \qquad M = S^{-1}RS^{-\mathsf{T}}.$$

Then $M \succeq (\omega_0/E_0)c_{\mathrm{line}}\nu_D I$ on the box. Moreover M is nonconstant on every neighborhood of zero whenever $c_{\mathrm{line}}, c_{\mathrm{tip}} \ge 0$ are not both zero. No positive constant-parameter cancellation is possible.

Proof. At each outer quadrature location, retain the x row of its configuration Jacobian. Stack these rows with the square roots of their positive outer weights into $F_x(q)$. The original spatial orientation is $(\sin\theta, \cos\theta)$, hence $D(0) = F_x(0)^{\mathsf{T}}F_x(0)$ and $D(q) \succeq F_x(q)^{\mathsf{T}}F_x(q)$. Every local or completed tangent summand is $W a_i \cos(a^{\mathsf{T}}q)$ with $W > 0$. The bound

$$|1 - \cos(a^{\mathsf{T}}q)| \le \tfrac{1}{2}\big(\tfrac{1}{10}\|a\|_1\big)^2$$

therefore gives the certified weighted error $\|F_x(q) - F_x(0)\|_F^2 \le \delta^2$. Let λ_0 be the exact Gershgorin lower bound of $D(0)$. Young's inequality, with $t = 1/100$, gives

$$D(q) \succeq \big((1-t)\lambda_0 - (t^{-1}-1)\delta^2\big)I = \mu_D I.$$

All quantities on the right are rational functions of the stored dyadic buffers; the displayed positive lower bound follows by exact comparison. Since $K_r \preceq r_+ I \preceq (r_+/\mu_D)D$, one has $G \preceq (f + r_+/\mu_D)D$. Congruence by S^{-1} proves the normalized bound; $T \succeq 0$ adds no loss.

For nonconstancy, set $I_0 = G(0)^{-1}$. All first configuration derivatives of D, T, G vanish at zero by parity. The coordinate second derivatives of their normalized traces are thus

$$h_{D,j} = \mathrm{tr}(D_{jj}I_0 - D(0)I_0G_{jj}I_0),$$
$$h_{T,j} = \mathrm{tr}(T_{jj}I_0 - T(0)I_0G_{jj}I_0).$$

Exact differentiation of the same inner and completed Gauss sums gives $h_{D,j} > 0$ and $h_{T,j} > 0$ for both $j = 1, 2$. Because $\mathrm{tr}\,M = (\omega_0/E_0)\,\mathrm{tr}((c_{\mathrm{line}}D + c_{\mathrm{tip}}T)G^{-1})$, its second derivative is strictly positive for any nonzero nonnegative coefficient pair. This excludes constant M and positive-parameter cancellation.

The units are $[D] = \mathrm{m}^3$, $[T] = \mathrm{m}^2$, $[c_{\mathrm{line}}] = \mathrm{N\,s/m^2}$ and $[c_{\mathrm{tip}}] = \mathrm{N\,s/m}$. Consequently R is dimensionless and the canonical dissipative force is $P_d' = -RG^{-1}P$. The box bound supplies true power coercivity only for a protocol that actually adopts these declared physical coefficients and this finite-quadrature law. The previous constant-damping data and control logs are not relabelled as mixed-law evidence.

For the historical numerical-study value $c_{\mathrm{line}} = 0.0006911873093836746\,\mathrm{N\,s/m^2}$ and $c_{\mathrm{tip}} = 0$, the resulting normalized damping lies between $0.0486272I$ and $0.0486618I$ on this box. The coefficient is a declared baseline inferred from an assumed damping ratio in the earlier beam study, rather than a measured fluid property. Outward evaluation at $q = 0$ and $q = (1/10, 1/10)$ gives a trace change exceeding 4.52267×10^{-8}. Hence every constant matrix incurs uniform operator error exceeding 1.13066×10^{-8} on these two points, since $|\,\mathrm{tr}\,B| \le 2\|B\|_2$. This quantifies the resolution required

of a subsequent identification experiment; it supplies no additional observation or fit.

Speed-quadratic relative drag. The speed-quadratic Morison term requires a separate velocity-dependent constitutive law. Let $v = q'$ and let integrable $k(s) \geq 0$ have units $\mathrm{kg/m^2}$, so the physical distributed force is $f_2 = -k(s)\|\dot{r}\|\dot{r}$ in a stationary fluid. Its dimensionless generalized damping is

$$\begin{aligned} R_2(q,v) &= \frac{\omega_0^2}{E_0}\int k(s)\|J(s,q)v\|J(s,q)^{\mathsf{T}}J(s,q)\,\mathrm{d}s, \\ P_2' &= -R_2(q,v)v, \\ v^{\mathsf{T}}R_2(q,v)v &= \frac{\omega_0^2}{E_0}\int k(s)\|J(s,q)v\|^3\,\mathrm{d}s \geq 0. \end{aligned} \tag{S288}$$

The result follows by substituting $\dot{r} = \omega_0 J v$ into $\int J^{\mathsf{T}} f_2\,\mathrm{d}s/E_0$. It also holds for a positive quadrature. The additional factor ω_0 relative to linear viscosity comes from the speed factor. Thus R_2 is symmetric positive semidefinite and preserves $H' = -v^{\mathsf{T}}(R + R_2)v + uy$. The linear distributed term retains its coercivity lower bound. If $V \geq 0$, $\|J\|_2 \leq \bar{J}$ and $G \succeq g_- I$ on a dimensionless energy guard $H \leq E_g$, with $G = SS^{\mathsf{T}}$, then

$$\|R_2\|_2 \leq \frac{\omega_0^2}{E_0}\left(\int k(s)\,\mathrm{d}s\right)\bar{J}^3\sqrt{2E_g/g_-}.$$

Moreover $\|S^{-1}R_2S^{-\mathsf{T}}\|_2 \leq \|R_2\|_2/g_-$. This supplies the normalized upper damping bound needed in the port-derivative estimate without differentiating the speed norm. A decoder for this law must accept velocity as well as configuration; a configuration-only matrix cannot represent its speed dependence in general. The required error estimates concern this real dissipative law and do not impose analyticity on it for the conservative KAM argument.

For a prescribed nonzero fluid velocity V_f, put $w = \dot{r} - V_f$. The rod power is instead

$$\int f_2^{\mathsf{T}}\dot{r}\,\mathrm{d}s = -\int k\|w\|^3\,\mathrm{d}s - \int k\|w\|w^{\mathsf{T}}V_f\,\mathrm{d}s.$$

The second integral is an external fluid-power contribution and can have either sign. Dividing this physical power by $E_0\omega_0$ gives its contribution to H'. It must enter the force or power uncertainty budget. Neither this velocity-dependent law nor a nonzero-flow experiment is identified by the preceding constant-normalized-damping data.

S3.26 A resolved distributed-viscosity reference pilot

Retain the distributed-viscosity reference of Section S3.25, with the declared historical coefficient $c_{\mathrm{line}} = 0.0006911873093836746\,\mathrm{N\,s/m^2}$, $c_{\mathrm{tip}} = 0$, and $u \equiv 0$. The conservative potential, stored finite-Gauss metric, canonical chart and magnetic port are unchanged. The coefficient specifies a numerical benchmark; it is not an experimentally calibrated fluid parameter.

A finite output class with a certified physical tail. Write $G = fD + K_r = SS^{\mathsf{T}}$, where f is the original stored translation factor and K_r is the stored rotary inertia. For $a = \omega_0 c_{\mathrm{line}}/(E_0 f)$, the normalized physical damping is

$$M_*(q) = a\big(I - S(q)^{-1}K_r S(q)^{-\mathsf{T}}\big). \tag{S289}$$

This identity avoids subtracting large entries of the distributed Gram. Let $t = G_{12}/G_{11}$ and $K = S^{-1}K_rS^{-\mathsf{T}}$. Its entries are

$$\begin{aligned} K_{11} &= (K_r)_{11}/G_{11}, \\ K_{12} &= \frac{(K_r)_{12} - (K_r)_{11}t}{\sqrt{\det G}}, \\ K_{22} &= \frac{(K_r)_{22} - 2(K_r)_{12}t + (K_r)_{11}t^2}{G_{22} - G_{12}t}. \end{aligned} \tag{S290}$$

On $|q_i| \leq 1/10$, outward polynomial evaluation of these positive denominators yields symmetric matrix polynomials P_2 and P_4, in $z = 10q$, such that

$$\begin{aligned} \sup_q \|M_*(q) - P_2(10q)\|_2 &< 1.087846 \times 10^{-10}, \\ \sup_q \|M_*(q) - P_4(10q)\|_2 &< 2.509287 \times 10^{-13}. \end{aligned} \tag{S291}$$

The retained coefficients are exact dyadic representatives. They are not assumed to be exact Taylor derivatives or supplied as training labels. In particular, P_2 lies in the quadratic symmetric output class of Section S3.24, now with the nonzero target remainder in (S291).

For completeness, the construction retains all bivariate coefficients through degree ten. A polynomial enclosure $p + [-e, e]$ on $[-1, 1]^2$ bounds its polynomial part by the coefficient ℓ^1 norm. Multiplication includes every discarded product and both polynomial–remainder cross terms. For a positive scalar graph $c(1 + z)$ with $\|z\|_\infty \leq \rho < 1$, the reciprocal and inverse square root use their binomial series. In both cases the absolute binomial coefficients are at most one, so truncation after order ten has remainder at most $\rho^{11}/(1 - \rho)$, multiplied by the appropriate power of c. The exact-buffer metric polynomial carries its

independent trigonometric remainder. Replacing retained interval coefficients by dyadic centers adds their radii to e; reducing to degree two or four adds all remaining coefficient magnitudes. Entry errors e_{11}, e_{12}, e_{22} give operator error at most $(e_{11}^2 + 2e_{12}^2 + e_{22}^2)^{1/2}$. This proves the uniform approximation bounds without differentiating an unstructured error.

Native trajectories and continuous observation enclosures. A newly created COMSOL Global Equations model implements this four-state finite-Gauss reference, with

$$q(0) = (19/200, 19/200),$$
$$P(0) = (-1/500, -3/2000).$$

Two fixed-step Dormand–Prince runs use internal steps $1/2000$ and $1/4000$ in dimensionless time and the same 401 output nodes over $0 \le \tau \le 1/5$. Position observations are native stored states; velocity observations are native stored time derivatives, converted from SI time. The observation files contain $\tau, q, v_{\mathrm{obs}}, u$; canonical momentum, force and power exports are kept separately for verification. No archived trajectory is relabelled.

An a posteriori enclosure validates a quintic Hermite reconstruction of each canonical trajectory against the new exact-real reference IVP. Its damping contribution is evaluated as

$$F_{P,d} = -aP + aK_r G^{-1} P.$$

With the previously certified positive matrix W, transport the old logarithmic growth bound using

$$\mu_W(DF_{\mathrm{new}}) \le \mu_W(DF_{\mathrm{old}}) + \|D(F_{\mathrm{new}} - F_{\mathrm{old}})\|_W.$$

The difference is $\delta F_P = (7/200 - a)P + aK_r G^{-1} P$. Its configuration and momentum derivative blocks are, respectively, $-aK_r G^{-1} G_j G^{-1} P$ and $(7/200 - a)I + aK_r G^{-1}$. Whole-box interval bounds give the new logarithmic growth upper bound 0.053001744. For a Hermite defect bound d_j over an interval of length h_j, the propagated radius satisfies

$$R_{j+1} \le e^{\ell h_j} R_j + d_j \frac{e^{\ell h_j} - 1}{\ell}.$$

The initial radius encloses the difference between the exact rational initial state and its native binary64 representation. The full canonical tube remains inside $|q_i|, |P_i| < 1/10$ over all 400 intervals of each run. The resulting bounds are

Step	W-radius	Nodal v error
$1/2000$	1.110687×10^{-13}	2.781662×10^{-13}
$1/4000$	1.469394×10^{-13}	3.676308×10^{-13}

The nodal velocity estimate additionally encloses $G(q_{\mathrm{obs}})v_{\mathrm{obs}} - P_{\mathrm{native}}$ and propagates this mismatch through the inverse metric and its configuration derivative. The certificate concerns the stated reduced reference IVP, with no continuum-rod or physical-calibration claim. A smaller internal step need not reduce this a posteriori bound, which also encloses rounded output derivatives and center-curve defects.

Identification scope. These bounds supply a target-class remainder and a validated observation interface. A residual-to-function error bound still requires a sufficiently excited design for the new reference. In particular, one trajectory with the original whole-window and four quarter-window family supplies at most eight independent scalar momentum equations: its whole-window row is the sum of the four quarter rows before positive row scaling. This cannot identify eighteen quadratic symmetric coordinates, or the twelve coordinates obtained by imposing the physical symmetry $M_*(-q) = M_*(q)$. Consequently, this pilot alone does not identify the nonconstant law or establish an advantage of complex over real networks. A prospective follow-up data-generation protocol specifies fourteen training trajectories and four held-out trajectories with distinct initial conditions. No fitted physical-damping model or new closed-loop result from that protocol is reported here.

S3.27 Derivative-labelled experiment and local analytic diagnostics

This subsection specifies the derivative-labelled reduced-model experiment. Native COMSOL joint identification uses the distinct weak-form protocol in Section S3.4. The reproducibility directory `deep_chnn_v11` contains the protocol, unchanged model inputs, all 13 pairs of energy/damping checkpoints, training histories, predictions and complete failed-check records. Its protocol SHA-256 begins `4f8beda2a60b75d5`; the full digest is recorded with each checkpoint. Source and data hashes were fixed before training; final checks verified their identity and that no evaluated checkpoint changed. Existing held-out data had been inspected during development. Only the additional initial-condition set was generated after all checkpoints were frozen.

Adam uses learning rate 0.003, cosine decay to 0.00003, and gradient norm clipping at 10. L-BFGS starts from the best Adam validation checkpoint, uses rate 0.5, strong-Wolfe line search and tolerances 10^{-10} for gradient and 10^{-12} for change. All runs use `float64`/`complex128`. The two constant damping scalars are mapped through softplus,

trained for 1200 Adam steps at rate 0.05, and selected by damping validation loss. The joint branch initializes its position head near that fitted constant damping and uses 2000 Adam steps at constant rate 0.001. Its conservative and dissipative field losses are weighted equally, each divided by its own training residual RMS squared. The training component losses and the summed validation criterion are retained; the minimum validation sum, including step zero, selects the final checkpoint.

For each test set the prespecified computational checks require total-field relative error below 0.01, nonlinear-field relative error below 0.5, conservative 1 s state error below 0.01, driven 2 s state error below 0.02, and maximum damping relative error below 0.05. Numerical consistency thresholds are 10^{-7} for the learned energy-balance error, 10^{-10} for the autograd/analytic-gradient difference, and 10^{-13} for checkpoint reload equality. These are declared engineering criteria for this experiment, not thresholds in a KAM theorem. Prior to optimization, 24 architecture checks and 9 independent joint-decoder checks passed, including state and parameter finite differences, mixed derivatives, PSD structure and the independent-velocity Rayleigh derivative.

Trajectory integration uses DOP853 with relative tolerance 10^{-10} and absolute tolerance 10^{-12}. The new reference trajectories use 2×10^{-12} and 2×10^{-14}, respectively. Every solver stage checks the box $[-0.6, 0.6]^4$; an excursion aborts that orbit and makes its full-trajectory metric missing. A 50000-evaluation and 60 s wall-time budget per orbit similarly preserves failures rather than averaging only successful trajectories. All trajectories in the reported experiment completed. All seven prior-informed models, including the joint extension, passed both test sets; the other six models failed the specified accuracy checks.

The five contrasts in the article are supplied for both test sets and five metrics, giving 150 individual seed-paired differences. Each record contains both operands. Common test metrics, rather than the differently normalized training objectives, define these differences. No noisy-label deep experiment or independent new COMSOL solve is included in this protocol.

Exact coefficient export and interval domain. After checkpoint freezing, the two validation-selected energy networks are exported by evaluating their effective affine coefficients once in binary64 and storing every coefficient as an exact hexadecimal float. These exported coefficients define a specific analytic function. `mpmath.iv` at 60 decimal digits encloses the row budgets, matrix norms and trigonometric margins in (S96) outward. Both exported functions satisfy the row constraints and have certified tube radii exceeding 5.58785×10^{-4} (selected GC_D) and 6.31070×10^{-4} (joint GC_D). This certificate concerns the exact exported coefficients; it does not enclose the discrepancy from an ideal real-arithmetic evaluation of the raw-parameter constraint map. Agreement with the Torch value and gradient is checked numerically, and directional derivatives through order four are cross-checked at 80 decimal digits.

Local derivatives and the remaining KAM calculation. At the origin the respective maximum gradient components are 5.01988×10^{-9} and 3.02530×10^{-9}, and the maximum Hessian-entry differences from the quadratic prior are 4.93963×10^{-8} and 1.65352×10^{-7}. The maximum third-order entries are 1.80472×10^{-5} and 1.35930×10^{-5}. The fourth-order tensor relative Frobenius errors, 0.059443 and 0.093885, compare the complete tensor with the analytic Python reference energy at its origin, $h_2 + 1 - \cos(d^{\mathsf{T}}X) - \frac{1}{2}(d^{\mathsf{T}}X)^2$. Its position block is $-d \otimes d \otimes d \otimes d$ and its other blocks vanish. This comparator is not the fitted COMSOL integral Hamiltonian H_F in Section S3.6.3. These values are local derivative diagnostics, not Birkhoff coefficients. Numerical equilibrium solves from zero give maximum displacements 1.73819×10^{-9} and 2.23893×10^{-9}; their Hessians are numerically positive definite. Neither solve is an interval existence or uniqueness proof.

Application of (S102) or (S104) requires an interval Taylor remainder or a uniform complex derivative error, together with the torsion, inverse and smallness conditions for the specified energy. The tube estimate establishes a domain bound, while the local jet quantifies derivative changes excluded by a fixed-quartic model. These diagnostics do not constitute a KAM certificate for the derivative-labelled networks considered in this subsection.

S4 Nonlinear stability of the learned magnetic model

S4.1 Normal form of the same magnetic cantilever

In the dimensionless chart of Section S1.3, write

$$\begin{aligned} h_\star &= h_2 + g(d^\top X), \\ h_2 &= \frac{1}{2}\sum_{j=1}^{n} \nu_j (X_j^2 + Y_j^2), \\ g(z) &= 1 - \cos z - \frac{z^2}{2}. \end{aligned} \tag{S292}$$

The magnetic term in this expression is the complete analytic function. In particular, its fourth-order part and its untruncated remainder are

$$h_{4,\star} = -\frac{(d^\top X)^4}{24},$$
$$T_{6,\star} = 1 - \cos(d^\top X) - \frac{(d^\top X)^2}{2} + \frac{(d^\top X)^4}{24}.$$

Proposition S61. Magnetic twist and analytic tail. *Let $I_j = (X_j^2 + Y_j^2)/2$, and let $A_\star$ denote the Hessian with respect to I of the angular average of $h_{4,\star}$. With $v = (d_1^2, \dots, d_n^2)^\top$,*

$$(A_\star)_{jj} = -\frac{d_j^4}{8},$$
$$(A_\star)_{ij} = -\frac{d_i^2 d_j^2}{4} \quad (i \neq j), \tag{S293}$$
$$A_\star = \frac{\operatorname{diag}(d_j^4) - 2vv^\top}{8}.$$

and

$$\det A_\star = (1-2n)8^{-n} \prod_{j=1}^{n} d_j^4. \tag{S294}$$

Thus every retained mode with nonzero tip-rotation participation contributes to a nonsingular twist. For $n \geq 2$, the twist has one negative and $n-1$ positive eigenvalues; positive definiteness is not required here. If $D = \sum_j |d_j|$ and $\mathbb{D}_r^{2n}$ is a complex Cartesian polydisc, then

$$\|T_{6,\star}\|_r \leq \frac{(Dr)^6}{720} \cosh(Dr). \tag{S295}$$

Proof. The angular averages are $\langle X_j^4 \rangle = 3I_j^2/2$ and $\langle X_i^2 X_j^2 \rangle = I_i I_j$. All terms with an odd exponent in some X_j have zero average. Substitution gives (S293). Factoring $\operatorname{diag}(d_j^2)$ on both sides gives a matrix congruent to $(I_n - 2\mathbf{1}\mathbf{1}^\top)/8$. Its eigenvalues are $1/8$ with multiplicity $n-1$ and $(1-2n)/8$, which proves the determinant and inertia statements. For the analytic tail, expand the cosine and use $(6+2k)! \geq 6!(2k)!$ and $|d^\top X| \leq Dr$.

For the two-mode, 32-element model, the magnetic-bias modes give $\nu \simeq (1, 6.152117504)$ and $d \simeq (0.22567077, 0.32179881)$. Consequently $\det A_\star \simeq -1.303708 \times 10^{-6}$ and $\sigma_{\min}(A_\star) \simeq 5.8064 \times 10^{-4}$. These quantities belong to the same cantilever model used for learning and shape reconstruction.

S4.2 Complete learned Hamiltonian and transfer of twist

The fitted family is

$$h_\eta = h_2 + \frac{b}{s_4} g(d^\top X) + \sum_{j=1}^{34} c_j \psi_j(X, Y), \tag{S296}$$

where $\eta = (b, c_1, \dots, c_{34})$, $a = b/s_4$, $s_4 = \|q_\star\|_2$ is the fixed quartic coefficient normalizer and the ψ_j span the coefficient-space complement of the target quartic direction. The class contains the complete target at $\eta_\star = (s_4, 0, \dots, 0)$; no cosine truncation error is assigned to the training labels. The polynomial part is represented by matched real or complex features. All bounds use the actually exported feature matrix, so numerical orthogonality of its columns is not a proof assumption.

Let $F(z) = \big(g(d^\top X)/s_4, \psi_1(z), \dots, \psi_{34}(z)\big)$, and put $D(z) = [\nabla F_0(z), \dots, \nabla F_{34}(z)] \in \mathbb{R}^{4\times 35}$. With $G_N = (4N)^{-1} \sum_i D(z_i)^\top D(z_i)$, $\lambda_{\min}(G_N) \geq \underline{\lambda} > 0$, empirical derivative root-mean-square loss r_N (averaged over all $4N$ scalar observations), and a certified derivative-label error with the same root-mean-square normalization bounded by ϵ_N, the deterministic estimate is

$$\|\eta - \eta_\star\|_2 \leq b_N := \frac{r_N + \epsilon_N}{\sqrt{\underline{\lambda}}}. \tag{S297}$$

This estimate follows from the smallest singular value of the stacked derivative design matrix and the triangle inequality.

The effective quartic coefficient vector is $q_\eta = (b/s_4)q_\star + Uc$, where U is the exported matrix of polynomial features. Let $\mathcal{A}q$ be the symmetric twist matrix obtained by angular averaging. Then the directly computable bound

$$\|A_\eta - A_\star\|_2 \leq \|\mathcal{A}[q_\star/s_4\, U]\|_{2\to F}\, b_N + \delta_{A,\mathrm{dec}} < \sigma_{\min}(A_\star) \tag{S298}$$

is sufficient for nonsingularity of the trained twist. The subscript F denotes the Frobenius norm on symmetric matrices. In the two-mode raw quartic basis, the previously derived bound $\|\mathcal{A}\Delta q\|_2 \leq \sqrt{19}\|\Delta q\|_2$ is also valid. A direct rational or interval check of A_η can be sharper than the sufficient loss-based bound. Here $\delta_{A,\mathrm{dec}}$ bounds the exported decoder's rounding error relative to its exact coefficient map; it is zero for exact arithmetic and is explicitly included in the numerical certificate. Specifically, the exported effective-quartic coefficient error δ_{dec} gives $\delta_{A,\mathrm{dec}} = \sqrt{19}\delta_{\mathrm{dec}}$.

Define the finite-feature analytic bound by

$$B_F(r)^2 = \left(\frac{(Dr)^4}{24 s_4}\cosh(Dr)\right)^2 + r^8 \sum_{j=1}^{34} \|\psi_j\|^2_{\mathrm{coeff},1}.$$

The same coefficient estimate gives a complete analytic perturbation bound,

$$\|h_\eta - h_\star\|_r \le b_N B_F(r) + \delta_{h,\mathrm{dec}}(r). \tag{S299}$$

This controls the entire learned function on a complex domain, including every higher-order magnetic term.

Suppose $\alpha_4 = \min_{0<|k|_1\le 4} |k\cdot\nu| > 0$. The exact quartic Lie transformation associated with the full effective $h_{4,\eta}$ gives

$$h_\eta \circ \Phi_\eta = \nu\cdot I + \tfrac{1}{2} I^\top A_\eta I + \mathcal{R}_{6,\eta}.$$

If C_4, C_Z, C_χ are the Cartesian coefficient norms of its quartic part, angular average and homological generator, then

$$\|\mathcal{R}_{6,\eta}\|_r \le 512 C_\chi (C_4 + C_Z) r^6 + \frac{|a|(2Dr)^6}{720}\cosh(2Dr). \tag{S300}$$

This holds when $32 C_\chi r^2 < 1$. The first term is the exact quartic Lie remainder; the second bounds $aT_{6,\star}\circ\Phi_\eta$ because the time-one flow maps the radius-r polydisc into the radius-$2r$ polydisc. This identity retains the full cosine rather than applying a theorem to an isolated quartic approximation.

With nonsingular A_η and fourth-order nonresonance, the ordinary Kolmogorov theorem applies on sufficiently small positive-action boxes (Pöschel 2001, Theorems A–B, arXiv:0908.2234v1). Indeed, their action width is $O(r^2)$, their integrable Hessian is A_η, and (S300) is $O(r^6)$. Choosing a Diophantine constant proportional to r^2 makes the analytic KAM smallness quotient $O(r^2)$; reducing its proportionality constant first leaves a positive-measure set of admissible frequencies. This establishes local KAM families for the complete target and trained cantilever Hamiltonians. The finite-amplitude result below is verified separately by an a posteriori theorem and is not inferred from this asymptotic statement.

S4.3 Validated invariant tori at finite physical amplitude

The local normal-form argument can be supplemented by a finite-amplitude certificate for the same complete Hamiltonian. We use the a posteriori theorem of Haro and Luque (2019); the equation and table numbering below refer to Theorem 3.1 and Appendix B, Tables 1–3, of its archived preprint, arXiv:1711.05515v1. The computation includes the analytic interpolation error and uses ball arithmetic, rather than identifying a small residual at finitely many sample points with an analytic norm.

The mathematical model certified here is specified exactly by the exported binary coefficients. In particular,

$$\nu = \left(1, \frac{6926668524826847}{1125899906842624}\right), \quad d = \left(\frac{4065323231059343}{18014398509481984}, \frac{2898505962069547}{9007199254740992}\right). \tag{S301}$$

The target uses $a = 1$ and zero quartic residual in (S296). Each trained case uses the exact dyadic value of its exported cosine multiplier and every exported quartic coefficient. This convention identifies the function being certified without hiding floating-point parameter uncertainty. It establishes a result for the exported two-mode reduction; independent finite-element and COMSOL comparisons do not themselves prove an infinite-dimensional KAM theorem for the continuum beam.

Theorem S62. Finite-amplitude tori of the target and trained models. *For the target Hamiltonian (S292), with parameters (S301), for each of the three primary exported GC Hamiltonians with training seeds 17, 43, and 91, and for the additional seed-17 fit at noise amplitude 10^{-8}, and for the GC model fitted to independent COMSOL position–velocity trajectories, there is a real analytic, embedded Lagrangian torus $K_\infty : \mathbb{T}^2 \to \mathbb{R}^4$ satisfying the invariance equation below, with the displayed frequency for the first five models:*

$$X_h(K_\infty) = DK_\infty \varpi, \quad \varpi = \frac{w_1}{2\pi}(1,\zeta), \quad w_1 = \frac{49999589341}{50000000000}. \tag{S302}$$

where, with $\phi = (1+\sqrt{5})/2$,

$$\zeta = \frac{1876278\phi + 592877}{304979\phi + 96369}. \tag{S303}$$

For the COMSOL-fitted model, the frequency is instead specified by

$$w_1 = \frac{99999143141}{100000000000}, \quad \zeta = \frac{974993\phi + 427310}{158480\phi + 69457}. \tag{S304}$$

It is chosen near that actual model's quartic normal-form frequency. Angles have period one; ϖ is measured in cycles per unit dimensionless time. Each torus lies within 3×10^{-41}, in the supremum norm on the complex strip of width $1/80$, of its computed finite Fourier polynomial. The reconstructed beam has nonzero millimetre-scale motion: the certified tip-displacement witness and the bound valid for every point of the centerline are given in Table S13.

Proof. We give the analytic estimates and specify the finite checks that close them. Vector norms are maximum norms. Following the source convention, analytic matrix norms are $\max_i\sum_j\sup_\theta|M_{ij}(\theta)|$, with componentwise suprema on $\mathbb{T}^2_\rho=\{\theta:|\operatorname{Im}\theta|_\infty<\rho\}$. The phase-space domain is $\mathbb{D}^4_{3/10}$.

Frequency and the cohomological equation. The continued fraction of ζ has prefix $[6;6,1,1,2,1,23,1,13,6,3]$ followed by an infinite tail of ones. For the COMSOL-fitted model the prefix is $[6;6,1,1,2,1,20,1,1,4,1,1,3,2]$, also followed by ones. All partial quotients in either case are at most 23. The bounded-partial-quotient inequality gives $|q\zeta-p|\ge1/(25|q|)$ for $q\neq0$. As $w_1>9/10$ and $2\pi<7$, this implies

$$|k\cdot\varpi|\ge\frac{\gamma}{|k|_1},\qquad\gamma=\frac{9}{1750},\qquad k\in\mathbb{Z}^2\setminus\{0\}. \tag{S305}$$

The case $k_2=0$ follows directly from $\varpi_1>9/70$. We use the weaker exponent $\tau_K=3$ in the a posteriori theorem. For a zero-mean analytic function v, Fourier division by $2\pi ik\cdot\varpi$ and the count $\#\{k:|k|_1=j\}=4j$ give, with $q=e^{-2\pi\delta}$,

$$\begin{aligned}\|\mathcal{R}_\varpi v\|_{\rho-\delta}&\le\frac{2}{\pi\gamma}\frac{q(1+q)}{(1-q)^3}\|v\|_\rho\\&\le\frac{e^{4\pi\rho}}{2\pi^4\gamma\delta^3}\|v\|_\rho<\frac{\|v\|_\rho}{\gamma\delta^3}.\end{aligned}$$

These inequalities hold for $0<\delta<\rho=1/40$, so the cohomological constant $c_R=1$ is justified directly. Reconstructing the continued-fraction convergent matrix verifies the stated frequency and Diophantine bound.

Global bounds and theorem constants. Write $h=h_2+ag(d^\top X)+p_4$, with quartic coefficient norm $C_4=\sum_\alpha|(p_4)_\alpha|$. The exact exported coefficients satisfy $|a|\le2$, $C_4<1/100$, $\|d\|_1<11/20$, and $\max\nu_j<7$. The full cosine and its derivatives then give, on $\mathbb{D}^4_{3/10}$,

$$\begin{aligned}c_{X_h,0}&=3,&c_{X_h,1}&=8,\\c_{X_h^\top,1}&=32,&c_{X_h,2}&=1.\end{aligned} \tag{S306}$$

The third quantity bounds the derivative of the *row-vector map* $X_h^\top$. It bounds the sum of the absolute Jacobian entries and is not the matrix norm of $(DX_h)^\top$. For example, its linear part is $2(\nu_1+\nu_2)<28$; its remaining bound is $2\|d\|_1^2(\cosh(3\|d\|_1/10)-1)+12C_4(3/10)^2<1$. For the second derivative, the row-sum tensor estimate $2\|d\|_1^3\sinh(3\|d\|_1/10)+24C_4(3/10)<1$ suffices. All geometric tensors are constant canonical tensors with norm one and zero derivatives.

Let $J=\left(\begin{smallmatrix}0&I\\-I&0\end{smallmatrix}\right)$, $L=DK$, $B=(L^\top L)^{-1}$, and $N=-JLB$. In the source convention the symplectic matrix is $\Omega=-J$. The torsion used in the theorem is therefore

$$T=-N^\top J\big(DX_h(K)N-D_\varpi N\big). \tag{S307}$$

There are no additional first integrals ($n=d=2$, canonical Case III). Choose $\rho=1/40$, $\rho_\infty=1/80$, $a_1=a_2=2$, $a_3=12$, $\delta=\rho/12$, and the iteration parameter $\nu_{\mathrm{it}}=1$. The bounds used with strict margins are

$$\begin{aligned}\|DK\|_\rho&<1<\sigma_K=2,\\\|DK^\top\|_\rho&<\tfrac32<\sigma_{K^\top}=2,\\\|B\|_\rho&<3<\sigma_B=4,\end{aligned}$$

$$\|\langle T\rangle^{-1}\|<10^5<\sigma_T=2\times10^5,\qquad\|K\|_\rho<\tfrac15.$$

The last inequality leaves a phase-space distance margin $1/10$. Substitution in Tables 1–3 gives exact rational constants with the following decimal approximations:

$$\begin{aligned}C_1&\simeq4.843521623\times10^{46},\\C_2&\simeq4.940319879\times10^{21},\\\frac{\gamma^4\rho^{12}}{C_1}&>8.6086\times10^{-76}.\end{aligned} \tag{S308}$$

The rational recurrence in those tables is evaluated using rational or ball arithmetic; rounded decimal values are used only for display. The source's C_{sym} follows from the Case III block identity in Remark 4.7, immediately after Eq. (4.39): $C_{\mathrm{sym}}=\max\{1,\sigma_B^2\}C_{\Omega_L}=192$, using $\sigma_B=4$ and $C_{\Omega_L}=12$. This uses C_{Ω_L} from the geometric identity in place of the $C_{\widetilde{\Omega}_L}$ printed in the corresponding row of Table 1.

Analytic residual and torsion checks. For each case the approximate embedding K is an exact dyadic finite Fourier polynomial supported on $|k_j|\le63$. Conjugate symmetry is checked exactly. Its derivative and wide-strip norms are bounded by $\sum_k|K_k|e^{2\pi s|k|_1}$ and the corresponding differentiated sums. To bound B, compare K with $K_0=(\frac{1}{10}\cos2\pi\theta_1,\frac{1}{10}\cos2\pi\theta_2,-\frac{1}{10}\sin2\pi\theta_1,-\frac{1}{10}\sin2\pi\theta_2)$. Put $e_L=\|D(K-K_0)\|$ and $e_{L^\top}=\|D(K-K_0)^\top\|$. As

$DK_0^\top DK_0 = (2\pi/10)^2 I$ exactly, the checked inequality

$$\|DK_0^\top\| e_L + e_{L^\top}\|DK_0\| + e_{L^\top} e_L < (2\pi/10)^2$$

provides a Neumann-series inverse bound for the analytic Gram matrix.

The residual $E = X_h(K) - DK\varpi$ is evaluated with 512-bit Arb balls on a 256×256 grid, including ball sine evaluations and all quartic terms. Let $\mathcal{I}_m E$ be its trigonometric interpolant, $\widehat{\rho} = 1/4$, and $M \geq \|E\|_{\widehat{\rho}}$, computed directly from the finite Fourier polynomial and the entire Hamiltonian. Fourier decay, the two-dimensional union bound for the omitted modes, and the alias sum give

$$\begin{aligned} \|E\|_\rho &\leq \|\mathcal{I}_m E\|_\rho + E_{\mathrm{tail}}, \\ E_{\mathrm{tail}} &= 8M \frac{1+q}{(1-q)^2} e^{-\pi(\widehat{\rho}-\rho)m}. \end{aligned} \tag{S309}$$

Here $q = e^{-2\pi(\widehat{\rho}-\rho)}$ and $m = 256$. Both the omitted Fourier tail and the aliased coefficients are included in the factor 8. The larger strip is used only to estimate the composition residual; the entire full-cosine Hamiltonian is defined there even when $K(\mathbb{T}^2_{\widehat{\rho}})$ leaves the smaller phase-space domain used by the KAM iteration.

For the mean torsion, the same Gram estimate on the strip $7/100$ proves $\|N\|, \|N^\top\| < 6$. Since $\|\varpi\|_1 < 2$, Cauchy's inequality on the strip $6/100$ gives $\|D_\varpi N\| < 1200$, and hence $\|T\| < 6(8 \cdot 6 + 1200) < 10^4$. An Arb trapezoidal average on 128^2 nodes is corrected by

$$10^4 \left[\left(\frac{1+z}{1-z} \right)^2 - 1 \right], \qquad z = e^{-2\pi(6/100)128}.$$

The derivative of B in this evaluation is calculated by differentiating its inverse identity, so no differentiated interpolation of N is used. Inverting the resulting interval 2×2 matrix proves the stated bound on $\langle T \rangle^{-1}$.

The resulting bounds in Table S13 satisfy

$$\frac{C_1 \|E\|_\rho}{\gamma^4 \rho^{12}} < 1. \tag{S310}$$

The cited theorem supplies the invariant Lagrangian torus and $\|K_\infty - K\|_{\rho_\infty} \leq C_2 \|E\|_\rho / (\gamma^2 \rho^6) < 3 \times 10^{-41}$. For completeness, embeddedness can also be checked quantitatively: writing $X_j - iY_j = 0.1 e^{2\pi i \theta_j} + \xi_j$, the verified C^0 and C^1 differences from K_0 make the phase map a degree-one map with derivative less than $1/2$ from the identity. It is a diffeomorphism. The same inequality holds after adding the KAM correction and its Cauchy derivative bound on a smaller strip.

Physical displacement. The exact exported shape matrix maps X to the Hermite nodal displacements and slopes. On an element of length ℓ, the cubic Bernstein control values are w_0, $w_0 + \ell\vartheta_0/3$, $w_1 - \ell\vartheta_1/3$, and w_1. For a control row b, its value on the invariant torus is bounded by $\sum_{j=1}^2 |b_j| (\sum_k |(X_j)_k| + 3 \times 10^{-41})$. The convex-hull property therefore bounds every centerline point, every phase, and all time. A ball evaluation at phase $(0, 0)$ gives the separate positive lower witness in the table. These bounds use the same exported shape map as the learning and beam reconstruction.

The additional low-noise fit connects the two certificates on one actual learned Hamiltonian. Its nonzero derivative MSE is $3.3866649 \times 10^{-17}$. The outward Gram lower bound 6.0167×10^{-8}, together with the label and decoder error budgets, gives a twist-error upper bound 2.81119×10^{-4}, smaller than the target singular-value lower bound 3.03081×10^{-4}. The fifth row of the table uses these same fitted coefficients. Thus the nonzero-loss twist certificate and the finite-amplitude KAM certificate concern the same fitted function; the latter additionally uses the full analytic invariance residual, as required by the theorem.

The sixth row certifies the actual GC function obtained from four independent COMSOL position–velocity trajectories by integral matching. The fit uses state increments and feature integrals, without reconstructed acceleration labels. Its exported cosine multiplier is $a = 1.022324882191665$; every exported quartic term is included in the analytic verification, using the fitted coefficients without re-estimation. The frequency in (S304) admits the same conservative $\gamma = 9/1750$, so the global bounds and theorem constants above remain unchanged. In this case the checked inverse averaged torsion is below 94762, within the required 10^5 margin.

This result certifies one specified nonresonant torus in each Hamiltonian at a useful physical amplitude. It does not attach a measured phase-space volume to these individual tori. Positive-measure families near the elliptic equilibrium follow from the separate normal-form argument above; the finite-amplitude theorem and the energy-based safety theorem have distinct hypotheses and conclusions.

S4.4 Positive-measure families around the certified tori

The nondegeneracy retained by the a posteriori iteration gives a further conclusion at the same finite-amplitude motion.

Corollary S63. KAM stability of the certified finite-amplitude tori. *Each of the six tori in Theorem S62, and the*

Table S13. Finite-amplitude certificates for the complete target and actual trained Hamiltonians. Displayed bounds are rounded outward. The lower witness is tip displacement at one phase; the upper bound covers the entire reconstructed centerline at all phases.

Model	$\|E\|_\rho$ upper	KAM quotient upper	Tip lower (mm)	Centerline upper (mm)
Target	$2.7310\,10^{-77}$	0.031724	1.8977905	1.8977964
GC17	$2.7372\,10^{-77}$	0.031796	1.8927460	1.8927517
GC43	$2.7538\,10^{-77}$	0.031988	1.8837514	1.8837573
GC91	$2.7285\,10^{-77}$	0.031694	1.9077295	1.9077354
GC17 (10^{-8} noise)	$2.7311\,10^{-77}$	0.031724	1.8977399	1.8977457
GC (COMSOL states)	$2.7330\,10^{-77}$	0.031746	1.8977374	1.8977433

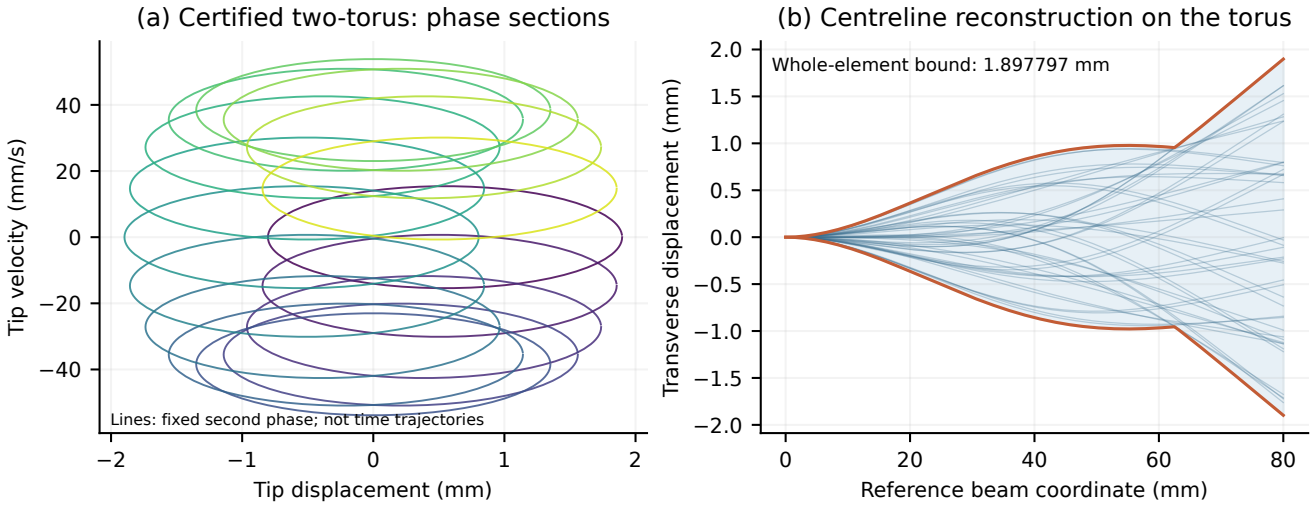


Figure S2. The certified target torus in the same SI beam coordinates. (a) Tip displacement and velocity along sampled sections with the second phase fixed; these curves are not time trajectories. (b) Sampled centerlines and a plotted phase envelope. The separate whole-element Bernstein/Fourier certificate bounds every centerline point on the corrected invariant torus by 1.897797 mm. Sampling illustrates the geometry; the analytic certificate supplies the all-phase bound.

additional integral-COMSOL torus in Section S4.5, is accumulated by positive-measure families of analytic Lagrangian invariant tori. In analytic symplectic tubular coordinates (θ, I) centered on that torus, their relative Lebesgue measure in $\mathbb{T}^2 \times \{|I|_\infty < r\}$ tends to one as $r \to 0$. This is an asymptotic statement about neighborhoods of the certified millimetre-amplitude torus; it does not specify a numerical tube thickness or a percentage in a preassigned experimental domain. In particular, for any prescribed fraction $p < 1$, a sufficiently thin tube has relative invariant-torus measure greater than p. The present statement is existential in the tube radius.

Proof. The inequalities controlling the inverse averaged torsion in the Haro–Luque iteration are uniform at every step: see their Eqs. (5.65)–(5.67) and the summation in Section 5.3. The strict margin used in (S308) therefore gives

$$\|\langle T_\infty \rangle^{-1}\| \le 2 \times 10^5.$$

This is nondegeneracy of the final invariant object, not only of its numerical approximation.

An analytic Lagrangian neighborhood chart Ψ can be chosen with $\Psi(\theta, 0) = K_\infty(\theta)$ and first derivative $D\Psi(\theta, 0) = [L_\infty, N_\infty]$. One construction starts with $K_\infty(\theta) + N_\infty(\theta)I$ and uses the relative analytic Moser correction. The forms agree on the zero section; the relative primitive and its correcting vector field are $O(|I|^2)$, so the correction preserves this first jet. In these coordinates invariance implies

$$h \circ \Psi = e + \varpi \cdot I + \tfrac{1}{2} I^\top A(\theta) I + O(|I|^3). \tag{S311}$$

The automatic-reducibility identity of Haro–Luque Section 2.4 is

$$DX_h(K_\infty)P - D_\varpi P = P \begin{pmatrix} 0 & T_\infty \\ 0 & 0 \end{pmatrix},$$

where $P = [L_\infty, N_\infty]$. Comparing this with the linearized vector field of (S311) gives $A = T_\infty$. More generally, changing the symplectic normal frame to $N_\infty + L_\infty S(\theta)$, with S symmetric, changes this block to $T_\infty - D_\varpi S$, whose average is the same. Hence $\overline{A} = \langle A \rangle$ is nonsingular independently of this choice of normal frame.

The quadratic normal form is justified without a formal-series convergence assumption. Solve the matrix

homological equation $D_\varpi S = A - \overline{A}$ with zero average by

$$S_k = \frac{A_k}{2\pi i k \cdot \varpi} \quad (k \neq 0), \qquad S_0 = 0.$$

Equation (S305) and analyticity make this series convergent on every strictly smaller angle strip; the cohomological bound established above controls it there. The exact time-one Hamiltonian flow of $\chi_2(\theta, I) = \frac{1}{2} I^\top S(\theta) I$ exists on a sufficiently thin tube and fixes $I = 0$. With the pullback convention $L_\chi F = \{F, \chi\}$, it changes the quadratic coefficient to $\overline{A}$ and leaves an analytic $O(|I|^3)$ remainder. The resulting normal form satisfies the hypotheses of Bounemoura (2016, Theorem 6, analytic case): the frequency is nonresonant, the averaged quadratic form is nonsingular, and the full remainder is analytic. That theorem gives the claimed density-one families. Symplectic coordinate changes preserve volume, and the constant factor relating the dimensionless chart to SI canonical coordinates cancels in relative measure.

This corollary establishes the local abundance of invariant tori around the certified motion. Section S4.6 defines their relative measure on a specified phase-space domain. For the integral-COMSOL fit, Appendix S11 supplements the local result with an explicit symplectic chart, a complete analytic remainder and a conservative quantitative bound on a fixed tube. The tube width and the resulting measure bound are reported together to distinguish this rigorous local estimate from the wider numerical sampling domain.

S4.5 A validated COMSOL state-to-KAM learning certificate

We add a controlled identification experiment for the same complete two-mode magnetic Hamiltonian. COMSOL independently integrates its four state equations for 64 initial conditions, with 17 state observations on each dimensionless interval $[0, T]$, $T \simeq 0.04$. This experiment is explicitly a two-mode modal ODE experiment; the earlier 64-DOF finite-element trajectories remain a separate validation. Only positions and velocities enter the new training labels. Neither acceleration reconstruction nor a solver tolerance is used as an error certificate.

For each observed orbit, fit a degree-eight polynomial curve $c_\ell : [0, T] \to \mathbb{R}^4$ to the observed states. Its coefficients are frozen as exact dyadic rationals, with its constant and linear coefficients corrected exactly so that both endpoints equal the exported observations z_ℓ^-, z_ℓ^+. This construction uses no evaluation of the physical acceleration. Let $Lz = X_{h_2}(z)$ and let $V(z) = [J\nabla F_0(z), \dots, J\nabla F_{34}(z)]$ be the nonlinear feature field of the same learned family. Define

$$\begin{aligned} D_\ell &= \frac{1}{T} \int_0^T V(c_\ell(\tau))\, d\tau, \\ y_\ell &= \frac{z_\ell^+ - z_\ell^-}{T} - \frac{1}{T} \int_0^T L c_\ell(\tau)\, d\tau. \end{aligned} \tag{S312}$$

The integral design and labels are therefore functions of the observed state curves, not instantaneous force-balance labels.

Proposition S64. Integral loss, dynamic defect and coefficient error. *Let $\eta_\star = (s_4, 0, \dots, 0)$ represent the declared full-cosine target, and let m be the number of observed curves. Put*

$$\begin{aligned} e_\ell &= \frac{1}{T} \int_0^T \big(\dot{c}_\ell - X_{h_\star}(c_\ell)\big)\, d\tau, \\ G_I &= \frac{1}{4m} \sum_{\ell=1}^{m} D_\ell^\top D_\ell, \\ r_I^2 &= \frac{1}{4m} \sum_{\ell=1}^{m} \|D_\ell \widehat{\eta} - y_\ell\|_2^2. \end{aligned} \tag{S313}$$

If $\lambda_{\min}(G_I) \geq \underline{\lambda} > 0$ and $(4m)^{-1} \sum_\ell \|e_\ell\|_2^2 \leq \epsilon_I^2$, then

$$\|\widehat{\eta} - \eta_\star\|_2 \leq b_I := \frac{r_I + \epsilon_I}{\sqrt{\underline{\lambda}}}. \tag{S314}$$

The curves need not be exact solutions of the physical differential equation. Their explicitly verified dynamic defect supplies the required error budget.

Proof. Exact endpoint matching and the fundamental theorem of calculus give $y_\ell - D_\ell \eta_\star = e_\ell$. The triangle inequality in the stacked, $1/(4m)$-normalized Euclidean norm gives $\|D(\widehat{\eta} - \eta_\star)\|_2 / \sqrt{4m} \leq r_I + \epsilon_I$. The Gram lower bound proves (S314).

The defect is independently evaluated against the explicitly declared physical target. Thus the proposition does not infer arbitrary unknown model error from a small training loss. It certifies consistency with this target using additional, verified physical information.

Validated integrals and numerical error. All polynomial feature integrals are evaluated by exact polynomial integration with Arb ball coefficients. For $\vartheta = d^\top(c_\ell)_X$, the magnetic feature uses the odd Taylor terms of $\sin\vartheta - \vartheta$ through degree 19, together with

$$\begin{aligned} S_{19}(\vartheta) &= \sum_{\substack{3 \leq j \leq 19 \\ j \text{ odd}}} \frac{(-1)^{(j-1)/2} \vartheta^j}{j!}, \\ |\sin\vartheta - \vartheta - S_{19}(\vartheta)| &\leq \frac{R^{21} \cosh R}{21!}, \quad |\vartheta| \leq R. \end{aligned} \tag{S315}$$

Table S14. Outward bounds for the independent COMSOL integral-learning experiment. The MSE has a strictly positive certified lower bound.

Quantity	Certified bound
Training MSE	$[3.6030, 3.6031]\,10^{-23}$
r_I	$\le 6.0026\,10^{-12}$
ϵ_I	$\le 6.8835\,10^{-12}$
$\lambda_{\min}(G_I)$	$\ge 2.9302\,10^{-7}$
$\|\widehat{\eta} - \eta_\star\|_2$	$\le 2.3805\,10^{-8}$
Twist error	$\le 1.0377\,10^{-7}$
$\sigma_{\min}(A_\star)$	$\ge 4.9032\,10^{-4}$
$\|h_{\widehat{\eta}} - h_\star\|_{0.3}$	$\le 1.3310\,10^{-9}$

The computed maximum remainder is below 4.8752×10^{-43}. The remainder is included in every affected feature integral and dynamic defect; the certified function remains the complete cosine model. The exact identity $y - D\eta_\star = e$ is also checked componentwise by independent ball evaluations.

The interval Gram inverse is computed at 256-bit precision. Since the exact Gram matrix is real symmetric and positive semidefinite, a validated inverse makes it positive definite and gives $\underline{\lambda} = 1/\|G_I^{-1}\|_\infty$. Decoder rounding is retained when the real readout coefficients are exported as a cosine multiplier and a quartic polynomial. With $B = [q_\star/s_4\ U]$, this gives

$$\begin{aligned} \|A_{\widehat{\eta}} - A_\star\|_2 &\le \sqrt{19}\big(\|B\|_2 b_I + \delta_{\mathrm{dec}}\big), \\ \|h_{\widehat{\eta}} - h_\star\|_r &\le b_I B_F(r) + \delta_{h,\mathrm{dec}}(r). \end{aligned} \tag{S316}$$

Here $\|B\|_2 \le \|B^\top B\|_\infty^{1/2}$ is verified from the exported feature matrix; exact numerical orthogonality is unnecessary. The analytic bound controls every derivative on a smaller polydisc by Cauchy's estimate:

$$\begin{aligned} &\|\partial^\alpha (h_{\widehat{\eta}} - h_\star)\|_{r'} \\ &\quad \le \alpha!\,(r - r')^{-|\alpha|}\,\|h_{\widehat{\eta}} - h_\star\|_r. \end{aligned}$$

The target is unchanged. The singular-value lower bound in Table S14 is a sharper bound for the same two-by-two symmetric matrix: $\sigma_{\min}(A_\star) \ge |\det A_\star|/\|A_\star\|_\infty$. The previously reported, smaller lower bound remains valid. The strict twist margin, fixed quadratic part and complete analytic tail therefore give local positive-measure KAM families for this actual fitted Hamiltonian by the normal-form argument above.

An independent bound on trajectory error. The observation curves deviate from every exported sample by at most 5.2538×10^{-14} and satisfy

$$\sup_\tau \|\dot{c}_\ell - X_{h_\star}(c_\ell)\|_2 \le 1.0973 \times 10^{-8}.$$

The linear modal field L is skew symmetric in these balanced coordinates. On the real box $|z|_\infty < 0.3$, the nonlinear part has Euclidean Lipschitz constant

$$\|d\|_2^2 \frac{(0.3\|d\|_1)^2}{2} < 0.002084 < 0.01.$$

For the exact physical solution z_ℓ with $z_\ell(0) = c_\ell(0)$, variation of constants and Gronwall's inequality therefore give

$$\begin{aligned} &\sup_{0 \le \tau \le T} \|z_\ell(\tau) - c_\ell(\tau)\|_2 \\ &\quad \le T e^{0.01T} \|\dot{c}_\ell - X_{h_\star}(c_\ell)\|_{\infty,2} \\ &\quad < 4.3907 \times 10^{-10}. \end{aligned}$$

The verified curve bound $|c_\ell|_\infty < 0.2000$ closes the bootstrap inside the radius-0.3 box. This is an a posteriori bound on the independently observed trajectories, not an interpretation of COMSOL's requested time-stepping tolerances.

The same fitted function at millimetre amplitude. The frozen network has cosine multiplier $a = 0.9999996888354415$. Its complete exported coefficients are also used in a separate, 512-bit a posteriori torus verification with the frequency in Eqs. (S302)–(S303) and the theorem constants of Theorem S62. The verified analytic invariance residual is at most 2.7310×10^{-77}, the KAM smallness quotient is at most 0.031724, and the inverse averaged torsion is below 80526. The invariant torus lies within 2.0894×10^{-41} of its exact Fourier candidate. The tip-displacement witness is at least 1.8977904 mm, and the whole reconstructed centerline is bounded by 1.8977962 mm for every phase and all time. Thus the nonzero integral-loss certificate and the finite-amplitude KAM certificate concern exactly the same independently trained function. The first proves coefficient, analytic and twist control; the second checks the additional invariance-residual condition needed at the stated finite amplitude.

The full-FE trajectory fit has an independent a posteriori torus certificate, but its integral observation matrix yields a coefficient bound above 0.025 before accounting for continuous-quadrature and model errors. This bound does not satisfy the coefficient condition used here. The independently observed two-mode trajectories provide the defect bounds needed for the complete integral-learning theorem; model-reduction errors remain part of the separate full-FE evaluation.

S4.6 Phase-space measure of the invariant family

The positive-measure family established above motivates a domain-specific measure of quasiperiodic initial states. A quantitative fraction requires a specified volume measure and initial-state domain. Following the measure formulation of Biasco and Chierchia (2018); Chierchia and Koudjinan (2021), let $U \subset \mathbb{R}^4$ be a bounded domain of positive Liouville volume for the identified two-degree-of-freedom Hamiltonian h, and let $\mathcal{K}_h$ denote the union of its analytic Lagrangian invariant tori carrying Diophantine quasiperiodic motion. Define

$$f_{\mathrm{KAM}}(U) = \frac{\mu_L(U \cap \mathcal{K}_h)}{\mu_L(U)}, \qquad \mu_L = \mathrm{d}X_1\,\mathrm{d}X_2\,\mathrm{d}Y_1\,\mathrm{d}Y_2. \tag{S317}$$

An explicitly certified subfamily provides a lower bound for this fraction. The complement of that subfamily can contain other regular motion; it is not thereby a chaotic set. The sampling confidence in a learning bound such as that of Chen et al. (2022) is a probability over training data, whereas (S317) measures initial states of one fixed fitted Hamiltonian.

For a symplectic chart Ξ and an action box D, set $U = \Xi(\mathbb{T}^2_{2\pi} \times D)$, where $\mathbb{T}^2_{2\pi} = (\mathbb{R}/2\pi\mathbb{Z})^2$. Then $\mu_L(U) = (2\pi)^2\,\mathrm{area}(D)$. The period-one angles used by the earlier torus certificate are changed to radians together with the inverse action rescaling, preserving the symplectic form. The constant volume factor in the SI canonical chart also cancels from the relative fraction. The radius and location of D must accompany every reported percentage.

Where the assumptions of Theorem S65 hold, the invariant-family lift to the 64-mode closed loop carries this low-dimensional parameter measure forward. It does not turn it into a positive fraction of the full 128-dimensional Lebesgue volume. Accordingly, the measure estimate in Appendix S11 and the numerical sampling experiment in Section S5.6 refer to the conservative low-mode Hamiltonian realized by the magnetic feedback, with their respective domains specified separately.

The explicit constants of Chierchia and Koudjinan (2021) yield a computable local measure bound after construction of a symplectic tube and estimation of the full analytic remainder. Appendix S11 gives that construction for the integral-COMSOL Hamiltonian. Its sufficient estimates certify an extremely thin transverse domain; the central motion remains at millimetre amplitude, but the resulting width does not quantify an experimentally resolvable initial-state tolerance. To examine a broader range of amplitudes, Section S5.6 samples a prescribed action domain and evaluates finite-time quasiperiodicity with independent integration and averaging checks.

S5 Reproducible learning and independent COMSOL validation

All experiments in this section use the clamped-free beam, static-field eigenvectors, complete magnetic cosine, and SI parameters defined above. The two-mode learning problem and its higher-order validation models therefore share the same mechanics. The reference energy is

$$h_\star(X,Y) = \frac{1}{2}\sum_{j=1}^{2}\nu_j(X_j^2+Y_j^2) + g(d^TX), \qquad g(s) = 1-\cos s - \frac{1}{2}s^2. \tag{S318}$$

Here $\nu = (1, 6.152117504)$ and $d = (0.225670773, 0.321798806)^T$; one dimensionless energy unit is 6.0×10^{-7} J and $\tau = 11.417287283t$. The tip displacement in metres is $0.013506054X_1 + 0.005470926X_2$. These mappings are used for every reported displacement, slope, and energy error, including the independent COMSOL comparison.

S5.1 Finite-feature identification and matched controls

The CHNN uses $z_j = (X_j + \mathrm{i}Y_j)/\sqrt{2}$, conjugate polynomial features, and a complex exponential feature for the full cosine. Its real energy readout is

$$h_\vartheta = h_2 + a\,g(d^TX) + m_4(X,Y)^TUc, \tag{S319}$$

where m_4 lists the 35 degree-four real monomials in (X_1, X_2, Y_1, Y_2). If v_4 is the coefficient vector of $-(d^TX)^4/24$, the 34 columns of U span its orthogonal complement. The feature $g/\|v_4\|_2$ and these complementary quartics give 35 fitted real parameters, avoid duplicating the leading cosine term, and retain the complete analytic cosine tail. The target corresponds to $a = 1, c = 0$ and is therefore in the feature class. The stored matrix U, rather than idealized exact orthogonality, is used in the certificate computations.

We compare real (R) and complex (C) implementations, each with either the fixed mechanical quadratic part (G) or a fitted general quadratic part (B). Thus GR and GC have 35 fitted real parameters, whereas BR and BC have 45. The two representations span the same real energy class, receive identical data, and use the same regularizer expressed in common real feature coordinates.

Both implementations evaluate their respective real or complex features. Because the representations are related by an invertible transformation, this comparison tests their numerical implementation within a common hypothesis class.

For each seed in $\{17, 43, 91\}$, we generate 12 conservative trajectories of 4 s from the two-mode Python Galerkin model, sampled at 129 times. Initial modal radii are drawn uniformly from $[0.03, 0.20]$ and $[0.03, 0.16]$, with independent uniform phases. The largest training tip slope is 0.0891 rad. Each component of the dimensionless vector-field label receives independent uniform noise in $[-10^{-6}, 10^{-6}]$; states are noiseless. Linear readout fitting minimizes derivative mean-square error with a 10^{-14} quadratic regularizer. Eight independent trajectories, generated with a separate seed and integrated for 16 s at 1601 output times, are held out in full. The reference solver is DOP853 with relative and absolute tolerances 2×10^{-10} and 2×10^{-12}. Tightening both tolerances by a factor of 100 changes the tip of the first held-out trajectory by at most 1.92×10^{-8} mm.

Table S15 and Fig. S3 show the benefit of fixing the known quadratic mechanics in this experiment, while matched real and complex fits agree to numerical precision. GC identifies $a = 1.00326 \pm 0.00154$ with nonzero training residuals. The energy error evaluates the true energy along the predicted and reference states; it is not an energy-conservation surrogate for trajectory accuracy. The small fitting errors reflect exact representation, accurate synthetic derivatives, and deliberately controlled label noise. They do not establish corresponding physical measurement accuracy.

We also continue one independent initial condition for 200 s, or 363 first-mode linear periods, using the three primary GC fits. The seed-level tip RMSE ranges from 1.38×10^{-4} to 4.03×10^{-4} mm; the largest modal phase error is 4.19×10^{-4} rad. The relative drift of each learned model's own energy remains below 5.11×10^{-9} with conservative integration at relative and absolute tolerances 2×10^{-11} and 2×10^{-13}. Figure S4 shows its Poincaré section, phase error, and numerical energy drift. These finite-time diagnostics assess the fitted dynamics; they do not estimate the measure of KAM tori or replace the analytic certificate.

S5.2 A nonzero-loss certificate for the fitted magnetic energy

The feature derivatives make the empirical Gram matrix and the complete quartic Taylor map computable. We repeat the same three fits at noise amplitudes 10^{-8} and 10^{-9}, in addition to the 10^{-6} experiments. Outward-rounded interval calculations enclose the derivatives of the analytic sine feature at the stored states, the Gram inverse residual, the observed loss, clean-label rounding, and the conversion from readout weights to the exported energy coefficients. The degree-17 sine series is enclosed with its degree-19 remainder.

For the target magnetic quartic, $A_{jj} = -d_j^4/8$ and $A_{12} = -d_1^2 d_2^2/4$. This twist is indefinite and nonsingular. Exact rational arithmetic on the stored dyadic d gives the conservative lower bound $\sigma_{\min}(A_\star) > 3.03081 \times 10^{-4}$. If G is the derivative Gram matrix, B the quartic coefficient map, ℓ the noisy derivative MSE, and η_{eff} the enclosed noise and arithmetic budget, the verified sufficient inequality is

$$\sqrt{19}\left(\|B\|_2 \frac{\sqrt{\ell} + \eta_{\text{eff}}}{\sqrt{\lambda_{\min}(G)}} + \delta_{\text{dec}}\right) < \sigma_{\min}(A_\star), \tag{S320}$$

where δ_{dec} encloses readout decoding roundoff. At $\eta = 10^{-8}$, Table S16 verifies this inequality with strictly positive losses for all seeds. Thus the data-dependent error bound itself transfers nondegeneracy to the actual learned Hamiltonians. The local KAM conclusions additionally use the nonresonance and analytic normal-form hypotheses proved for this same model. This table alone does not certify a finite experimental oscillation radius. For the primary 10^{-6} fits, this sufficient error bound is too loose; their directly computed twist singular values are nevertheless between 5.78×10^{-4} and 5.93×10^{-4}.

S5.3 Learning dissipation and opening the magnetic input port

The imaginary readout is the nonnegative Rayleigh potential $\mathcal{D}_\gamma(v) = \gamma \sum_j v_j^2/(2\nu_j)$, with $v = h_{\vartheta,Y}$. For each seed, eight additional damped, unforced 5 s trajectories provide noisy derivative labels. Nonnegative least squares fits the single γ using the already fitted conservative gradient. In particular, the decoder retains the learned momentum-dependent quartics:

$$\begin{aligned} X' &= h_{\vartheta,Y}, \\ Y' &= -h_{\vartheta,X} - \gamma\,\mathrm{diag}(\nu^{-1}) h_{\vartheta,Y} \\ &\quad + u\,d\cos(d^T X). \end{aligned} \tag{S321}$$

The fitted dimensional mass-drag coefficients are 0.555585710, 0.555587102, and 0.555584024 s^{-1}, compared with 0.555585237 s^{-1} for the generator. On six independent driven trajectories with amplitudes 4–8 mT and frequency ratios 0.65–1.20 relative to the biased first

Table S15. Conservative held-out errors, mean ± sample standard deviation over three training seeds. Each seed is evaluated on all eight independent 16 s trajectories. These are errors within the declared simulated model.

Model	Tip RMSE (mm)	Slope RMSE (mrad)	Energy RMSE (nJ)
BR	$(2.28\pm0.75)10^{-5}$	$(5.80\pm1.20)10^{-4}$	$(3.21\pm0.51)10^{-6}$
BC	$(2.28\pm0.75)10^{-5}$	$(5.80\pm1.20)10^{-4}$	$(3.21\pm0.51)10^{-6}$
GR	$(1.20\pm0.71)10^{-5}$	$(3.90\pm1.63)10^{-4}$	$(3.14\pm0.71)10^{-6}$
GC	$(1.20\pm0.71)10^{-5}$	$(3.90\pm1.63)10^{-4}$	$(3.14\pm0.71)10^{-6}$

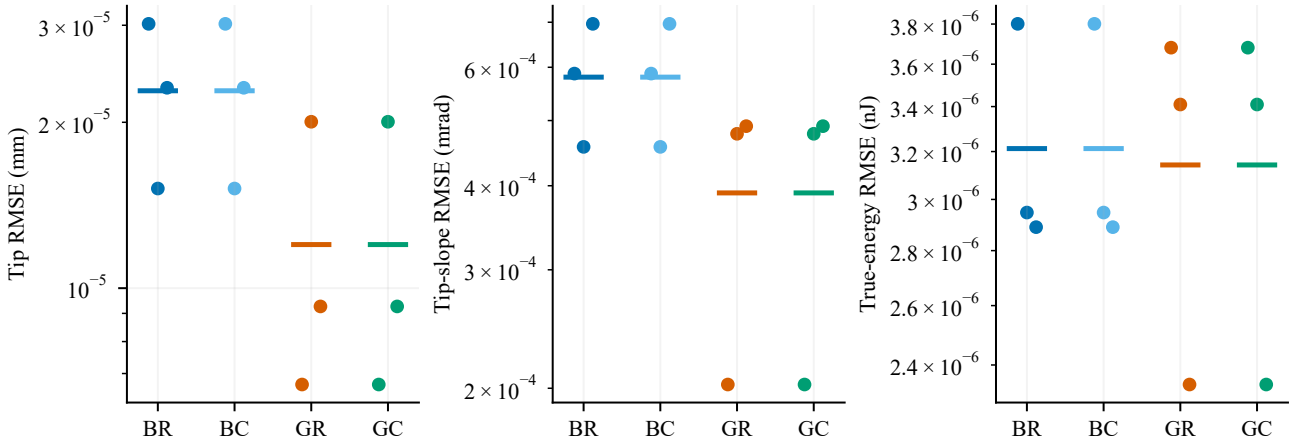


Figure S3. Identical full-cosine beam data and real parameter budgets for the four learning controls. Points are individual seeds and horizontal marks are means. The quadratic mechanics prior changes the result; the matched real/complex representation changes it only at numerical precision.

frequency, the three seed-level tip RMSEs are 8.83×10^{-5}, 1.37×10^{-4}, and 6.40×10^{-5} mm. Closing the learned drag port increases the tip RMSE to approximately 3.01 mm. The integrated work balance is

$$h_\vartheta(\tau)-h_\vartheta(0)=\int_0^\tau u\cos(d^TX)d^Th_{\vartheta,Y}\,ds - \int_0^\tau \gamma\sum_j h_{\vartheta,Y_j}^2/\nu_j\,ds. \tag{S322}$$

Its largest numerical residual across these cases is 4.41×10^{-17} J. These driven tests assess the port model, rather than autonomous KAM persistence. Some stress cases exceed the channel clearance and are free-motion continuations; safe operating trajectories are evaluated separately with the physical energy bound and input governor.

S5.4 Independent COMSOL assembly and nonlinear time integration

COMSOL Multiphysics 6.3 independently assembles the same 32-element clamped-free beam and solves its nonlinear magnetic-tip response. The relative Frobenius differences between the exported and Python consistent-mass matrices are 7.75×10^{-15} for mass and 3.05×10^{-14} for bending stiffness. The nonlinear held-out COMSOL trajectory starts from rest, uses the same mass drag, and applies

$$B_y(t)=0.006\,[1-\exp(-(t/0.25)^2)]\sin(\Omega t)\quad \mathrm{T} \tag{S323}$$

with $\Omega=9.4449490256$ rad/s for 8 s, relative tolerance 10^{-7}, and 2001 output samples. The exported nodal displacement, rotation, velocity, and angular velocity are evaluated directly, with no fitting to this driven trajectory. Its peak tip displacement is 3.09032 mm. A separate COMSOL solve with relative tolerance 10^{-8} changes the tip by 4.59×10^{-5} mm RMSE and reduces its discrepancy against the full Python FE solution to 1.03×10^{-5} mm RMSE. The original held-out dataset is used for the primary comparison below.

The nested modal comparison in Table S17 separates reduction error from learning error. Against the exact two-mode response, GC contributes only 1.81×10^{-6}–3.48×10^{-6} mm tip RMSE for this input. Against COMSOL, both have approximately 0.0156 mm RMSE. The two-mode maximum slope discrepancy is 3.69 mrad; its true stored-energy RMSE is 0.132 nJ. These discrepancies decrease with modal enrichment. Figure S6 shows the response and both error scales. The primary training data were generated by the two-mode Python Galerkin model; the COMSOL time history is an independent held-out validation dataset. Neither dataset is a physical hardware measurement.

S5.5 COMSOL-derived training data and an independent integral check

We generate a separate conservative COMSOL dataset comprising four 4 s training trajectories and two independent

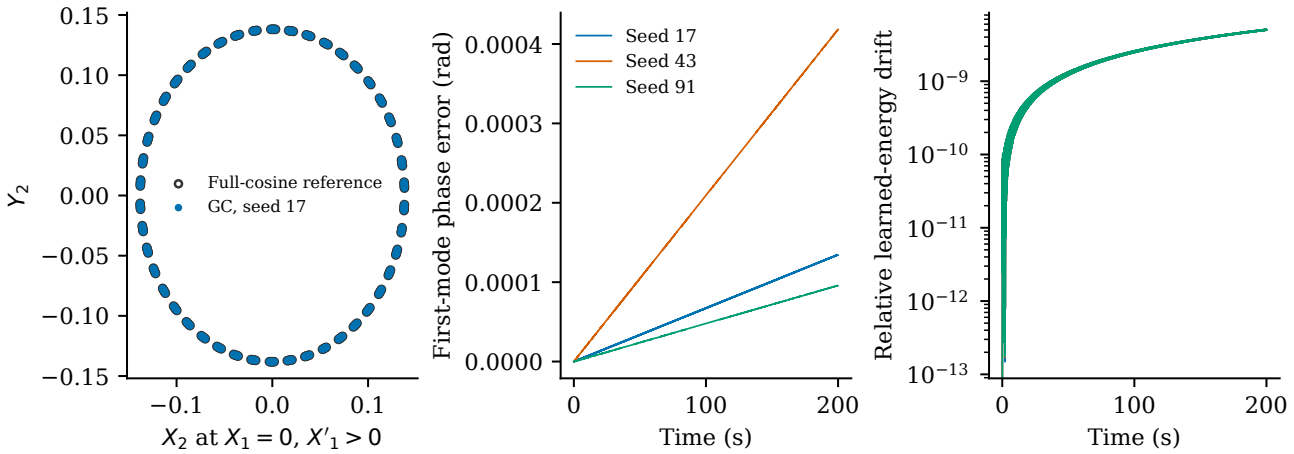


Figure S4. Conservative 200 s continuation of an independent initial state. Left: 364 upward crossings of $X_1 = 0$ for the reference and GC17. Middle: unwrapped first-mode phase errors of the three learned energies. Right: each learned energy's relative integration drift. All ports are closed for these conservative diagnostics.

Table S16. Enclosed loss-to-twist checks at noise amplitude 10^{-8}. All twist-error upper bounds are below 3.03081×10^{-4}.

Seed	Noisy MSE	Gram eigenvalue lower bound	Twist-error upper bound
17	3.387×10^{-17}	6.0167×10^{-8}	2.8112×10^{-4}
43	3.275×10^{-17}	2.2734×10^{-7}	1.4374×10^{-4}
91	3.283×10^{-17}	2.0968×10^{-7}	1.4973×10^{-4}

8 s test trajectories, sampled every 0.004 s. Initial displacements and velocities excite both retained modes. The remaining 62 modes receive the static response to the nonlinear magnetic residual and its compatible first time derivative; all 64 modes remain active during integration. This initial tail condensation is not assumed to be an exact nonlinear invariant manifold. Native solver-vector checks confirm that the exported initial displacement and velocity are correct; small changes introduced by consistent initialization are handled by starting each predictor from the actual exported state.

The derivative-label construction requires care. At the stored output times, the exported second time derivatives differentiate a time interpolant and show a substantial lag relative to the velocity evolution. For example, the first trajectory has a dimensionless modal equation-residual RMS of 4.58×10^{-2} when those derivatives are used directly. We therefore reconstruct instantaneous acceleration from the exported COMSOL mass and stiffness matrices:

$$\ddot{q} = M_{\mathrm{C}}^{-1} \left[-K_{\mathrm{C}} q - \mu B_0 \sin(\theta_{\mathrm{tip}}) e_\theta \right]. \tag{S324}$$

The training states and velocities therefore come from the COMSOL time solver, while the acceleration labels use known physical force balance. This is surrogate identification for the declared mechanical model with force-balance labels. The unresolved-mode force is included: its largest component is 3.36×10^{-9} in dimensionless units for the checked training trajectory, and the remaining reconstructed equation-residual RMS is 1.81×10^{-10}.

These COMSOL-based models are fitted separately from the primary Python-trained GC models. Three seeds vary the additional componentwise label noise, bounded by 10^{-6}, while using the same four COMSOL trajectories. The real/complex and prior/no-prior controls use the same parameter budgets as before.

An independent check uses only the COMSOL displacement and velocity samples. A nine-point closed Newton–Cotes rule integrates the vector-field features on nonoverlapping 0.032 s windows, and the observed state increments provide the regression targets. No acceleration or force-balance label enters this integral fit. With the initial solver settings (relative tolerance 10^{-6} and scaled absolute tolerance 10^{-5}), the integral residual is too large to resolve the weak magnetic correction reliably: on the first trajectory, its RMS is 2.16 times the nonlinear magnetic signal. An unconstrained 35-feature integral fit then yields $a = 1.437$, illustrating that a good fit to a numerically dispersed trajectory need not recover the physical magnetic coefficient. A stricter COMSOL solve, with relative tolerance 10^{-8} and scaled absolute tolerance 10^{-7}, reduces that ratio to 0.112 on the same trajectory. This independent data-quality check resolves time-integration error that the force-balance loss alone does not detect.

All six trajectories were then integrated with these stricter tolerances. The refined runs use generalized-alpha

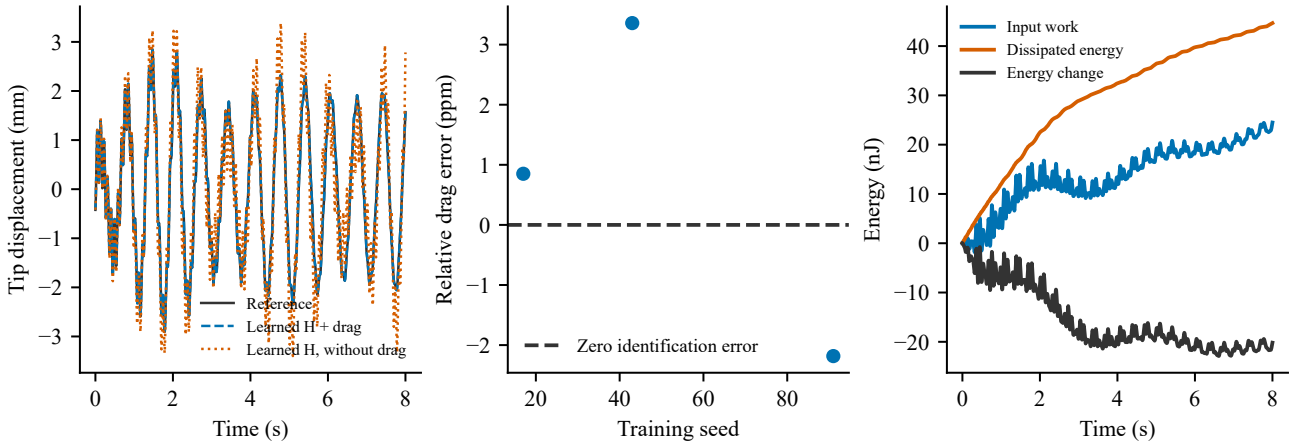


Figure S5. The same learned conservative energy with an identified nonnegative drag readout and magnetic input port. Left: an independent driven trajectory and the drag ablation. Middle: signed relative errors in the drag coefficient. Right: supplied work, dissipated energy, and stored-energy change for that trajectory.

Table S17. Independent nonlinear COMSOL trajectory comparison. Modal models use the common static-field chart. GC includes learned drag; its range covers the three primary training seeds.

Predictor	Tip RMSE (mm)	Maximum tip error (mm)
2-mode exact cosine	0.0156168	0.0226548
4-mode exact cosine	0.00407422	0.00592596
8-mode exact cosine	0.00102484	0.00152782
64-mode Python FE	0.00005614	0.00011835
2-mode learned GC	0.0156171–0.0156177	0.0226545–0.0226549

integration with high-frequency spectral radius zero and backward-Euler consistent initialization. The four refined training trajectories provide 500 integral windows, or 2000 scalar equations, for the 35 real readout parameters. The resulting GC model has $a = 1.0223248822$ and a nonzero averaged-derivative integral RMS residual of 4.0524×10^{-7}. The corresponding residuals of the quadratic prior alone and the physical full-cosine target are 3.9099×10^{-6} and 6.6454×10^{-7}; the nonlinear magnetic signal RMS is 3.6601×10^{-6}. The fitted quartic twist has minimum singular value 5.1166×10^{-4}. The conservative weights were frozen before evaluating the two independent refined test trajectories.

Table S18 and Figure S7 report the held-out errors. The integral fit partly absorbs the remaining time-solver dispersion: its agreement with COMSOL is better than that of the exact closed two-mode target, despite a 2.23% error in the cosine multiplier. These errors quantify identification against simulated observations. The requested solver tolerances do not supply a rigorous defect budget for applying the loss-based certificate to these integral residuals. The separate two-mode short-trajectory experiment in Section S4.5 supplies a verified integral defect budget and the corresponding loss-based certificate for its own frozen fitted function.

For comparison, the separately fitted models using the known-force labels in (S324) recover $a = 1.0000690$ without added label noise. With the three added-noise seeds, the prior-informed GC models give 0.25347–$0.25940\,\mu$m pooled tip RMSE against the same refined COMSOL tests; their additional error relative to the closed two-mode trajectories is 0.00364–$0.00771\,\mu$m. Matched real and complex features again give coincident results to numerical precision.

Finally, the position–velocity fit is used in the independent a posteriori proof of Section S4.3. Its complete cosine and all fitted quartic coefficients yield an analytic residual below 2.7330×10^{-77} and a theorem smallness quantity below 0.031746. The resulting exact invariant torus has a tip-displacement witness above 1.8977374 mm and a whole-centreline upper bound below 1.8977433 mm. This certificate concerns the actual fitted Hamiltonian and does not require its coefficients to equal the physical target. The same frozen conservative readout is paired with a nonnegative drag coefficient refitted on the existing Python damped-beam trajectories, $\gamma = 0.0486617702$ for seed 17. Its independent driven COMSOL tip RMSE is 0.015622 mm. Section S7.4 then uses this same energy, full gradient and Hessian, and drag readout to generate the magnetic tracking input.

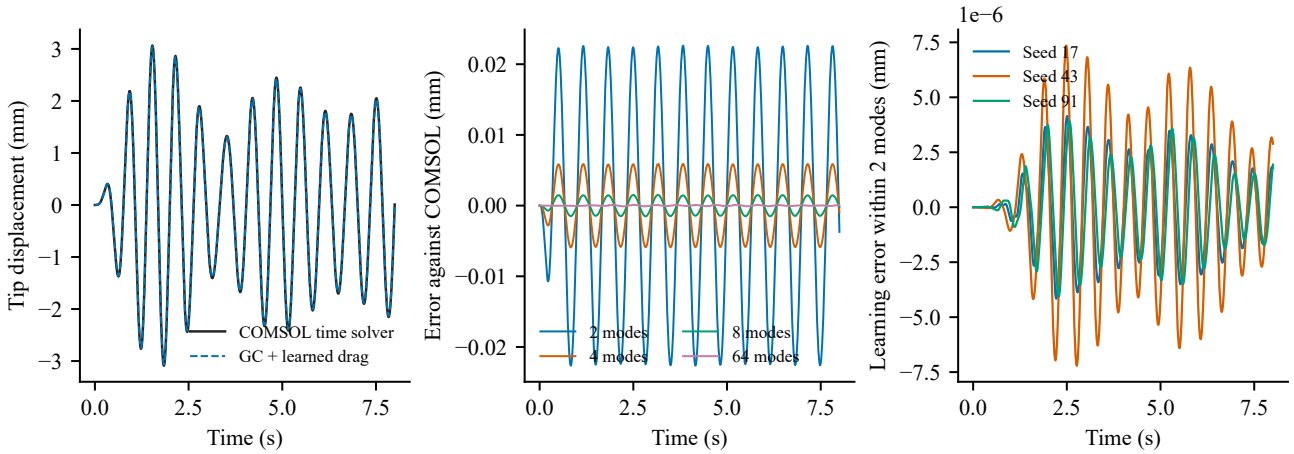


Figure S6. Independent nonlinear COMSOL time integration. Left: COMSOL and the learned two-mode GC response with learned drag. Middle: errors of the nested exact-cosine modal models against COMSOL. Right: the additional GC identification error within the two-mode model. Reduction error is the larger contribution in this validation.

Table S18. Independent refined COMSOL tests of the GC model identified only from displacement and velocity increments. Predictions start from each actual exported initial state. Energy errors use the physical stored-energy function evaluated along the learned trajectory.

Test	Tip RMSE (μm)	Max. tip error (μm)	Slope RMSE (mrad)	Energy RMSE (nJ)	Exact 2-mode tip RMSE (μm)
1	0.14836	0.39240	0.008677	0.0002067	0.24172
2	0.05140	0.17435	0.002411	0.0002253	0.27459

S5.6 Finite-time fraction in an operating action annulus

We examine the phase-space fraction defined in Section S4.6 on a domain spanning millimetre-scale oscillations. The Hamiltonian is identified from the two-mode COMSOL position–velocity data in Section S4.5. In its canonical coordinates, set

$$\begin{aligned} I_j &= \tfrac{1}{2}(X_j^2 + Y_j^2), \quad j = 1, 2, \\ \phi_j &= \operatorname{atan2}(-Y_j, X_j). \end{aligned} \tag{S325}$$

and prescribe, before classifying any orbit,

$$\mathcal{D}_{\mathrm{op}} = [0.0032, 0.0072]^2 \times \mathbb{T}^2_{2\pi}. \tag{S326}$$

Both harmonic modal radii therefore lie in $[0.08, 0.12]$, surrounding the radius 0.1 of the previously certified $1.898\,\mathrm{mm}$ torus. The corresponding harmonic tip envelope $|\ell_1|\sqrt{2I_1} + |\ell_2|\sqrt{2I_2}$ ranges from 1.518 to $2.277\,\mathrm{mm}$. The canonical polar Jacobian has absolute determinant one, so independent uniform actions and phases sample normalized four-dimensional Liouville volume:

$$\mu_L(\mathcal{D}_{\mathrm{op}}) = (2\pi)^2(0.004)^2 = 6.3165468 \times 10^{-4}. \tag{S327}$$

We draw $N = 1024$ independent points with PCG64 seed 20261002; the domain, initial points, observables, and decision thresholds are specified before classification. These points are independent of the training observations and the certified-torus phase grid.

Following weighted Birkhoff averaging (Das et al. 2017; Duignan and Meiss 2023), we compare consecutive windows using the smooth bump $b(s) = \exp[-1/(s(1-s))]$ for $0 < s < 1$, with $b = 0$ otherwise. The normalized weighted mean is

$$W_T(F; z) = \frac{\int_0^T b(t/T) F(\Phi_h^t z)\, dt}{T \int_0^1 b(s)\, ds}, \tag{S328}$$

where Φ_h^t denotes the flow of the fitted Hamiltonian h. The computations use normalized sampled sums of this weight; integration and quadrature refinements are reported below. Eight observables are used: $X_j/0.1$, $Y_j/0.1$, $I_j/0.005$, $X_1X_2/0.01$, and $Y_1Y_2/0.01$. We also average the polar angular velocities

$$\dot{\phi}_j = \frac{X_j \partial_{X_j} h + Y_j \partial_{Y_j} h}{X_j^2 + Y_j^2}. \tag{S329}$$

For each $T \in \{512, 1024, 2048, 4096\}$, let $e_F(T)$ be the largest absolute difference of the eight weighted means on $[0, T]$ and $[T, 2T]$, and let $e_\omega(T)$ be the largest relative difference of the two weighted frequencies. A candidate passes the principal test when $e_F, e_\omega \le 10^{-8}$ at both $T = 2048$ and 4096. It must also keep both sampled actions above 3.2×10^{-4}, have sampled relative energy drift below 10^{-8},

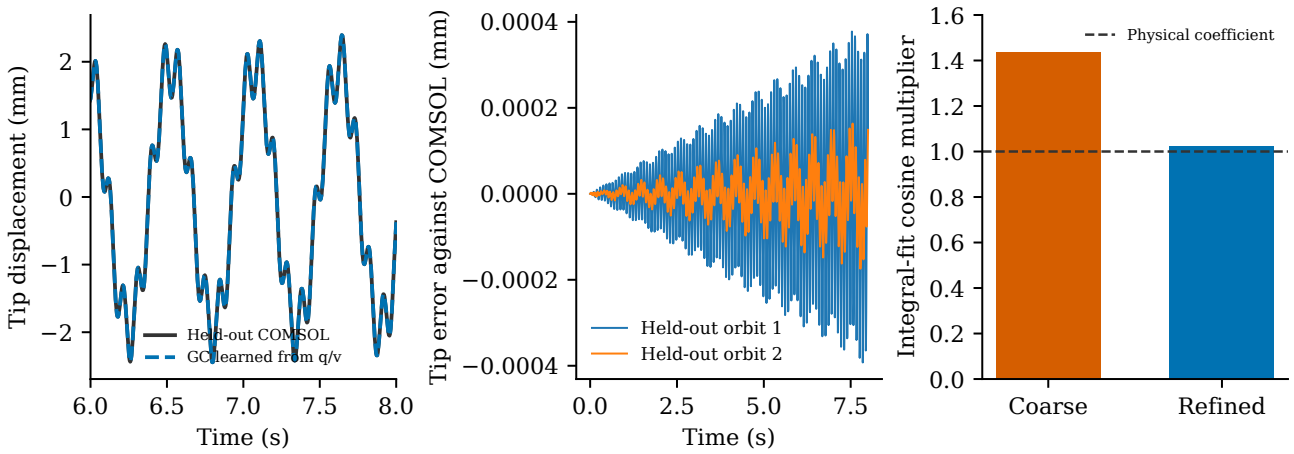


Figure S7. Identification using full-FE COMSOL displacement and velocity outputs only. Left: independent test response. Middle: prediction errors for both held-out trajectories. Right: the fitted cosine multiplier under the two declared COMSOL tolerance settings; the physical value is one. The refined coefficients are held fixed during the subsequent evaluations.

and satisfy the finite-order screen

$$\min_{\substack{0<|k|_1\le 64\\ k\in\mathbb{Z}^2 \text{ primitive}}} \frac{|k\cdot\overline{\omega}|}{|k|_1\|\overline{\omega}\|_\infty} > 10^{-7}, \tag{S330}$$

where $\overline{\omega}$ uses the two longest consecutive windows. The total simulated time is $8192/\omega_0 = 717.508\,\mathrm{s}$. Full cosine and momentum-dependent quartic terms are retained.

Step refinement resolves integration error in the diagnostics. Fixed-step eighth-order DOP853 at $\Delta\tau = 1/16$ produces relative energy drift up to 3.94×10^{-7} and fails the numerical validity test. At $\Delta\tau = 1/32$, all points pass the principal eight-digit criterion and the energy test, but the consecutive-window averages fail the ten-digit criterion. Refining all 1024 trajectories to $\Delta\tau = 1/64$, with the same initial points and thresholds, resolves the ten-digit criterion. The largest relative energy drift then falls to 6.11×10^{-12}; $e_F(4096) \le 4.49\times10^{-12}$ and $e_\omega(4096) \le 3.63\times10^{-14}$. Halving the step again for the preselected first 64 points changes no six-, eight-, or ten-digit labels, with maximum relative frequency change 2.40×10^{-14}. Halving the averaging output interval on identical saved trajectories changes the weighted means by at most 3.82×10^{-14}. An independent adaptive solver, phase points from the certified torus's Fourier approximation, an exactly resonant oscillator, and regular/chaotic two-wave examples provide additional checks. In particular, the resonant oscillator passes the averaging test but is rejected by (S330).

The refined ensemble gives $1024/1024$ candidates at each of the six-, eight-, and ten-digit thresholds. For the specified finite-time classifier C,

$$\widehat{p}_C = \frac{1}{1024}\sum_{i=1}^{1024} \mathbf{1}_C(z_i) = 1, \qquad \mathrm{CI}_{95\%}(p_C) = [0.996404, 1]. \tag{S331}$$

using the two-sided exact Clopper–Pearson interval. The smallest normalized finite-order detuning is 2.113×10^{-4}; the largest sampled tip displacement is $2.269\,\mathrm{mm}$. Figure S8 shows the fixed-domain sample and the integration and threshold checks. This interval quantifies only random initial-condition sampling for C. It does not bound classification error, unresolved high-order resonance, sticky chaos, or physical model error. Thus (S331) is an empirical finite-time candidate fraction in the operating domain. Its 99.6404% lower confidence endpoint concerns the classifier probability, whereas Appendix S11 establishes a rigorous KAM-volume bound on a separate, narrower analytic domain.

S5.7 State-only identification beyond the nominal hypothesis class

The nominal full-cosine experiment places its target inside the chosen feature class. We additionally test unknown material and magnetic parameters, state noise, and two omitted physical effects. All training observations in this experiment are generated by the full 32-element, 64-DOF Python FE plant; the predictors retain the same four nominal static-field modes. Let $e = E/E_{\mathrm{ref}}$ and $m = \mu/\mu_{\mathrm{ref}}$. The local applied field is

$$B_x = B_0 - G(x-L), \qquad B_y = B_y(t) + Gy. \tag{S332}$$

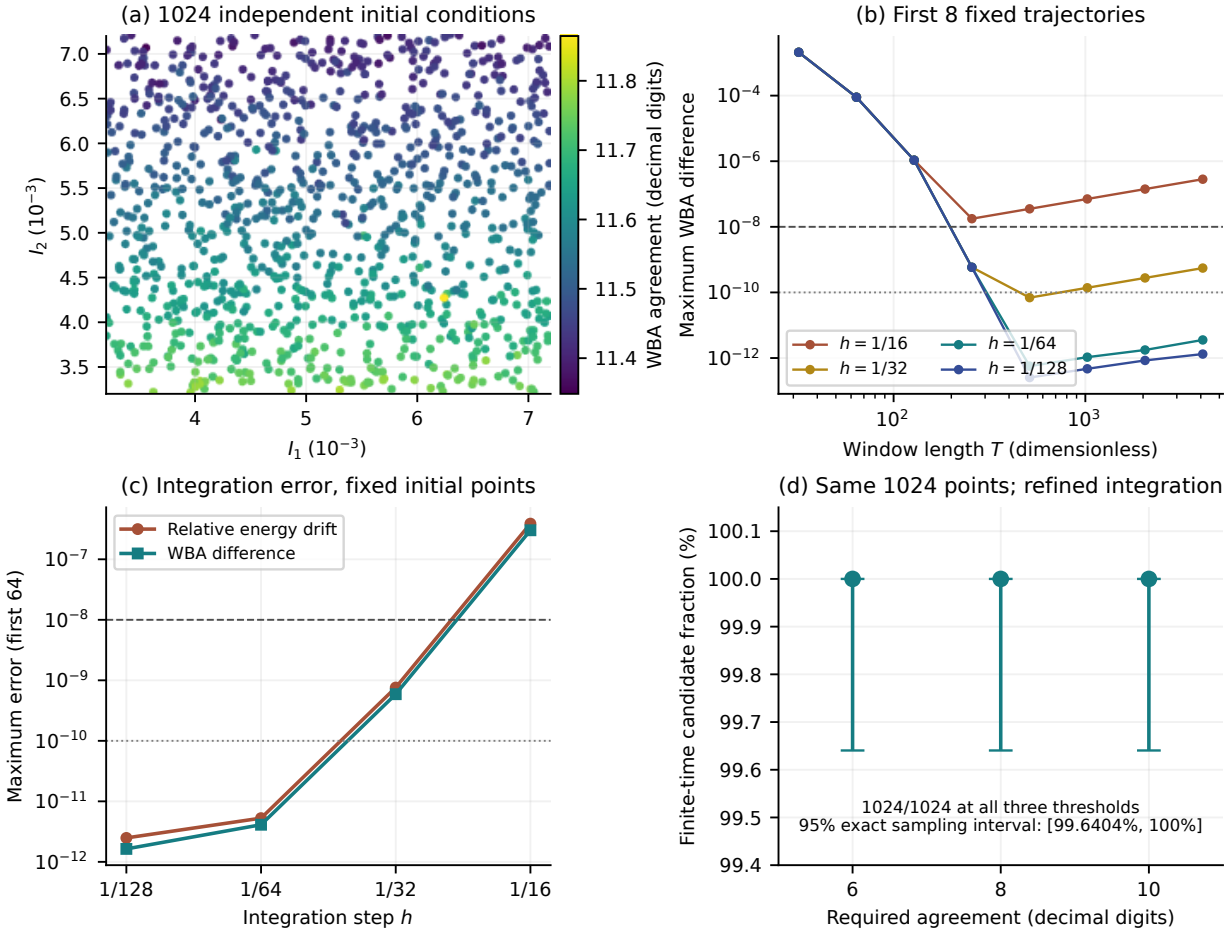


Figure S8. Finite-time quasiperiodicity diagnostics for the fitted Hamiltonian. (a) Uniform action sampling; angles are independently uniform. (b) Consecutive-window agreement on the first eight fixed trajectories; the long-window floor reflects integration error and roundoff. (c) Step refinement on the first 64 fixed initial points identifies the coarse integration error. (d) Candidate fractions and exact sampling intervals at three thresholds, using the same 1024 points with refined integration. Candidate status describes finite-time agreement of the specified observables.

The field is curl-free and divergence-free. Under the declared linear-beam kinematics, $x_{\mathrm{tip}} = L$, so its additional potential is $-\mu G w(L)\sin\vartheta$. It supplies both the transverse tip force $\mu G\sin\vartheta$ and the corresponding moment $\mu G w(L)\cos\vartheta$. The axial force $-\mu G\cos\vartheta$ is balanced by the axial constraint reaction and does no virtual work in this specialization. We also add the still-fluid Morison drag density $-c_2|w_t|w_t$, with dissipation potential $c_2\int_0^L |w_t|^3 ds/3$.

In the fixed dimensionless chart, the conservative energy is

$$h = \tfrac{1}{2}Y^{\mathsf{T}}\,\mathrm{diag}(\nu)Y + \tfrac{e}{2}X^{\mathsf{T}}A_{\mathrm{el}}X + m(1-\cos\vartheta) - g\frac{G_{\mathrm{ref}}}{B_0}(\ell^{\mathsf{T}}X)\sin\vartheta, \qquad \vartheta = d^{\mathsf{T}}X, \tag{S333}$$

where $A_{\mathrm{el}} = \mathrm{diag}(\nu) - dd^{\mathsf{T}}$, $G_{\mathrm{ref}} = 0.5$ T/m, and $g = mG/G_{\mathrm{ref}}$. The five fitted readouts are $(e, m, g, \alpha/\alpha_{\mathrm{ref}}, c_2/c_{2,\mathrm{ref}})$, with $c_{2,\mathrm{ref}} = 0.02$ kg/m^2. The elastic and magnetic strengths and both dissipative coefficients are constrained to be nonnegative; the gradient may have either sign. The kinetic metric is known. Complex features evaluate the polynomial and exponential terms through $z = (X + iY)/\sqrt{2}$ and its conjugate.

Each of three parameter settings provides six independent 3 s training trajectories and three 4 s tests with different initial conditions and multisine inputs. The two changed settings use respectively $(e, m, \alpha/\alpha_{\mathrm{ref}}, G, c_2) = (0.85, 1.20, 1.40, 0.5, 0.02)$ and $(1.20, 0.85, 0.70, -0.4, 0.012)$ in SI units where dimensional. The first four modes are excited, and the remaining initial coordinates are statically condensed with zero initial velocity. This construction does not assume an invariant modal manifold. Samples are separated by 1 ms. On overlapping 32 ms windows, multiplication by $\psi = \sin^4(\pi s)$ and integration by parts gives

$$-\int \psi' Y\, d\tau = \int \psi F_\theta(X, Y, u)\, d\tau. \tag{S334}$$

Table S19. Mean tip RMSE over three independent full-FE tests in μm. The noisy entries average three independent training-noise seeds; the clean fit is counted once. The last control has a freely learned quadratic potential.

Case	Noise	Omitted	Expanded	Free H_2
Nominal	0	2.415	2.415	2.417
Nominal	0.5%	2.833	2.807	2.996
Soft/$+G$	0	182.988	9.829	7.258
Soft/$+G$	0.5%	183.028	9.757	7.216
Stiff/$-G$	0	166.496	2.121	1.978
Stiff/$-G$	0.5%	166.916	2.841	2.808

The endpoints of ψ vanish. Fitting therefore uses only positions, velocities and the applied field; neither accelerations nor reconstructed force-balance labels enter the regression.

We compare the expanded CHNN with a three-readout model that re-identifies elasticity, magnetic strength and linear drag but omits the gradient and quadratic drag. A further control freely learns the ten entries of the symmetric quadratic potential while retaining the analytic magnetic features and dissipative heads, giving fourteen readouts. Thus every comparison uses four modes. Matched real and complex realizations agree in effective coefficients to 1.31×10^{-11} or better. The experiment does not attribute an accuracy advantage to complex arithmetic.

Independent Gaussian noise is added to the observed modal positions and velocities at 0.1% and 0.5% of each coordinate's training standard deviation, with dimensionless floors 10^{-4} for X and 10^{-5} for Y. The modal projections are assumed observable; this test does not validate a particular sensor layout. At 0.5%, the reconstructed training-tip noise RMS is 3.80, 5.72 and 3.33 μm for the three settings. No noise is added to future reference labels. Figure S9 and Table S19 show that adding the missing physics resolves the main model-class error. The free-quadratic model can absorb reduction error and is slightly more accurate in the two changed settings; a universal advantage for the physical prior is not inferred. For the soft setting, the expanded clean fit identifies $e = 0.849522$, $m = 1.199860$, $G = 0.500272$ T/m, $\alpha = 0.753909$ s^{-1} and $c_2 = 0.020832$ kg/m^2. The last two values differ from their full-plant values by approximately 3.1% and 4.2%. Thus accurate state prediction is not presented as exact parameter recovery. A four-mode predictor supplied with the true soft-setting parameters itself has 10.894 μm mean tip RMSE, showing the retained influence of reduction error.

An independent COMSOL solve tests the changed soft-rod setting from rest for 4 s, including the same gradient force, magnetic moment and distributed quadratic drag. Against the separate full-FE Python solve, the tip RMSE is 0.000189 μm and the maximum difference is 0.000623 μm; the input-field difference is below 2.73×10^{-17} T. The learning coefficients were frozen before this COMSOL test and are not refitted. The omitted-physics, expanded and free-quadratic predictors obtain respectively 190.055, 9.644 and 6.946 μm tip RMSE. This comparison checks an independently implemented altered mechanical model, rather than only another trajectory of the original exact-in-class target.

S5.8 Causal adaptation to slowly changing unknown parameters

To test time variation separately from recalibration between settings, we generate a 29 s full-FE trajectory with a smooth drift. Define $s = \mathrm{clip}_{[0,1]}((t - 4\,\mathrm{s})/(24\,\mathrm{s}))$ and $a = (1 - \cos \pi s)/2$. The hidden plant has $e = 0.85 + 0.20a$, $m = 1.20 - 0.20a$ and $\alpha/\alpha_{\mathrm{ref}} = 1.40 - 0.40a$, while $G = 0.5$ T/m and $c_2 = 0.02$ kg/m^2 remain fixed. Multisine excitation continues throughout. At cutoffs $8, 12, 16, 20, 24, 28$ s, each estimator receives only the preceding 4 s of positions, velocities and known input. It then predicts the next 1 s from the currently observed state. The coefficients are frozen before future state labels are accessed.

A constant sliding-window fit and a locally affine coefficient fit use the same weak identity (S334); the latter adds features multiplied by $(t - t_{\mathrm{cut}})/(4\,\mathrm{s})$. Linear constraints preserve the nonnegative readouts over both the fitting interval and the forecast interval. The known future input is supplied to all predictors, but future states and true parameter values are withheld. Across the six forecasts, the mean next-second tip RMSE is 387.483 μm for the initial fit frozen at 4 s, 59.217 μm for causal constant refitting and 9.867 μm for causal affine adaptation. The affine errors range from 2.701 to 17.295 μm. Constant refitting is better at the final cutoff, where the physical drift stops; this limitation of extrapolating a local trend is retained. These are six forecasts on one declared drift trajectory, not independent patient experiments or a general online-learning theorem.

For this nonautonomous plant, the physical energy ledger includes $E_0[\dot{e} X^{\mathsf{T}} A_{\mathrm{el}} X/2 + \dot{m}(1 - \cos\vartheta) - \dot{g}(G_{\mathrm{ref}}/B_0)(\ell^{\mathsf{T}} X)\sin\vartheta]$ in addition to magnetic input power and fluid losses. Its maximum numerical residual is 6.06×10^{-14} J. The stationary conservative KAM certificates are not transferred to these driven, dissipative, time-dependent trajectories.

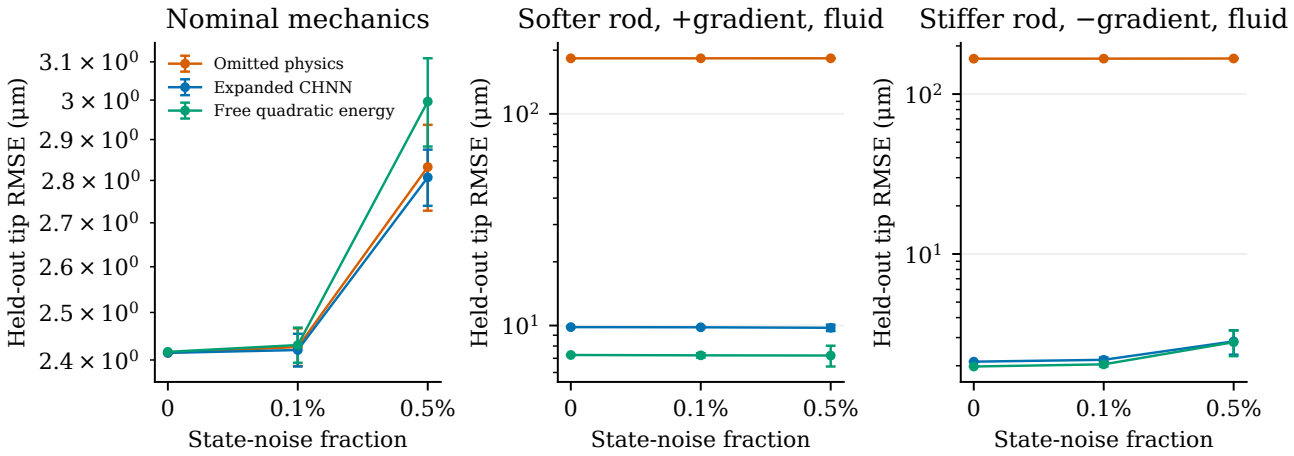


Figure S9. State noise and physics outside the nominal hypothesis class. Each point averages three independent held-out trajectories. Bars show the sample standard deviation across three training-noise seeds; the clean fit is counted once. The higher-error model still re-identifies its retained parameters.

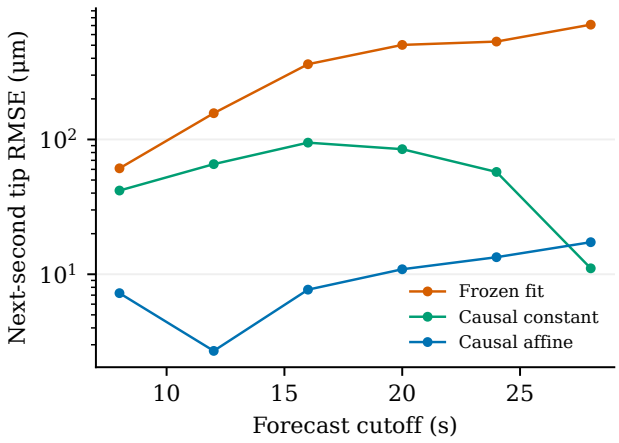


Figure S10. One-second forecasts made using only past data under slow material and magnetic drift. The frozen, constant and affine predictors share the same current observed state and prescribed future input.

S5.9 Forced-response stability and clearance boundaries

To distinguish resonance amplification from a dynamical loss of stability, we additionally scan the same nominal beam under a prescribed transverse field. This experiment concerns attracting responses of a driven, damped system; it does not measure destruction of the conservative KAM torus. The physical parameters, frozen static-field modes and mass-drag coefficient $\alpha = 0.5555852368\,\mathrm{s}^{-1}$ are unchanged. No field-gradient input, learned-model substitution, governor or contact force is used. In physical time, the retained modal equations are

$$\begin{aligned}\dot{X}_j &= \omega_j Y_j,\\ \dot{Y}_j &= -\omega_j X_j - \alpha Y_j \\ &\quad + \omega_0 d_j[\vartheta - \sin\vartheta + u(t)\cos\vartheta],\end{aligned} \tag{S335}$$

where $\vartheta = d^{\mathsf{T}}X$, $u = B_y/B_0$, $B_0 = 30$ mT and $\omega_0 = 11.417287283\,\mathrm{rad\,s}^{-1}$.

Fixed protocol and stability diagnostics. Independent runs use $B_y = A[1-\exp(-(t/(0.25\,\mathrm{s}))^2)]\sin(r\omega_0 t)$. The amplitude set is $A \in \{1,2,4,6,10,15,20\}$ mT. The frequency set contains the 0.02 grid over $[0.70, 1.30]$ and the 0.005 grid over $[0.94, 1.06]$, giving 49 distinct ratios. For each pair, the initial low state is either zero or $\pm(0.18, 0.04, 0, 0)$, giving 1029 independent integrations. Increasing- and decreasing-frequency continuations add 686 runs. Each continuation changes frequency at the zero-field phase and inherits the previous endpoint; the ramp is applied only at its first frequency. Each run discards 128 forcing periods and measures the next 64, for a total duration of 81.28–150.95 s. Fixed-step DOP853 uses 256 steps per forcing period in the two-mode grid.

We integrate the analytic variational equation simultaneously, renormalize one tangent vector every period, and average its logarithmic growth over the last 64 periods. These conditional finite-time Lyapunov exponents hold the prescribed forcing phase fixed. We also integrate the complete 4×4 fundamental matrix over a further period with 512 steps, and calculate the Floquet multipliers of every converged two-mode orbit. The nontrivial Jacobian block is

$$D_X\dot{Y} = -\operatorname{diag}(\omega_j) + \omega_0[1 - \cos\vartheta - u\sin\vartheta]dd^{\mathsf{T}}. \tag{S336}$$

The identity $\det\mathcal{M} = \exp(-2\alpha T)$ for its monodromy matrix supplies an independent numerical check.

All 1715 runs converge to forcing-periodic responses. The largest single-period canonical residual is 6.65×10^{-11}; the maximum endpoint differences among the three initial states and between the two continuation directions are

2.31×10^{-11} and 4.01×10^{-11}, respectively. Conditional exponents in the independent grid range from -0.28046 to $-0.27044\,\mathrm{s}^{-1}$. Floquet spectral radii range from 0.803809 to 0.889055, with the determinant identity accurate to 1.41×10^{-14}. Fifteen cases at amplitudes 2, 6 and 20 mT and frequency ratios 0.8, 0.95, 1, 1.05 and 1.2 retain periodic recurrence and negative conditional exponents when both the integration time and steps per period are doubled, and separately with four and eight modes. The latter use 1024 and 4096 steps per period. Time refinement changes the sampled two-mode steady peak by at most 3.81×10^{-7} mm. An independently written adaptive DOP853 implementation also agrees in canonical state to 3.13×10^{-13} on three twelve-period checks.

These are numerical local-stability and basin checks at specified points, not a proof over a continuous parameter set. Of the 1029 independent runs, 550 keep their sampled whole-beam slope bound below 0.1 rad throughout startup and measurement. We use this value as a conservative reporting screen for the small-slope model, not as a calibrated physical validity threshold. Large-amplitude rows are retained in the data and marked in Figure S11, but do not establish physical device performance. No resolved multistability or chaos is detected in this scan; neither a chaos threshold nor conservative-torus breakup is inferred from it.

Full-model resonance and clearance contours. To resolve the operational boundary without high-mode truncation, we separately calculate periodic responses using all 64 frozen FE modes. For $f = \vartheta - \sin\vartheta + u\cos\vartheta$ with Fourier coefficients f_k and $\Omega = r\omega_0$, the exact linear modal response is

$$X_{j,k} = \frac{\omega_0 \omega_j d_j f_k}{\omega_j^2 - (k\Omega)^2 + \mathrm{i}\alpha k\Omega}. \tag{S337}$$

Iterating this response with the nonlinear force gives a numerical periodic solution; its full nonlinear ODE residual is checked after reconstructing $\vartheta = d^{\mathsf{T}}X$. Amplitude root-finding then locates a 4 mm peak-displacement crossing at 31 of 35 frequency ratios; the remaining four ratios, 1.225–1.30, have no such crossing below 20 mT on the calculated branch. Cubic-Hermite Bernstein bounds cover every centreline point at each sampled phase. At every reported crossing these spatial upper bounds equal the actual tip peak to 8.9×10^{-16} mm, so the contour also locates a tip reaching 4 mm in the numerical periodic solution.

The steady contour has a numerical minimum near $r = 0.999245$ ($\Omega = 11.408664\,\mathrm{rad\,s}^{-1}$), at $A = 1.915911$ mT and a spatial slope bound of 0.06750 rad. Thus a 20 mT actuator limit alone cannot ensure clearance near resonance.

Table S20. Numerical 4 mm crossing amplitudes in mT. Steady crossings use all 64 modes. Startup crossings use eight modes, zero initial state, the stated 0.25 s ramp, and 192 forcing periods. They are protocol-dependent response contours, not certified safe-field limits.

Ω/ω_0	Steady	Startup
0.8	13.6362	8.6513
0.9	7.5005	5.1882
1.0	1.9168	1.9169
1.1	8.7909	5.7662
1.2	18.5936	10.4455

At five representative crossings, increasing the Fourier phase grid from 1024 to 4096 changes the peak by at most $0.00526\,\mu$m. Comparisons with 2, 4, 8, 16 and 32 modes are retained: at these same inputs the two-mode peak differs from the full FE peak by up to $67.73\,\mu$m, whereas the eight-mode discrepancy is below $4.18\,\mu$m. Independent full-FE Radau integration from the computed periodic states at $r = 0.8, 1, 1.2$ agrees in tip position to 1.11×10^{-12} m over one period.

Startup overshoot imposes a different contour. The eight-mode startup roots in Table S20 use 512 observation phases per cycle; doubling time, integration resolution and observation resolution at $r = 0.8, 1, 1.2$ changes the peak by at most $0.0678\,\mu$m. Full-FE integration from rest over the first 32 periods, combined with the separately computed periodic response, changes the checked peak by at most $2.44\,\mu$m. At $r = 0.8$, the approximately 8.6513 mT request has a 2.5411 mm full-FE steady amplitude but reaches 4.0020 mm during startup. This is a numerical overshoot example with no wall-contact force. The continuous-time input-work governor of Section S7.1 remains the proved clearance mechanism; the response contours explain why both frequency and startup must enter a practical operating choice.

S6 Preserving KAM dynamics through magnetic actuation

S6.1 Magnetic feedback that preserves the learned KAM dynamics

A spatial magnetic-field gradient supplies a tip-force input that, together with the transverse field, preserves the certified conservative dynamics despite physical damping. Both inputs act through the magnetic dipole on the same finite-element beam.

Let a_j be the physical mass-normalized modal positions of the complete 64-mode FE model, with velocities v_j,

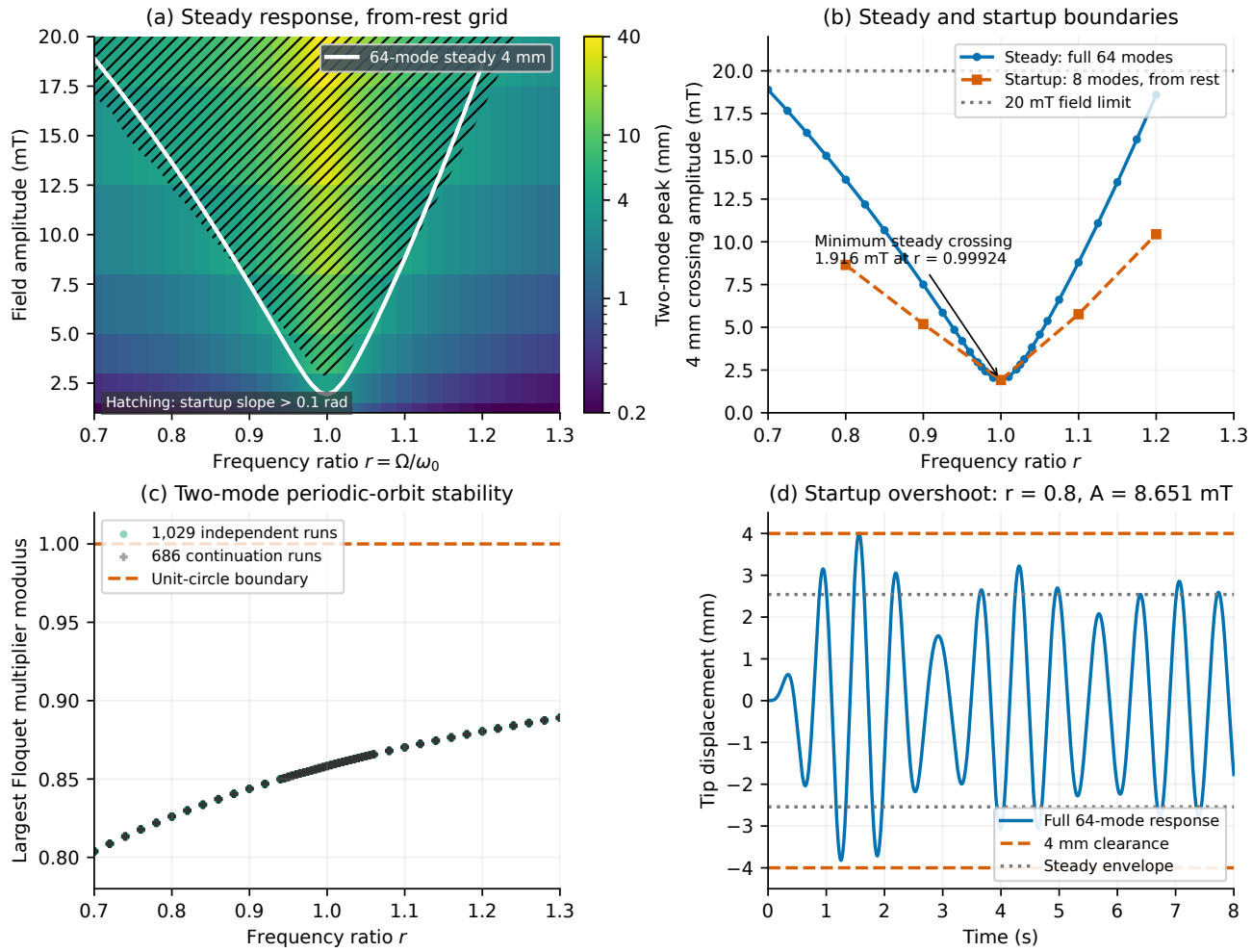


Figure S11. Driven-response and clearance study of the same magnetic beam. (a) Two-mode steady peak on the from-rest grid; hatching marks sampled startup slopes above the 0.1 rad reporting screen. The white line is the independently computed full-FE steady 4 mm contour. (b) Full-FE steady and eight-mode startup contours under the fixed ramp. (c) Floquet multipliers of the two-mode periodic responses from all independent and continuation runs. (d) Full-FE startup overshoot near the eight-mode numerical crossing. Stability of an attracting forced periodic response is distinct from conservative KAM-torus persistence.

frequencies ω_j, tip-displacement participation b_j, and tip-rotation participation t_j. The numerical theorem uses the exported modal coefficients and Hermite shape map as the exact definition of the computational model. The diagonalization residual relative to the assembled nodal matrices is quantified below. Write $\vartheta = \sum_j t_j a_j$ and $E_0 = \mu B_0$. With transverse tip force F and transverse field B_y, the modal equations are

$$\begin{aligned} \ddot{a}_j + \alpha \dot{a}_j + \omega_j^2 a_j &= b_j F + t_j U, \\ U &= E_0(\vartheta - \sin\vartheta) + \mu B_y \cos\vartheta. \end{aligned} \tag{S338}$$

The frequencies already include the linear stiffness from B_0. For the first two modes define

$$A_{\mathrm{mag}} = \begin{pmatrix} b_1 & t_1 \\ b_2 & t_2 \end{pmatrix}. \tag{S339}$$

An interval enclosure of its determinant excludes zero.

Let $h(X, Y)$ be the fitted Hamiltonian used by the KAM certificate, including momentum-dependent quartics. Set $a_i = S_i X_i$ and recover Y from

$$h_Y(X, Y) = \mathrm{diag}(S_i)^{-1} v_{1:2}/\omega_0. \tag{S340}$$

This implicit velocity-to-momentum map cannot be replaced by $Y_i = v_i/(S_i\omega_0\nu_i)$ when the learned momentum quartics are retained. The verified lower bound on h_{YY} below makes the map locally invertible and injective in Y on the stated real box. Define

$$a^{\mathrm{des}}_{i,tt} = S_i\omega_0^2\big(h_{YX}h_Y - h_{YY}h_X\big)_i, \tag{S341}$$

$$\begin{pmatrix} F \\ U \end{pmatrix} = A_{\mathrm{mag}}^{-1}\left(a^{\mathrm{des}}_{1:2,tt} + \alpha v_{1:2} + \mathrm{diag}(\omega_1^2, \omega_2^2) a_{1:2}\right). \tag{S342}$$

Consequently $\dot{X} = \omega_0 h_Y$ and $\dot{Y} = -\omega_0 h_X$ exactly where the implicit inverse is defined. The proof follows by differentiating (S340): the specified acceleration cancels the h_{YX} term and invertibility of h_{YY} gives the second canonical equation.

The physical commands realizing (S342) are

$$B_y = \frac{U - E_0(\vartheta - \sin\vartheta)}{\mu\cos\vartheta}, \qquad G = \frac{F}{\mu}\begin{pmatrix} -\sin\vartheta & \cos\vartheta \\ \cos\vartheta & \sin\vartheta \end{pmatrix}. \tag{S343}$$

Use $B(r) = B_{\mathrm{tip}} + G(r - r_{\mathrm{tip}})$ locally, with $B_{\mathrm{tip}} = (B_0, B_y)$. The matrix G is symmetric and traceless, so this local planar field satisfies the source-free curl and divergence constraints. For dipole $m = \mu(\cos\vartheta, \sin\vartheta)$ it gives $Gm = (0, F)$, while the tip torque is $\mu(B_y\cos\vartheta - B_0\sin\vartheta)$. Moreover $\|G\|_2 = |F|/\mu$. The applied mechanical power is $F\dot{w}(L) + \mu B_y\cos\vartheta\,\dot{\vartheta}$, in addition to the static magnetic energy already included in the Hamiltonian. The construction assumes these local field and gradient commands are available and the required modal and tip states are observed.

S6.2 Lifting a certified torus to the complete controlled FE model

Theorem S65. Invariant-torus lift under physical magnetic feedback. *Suppose a real analytic embedded two-torus K_∞ of h has frequency $\Omega \in \mathbb{R}^2$ in radians per physical second. Suppose h_{YY} is positive definite on a neighbourhood of this torus with convex Y-fibres for each X, A_{mag} is invertible, $\alpha > 0$, and the complete lifted solution has $|\vartheta| < \pi/2$. Then the full FE closed loop (S338)–(S343) has an embedded invariant two-torus projecting to K_∞. For a fixed low-mode orbit on this torus, each high-mode solution converges exponentially to a unique quasiperiodic response, provided the feedback remains in its stated field and state domain.*

Proof. The first two canonical modes follow h exactly. Hence F and U in (S342) depend only on their torus phase, even though the physical field command also uses the complete tip angle. For every $j \geq 3$, the remaining dynamics are the stable linear equation (S338), forced by analytic quasiperiodic functions. If F_k, U_k denote their Fourier coefficients in the convention $e^{ik\cdot\phi}$, the unique response graph is

$$a^*_{j,k} = \frac{b_j F_k + t_j U_k}{\omega_j^2 - (k\cdot\Omega)^2 + i\alpha(k\cdot\Omega)}. \tag{S344}$$

For real $k\cdot\Omega$, this denominator never vanishes: at zero frequency its real part is $\omega_j^2 > 0$, and at nonzero frequency its imaginary part is nonzero. Its inverse is uniformly bounded over real frequencies. Exponential Fourier decay is therefore preserved on a possibly smaller analytic strip. The velocity graph is obtained by multiplication by $ik\cdot\Omega$ and is analytic on a smaller strip. Positive definiteness on each convex Y-fibre makes $Y \mapsto h_Y(X, Y)$ strictly monotone and injective. Thus the physical low-mode positions and velocities preserve the injective canonical embedding. Differences in the high modes obey the homogeneous damped equations and decay exponentially.

The lifted closed loop combines Hamiltonian low-mode motion with contracting high-mode fibres. The contraction leaves each initial low-mode torus invariant; it does not select a unique torus or make the full controlled system Hamiltonian. Each lifted two-torus has zero volume in the full 128-dimensional phase space. The KAM-family measure therefore refers to the conservative low-mode coordinates.

All-phase response and startup bounds. We bound both the response graph and the startup with all high modal positions and velocities initially zero. Write $h = h_2 + R$, and let the certified torus satisfy $\|K_\infty - K_0\|_\infty \leq \epsilon_K$, where $K_0 = 0.1(\cos\phi_1, \cos\phi_2, -\sin\phi_1, -\sin\phi_2)$. The first two components of the force demand can be written as

$$r_i = S_i\{\omega_0^2[(R_{YX}h_Y)_i - \nu_i R_{X_i} - (R_{YY}h_X)_i] + \alpha\omega_0 h_{Y_i} + (\omega_i^2 - \omega_0^2\nu_i^2)X_i\}, \tag{S345}$$

This identity cancels the large linear restoring terms before bounding. The linear part on K_0 consists of just the two known physical frequencies. Each is bounded using its actual damped denominator in (S344). The remaining terms are bounded on $|X_i|, |Y_i| \leq 0.1 + \epsilon_K$ using every fitted quartic coefficient and the complete trigonometric remainder bounds $|\sin z - z| \leq |z|^3/6$ and $|1 - \cos z| \leq z^2/2$ for real z.

For $\beta = \alpha/2$ and $\omega_{d,j} = \sqrt{\omega_j^2 - \beta^2} > 0$, the impulse response $g_j(t) = e^{-\beta t}\sin(\omega_{d,j}t)/\omega_{d,j}$ obeys

$$\|g_j\|_{L^1(0,\infty)} \leq \frac{2}{\alpha\omega_{d,j}}, \qquad \|g_j'\|_{L^1(0,\infty)} \leq \frac{2}{\alpha} + \frac{1}{\omega_{d,j}}. \tag{S346}$$

These estimates include the entire nonlinear forcing remainder and its Fourier tail. Let P_j, V_j be the resulting all-phase bounds on the graph's position and velocity. Starting high mode j at zero adds at most

$$T_j = P_j + \frac{V_j + \beta P_j}{\omega_{d,j}} \tag{S347}$$

to its position envelope. This follows from the exact homogeneous solution initialized by $(-a_j^*(0), -\dot{a}_j^*(0))$. The bounds sum all 62 high modes and use the cubic Bernstein control rows of every Hermite element to convert these modal envelopes into bounds on every centreline point and every phase.

For the two-mode COMSOL integral fit in Section S4.5, which satisfies the nonzero-loss and finite-amplitude KAM conditions, 256-bit Arb arithmetic encloses $h_{YY} \succeq 0.9999999983I$ on the real 0.3 box. The complete response graph has centreline amplitude below 1.922696 mm. Including the zero-high-mode startup yields an all-time bound below 1.970263 mm, strictly inside the 4 mm net corridor. The latter envelope gives $|\vartheta| < 0.155261$ and $\cos\vartheta > 0.987971$, which closes the feedback-domain bootstrap. The required transverse field is below 0.921665 mT and the gradient spectral norm below $0.02572538\,\mathrm{T\,m^{-1}}$ at every phase and throughout that startup. These bounds validate feasibility under a 20 mT transverse field limit and a $1\,\mathrm{T\,m^{-1}}$ gradient limit; they are derived bounds, not the extrema of a simulated time history. Applying the same computation to the fit trained on complete COMSOL FE trajectories in Section S5.5 gives graph and zero-high-state startup bounds of 1.937308 mm and 2.014078 mm, respectively.

The theorem applies to initial low states on the exact certified torus. The numerical Fourier initial state approximates it within the certified KAM correction, and finite-time integration tests the feedback implementation. The theorem uses the exported modal FE coefficients and fitted weights; independent COMSOL comparisons assess the corresponding nodal mechanical implementation. To make the modal definition explicit, if Φ is the stored full shape basis, M, K_b are the independently assembled nodal matrices and $D_\omega = \mathrm{diag}(\omega_j)$, outward Arb products give

$$\|\Phi^{\mathsf{T}} M \Phi - I\|_\infty < 1.304685 \times 10^{-14}, \quad \text{(S348)}$$

$$\|D_\omega^{-1}\Phi^{\mathsf{T}} K_b \Phi D_\omega^{-1} - I\|_\infty < 3.000518 \times 10^{-9}. \quad \text{(S349)}$$

These are relative kinetic- and stiffness-energy form bounds. They quantify the relationship between the frozen modal model and the original assembled FE representation; they are not used to assert an exact conjugacy or an unproved persistence result for perturbed matrices. The two-mode shape map used for identification differs from the reassembled map by at most 1.258×10^{-14} m, and the reference frequency ω_0 is unchanged.

S6.3 Full-model implementation and frequency preservation

The feedback is integrated on all 64 modes for 12 s, using the two-mode COMSOL integral fit certified in Section S4.5. The initial low state is the phase-zero Fourier candidate and all higher modal positions and velocities are zero. A separate high-accuracy integration of the four-dimensional learned Hamiltonian supplies an implementation reference. The plant is actuated through the physical tip force and moment. The maximum canonical state difference is 1.54×10^{-10}, the relative learned-energy change is below 2.64×10^{-9}, and the full physical work-balance residual is below 6.06×10^{-17} J. The sampled field and gradient peaks are 0.890325 mT and 0.0256844 T/m. Figure S12 compares this trajectory with the same initial state under passive damping, for which the learned energy decreases by 99.8726%. The numerical trajectory is an implementation check; the all-time envelopes come from the preceding interval proof. An independent implementation integrates the low Hamiltonian separately and propagates each of the 62 high damped oscillators by its exact exponential response to piecewise-linear force demands. Force-grid steps of 1, 0.5 and 0.25 ms give maximum tip differences from the full-state integration of 1.073, 0.277 and 0.106 nm, respectively. Every high-mode homogeneous transient is retained. This check separates time-step error from modal truncation and supports the full-state implementation. The resulting differences measure numerical convergence; the rigorous bounds are supplied by the interval proof.

Independent online COMSOL feedback. The complete nodal beam is also solved with the CHNN feedback inside COMSOL. The first two modal coordinates and velocities are projected from its current beam state; two global algebraic equations enforce the learned Legendre inverse. The physical tip force and magnetic moment are recomputed online. This experiment is distinct from the prescribed-field S-path replay. Generalized-alpha integration uses $\rho_\infty = 0$, a nonlinear relative tolerance of 10^{-8} and fixed steps. Table S21 reports four 4 s refinements and compares them at common sampled times with the independent full-state Python run. At the finest step, the maximum relative change in the learned energy is 0.0002958. These numerical errors are reported separately from the analytic KAM and all-time amplitude certificates. Adaptive stepping stalled at very small steps in two preliminary runs; fixed stepping completed each refinement.

The two certified physical angular frequencies are approximately 11.41719351 and 70.24034116 rad/s. Their

Table S21. Online COMSOL feedback versus independent Python integration. Errors are in μm; no step-size tolerance is interpreted as a rigorous error bound.

Step (ms)	Tip RMSE	Maximum tip difference
1	147.852	355.845
0.5	41.356	99.176
0.25	12.006	28.161
0.125	3.848	8.658

exact specification and all-integer Diophantine bound follow from the KAM certificate under the stated time normalization. Magnetic feedback therefore preserves a certified nonresonant frequency pair while compensating damping. The complete nonlinear command contains the harmonics determined by the torus; it is not approximated by two sinusoidal inputs when invoking the lifting theorem. The rigorous state and command envelopes provide a simultaneous amplitude check. The operating-domain fraction is examined separately in Section S5.6.

S7 Clearance guarantees and magnetic safety control

S7.1 A whole-beam clearance certificate for the same magnetic model

The conservative normal form and the safety calculation use the same clamped-free beam, static field, magnetic dipole, and energy. Safety is a statement about the driven physical equations, with their nonnegative drag; it does not require a driven trajectory to lie on a conservative KAM torus. For the planar, small-slope model, write $w(s,t)$ for transverse displacement, $m=\rho A$ for line density, $c_d\ge 0$ for distributed viscous drag, and $\vartheta = w_s(L,t)$ for the tip slope used in the dipole potential. The equations and natural end conditions are

$$mw_{tt}+c_dw_t+EIw_{ssss}=0, \qquad 0<s<L, \tag{S350}$$

$$w(0,t)=w_s(0,t)=0, \qquad w_{sss}(L,t)=0, \tag{S351}$$

$$EIw_{ss}(L,t)=\mu\{B_y(t)\cos\vartheta - B_0\sin\vartheta\}. \tag{S352}$$

The end moment is the derivative of the same cosine potential used to generate the training trajectories. In particular, the varying transverse field is an input port, not an additional autonomous Hamiltonian perturbation. For a sufficiently regular solution satisfying the mechanical power identity, the stored energy relative to the aligned static-field equilibrium is

$$E(t)=\frac{1}{2}\int_0^L \{mw_t^2+EIw_{ss}^2\}\,\mathrm{d}s+\mu B_0(1-\cos\vartheta), \tag{S353}$$

$$\dot E=-c_d\int_0^L w_t^2\,\mathrm{d}s+\mu B_y\cos\vartheta\,\dot\vartheta. \tag{S354}$$

All inequalities below also hold for the conforming cubic-Hermite finite element (FE) system and any of its Galerkin subspaces, because their energies are the restrictions of this expression and retain the same input power. In the balanced coordinates of the learning experiment, $E=\mu B_0 h_*$ for the physical target energy, $\vartheta = d^{\mathsf T}X$, and the port force in the Y equation is $d(B_y/B_0)\cos(d^{\mathsf T}X)$.

Proposition S66. Clearance and field-variation budget. *Assume $B_0>0$, $c_d\ge 0$, and an absolutely continuous applied field B_y on $[0,T]$. Define*

$$S_T=\sup_{0\le t\le T}|B_y(t)|+|B_y(0)| +\int_0^T|\dot B_y(t)|\,\mathrm{d}t, \tag{S355}$$

$$C_T=\mu\sqrt{\frac{2L}{EI}}S_T, \tag{S356}$$

$$R_T=\frac{C_T+\sqrt{C_T^2+4E(0)}}{2}. \tag{S357}$$

Then, throughout $[0,T]$,

$$E(t)\le R_T^2,\qquad |w(s,t)|\le\sqrt{\frac{2s^3}{3EI}}R_T,\qquad |w_s(s,t)|\le\sqrt{\frac{2s}{EI}}R_T. \tag{S358}$$

For a straight corridor with wall half-width a and rod radius r, let $c=a-r>0$. A sufficient condition for positive clearance of the whole rod is $R_T<\sqrt{E_c}$, where

$$E_c=\frac{3EIc^2}{2L^3}. \tag{S359}$$

For $E(0)<E_c$, the following directly checkable operating condition is equivalent to this sufficient energy condition:

$$S_T<\frac{E_c-E(0)}{\mu\sqrt{2LE_c/EI}}. \tag{S360}$$

These conclusions hold for every nonnegative drag coefficient, including zero.

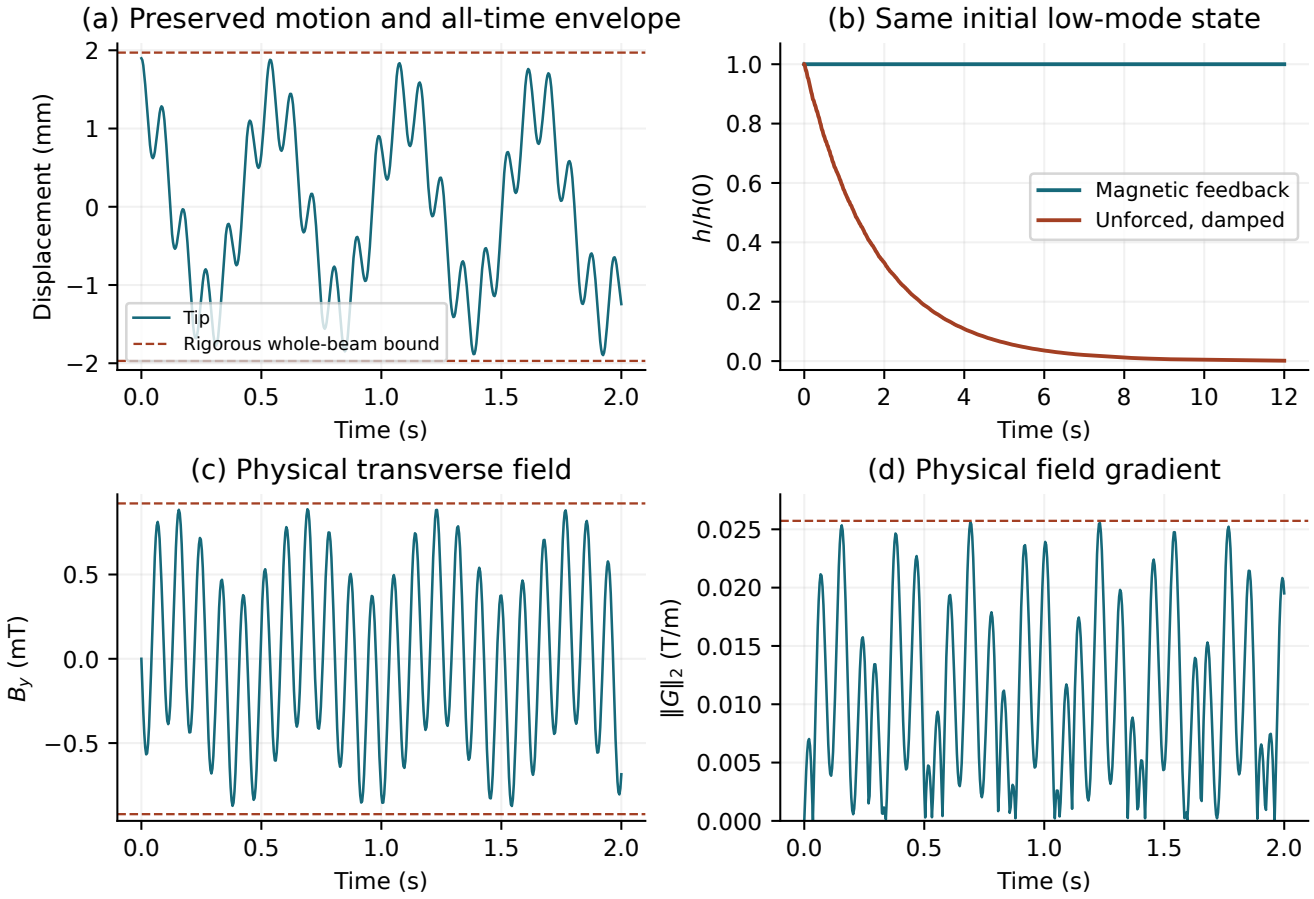


Figure S12. Magnetic preservation of the COMSOL-integral learned Hamiltonian on the complete exported modal FE plant. The higher modes start at zero and undergo a decaying transient toward their unique quasiperiodic response graph. Dashed bounds in (a), (c) and (d) are validated all-phase bounds, including that startup. The field and gradient in (c,d) jointly realize physical tip force and torque. Panels (a,c,d) show the first 2 s; panel (b) shows the complete 12 s run.

Proof. Since $B_0 > 0$, the static magnetic energy is nonnegative and $\|w_{ss}\|_{L^2}^2 \le 2E/EI$. The clamped conditions imply $w(s) = \int_0^s (s-\xi) w_{ss}(\xi)\,\mathrm{d}\xi$ and $w_s(s) = \int_0^s w_{ss}(\xi)\,\mathrm{d}\xi$. Cauchy–Schwarz gives the two spatial inequalities in (S358) with $\sqrt{E}$ in place of R_T. Write $D_c(t) = \int_0^t c_d \|w_t\|_{L^2}^2\,\mathrm{d}\tau$. Integrating (S354) by parts in the input term gives

$$\begin{aligned} E(t) + D_c(t) - E(0) &= \mu B_y(t)\sin\vartheta(t) \\ &\quad - \mu B_y(0)\sin\vartheta(0) \\ &\quad - \mu \int_0^t \dot{B}_y(\tau)\sin\vartheta(\tau)\,\mathrm{d}\tau. \end{aligned}$$

Set $M_T = \sup_{[0,T]} \sqrt{E(t)}$. Using $|\sin\vartheta| \le |\vartheta| \le \sqrt{2L/EI}\, M_T$ and discarding the nonnegative dissipation yields $M_T^2 \le E(0) + C_T M_T$. The positive root is R_T. The entire radius-r tube about the centerline remains between the walls when $\sup_s |w(s,t)| < a - r$. Solving $R_T < \sqrt{E_c}$ gives (S360).

For a nonuniform planar corridor, let $c(s) > 0$ be the Euclidean distance from the undeformed point $(s, 0)$ to the wall, less the rod radius. The distance-to-wall function is one-Lipschitz, so $|w(s)| < c(s)$ keeps the radius-r tube clear. The same argument replaces E_c by $\inf_{0<s\le L} 3EIc(s)^2/(2s^3)$. For an inclined or curved wall, a local vertical half-width is not a substitute for this Euclidean clearance. This sufficient condition applies to the specified corridor geometry; vessel-specific use requires its own wall data. The continuum inequality applies directly to solutions with the stated initial-energy and applied-field bounds, and holds separately for conforming FE solutions. It therefore requires neither a two-mode reconstruction error estimate nor an identification of an FE trajectory with a continuum solution.

For the shared SI parameters, choose $a = 4.6$ mm and $r = 0.6$ mm, so that $c = 4$ mm. The resulting E_c is 23.8565 nJ. For a monotone field ramp from zero to A followed by a hold, $S_\infty = 2A$. With zero initial energy, a sufficient

condition is

$$A < \frac{\sqrt{3}\,EIc}{4\mu L^2} = 6.8868\ \mathrm{mT}. \tag{S361}$$

For $B_y = A\rho(t)\sin(\Omega t)$ with $\rho(t) = 1 - \exp[-(t/0.25\,\mathrm{s})^2]$, $S_T \le A(2 + \Omega T)$, because ρ increases from zero and is at most one. Thus amplitude, frequency and operating duration enter an explicit joint condition by substituting this bound in (S360). For $T = 8$ s, $\Omega = 9.444949$ rad/s and $E(0) = 0$, its sufficient amplitude limit is approximately 0.1776 mT. This conservative open-loop test does not certify the default 6 mT sinusoid, even when a simulated trajectory has positive clearance.

S7.2 A magnetic-input governor with a full-model energy guarantee

The field-variation test can be strengthened for feedback operation by measuring the actual mechanical input port. Let $\overline{E}_0 \ge E(0)$ be a valid initial-energy bound and integrate

$$W(t) = \overline{E}_0 + \int_0^t \mu B_y \cos\vartheta\,\dot{\vartheta}\,\mathrm{d}\tau. \tag{S362}$$

Equation (S354) gives $E(t) \le W(t)$ without measuring or estimating the dissipated energy. Given $E(0) \le \overline{E}_0 < E_{\max} < E_c$, a bounded requested field $|B_{\mathrm{req}}| \le B_{\max}$, and $\kappa > 0$, choose the applied field as the closest value to B_{req} satisfying

$$\begin{aligned} |B_y| &\le B_{\max}, \\ a_p B_y &\le \kappa(E_{\max} - W), \\ a_p &= \mu\cos\vartheta\,\dot{\vartheta}. \end{aligned} \tag{S363}$$

This is a scalar projection onto an interval. Whenever the requested input violates the work constraint and $a_p \neq 0$, the applied field is $B_y = \kappa(E_{\max} - W)/a_p$; otherwise it is unchanged. On $W \le E_{\max}$, zero field is feasible, and this projection never increases the requested field amplitude.

Proposition S67. Port-work safety invariant. *Suppose the actual applied field and actual tip slope and slope rate are available to* (S362)–(S363)*, and the field command is applied without unbudgeted error or delay. Every closed-loop solution satisfying the mechanical power identity obeys*

$$\begin{aligned} E(t) &\le W(t) \le E_{\max}, \\ \sup_s |w(s,t)| &\le \sqrt{\frac{2L^3 E_{\max}}{3EI}} < c. \end{aligned} \tag{S364}$$

Consequently the entire rod retains positive wall clearance for all times on which the stated beam model and port measurements apply, for any $c_d \ge 0$.

Proof. The power identity gives $W - E = \overline{E}_0 - E(0) + \int_0^t c_d\|w_t\|_{L^2}^2\,\mathrm{d}\tau \ge 0$. The input constraint yields $\dot{W} \le \kappa(E_{\max} - W)$, hence $E_{\max} - W(t) \ge (E_{\max} - \overline{E}_0)\exp(-\kappa t) \ge 0$. The spatial estimate proved above completes the result.

The guarantee uses the full physical input power, including the work associated with unobserved high modes. A learned model supplies the requested field. Accurate actual-port measurements and ideal command application are assumed; sensor, actuation and delay errors require additional work-error budgets. The projection constrains field amplitude and accumulated work, with no imposed slew-rate bound. Because dissipated energy is not credited back, the observer can restrict the request even when stored energy has decreased. Tracking performance is therefore evaluated separately under this conservative safety constraint.

A bounded port-measurement error can be included without assuming an exact learned state. If $|a_p - \widehat{a}_p| \le \delta_a(t)$, replace the observer derivative and its input constraint by

$$\begin{aligned} \dot{W} &= \widehat{a}_p B_y + \delta_a|B_y| \\ &\le \kappa(E_{\max} - W). \end{aligned} \tag{S365}$$

Actual input power is at most this observer derivative, so the same proof applies. Zero field remains feasible and the scalar feasible set is convex. Here B_y must be the actual applied field; an uncertain command-to-field map requires a further bound on actual input power. The numerical tests below use $\delta_a = 0$ and do not experimentally validate a sensor-error bound.

S7.3 Full-FE safety, violation and truncation checks

All tests use the shared 32-element consistent-mass FE beam, retain all 64 unconstrained displacement/slope degrees of freedom, and integrate the exact cosine tip potential. The coordinates are the complete 64-mode static-field basis; this is a change of coordinates, not a modal truncation. The default drag is $C = \alpha M$ with $\alpha = 0.55558524\ \mathrm{s}^{-1}$. Unless stated otherwise, trajectories start from rest at the straight equilibrium. The sinusoidal tests use the frequency $\Omega = 9.444949$ rad/s and the 0.25 s ramp above; the monotone tests omit the sine factor. The integration interval is 8 s. The continuous-time claims come from Propositions S66 and S67; cubic-Hermite spatial extrema are evaluated exactly at each saved time for the numerical diagnostics.

Table S22 separates numerical maxima from continuous-time certificates. The 6 mT monotone ramp satisfies an all-time displacement bound of 3.48494 mm, including zero drag. The default 6 mT sinusoid has a numerical peak of 3.09031 mm, but fails the conservative field-variation test; that peak alone is not a safety proof. Increasing its amplitude to 10 mT produces a 5.14478 mm free-motion peak and crosses the 4 mm boundary. The continuation after crossing is a diagnostic of a violated constraint, not a simulation of wall contact.

With $E_{\max} = 0.95E_c = 22.66365$ nJ and $\kappa = 5\ \mathrm{s}^{-1}$, the work governor applied to the same 10 mT request has a rigorous whole-centerline bound of 3.89872 mm and thus a minimum guaranteed clearance of 0.10128 mm. Its numerical peak is 3.14362 mm with the default drag and 3.55853 mm with zero drag. The same energy cap also implies $\sup_s |w_s| \le 0.08441$, consistent with the declared small-slope regime for the certified trajectories. Tightening both relative and absolute integration tolerances tenfold for the full-FE governed case changes the tip trajectory by at most 8.19×10^{-11} m; the refined power-balance residual is 3.86×10^{-18} J (1.74×10^{-10} relative to the work scale). The governor changes the requested field at 55.37% of saved times in this oscillatory case.

For the 6 mT sinusoid, the maximum tip discrepancies of the 2-, 4-, and 8-mode reconstructions against the full 64-mode FE trajectory are 22.7168, 5.9661, and 1.5156 μm, respectively (0.73510%, 0.19306%, and 0.04904% of the full-FE peak). These are measured truncation errors for this input, not rigorous bounds for all neglected modes or all trajectories. The full-continuum energy certificate does not substitute any of them for an unknown error constant. An independent COMSOL HermitianBeam integration of the same 32-element consistent-mass model gives a default-sinusoid peak of 3.09032 mm; its largest tip difference from the present integration is 0.11835 μm. Tightening the COMSOL relative tolerance to 10^{-8} reduces the largest tip difference to 0.02160 μm. Independent COMSOL runs also reproduce the 6 mT monotone-ramp safe case and 10 mT sinusoidal violation: their peaks are 0.924046 and 5.144787 mm, with maximum cross-solver tip discrepancies of 0.00663 and 0.03489 μm, respectively. This cross-check concerns the same mathematical model and does not calibrate material or actuation parameters against hardware.

The closed-loop governor was also implemented inside COMSOL, coupling a global ODE for $W/E_{\max}$ to the nonlinear magnetic end moment. This run used the generalized-α time integrator with $\rho_\infty = 0$, relative tolerance 10^{-5}, and a maximum internal step of 0.002 s; the 2001 saved outputs are spaced by 0.004 s. Its full-FE centerline peak is 3.143643 mm, leaving a sampled net clearance of 0.856357 mm. The maximum tip discrepancy from the refined Python trajectory is 0.582073 μm, or 0.018516% of its peak. The applied field remains below 9.816986 mT. The COMSOL-integrated work observer peaks at 22.253032 nJ, or $0.9818823E_{\max}$; $E \le W \le E_{\max}$ and the magnetic-power projection hold at every saved time. Its maximum work-observer discrepancy from Python is 2.263×10^{-12} J.

For an independent energy ledger, reconstruct E from all displacement and velocity DOFs and compute

$$D_h(t) = \int_0^t \alpha \dot{q}^{\mathsf{T}} M \dot{q}\, \mathrm{d}\tau,$$
$$R_E(t) = E(t) - E(0) - [W(t) - W(0)] + D_h(t).$$

Simpson quadrature on the saved outputs gives $\max_t |R_E(t)| = 5.796 \times 10^{-12}$ J after rounding up, or 2.605×10^{-4} of the work scale. The Simpson–trapezoid difference for that dissipation integral is 2.424×10^{-13} J. This COMSOL run uses high-frequency numerical damping and a looser tolerance than the refined Python integration; its ledger residual includes time-integration and output-quadrature errors. The separately integrated COMSOL work state is used for the observer comparison, rather than replacing it by a coarse-output input power integral. These numerical checks corroborate the implementation; the continuous-time clearance statement remains the analytic invariant in Proposition S67.

S7.4 The learned CHNN generates a feasible magnetic tracking request

To evaluate learned magnetic tracking, the fitted GC model with seed 17 and its identified dissipation coefficient generate a nominal magnetic field for the full FE plant. Let $z = (X_1, X_2, Y_1, Y_2)$ denote the observed projection onto the first two static-field modes, $\ell^{\mathsf{T}} X$ its reconstructed tip displacement, and h_θ the fitted energy, including all learned mixed and momentum-dependent quartic coefficients. Define, in the dimensionless time $\tau = \omega_0 t$,

$$y = \ell^{\mathsf{T}} X, \qquad y' = \ell^{\mathsf{T}} h_{\theta,Y}, \tag{S366}$$
$$f_0 = \ell^{\mathsf{T}} h_{\theta,YX} h_{\theta,Y} - \ell^{\mathsf{T}} h_{\theta,YY}(h_{\theta,X} + D_\theta h_{\theta,Y}), \tag{S367}$$
$$g_0 = \ell^{\mathsf{T}} h_{\theta,YY} d \cos(d^{\mathsf{T}} X), \tag{S368}$$
$$D_\theta = \widehat{\gamma}\ \mathrm{diag}(\nu)^{-1}. \tag{S369}$$

Table S22. Safety tests on the same full 32-element, 64-DOF FE beam. Numerical maxima are cubic-Hermite spatial extrema at 2001 saved times over 8 s. Bounds are continuous-time sufficient bounds from the stated propositions. "No" means the sufficient test does not certify the case. All fields use a 0.25 s ramp; sine frequency is 9.444949 rad/s. Except the stated $E_c/4$ case, initial energy is zero.

Applied protocol	Peak (mm)	Bound (mm)	Certified
6 mT ramp/hold	0.92405	3.48494	Yes
3 mT ramp, $E(0) = E_c/4$	2.06222	3.05276	Yes
0.15 mT sine	0.07731	3.37863	Yes
6 mT sine	3.09031	135.145	No
10 mT sine	5.14478	225.242	No
Governed 10 mT sine	3.14362	3.89872	Yes
Governed 10 mT, zero drag	3.55853	3.89872	Yes
6 mT ramp, zero drag	0.95405	3.48494	Yes

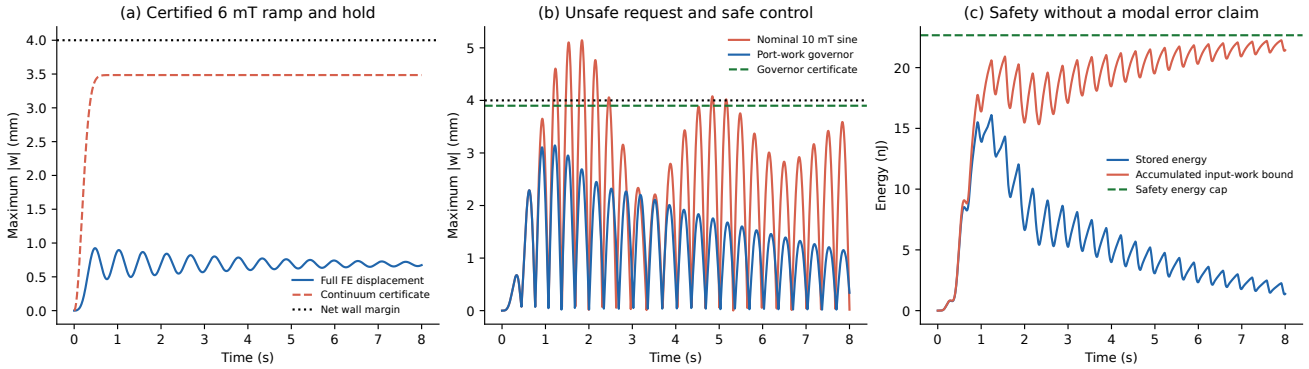


Figure S13. Whole-beam safety for the shared full FE plant, with zero initial energy and default drag. (a) A 6 mT monotone ramp and its all-time continuum energy certificate. (b) The 10 mT ramped sinusoidal request at 9.444949 rad/s violates the 4 mm net corridor margin; the actual port-work governor preserves it with an analytically guaranteed bound. (c) Stored energy and accumulated input-work bound for the governed case. The wall half-width is 4.6 mm and rod radius 0.6 mm. These geometric constraints define the simulated corridor; patient-specific clearance requires a corresponding geometry and parameter model.

Where $g_0 \neq 0$, the nominal request is

$$a_r = r'' - k_p(y - r) - k_d(y' - r'), \tag{S370}$$

$$u_* = (a_r - f_0)/g_0, \tag{S371}$$

$$B_{\mathrm{req}} = B_0 \,\mathrm{clip}_{[-b,b]}(u_*), \tag{S372}$$

$$b = B_{\max}/B_0. \tag{S373}$$

The primes on r denote derivatives in τ. We use $k_p = 16/\omega_0^2$, $k_d = 8/\omega_0$, and $\widehat{\gamma} = 0.0486617964$. Both Hessian blocks are evaluated from the fitted model; they are not replaced by their quadratic-prior values. The plant receives only the scalar B_y through its actual tip moment, after the work-governor projection. It receives no independently assigned modal forces.

The requested SI tip trajectory is $y_{\mathrm{ref}}(t) = 2\,\mathrm{mm}\,(10s^3 - 15s^4 + 6s^5)$ for $s = t/(4\,\mathrm{s}) \in [0, 1]$, followed by a hold through 8 s, and $r(\tau) = y_{\mathrm{ref}}(\tau/\omega_0)$. The field limit is $B_{\max} = 20$ mT for this separate tracking protocol. This limit follows from the static input authority: a 2 mm equilibrium requires $\vartheta = 2y_{\mathrm{tip}}/L = 0.05$ and

$$B_y^{\mathrm{eq}} = \frac{EI\vartheta/(\mu L) + B_0 \sin\vartheta}{\cos\vartheta} \simeq 17.43\ \mathrm{mT}. \tag{S374}$$

The 10 mT sinusoidal safety tests and the 2 mm tracking experiment thus use different input amplitudes, with the tracking limit set by the required equilibrium field.

On the full 64-DOF plant, the actual tip has an RMS tracking error of 33.734 μm and a maximum error of 47.069 μm. The final tip position is 2.03515 mm and the largest centerline displacement is 2.03838 mm. The applied field peaks at 17.75695 mT; saturation and governor intervention occur at 0% of saved times. Thus the feasible learned request passes through the filter unchanged, and the separate unfiltered run is identical. The input-work bound peaks at 9.05293 nJ, below the 22.66365 nJ cap, and the same continuous-time safety invariant applies. The minimum absolute input gain along this run is 0.0138678 m, so the division in (S373) does not approach a singularity. The physical power-balance residual is 3.53×10^{-20} J. This tracking result assumes access to the two-mode projection of the full state. Extending it to incomplete sensing requires a state observer and a corresponding tracking analysis.

A second run uses the CHNN identified by integral fitting of full-FE COMSOL displacement and velocity exports, with cosine multiplier 1.0223248822. Its complete gradient

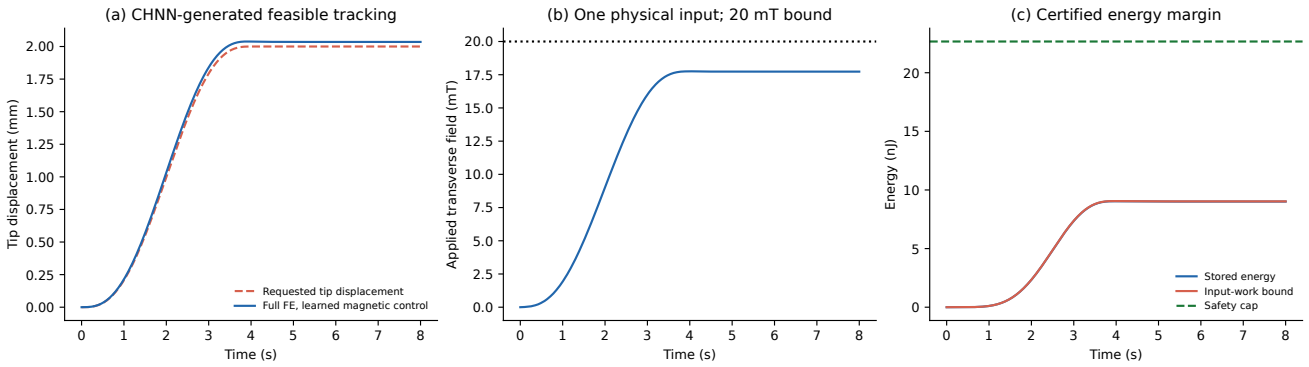


Figure S14. The fitted GC17 CHNN and identified drag generate a single bounded transverse magnetic input for the same full 32-element FE beam. (a) A feasible 2 mm tip request over 4 s and the actual full-FE tip motion. (b) The applied field remains below the declared 20 mT limit. (c) Stored energy and the actual-port work observer stay below the safety cap. The controller uses the exact first-two-mode projection available in this simulation; the safety observer uses full-model tip power. No modal actuation authority or unmeasured-state observer is assumed. The governor is inactive for this feasible tracking request; its active effect is tested separately in Figure S13.

and Hessian, including learned momentum quartics, generate the same 2 mm request on the same full-FE Python plant. For this conservative fit, a separate PSD drag identification on the existing Python-generated damped beam trajectories gives $\widehat{\gamma} = 0.0486617702$; this dissipative training source is distinct from the COMSOL conservative data. The tip RMSE is $33.726\ \mu$m, its maximum error is $47.065\ \mu$m, and the largest centerline displacement is 2.038365 mm. The applied field peaks at 17.756854 mT and the work observer at 9.052842 nJ. Saturation and governor intervention are again zero at the saved times, and the physical power-balance residual is 3.53×10^{-20} J. This COMSOL-trained model and the Python-trained GC17 model use the same physical plant. The full-state projection and actual-port measurement assumptions are unchanged, and the same continuum clearance guarantee applies.

Finally, the single uniform-field dipole supplies one boundary moment. In static equilibrium with no distributed transverse loading, $w_{ssss} = 0$ and the free-tip shear condition imply $w(s) = \vartheta s^2/(2L)$. This one-parameter family cannot realize an arbitrary S-shaped static centerline. The moving-base endpoint task in Section S7.6 instead follows an S-shaped tip history using axial base feed and magnetic bending. The governor preserves the corridor for feasible magnetic requests. Prescribing more general static shapes requires additional actuation authority, such as distributed magnetic moments or independently controlled loads.

S7.5 Finite initialization and port-error tolerances

The transverse action half-width 10^{-220} in Appendix S11 quantifies local invariant-torus measure. For operation, we instead construct a finite energy set and a robust input-work constraint on the same full beam. This construction accepts an ordinary finite-precision initial state, explicit sensing errors and bounded command errors. It preserves clearance when the safety filter changes the KAM-realizing request; exact quasiperiodic motion is then no longer asserted.

Physical object and a measurable initialization set. For this calculation the mathematical plant is the frozen, symmetric nodal 64×64 matrices M, K supplied with the independent FE comparison. All of their exported binary entries are treated as exact. This retains the complete physical energy

$$\begin{aligned} E(q,v) &= \tfrac{1}{2} v^{\mathsf{T}} M v + \tfrac{1}{2} q^{\mathsf{T}} K q + E_0(1-\cos\vartheta), \\ \dot{E} &= -\alpha v^{\mathsf{T}} M v + F v_w + \mu B \cos\vartheta\, v_\vartheta, \end{aligned} \tag{S375}$$

where $v = \dot{q}$, $\vartheta = e_\vartheta^{\mathsf{T}} q$, $v_w = e_w^{\mathsf{T}} v$, $v_\vartheta = e_\vartheta^{\mathsf{T}} v$, $E_0 = \mu B_0$, and F, B are the actual transverse tip force and field. The same two physical ports appear in (S343). The use of nodal matrices here includes their small diagonalization discrepancy relative to the frozen modal plant in Theorem S65; no exact conjugacy between these two finite-precision definitions is required for the work bound.

Let b_r list the four cubic Bernstein control rows on every Hermite element, including the fixed zero base coordinates, and define

$$C_w = \max_r b_r K^{-1} b_r^{\mathsf{T}}, \qquad E_c = \frac{c^2}{2C_w}, \qquad c = 4\,\mathrm{mm}. \tag{S376}$$

The Bernstein convex-hull property and Cauchy–Schwarz in the K metric give $\sup_s |w(s)| \le \sqrt{2C_w E}$ for every FE

state, including all higher modes. Thus $E < E_c$ is a whole-centreline clearance condition. Using exact Hermite coefficients $\ell/3 = 1/1200$ m, 256-bit ball matrix inverses and positive LDL pivots gives $C_w < 335.338810137\,\mathrm{m\,N^{-1}}$ and $E_c > 23.85646921\,\mathrm{nJ}$. This discrete certificate agrees with the continuum compliance while avoiding an unbudgeted matrix-rounding step.

Set $K_b = K + E_0 e_\vartheta e_\vartheta^{\mathsf{T}}$ and

$$Q(q,v) = \tfrac{1}{2} v^{\mathsf{T}} M v + \tfrac{1}{2} q^{\mathsf{T}} K_b q. \tag{S377}$$

For full-state estimates $(\widehat{q}, \widehat{v})$, require the initialization error to satisfy $Q(q-\widehat{q}, v-\widehat{v}) \le \mathcal{E}_{\mathrm{init}}$. Since $1-\cos\vartheta \le \vartheta^2/2$ and $\sqrt{Q}$ is a norm,

$$E(q,v) \le Q(q,v) \le \overline{E}_0 := \left[\sqrt{Q(\widehat{q},\widehat{v})} + \sqrt{\mathcal{E}_{\mathrm{init}}}\right]^2. \tag{S378}$$

This specifies a nonzero finite-dimensional admissible initialization region with nonempty interior, without deciding membership in an exact invariant torus. The energy-error bound is a full-state condition; tip sensing alone does not bound unobserved bending or kinetic energy.

Sensing and execution error envelopes. Let $u = (u_F, u_B)$ be the commanded ports and suppose that, throughout the specified interval,

$$\begin{aligned} F &= (1+g_F)u_F + e_F, \\ B &= (1+g_B)u_B + e_B, \\ |g_i| &\le \eta_i, \qquad |e_i| \le \epsilon_i. \end{aligned} \tag{S379}$$

Assume available estimates obey $|v_w - \widehat{v}_w| \le \delta_v$, $|\vartheta - \widehat{\vartheta}| \le \delta_\vartheta$ and $|v_\vartheta - \widehat{v}_\vartheta| \le \delta_\omega$. Define the estimated power coefficients and their error bounds by

$$\begin{aligned} \widehat{a}_F &= \widehat{v}_w, & \Delta_F &= \delta_v, \\ \widehat{a}_B &= \mu\cos\widehat{\vartheta}\,\widehat{v}_\vartheta, & \Delta_B &= \mu(\delta_\omega + |\widehat{v}_\vartheta|\delta_\vartheta). \end{aligned} \tag{S380}$$

The cosine is one-Lipschitz on the real line, hence $|a_i - \widehat{a}_i| \le \Delta_i$. Both sensing error and relative execution error are covered by the convex, positively homogeneous function

$$P^+(u) = \sum_{i\in\{F,B\}} \left\{\widehat{a}_i u_i + [\Delta_i + \eta_i(|\widehat{a}_i| + \Delta_i)]|u_i|\right\}. \tag{S381}$$

In particular, $P^+(0) = 0$. There is no division by a measured velocity, so a zero or incorrectly signed noisy velocity creates no singularity.

The independent additive errors need a separate energy allowance. For $\alpha > 0$, write $m_F = e_w^{\mathsf{T}} M^{-1} e_w$ and $m_\vartheta = e_\vartheta^{\mathsf{T}} M^{-1} e_\vartheta$, and put

$$p_e = \frac{(\epsilon_F\sqrt{m_F} + \mu\epsilon_B\sqrt{m_\vartheta})^2}{4\alpha}, \qquad R_T = T p_e. \tag{S382}$$

For time-dependent bounds one may instead integrate their corresponding $p_e(t)$. With $r = e_w e_F + \mu e_\vartheta e_B \cos\vartheta$, completion of the square gives

$$r^{\mathsf{T}} v - \alpha v^{\mathsf{T}} M v \le \frac{r^{\mathsf{T}} M^{-1} r}{4\alpha} \le p_e. \tag{S383}$$

All quantities in these equations use SI units: P^+, p_e are powers, and Q, E, R_T are energies. No dissipated energy needs to be measured.

Proposition S68. Robust finite-horizon work and clearance. *Fix $T > 0$ and $\overline{E}_0 < C := E_{\max} - R_T < E_c - R_T$. Initialize $W(0) = \overline{E}_0$ and integrate $\dot{W} = P^+(u)$. Clip a requested force and field to symmetric command limits, then apply $u = \sigma u_{\mathrm{req}}$, with $\sigma = 1$ if $P^+(u_{\mathrm{req}}) \le \kappa(C-W)$, and otherwise*

$$\sigma = \frac{\kappa(C-W)}{P^+(u_{\mathrm{req}})}, \qquad \kappa > 0. \tag{S384}$$

For solutions satisfying the mechanical power identity and the stated error envelopes, throughout $0 \le t \le T$,

$$\begin{aligned} E(t) &\le W(t) + t p_e \le C + R_T = E_{\max}, \\ \sup_s |w(s,t)| &\le \sqrt{2 C_w E_{\max}} < c. \end{aligned} \tag{S385}$$

Zero command remains feasible, even with nonzero sensing errors. To enforce actual actuator limits $F_{\lim}, B_{\lim}$, use command limits $(F_{\lim} - \epsilon_F)/(1+\eta_F)$ and $(B_{\lim} - \epsilon_B)/(1+\eta_B)$, respectively.

Proof. The two non-additive port powers are at most $P^+(u)$, by (S380) and (S379). Equation (S383) bounds the remaining power and damping together. Thus $\dot{E} \le \dot{W} + p_e$, and integration with (S378) proves $E \le W + t p_e$. Positive homogeneity makes the radial choice satisfy $\dot{W} \le \kappa(C-W)$; consequently $C - W(t) \ge (C - W(0))e^{-\kappa t} > 0$. The spatial compliance bound completes the proof. The amplitude limits follow directly from $|F| \le (1+\eta_F)|u_F| + \epsilon_F$ and its field counterpart. If additive errors vanish, $R_T = 0$ and the same argument holds for all time and for any $\alpha \ge 0$. A constant nonzero additive error allowance has the stated finite horizon; it is not an all-time KAM or clearance assertion.

Sampled sensing and delayed execution. The sensing bounds must cover the age of the data, not only sensor quantization. For a sample at t_k used for an interval of length h_k, let the maximum age including sensing latency be ℓ_k. If

valid bounds $|\dot{v}_w| \le A_w$, $|v_\vartheta| \le V_\vartheta$ and $|\dot{v}_\vartheta| \le A_\vartheta$ hold on that interval, replace

$$\delta_v \mapsto \delta_v + A_w \ell_k, \quad \delta_\vartheta \mapsto \delta_\vartheta + V_\vartheta \ell_k, \quad \delta_\omega \mapsto \delta_\omega + A_\vartheta \ell_k. \tag{S386}$$

Compute P_k^+ with this interval envelope. A fully discrete sufficient rule is to hold u_k such that $h_k P_k^+(u_k) \le C - W_k$ and set $W_{k+1} = W_k + h_k P_k^+(u_k)$. Linear interpolation of W proves the same within-interval energy bound; $u_k = 0$ remains feasible. This rule does not treat a value at one sample as an interval guarantee.

Similarly, if a continuously specified command has a verified slew bound $|\dot{u}_i| \le S_i$ and the applied copy is delayed by at most d_i, including a zero-order-hold age when applicable, it suffices to use

$$\epsilon_i \ge \epsilon_{i,0} + (1 + \eta_i) S_i d_i \tag{S387}$$

in the additive reserve. This is an explicit joint delay/slew budget. The radial governor alone does not impose a slew limit. A controller that also enforces slew must verify its feasibility, or use the discrete interval-power rule above. The numerical experiment below uses continuous bounded port estimates and the declared direct execution-error envelope; it does not identify a hardware latency or an unmeasured acceleration bound.

A nonzero SI budget next to the finite-amplitude reference. The reference $(\widehat{q}, \widehat{v})$ is the frozen nodal reconstruction of phase $(0, 0)$ of the computed Fourier candidate for the integral-COMSOL fit, with the learned velocity h_Y, rather than an assumed exact torus point. Its quadratic energy is below 21.45855219 nJ. For $\mathcal{E}_{\mathrm{init}} = 5$ pJ, ball evaluation of (S378) gives $\overline{E}_0 < 22.11866366$ nJ. This full-state ellipsoid implies, among other coordinate bounds, an initial tip-displacement error below 56.006 μm and tip-angle error below 1.19851 mrad. It also implies mass-weighted RMS velocity error below 0.316981 mm/s, where the squared RMS is $\delta v^{\mathsf{T}} M \delta v / (\rho A_c L)$. These are projections of one energy ellipsoid; their endpoints cannot be independently combined as a Cartesian error box.

The declared actual limits are 20 mT and 20 μN; the latter corresponds to a gradient norm 1 T/m in the ideal local magnetic realization. The command bounds reserve the relative and additive errors before clipping. Matrix ball arithmetic, rather than a two-mode estimate of unobserved velocity, gives $m_F < 4.452635925\,10^6$ and $m_\vartheta < 3.940922387\,10^{13}$ in their respective SI inverse-mass units, which establish the work reserve in the table.

Table S23. One sufficient operational budget for the full nodal FE plant. Bounds are deterministic hypotheses; they are not measured sensor specifications.

Quantity	Value or outward bound
Horizon T	4 s
Energy cap $E_{\max}$	22.66 nJ
Initialization error $\mathcal{E}_{\mathrm{init}}$	5 pJ
Angle / angular-rate error	1 mrad / 0.01 rad/s
Tip-velocity error	0.1 mm/s
Relative force and field execution error	1% each
Additive force / field error	0.5 nN / 0.05 μT
Additive work reserve R_T	< 0.09677917 nJ
Available commanded-work cap C	> 22.56322083 nJ
Initial commanded-work margin	> 0.44455717 nJ
Whole-centreline displacement bound	< 3.898405 mm
Minimum net wall clearance	> 0.101595 mm

Full-model numerical check. The reproducibility directory `reviewer_revision_20261008/robustness` contains the complete input matrices, learned coefficients, finite initial reference, original-file hashes, code and trajectories. The physical integrator retains $\Phi^{\mathsf{T}} M \Phi$ and $\Phi^{\mathsf{T}} K_b \Phi$ in full, so it integrates the frozen nodal plant in a well-scaled modal basis instead of discarding diagonalization residuals. The requesting controller uses the same fitted h_Y, h_{YX}, h_{YY} and velocity-to-momentum inverse as (S342), with an additional bounded low-state perturbation. Independent deterministic sinusoids supply the stated port-sensor, gain and additive execution errors. The full-state initialization error is verified by direct Arb matrix products, including the modal-to-nodal transformation.

The initialized energy is 21.743724 nJ, with a 46.354 μm tip position error and a 0.347654 mm/s tip velocity error; its combined initial error energy is below 4.951 pJ. Over 4 s the saved-time Bernstein centreline bound is below 1.944145 mm and the physical energy is below 21.744899 nJ. The work filter is active at 97.8011% of the 2001 saved times, illustrating its conservative use of work without crediting dissipated energy. Its activation changes the desired Hamiltonian trajectory. Clearance follows from Proposition S68, while invariant-torus persistence remains the separate conservative and ideal-feedback result.

The Radau run uses relative tolerance $2\,10^{-8}$ and componentwise absolute tolerances $2\,10^{-13}$ for modal positions, $2\,10^{-11}$ for modal velocities and $2\,10^{-13}$ for normalized energy states. Simultaneously reducing them tenfold changes the saved tip trajectory by less than 0.010 μm and the energy by less than $3.2\,10^{-15}$ J; the refined energy-ledger residual is below $3\,10^{-16}$ J. These are numerical implementation checks, not a validated enclosure of the integrated trajectory. The continuous-time clearance statement uses the analytic work proof and its independently validated initialization and parameter bounds.

S7.6 A physically actuated S-shaped endpoint path

The moving clamped base permitted by the distributed formulation supplies an axial feed coordinate $b(t)$, while the same single transverse magnetic field bends the catheter. In the Cartesian small-slope model, the centreline is

$$\begin{aligned} r(s,t) &= \big(b(t)+s,\, w(s,t)\big), \\ r_{\mathrm{tip}}(t) &= \big(b(t)+L,\, w(L,t)\big). \end{aligned} \tag{S388}$$

The axial feed and magnetic bending together prescribe the endpoint history. The material length remains 80 mm throughout; advancing a varying length through a guide is a different boundary-value problem.

The base starts at rest. Over the first second its speed is $\dot b = v_b(10\xi^3 - 15\xi^4 + 6\xi^5)$, where $\xi = t/(1\,\mathrm{s})$ and $v_b = 4\,\mathrm{mm\,s^{-1}}$; afterwards $\dot b = v_b$. The transverse request is zero until 2 s. During the subsequent 12 s stroke, put $\sigma = (t - 2\,\mathrm{s})/(12\,\mathrm{s})$ and prescribe

$$y_r(t) = \frac{8A}{3\sqrt{3}}\left(\frac{1}{2}\sin 2\pi\sigma - \frac{1}{4}\sin 4\pi\sigma\right), \quad A = 1.5\,\mathrm{mm}. \tag{S389}$$

Outside this interval $y_r = 0$. Its position, velocity and acceleration join continuously to zero, and its two lateral extrema are $\pm A$. The resulting S-shaped endpoint stroke advances 48 mm axially. The base acceleration is at most $7.5\,\mathrm{mm\,s^{-2}}$ and ends before any bending command is applied. Consequently the beam is exactly straight during axial acceleration and the base has zero acceleration throughout the bending stroke. There is no axial fluid load, guide friction, or imposed axial prestress in this test.

Base work and clearance. Let $m_b = \rho A_c L$ denote the rod mass, with A_c its cross-sectional area. Orthogonality of the axial and transverse Cartesian velocities gives

$$E_{\mathrm{total}} = E_\perp + \frac{1}{2}m_b\dot b^2, \qquad P_b = F_b\dot b = m_b\ddot b\dot b. \tag{S390}$$

Here $E_\perp$ is the same transverse kinetic, bending and static-field energy used in the clearance theorem. The chosen timing also avoids transverse geometric-stiffness effects from an accelerating bent beam: the axial inertial load acts only while $w = 0$. During bending, axial translation is force-free in the stated model. Thus

$$\dot E_{\mathrm{total}} = \mu B_y \cos\vartheta\,\dot\vartheta - \alpha\dot q^{\mathsf T} M\dot q + P_b. \tag{S391}$$

The base work is $W_b = m_b\dot b^2/2$ and is accounted for separately from the magnetic input budget. The straight, open-ended corridor is invariant under axial translation. Its lateral noncollision proof therefore applies to $E_\perp$ and the actual magnetic port, while the separate base-energy balance is retained. For the same 4 mm net clearance and $0.95E_c$ transverse work cap, the continuous-time centreline bound remains 3.898718 mm. This statement would require additional work and geometry terms for curved walls, axial contact, variable insertion length or axial drag.

Learned control and comparisons. The conservative readout is the integral fit to full-FE COMSOL position–velocity data in Section S5.5. It has cosine multiplier 1.0223248822 and includes all fitted quartic terms; its invariant torus is certified independently. The conservative coefficients and the nonnegative drag readout are held fixed throughout this experiment. The controller evaluates its drift and magnetic input gain from the complete learned gradient and Hessian, as in the preceding tracking construction. For this experiment the feedback error uses the actual full-beam tip position and velocity; the first two modal projections supply the learned drift and gain. The request is clipped to $|B_y| \le 20$ mT and passed through the same physical input-work governor. Every plant run integrates all 64 FE degrees of freedom. The imposed base motion has the same force and speed budget in every comparison.

The known-physics static inverse has the smallest tracking error in Table S24 for this slow, weakly nonlinear nominal trajectory. The learned controller also follows the S-shaped endpoint path, with accuracy depending on the measured output: full-tip feedback reduces the CHNN error relative to its two-mode reconstruction while using the same fitted energy and physical input.

The learned controller uses a peak field of 12.9177 mT and a sampled peak field slew of $10.4268\,\mathrm{mT\,s^{-1}}$. Its largest centreline displacement is 1.48892 mm, leaving 2.51108 mm of net lateral clearance at the saved times; the largest slope is 0.03718 rad. The recorded work-based energy estimate peaks at 4.84997 nJ, below the 22.66365 nJ cap, and neither field saturation nor governor intervention occurs. The base's maximum kinetic energy is 0.796205 nJ, its peak ideal inertial force is $0.746442\,\mu$N, and its peak power is 1.90256 nW. Numerical total-energy balance includes this base work and has maximum residual 2.44×10^{-18} J. Tightening integration tolerances by a factor of ten and halving the maximum step changes the tip by at most 3.14×10^{-12} m and the commanded field by at most 1.47×10^{-10} T at saved times.

Independent COMSOL field replay. An additional COMSOL solve integrates the same full beam for 16 s using the saved learned-controller field, with piecewise-cubic

Table S24. S-path tracking on the full FE plant. RMS errors use the 12 s stroke; maxima use the full 16 s simulation. All baselines share the same trajectory, physical input and field bound.

Controller	RMS (μm)	Max. (μm)
CHNN, measured tip feedback	6.419	12.907
Quadratic model, measured tip feedback	6.207	12.476
Known-physics static inverse feedforward	4.124	8.005
CHNN, modal-output feedback	25.308	41.544

field interpolation, generalized-alpha integration, relative tolerance 10^{-8} and maximum step 0.002 s. The prescribed field is the controller output from the preceding simulation. Its tip differs from the Python result by 1.312×10^{-10} m RMS and at most 4.038×10^{-10} m, and its S-stroke tracking RMSE is $6.419\,\mu$m. COMSOL independently integrates the mechanical response to this prescribed command; the controller is not recomputed online in this replay. Axial translation is reconstructed using the same analytic base protocol. Power balance reconstructed by trapezoidal integration of the 0.004 s output samples has a 1.244×10^{-14} J residual; this diagnostic is distinguished from the much tighter internal Python work-state integration and is not used as a rigorous solver-error bound.

S8 Structure-preserving time integration

For a real smooth Hamiltonian, implicit midpoint is

$$x_{k+1} = x_k + hJ\nabla H\big((x_k + x_{k+1})/2\big). \tag{S392}$$

Let $A = JH''(\bar{x})$. Differentiation gives $D\Phi_h = (I - hA/2)^{-1}(I + hA/2)$. Since $A^{\mathsf{T}}J + JA = 0$, this Cayley transform satisfies $D\Phi_h^{\mathsf{T}} J D\Phi_h = J$ whenever the implicit equation defines a differentiable local map. The proof assumes the nonlinear step is solved exactly; residual tolerances add a numerical defect. Explicit RK4 is not symplectic in general. For the separable physical Hamiltonian, the kick–drift–kick Störmer–Verlet map is also symplectic, because each substep is the exact flow of the kinetic or potential Hamiltonian. A general learned momentum-dependent quartic no longer admits that separable split, so the same formula cannot be reused without justification.

Energy error and symplecticity are distinct diagnostics. Finite-time energy plots are evaluated against the specified reference Hamiltonian and accompanied by a step-size check. A small integration error does not certify a Diophantine inequality, nor does a positive finite-time Lyapunov estimate alone establish asymptotic chaos. These facts determine the labels used in the numerical figures.

S9 The fourth-order central generating-function scheme

Let $f = J\nabla H$, $A = H''$ and define a central generating function

$$S_h(x) = hH(x) + \frac{h^3}{24} f(x)^{\mathsf{T}} A(x) f(x). \tag{S393}$$

The map is specified implicitly by

$$x_+ - x_- = J\nabla S_h\big((x_- + x_+)/2\big). \tag{S394}$$

The central generating-function construction and fourth-order formula are classical (Feng et al. 1989, Theorem 24, p. 90; formula on p. 95); geometric integration is developed systematically by Hairer et al. (2006). Here every derivative is computed from the same learned scalar Hamiltonian. The identity

$$J\nabla(f^{\mathsf{T}} A f) = f''[f,f] - 2(f')^2 f \tag{S395}$$

follows by differentiating both occurrences of f, using $f' = JA$ and $J^{\mathsf{T}} = -J$. It gives the displacement implemented in the reproducibility code:

$$g_h = hf + \frac{h^3}{24}\big(f''[f,f] - 2(f')^2 f\big). \tag{S396}$$

The Hessian S_h'' is symmetric, so the midpoint Cayley-transform argument in Supplement S8 proves exact symplecticity of the locally defined map. Since $S_{-h} = -S_h$, interchanging its endpoints proves reversibility.

To verify order, expand an exact orbit about its temporal midpoint x_0. Its endpoint midpoint m and endpoint difference satisfy

$$m = x_0 + \tfrac{1}{8}h^2 f'(x_0) f(x_0) + O(h^4), \tag{S397}$$

$$x_+ - x_- = hf(x_0) + \tfrac{1}{24}h^3\big(f''[f,f] + (f')^2 f\big)(x_0) + O(h^5). \tag{S398}$$

Substituting $x_0 = m - \tfrac{1}{8}h^2 f'(m) f(m) + O(h^4)$ gives $x_+ - x_- = g_h(m) + O(h^5)$, proving fourth order under the required smoothness and local solvability assumptions.

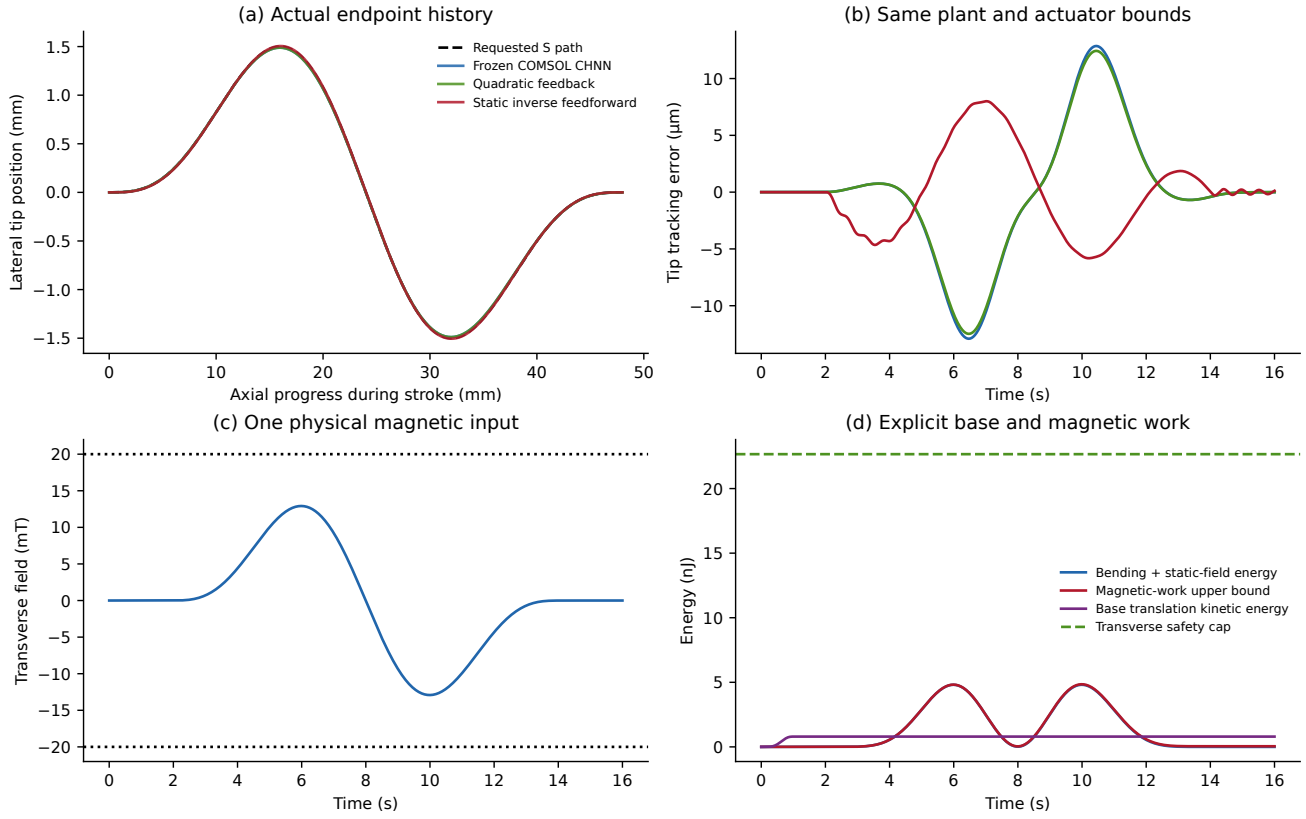


Figure S15. A realized S-shaped endpoint path generated by axial base feed and one transverse magnetic input. The controller uses the fitted COMSOL-trained CHNN on the full FE plant. Axial translation work is explicitly included in the energy balance. The matched controls reveal that known-physics static inverse feedforward is more accurate on this particular slow nominal path.

S9.1 Integration accuracy for the learned magnetic energy

We implement (S394) for the Python-trained GC17 Hamiltonian, retaining its complete cosine and all mixed and momentum-dependent quartics. Polynomial and trigonometric derivatives through order four are evaluated analytically from the fitted energy; they supply both the central displacement and its Newton Jacobian. This test closes all forcing and damping ports. Starting at $(X_1, X_2, Y_1, Y_2) = (0.1, 0.06, 0, 0.02)$, we integrate to $\tau = 10$ and compare against DOP853 with relative and absolute tolerances 3×10^{-14} and 3×10^{-16}. Tightening the reference tolerances from 2×10^{-13} and 2×10^{-15} changes the sampled reference by at most 8.32×10^{-14} per state component. The endpoint errors in Table S25 approach fourth order for Central4 and second order for implicit midpoint.

Table S25. Conservative integration of the GC17 energy. Errors are Euclidean endpoint errors in the balanced dimensionless state; orders compare successive step halvings.

h	Midpoint error	Order	Central4 error	Order
0.08	7.1351×10^{-2}	—	1.8346×10^{-3}	—
0.04	1.9383×10^{-2}	1.8802	1.1784×10^{-4}	3.9606
0.02	4.8966×10^{-3}	1.9849	7.4149×10^{-6}	3.9902
0.01	1.2265×10^{-3}	1.9972	4.6422×10^{-7}	3.9976

The largest implicit-step residual in the infinity norm is 2.00×10^{-15}, with at most three Newton evaluations per step. For each tested step size, differentiating the converged implicit equation gives its one-step tangent map $D\Phi_h$. The largest measured defect $\|D\Phi_h^{\mathsf{T}} J D\Phi_h - J\|_\infty$ is 3.34×10^{-16} for Central4 and 2.24×10^{-16} for midpoint. An independent fourth-order central-difference evaluation of the solved step map agrees with the implicit tangent to 1.39×10^{-12}; its largest symplectic defect is 2.75×10^{-12}. Thus both methods preserve the symplectic structure to the measured implicit-solve and arithmetic accuracy, while their convergence orders differ. The test assesses numerical integration of the complete momentum-dependent Hamiltonian; the KAM statement follows separately from its analytic certificate.

S10 Complex-gradient descent with a strict descent margin

Theorem 3 in the published version of Zhang et al. (2022) motivates complex-gradient optimization, but a stationary-point conclusion requires a strict descent margin. The following convention-explicit statement also applies to a smooth CHNN regression objective.

Proposition S69. Complex descent. *Let $\mathcal{L} : \mathbb{C}^P \simeq \mathbb{R}^{2P} \to \mathbb{R}$ be bounded below and have an L-Lipschitz real gradient on every segment traversed by the iterates. For $w = a + ib$, define $G(w) = \nabla_a \mathcal{L} + i\nabla_b \mathcal{L} = 2\partial\mathcal{L}/\partial\bar{w}$. Choose a constant $\rho \in \mathbb{C}$ with $\kappa = \operatorname{Re}\rho - (L/2)|\rho|^2 > 0$, and update $w_{t+1} = w_t - \rho G(w_t)$. Then*

$$\mathcal{L}(w_{t+1}) \le \mathcal{L}(w_t) - \kappa\|G(w_t)\|_2^2, \qquad \lim_{t\to\infty} \|G(w_t)\|_2 = 0,$$

and

$$\min_{0\le t<T} \|G(w_t)\|_2^2 \le \frac{\mathcal{L}(w_0) - \inf \mathcal{L}}{\kappa T}.$$

Proof. Under realification, multiplication by $\rho = u + iv$ is $uI + vJ_c$, with $J_c^\top = -J_c$ and $J_c^\top J_c = I$. Hence the update has inner product $-u\|G\|^2$ with the real gradient and squared norm $|\rho|^2\|G\|^2$. The real descent lemma gives the first inequality. Summing it and using the lower bound for $\mathcal{L}$ gives $\kappa\sum_{t\ge 0}\|G(w_t)\|^2 \le \mathcal{L}(w_0) - \inf\mathcal{L}$. Square summability proves convergence of the gradient norm and the minimum bound. The result asserts convergence to stationarity in gradient norm, not convergence to a global minimum or to a particular parameter vector.

For a real step this condition is $0 < \rho < 2/L$ under the stated convention. The case $\rho = 0$, allowed by a non-strict printed step-size interval, cannot imply gradient convergence. Nonsmooth complex activations also need separate treatment; a smooth polynomial CHNN satisfies the differentiability requirements, with a Lipschitz gradient bound on any bounded iterate region. This proposition concerns smooth gradient descent. The finite-feature readouts use linear least squares; the deep extension uses the optimizers reported with its experiment. Neither Adam nor L-BFGS convergence follows from this fixed-step proposition. On a bounded parameter region with a uniform pole margin, the complex $\tanh$ derivative-regression objective is smooth and admits finite local Lipschitz bounds; a global iterate-region and step-size verification would still be required for applying this proposition to a training run. This optimization proposition is independent of the identification-error and a-posteriori KAM results: those results do not assume that the fitted parameters are stationary or globally optimal.

S11 An explicit local KAM measure estimate

S11.1 A fixed-domain bound and its scope

This appendix quantifies a sufficient local measure bound for the Hamiltonian identified from COMSOL integral observations. The construction uses the exact finite-amplitude torus and bounds the complete transformed Hamiltonian, including the action-domain remainder. The resulting width illustrates the conservatism of the explicit constants in Chierchia and Koudjinan (2021).

Theorem S70. Explicit finite-amplitude tubular measure. *Let h be the fitted Hamiltonian certified in Section S4.5, and let K_∞ be its exact invariant torus. There is an explicitly constructed analytic symplectic chart $\Xi = \Psi \circ \Phi_\chi$, specified in Sections S11.2–S11.3, that fixes this torus at zero action. For the fixed domain*

$$U_w = \Xi\big(\mathbb{T}^2_{2\pi} \times (-w, w)^2\big), \qquad w = 10^{-220}, \tag{S399}$$

the KAM fraction satisfies

$$f_{\mathrm{KAM}}(U_w) \ge 0.9669383. \tag{S400}$$

The centre torus is the previously certified motion with the 1.8977904 mm tip witness and the 1.8977962 mm all-phase centreline bound. The width w is a dimensionless action half-width, not a displacement amplitude.

Proof. The explicit chart and a convergent quadratic normalization give

$$h \circ \Xi = e + \omega \cdot I + \tfrac{1}{2} I^\top \overline{A} I + R_3, \qquad \omega = 2\pi\varpi.$$

The outward bounds are $\|\overline{A}\|_\infty \le 3$, $\|\overline{A}^{-1}\|_\infty \le 5100$ and $\|R_3\|_{2w,1/128} \le 4 \times 10^{51} w^3$. Use the global KAM theorem with real action radius w, complex radius $2w$, final angle strip $1/256$, frequency exponent $\tau = 2$, boundary allowance $\rho = w/500$ and $\alpha = w/(1000 \times 5100)$. The dimensionless smallness quantity is at most 3.1212×10^{-155}, below the theorem threshold 7.6577×10^{-155}. The geometric estimate in Section S11.4 gives

$$\begin{aligned} 1 - f_{\mathrm{KAM}}(U_w) &\le \frac{25}{16}\left(\frac{4\rho}{w} + \frac{8\alpha L\zeta(2)}{w}\right) \\ &= \frac{25}{16}\,(0.008 + 0.008\zeta(2)) < 0.0330617, \end{aligned} \tag{S401}$$

where $L = 5100$. The two terms account for the boundary layer and all integer resonance strips, respectively. Symplecticity transfers this relative measure to U_w without a density correction.

The width in Table S26 quantifies the conservatism of this general theorem. The domain is centred on the exact millimetre-amplitude torus, but its thickness is far below any operational resolution. In particular, the computed Fourier approximation has a certified canonical-embedding error

Table S26. Outward bounds for the explicit finite-amplitude tube. The smallness quotient is the tested perturbation divided by the sufficient threshold; failure of that test does not imply chaos.

Quantity	Bound or value
Initial chart action radius	10^{-3}
Quadratic-normalization radius	10^{-17}
Certified tube half-width w	10^{-220}
$\|\overline{A}\|_\infty$, $\|\overline{A}^{-1}\|_\infty$	$3,5100$
Smallness quotient at $w = 10^{-220}$	< 0.407586
Same quotient at $w = 10^{-219}$	> 4.0758
Certified fraction in U_w	$> 96.6938\%$

upper bound below 2.1×10^{-41}. This available guarantee cannot resolve membership in a tube whose transverse distance is bounded by $41w = 4.1 \times 10^{-219}$; it is not an estimate of the actual embedding error. The domain is mathematically specified through K_∞; this certificate does not permit an experimental initial state to be assigned to that tube from the finite-precision approximation. It supplies a concrete fixed-domain measure result while leaving the much wider operating-domain question to a separate calculation. Section S5.6 reports that wider-domain numerical diagnostic, with its own error checks and interpretation.

S11.2 An explicit symplectic tube at the certified physical amplitude

Let K_∞ be the exact torus supplied by the integral-COMSOL certificate, with its phase normalization fixed by that construction. Write $t = 2\pi\theta$, $k(t) = K_\infty(t/(2\pi))$, and $\omega = 2\pi\varpi$. Thus $D_\omega k = X_{h_\eta}(k)$, and the already verified Diophantine inequality becomes

$$|\omega \cdot \ell| \geq \gamma_0 |\ell|_1^{-3}, \qquad \gamma_0 = 2\pi \frac{9}{1750}, \quad 0 \neq \ell \in \mathbb{Z}^2. \tag{S402}$$

The exponent 3 here concerns this fixed central torus. The family measure estimate below uses the separate exponent $\tau = 2$.

For each coordinate plane, define

$$\begin{aligned} j_a(t) &= \tfrac{1}{2}(k_{X_a}^2 + k_{Y_a}^2), \\ F_a(t) &= t_a + \frac{\log(1+u_{a,+}) - \log(1+u_{a,-})}{2i}, \\ u_{a,\pm}(t) &= 10(k_{X_a} \mp i k_{Y_a}) e^{\mp i t_a} - 1. \end{aligned} \tag{S403}$$

The logarithms use the branches at 1. On the real torus, F_a is the phase of $k_{X_a} - ik_{Y_a}$ and $j_a > 0$. Set $B(t) = DF(t)^{-\mathsf{T}}$ and define

$$\begin{aligned} J(t,I) &= j(t) + B(t)I, \\ \Psi_{X_a}(t,I) &= \sqrt{2J_a(t,I)}\cos F_a(t), \\ \Psi_{Y_a}(t,I) &= -\sqrt{2J_a(t,I)}\sin F_a(t). \end{aligned} \tag{S404}$$

Equivalently, each pair (k_{X_a}, k_{Y_a}) is multiplied by $\sqrt{1 + (BI)_a/j_a}$; this expression is convenient for interval evaluation and fixes the square-root branches.

This is an exact symplectic chart. Indeed, polar coordinates have canonical one-form $J \cdot dF$, and

$$\Psi^*(J \cdot dF) = j(t) \cdot DF(t)dt + I \cdot dt.$$

The first term is closed because k is Lagrangian. Taking the exterior derivative gives $\sum_a dI_a \wedge dt_a$. This argument requires closedness, not exactness, of the first term. Since DF is uniformly close to the identity, its real lift is a global diffeomorphism of $\mathbb{T}^2$; positive J then makes Ψ injective on its real tube. Moreover $\Psi(t,0) = k(t)$.

The invariant-torus estimates and outward-rounded interval arithmetic give the following upper bounds:

$$\begin{gathered} \|k - k_0\|_{1/16} < 7\,10^{-6}, \\ k_0 = 0.1(\cos t_1, \cos t_2, -\sin t_1, -\sin t_2), \\ \|u_\pm\|_{1/16} < 1/5000, \qquad \|j - 0.005\mathbf{1}\|_{1/16} < 3\,10^{-6}, \\ \|F - t\|_{1/16} < 0.000201, \\ \max\{\|DF - \mathrm{Id}\|_\infty, \|DF - \mathrm{Id}\|_1\}_{1/32} < 0.013, \\ \|B\|_{1/32,\infty} < 1.014. \end{gathered} \tag{S405}$$

Here a subscript is the complex angular strip width in radians; vector norms are maximum norms and matrix norms are induced norms. The first bound includes the a posteriori correction to the numerical embedding. For example, Cauchy's estimate bounds each entry of $D(F - t)$ by $0.000201/(1/32)$, so both row and column sums are below 0.013. For $R_0 = 10^{-3}$, $|I|_\infty < R_0$ therefore implies

$$\begin{gathered} |(BI)_a/j_a| < \frac{1.014\,10^{-3}}{0.005 - 3\,10^{-6}} < \tfrac{1}{4}, \\ \|\Psi\|_{1/32,R_0} < 0.12. \end{gathered}$$

All logarithms, square roots and inverses above are thus holomorphic on the declared domain.

Write $h_\eta = h_2 + h_{\mathrm{nl}}$. The full cosine and the exact exported quartic coefficients give

$$\|h_{\mathrm{nl}} \circ \Psi\|_{1/32,R_0} < 10^{-6}, \qquad \|h_\eta \circ \Psi\|_{1/32,R_0} < 0.05. \tag{S406}$$

The second estimate uses the identity $h_2 \circ \Psi = \nu \cdot (j + BI)$, whose norm is less than $7.153(0.005003 + 0.001014) < 0.044$. Invariance and symplecticity give the exact expansion

$$h_\eta \circ \Psi = e + \omega \cdot I + \tfrac{1}{2} I^{\mathsf{T}} A(t) I + O(|I|^3). \tag{S407}$$

Only $h_{\mathrm{nl}} \circ \Psi$ contributes to A: Cauchy bounds on its diagonal and off-diagonal second derivatives give $\|A\|_{1/32,\infty} \le 3$.

The averaged matrix $\overline{A}$ is nonsingular. To check the normalization, the tangent frame with radian angles is the certified cycle-angle frame divided by 2π; a symplectically dual normal frame is multiplied by 2π. Replacing that normal frame by $\partial_I \Psi(t, 0)$ adds a symmetric tangential shear. Its contribution to the torsion is a directional derivative with zero angular average. Consequently

$$\overline{A} = (2\pi)^2 \langle T_\infty \rangle, \qquad \|\overline{A}^{-1}\|_\infty \le \frac{2\,10^5}{(2\pi)^2} < 5100. \tag{S408}$$

The inverse-torsion bound applies to the exact final torus through the uniform a posteriori iteration.

S11.3 A complete analytic remainder on a fixed tube

Let S be the symmetric, zero-average solution of $D_\omega S = A - \overline{A}$. Its Fourier coefficients for $\ell \neq 0$ are $S_\ell = A_\ell/(i\omega \cdot \ell)$. Equation (S402) and the angular strip loss $\delta = 1/64$ imply

$$\begin{aligned} \|S\|_{1/64,\infty} &\le \frac{3}{\gamma_0} 4 \sum_{m \ge 1} m^4 e^{-m/64} \\ &\le \frac{288 e^{1/64} 64^5}{\gamma_0} < 10^{13} =: T. \end{aligned} \tag{S409}$$

There are $4m$ integer vectors of one-norm m. The second inequality uses $\sum_{m \ge 1} m^4 e^{-\delta m} \le e^\delta \Gamma(5) \delta^{-5}$, obtained by integrating on each interval $[m, m+1]$.

Use the exact time-one flow Φ_χ of $\chi(t, I) = \frac{1}{2} I^{\mathsf{T}} S(t) I$, not a truncated Lie series. Fix $R_1 = 10^{-17}$. On the intermediate strip $3/256$, Cauchy's estimate from the strip $1/64$ gives $\|\partial_{t_a} S\|_\infty \le 256 T$. As long as $|I|_\infty < 2R_1$, the flow satisfies

$$\begin{aligned} |\dot{t}|_\infty &\le 2TR_1 = 2\,10^{-4} < 1/256, \\ |\dot{I}|_\infty &\le 1024 T R_1^2 = 0.1024 R_1 < R_1. \end{aligned}$$

These strict bounds close the continuation argument from the initial strip $1/128$ and action polydisc R_1: throughout the unit time interval, the angle stays inside $3/256$ and the action inside $2R_1 < R_0$. The flow is holomorphic on the initial domain and symplectic on its real part.

With the pullback convention $L_\chi H = \{H, \chi\}$, $\{\omega \cdot I, \chi\} = -D_\omega \chi$. Hence the exact transformed Hamiltonian has the form

$$h_\eta \circ \Psi \circ \Phi_\chi = e + \omega \cdot I + \tfrac{1}{2} I^{\mathsf{T}} \overline{A} I + \mathcal{R}_3, \tag{S410}$$

where $\mathcal{R}_3$ starts at total action degree three. The complete left-hand side remains bounded by 0.05 on the strip $1/128$ and radius R_1. Cauchy's coefficient estimate in the two action variables therefore yields, for $0 < r \le R_1/2$,

$$\begin{aligned} \|\mathcal{R}_3\|_{1/128,r} &\le 0.05 \sum_{m \ge 3} (m+1)(r/R_1)^m \\ &\le \frac{0.5}{R_1^3} r^3 = 5\,10^{50} r^3. \end{aligned} \tag{S411}$$

This bound includes the full magnetic cosine, all fitted quartics, the exact chart and the exact normalization flow.

S11.4 A quantitative measure bound with all resonances included

The action-domain dependence of KAM measure estimates is central in Biasco and Chierchia (2018). We use the fully quantified formulation of Chierchia and Koudjinan (2021, Theorems 1 and 3, Eqs. (4), (9), (13)–(14)). Consider a Hamiltonian $K(I) + \mathcal{R}(I, t)$ on $B_R(I_c) \times \mathbb{T}^2_s$, with $K = e + \omega \cdot I + \frac{1}{2} I^{\mathsf{T}} A I$, real symmetric invertible A, and $\|\mathcal{R}\|_{R,s} \le E$. Let $M \ge \|A\|_\infty$, $L \ge \|A^{-1}\|_\infty$, $\vartheta = ML$, and $\epsilon = ME/\alpha^2$. For $0 < \rho < r \le R$, $0 < s_* < s \le 1$, and $\tau > 1$, the sufficient conditions are

$$\alpha \le c_0 \rho / L, \qquad \epsilon \le \epsilon_* := \frac{(s - s_*)^{7(\tau+1)+10}}{c_* \vartheta^6}. \tag{S412}$$

Here c_0, c_* are exactly the constants in their Appendix A, Remark A.4, Eq. (A.38), evaluated for $d = 2$ and the chosen τ. Upper bounds on M, L, E, c_*, and a lower bound on c_0, are used in the appropriate directions. The normalization is $H = K + \varepsilon P$ with $\varepsilon \|P\| \le E$.

Take the centre set in that theorem to be the singleton $\{I_c\}$. The real domain is precisely $D = B_r(I_c)$, the inner domain is $B_{r-\rho}(I_c)$, and the boundary layer in Theorem 3 is $B_{r+\rho}(I_c) \setminus B_{r-\rho}(I_c)$, of area $16 r \rho$. Let $\mathcal{K} \subset D \times \mathbb{T}^2$ be the certified KAM set. The Lipschitz estimate in their Eq. (9) is at most $1/8$, so the factor multiplying the bad action measure in Eq. (14) is bounded by $(25/16)(2\pi)^2$.

For completeness the remaining estimate includes every integer resonance. For $m = |\ell|_1$, the strip

$$|\ell \cdot (\omega + AI)| < \alpha/m^\tau$$

has intersection with D of area at most $4r\alpha/(m^\tau \|A\ell\|_\infty)$. Since $\|A\ell\|_\infty \geq m/(2L)$ and there are $4m$ vectors on the shell, summing over all shells gives

$$\mathrm{meas}(R_\alpha) \leq 32 r\alpha L \zeta(\tau).$$

Counting both signs of each resonance gives a conservative upper bound. Dividing Theorem 3's bound by $\mathrm{meas}(D \times \mathbb{T}^2) = 4r^2(2\pi)^2$ proves

$$\frac{\mathrm{meas}(\mathcal{K})}{4r^2(2\pi)^2} \geq 1 - \frac{25}{16}\left(\frac{4\rho}{r} + \frac{8\alpha L\zeta(\tau)}{r}\right). \tag{S413}$$

The estimate therefore includes resonances of every integer order.

Apply this result to (S410), with

$$\begin{gathered} I_c = 0, \quad r = w, \quad R = 2w, \\ s = 1/128, \quad s_* = 1/256, \quad \tau = 2, \\ M = 3, \quad L = 5100, \quad \vartheta = 15300, \\ E = 4\,10^{51} w^3, \\ \rho = w/500, \qquad \alpha = w/(1000L). \end{gathered} \tag{S414}$$

Any specified positive $w \leq R_1/4$ satisfying

$$4\,10^{57} M L^2 w \leq \frac{256^{-31}}{c_* \vartheta^6} \tag{S415}$$

passes the smallness condition. The first condition in (S412) follows already from $c_0 > 1$. For every such fixed w, define the physical canonical tube

$$U_w = (\Psi \circ \Phi_\chi)(\mathbb{T}^2 \times B_w(0)). \tag{S416}$$

It is an injective symplectic image of a declared domain, so its Liouville volume is $4w^2(2\pi)^2$. Equations (S413) and (S414) give the explicit bound

$$\begin{aligned} \frac{\mu_L(U_w \cap \mathcal{K}_{h_\eta})}{\mu_L(U_w)} &\geq 1 - \frac{25}{16}\left(\frac{4}{500} + \frac{8}{1000}\frac{\pi^2}{6}\right) \\ &> 0.9669383. \end{aligned} \tag{S417}$$

The numerical certificate takes the exact value $w = 10^{-220}$. It encloses ϵ_* above $7.6577\,10^{-155}$ and the smallness ratio ϵ/ϵ_* below 0.40759, verifying (S415) with outward rounding.

The transverse size of this particular tube is also bounded. For any real point of U_w, write $(t', I') = \Phi_\chi(t, I)$. The same flow estimate on the smaller action domain gives $|I'|_\infty < 2w$. Comparison with the central point $k(t')$, rather than with a fixed phase label before the flow, eliminates the irrelevant phase shift. For real $q > -1$, $|\sqrt{1+q} - 1| = |q|/(1 + \sqrt{1+q}) \leq |q|$; hence

$$\begin{aligned} \|\Psi(t', I') - k(t')\|_\infty &\leq \frac{0.100007\,(1.014)\,2w}{0.005 - 3\,10^{-6}} \\ &< 41w = 4.1\,10^{-219}. \end{aligned} \tag{S418}$$

This is a bound in the dimensionless Cartesian canonical variables; the SI displacement bound follows by applying the fixed modal reconstruction matrix. The millimetre value describes the amplitude of the central motion, not the thickness in (S418).

The fraction concerns the conservative Hamiltonian on a four-dimensional symplectic tube. The constant SI scaling cancels from the ratio. As quantified in Section S11.1, the transverse width is far below operational resolution and the available embedding-error guarantee. The exact set is mathematically defined, but membership cannot be resolved from the computed torus approximation. The lift to the damped closed loop carries only the low-mode parameter measure specified in Section S4.6.

References

Abbott JJ, Diller E and Petruska AJ (2020) Magnetic methods in robotics. *Annual Review of Control, Robotics, and Autonomous Systems* 3: 57–90. doi: 10.1146/annurev-control-081219-082713.

Antman SS (2005) *Nonlinear Problems of Elasticity*. 2nd ed. Springer. doi: 10.1007/0-387-27649-1.

Biasco L and Chierchia L (2018) Explicit estimates on the measure of primary KAM tori. *Annali di Matematica Pura ed Applicata* 197(1): 261–281. doi: 10.1007/s10231-017-0678-8. https://arxiv.org/abs/1612.01903.

Bounemoura A (2016) Non-degenerate Liouville tori are KAM stable. *Advances in Mathematics* 292: 42–51. doi: 10.1016/j.aim.2016.01.012. Preprint version used: https://arxiv.org/abs/1412.0509v1.

Bounemoura A (2020) Positive measure of KAM tori for finitely differentiable Hamiltonians. *Journal de l'École polytechnique – Mathématiques* 7: 1113–1132. doi: 10.5802/jep.137.

Chen Y, Matsubara T and Yaguchi T (2022) KAM theory meets statistical learning theory: Hamiltonian neural networks with non-zero training loss. *Proceedings of the AAAI*

Conference on Artificial Intelligence 36(6): 6322–6332. doi: 10.1609/aaai.v36i6.20582.

Chierchia L and Koudjinan CE (2021) V. I. Arnold's "Global" KAM theorem and geometric measure estimates. *Regular and Chaotic Dynamics* 26(1): 61–88. doi: 10.1134/S1560354721010044. Preprint version used: https://arxiv.org/abs/2010.13243v1.

Cranmer M, Greydanus S, Hoyer S, Battaglia P, Spergel D and Ho S (2020) Lagrangian neural networks. *ICLR Workshop on Integration of Deep Neural Models and Differential Equations*. https://arxiv.org/abs/2003.04630.

Das S, Saiki Y, Sander E and Yorke JA (2017) Quantitative quasiperiodicity. *Nonlinearity* 30(11): 4111–4140. doi: 10.1088/1361-6544/aa84c2.

Desai SA, Mattheakis M, Sondak D, Protopapas P and Roberts SJ (2021) Port-Hamiltonian neural networks for learning explicit time-dependent dynamical systems. *Physical Review E* 104: 034312. doi: 10.1103/PhysRevE.104.034312.

Duignan N and Meiss JD (2023) Distinguishing between regular and chaotic orbits of flows by the weighted Birkhoff average. *Physica D: Nonlinear Phenomena* 449: 133749. doi: 10.1016/j.physd.2023.133749.

Feng K, Wu HM, Qin MZ and Wang DL (1989) Construction of canonical difference schemes for Hamiltonian formalism via generating functions. *Journal of Computational Mathematics* 7(1): 71–96. https://computmath.cjoe.ac.cn/jcm/EN/Y1989/V7/I1/71.

Francescon V, Murasovs N, Lloyd P, Onaizah O, Chathuranga DS and Valdastri P (2025) Closed-loop shape-forming control of a magnetic soft continuum robot. *IEEE Robotics and Automation Letters* 10(6): 6071–6078. doi: 10.1109/LRA.2025.3565124.

Greydanus S, Dzamba M and Yosinski J (2019) Hamiltonian neural networks. In: *Advances in Neural Information Processing Systems*, vol. 32. https://arxiv.org/abs/1906.01563.

Hairer E, Lubich C and Wanner G (2006) *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*. 2nd ed. Springer. doi: 10.1007/3-540-30666-8.

Haro À and Luque A (2019) A-posteriori KAM theory with optimal estimates for partially integrable systems. *Journal of Differential Equations* 266(2–3): 1605–1674. doi: 10.1016/j.jde.2018.08.003. Preprint version used: https://arxiv.org/abs/1711.05515v1.

Hornik K (1991) Approximation capabilities of multilayer feedforward networks. *Neural Networks* 4(2): 251–257. doi: 10.1016/0893-6080(91)90009-T.

Kim Y, Parada GA, Liu S and Zhao X (2019) Ferromagnetic soft continuum robots. *Science Robotics* 4(33): eaax7329. doi: 10.1126/scirobotics.aax7329.

NASA (2025) Risk of spaceflight associated neuro-ocular syndrome (SANS). NASA Human Research Program risk page (updated 16 May 2025; accessed 7 October 2026).

Pöschel J (2001) A lecture on the classical KAM theorem. In: *Proceedings of Symposia in Pure Mathematics*, vol. 69, pp. 707–732. doi: 10.1090/pspum/069/1858551. Preprint version used: https://arxiv.org/abs/0908.2234v1 (Version 1.3, 2009).

Simo JC (1985) A finite strain beam formulation. The three-dimensional dynamic problem. Part I. *Computer Methods in Applied Mechanics and Engineering* 49(1): 55–70. doi: 10.1016/0045-7825(85)90050-7.

Su H and Qin M (2003) Symplectic schemes for Birkhoffian system. Preprint, https://arxiv.org/abs/math-ph/0301001v1.

Tong D, Hao Z, Li J, Sun B, Liu M, Wang L and Huang W (2025) Real-time simulation enabled navigation control of magnetic soft continuum robots in confined lumens. *Journal of the Mechanics and Physics of Solids* 203: 106198. doi: 10.1016/j.jmps.2025.106198.

Trabelsi C, Bilaniuk O, Zhang Y et al. (2018) Deep complex networks. In: *International Conference on Learning Representations*. https://arxiv.org/abs/1705.09792.

Zhang SQ, Gao W and Zhou ZH (2022) Towards understanding theoretical advantages of complex-reaction networks. *Neural Networks* 151: 80–93. doi: 10.1016/j.neunet.2022.03.024. Published version used for theorem numbering.

Zhong YD, Dey B and Chakraborty A (2020) Dissipative SymODEN: Encoding Hamiltonian dynamics with dissipation and control into deep learning. In: *ICLR Workshop on Integration of Deep Neural Models and Differential Equations*. https://arxiv.org/abs/2002.08860.